



Internal Use Only

MOBILE PHONE SERVICE MANUAL

CAUTION

BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS"
IN THIS MANUAL

MODEL : LG-F400L/S/K

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가. 제품의 특징점



1. 제품의 특징점

제1조 제품의 명칭

본 제품은 LTE 방식의 무선 접속 Algorithm을 적용한 한국향 Smart Phone으로 모델명은 F400L/S/K이라 칭한다.

제2조 목적,제품 개요

□ 목적

디지털 휴대폰 기능과 위치 정보 제공을 위한 GPS 기능을 내장하고, 5.5" TFT LCD를 통해 Android JB+을 지원하는 Smart phone으로 DOP type이다.

2.1M 전면 Camera 와 후면 13M Camera 를 내장하고 있다. 통신 방식으로는 LTE를 지원하며, Roaming Band인 GSM, WCDMA를 지원한다.

근거리 무선통신방식으로 Bluetooth 4.0, IEEE 802.11 2.4G b/g/n/ac and 5G 11a/n/ac, NFC를 지원하는 한국향 내수 모델이다.

□ 제품개요

1. Phone Type : DOP type
2. Main Chipset : MSM8974AC,
Memory(24Gbits LPDDR3 SDRAM -POP, 32GB eMMC)
3. LCD : 5.5" QHD, 1440 x 2560 pixels, 16.7M Colors, TFT
4. Touch screen input : Capacitive
5. OS : Android 4.4.2(Kitkat)
6. Camera: Built-in Camera (13MP OIS with flash, 2.1M VT CMOS Image sensor)
7. PMIC 적용: 전원 control, Flash Driver, Charger controller, Battery 보호 기능 등
8. Voice Recording/Memo/Playback, Voice Recognition System
9. Web Browser: Browser
10. 내장 32GB eMMC
11. Bluetooth 4.0 BLE 지원
12. WLAN 2.4GHz 802.11 b/g/n/ac , 5GHz 11 a/n/ac
13. NFC 지원
14. TDMB 지원
15. Slimport : 최대1080p 60Hz 지원
16. Audio zooming
17. Dual Recoding
18. Q-slide 기능 지원
19. 무선 충전 지원[무선 충전 Battery cover 사용 시(별매)]
20. Q-Remote 기능 지원

제3조 적용범위

1. Phone : F400L/S/K (w/ 3000mAh(typ. 착탈형) battery)
2. Adapter (TA)
3. Manual
4. USB cable
5. Stereo headset
6. Battery

제4조 용어의 정의

1. **MSM** : 멀티미디어 처리 및 Phone 전체 동작 제어를 위한 **Application Processor**
2. **Memory** : Protocol, User Interface등의 **Software**를 저장함.
3. **Alert부** : Call Paging, 사용자 오 동작 등의 지원을 위한 기능 수행함.
4. **LCD Display, Keypad부** : Handset의 동작 상태, 사용자 정보 등을 입력, 표시해 주기 위한 기능을 수행함.
5. **Micro USB port**: Battery 충전, Data 통신을 위한 외부접속 Port.
6. **Band Pass Filter (BPF)** : 일정 Band 내의 신호만을 통과시키고 나머지는 제거하는 Filter
7. **CAI(Common Air Interface)** : 호 처리, 핸드오프 등 단말기의 핵심 알고리즘
8. **Dual Tone Multi-Frequency(DTMF) Signaling** : Low Frequency와 High Frequency의 조합으로 동시에 전송되는 signal
9. **Frame Error Rate(FER)** : 총 수신 프레임 대 Error Frame의 비
10. **Low Noise Amplifier (LNA)** : 수신된 신호를 낮은 noise만을 add 시키고 원하는 신호를 증폭시키는 증폭기.
11. **Time & Control Processor(TCP)** : GPS수신기로부터 제공되는 신호를 이용하여 기지국내의 모든 프로세서에 Clock 제공
12. **Direction Conversion** : 수퍼 헤테로다인 방식의 반대되는 개념. 즉 IF(중간주파수)를 사용하지 않고 반송파(carrier)를 기저대역(baseband)로 곧바로 끌어내리고 올리는 방식.
13. **Voice Mail Service(VMS)** : 음성사서함 서비스. (전화를 통해 음성메시지를 저장 또는 전달하는 서비스)
14. **Code Division Multiple Access(CDMA)** : 부호분할 다중 접속방식
15. **BGA (Ball Grid Array)** : 반도체 실장기술에서 프린트배전기판의 뒷면에 리드대신 구형의 볼을 이용하는 실장형태의 한가지
16. **Vocoder** : 음성신호를 음원 및 공명계의 특성을 나타내는 몇 개의 파라미터로 분해 코드화하는 디바이스
17. **Baseband** : 고주파 반송파를 down converting한 최종적 신호처리밴드, 즉 실제 정보를 담고 있는 음성,영상,데이터 등의 최종 원천신호주파수 영역을 말함.

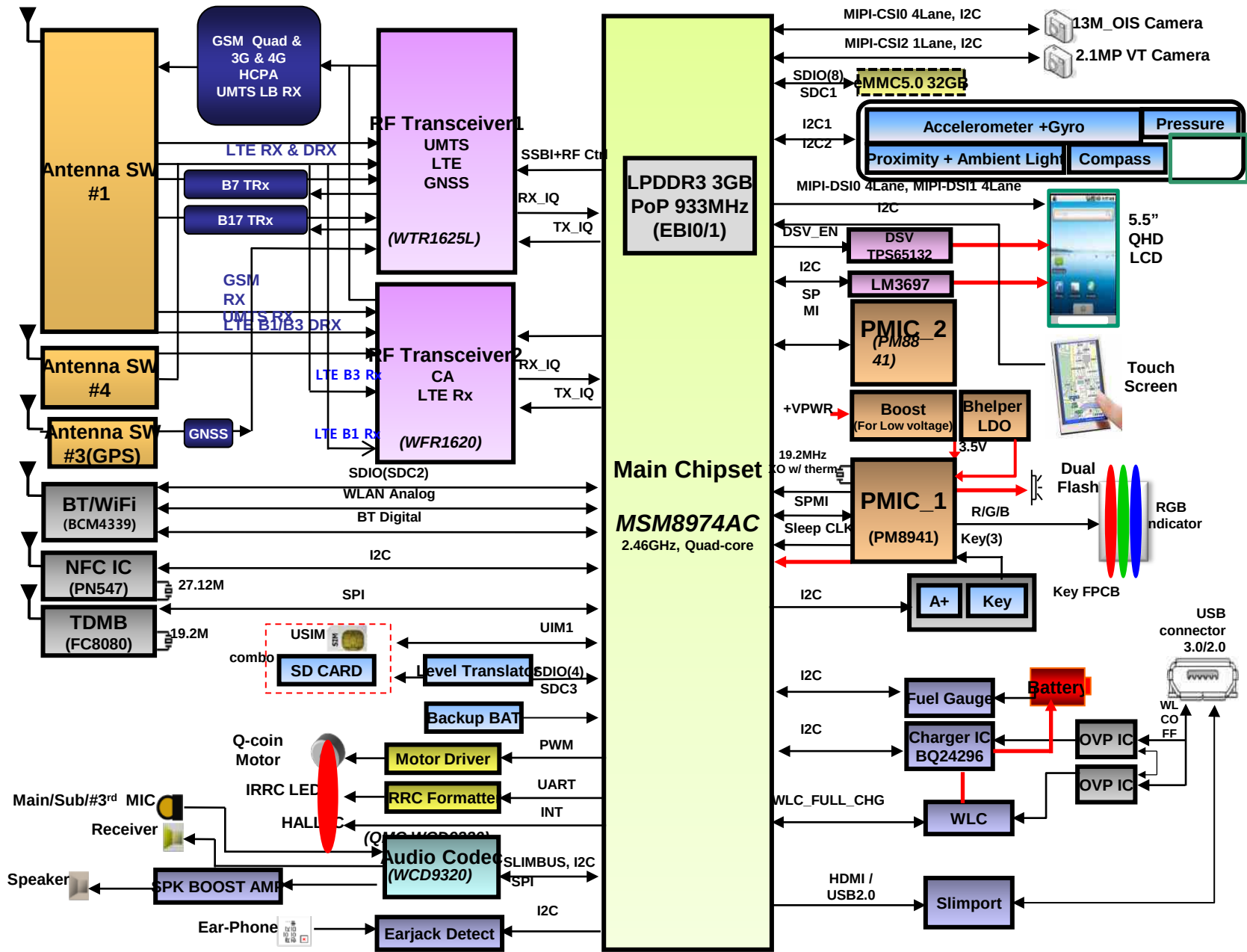
- 18. **Watchdog** : 예상치 못한 하드웨어나 소프트웨어의 비 정상적 동작 시 재 부팅시키는 역할
- 19. **CODEC** : 음성을 Digital화하는 Coder와 Digital신호를 음성으로 바꾸는 Decoder의 합성어
- 20. **PC Sync(PC synchronization)** : PC 와 Phone 의 자료를 교환하고 동기화 할 수 있는 기능
- 21. **AGC(Automatic Gain Control)** : 입력신호에 따라 회로의 이득을 조절하여 출력을 정하게 유지하는 회로
- 22. **SMS (Short Message Services)** : 비교적 짧은 길이 (80byte)의 데이터를 전송하는 서비스
- 23. **RSSI (Received Signal Strength Indicator)** : 수신 신호 전력 세기의 표시
- 24. **WLAN (Wireless Local Area Network)** : 오피스, 가정, 상가 등 옥내 또는 옥외 환경에서 무선으로 구축된 네트워크 환경. 일반적으로 30~150m 정도의 거리에서 무선으로 1~54Mbps의 데이터를 고속으로 전송 가능
- 25. **MMS (Multimedia Messaging System)** : 3세대 이동통신 서비스의 기본 요소로서 영상, 음악, 음성, 동영상 등 다양한 형식의 데이터를 주고받을 수 있는 메시징 시스템
- 26. **H.263** : H.261보다 낮은 데이터 전송 속도의 영상 회의나 텔레비전 전화 실현을 목적으로 하는 컬러 영상 신호의 디지털 부호화 기술을 규정한 ITU-T 권고
- 27. **LTE (Long Term Evolution)** :HSDPA(고속하향패킷접속)보다 12배 이상 빠른 고속 무선 데이터 패킷통신 규격(채널 대역폭은 1.25~20MHz이며, 20MHz 대역폭을 기준으로 하향링크의 최대 전송속도는 100Mbps, 상향링크의 최대 전송속도는 50Mbps)
- 28. **SIM (Subscribe Identity Module)** : 이동전화기에서 사용할 수 있는 카드 형태의 모듈로 가입자의 인증, 과금, 보안 기능 등 다양한 서비스를 제공할 수 있도록 개인 정보를 저장한 칩
- 29. **HDMI (High-Definition Multimedia Interface)** : 비압축 방식의 디지털 비디오/오디오 인터페이스 규격
- 30. **Slim port** : Micro USB port(전용 젠더 사용 必)를 이용한 고속 데이터 전송 기술
- 31. **NFC(Near Field Communication)** : 비 접촉식 근거리 무선통신

제5조 제품구성

1. 제품 전체 외관도



2. 제품 구성 블록도



3. 제품의 기본구성

항목	Type/Spec	기능/구성 설명	제 조업체
1. 본체	DOP Type (w/ 3000mAh(typ) 3.8V,Li-ion 배터리	F400L/S/K Set	LG전자
2. Adapter (TA)	Input : AC100 ~ 240V, 50-60Hz, Output : DC 5V, 1.8A		선린/동도
3. User Manual		사용자 Manual	LG전자
4. USB cable	Micro USB	USB2.0, 1.8A	CRESYN
5. Stereo Headset	3.5Φ, 4극, 3Key	Ear-Mic Stereo Headset	i-sound
6. Battery	(3000mAh(typ) 3.8V,Li-ion 배터리	추가 배터리	LG화학 / 테크노힐

4. Option 구성

항목	Type/Spec	기능/구성 설명
1. Bluetooth Headset		
2. Bluetooth Hands-Free kit		
3. Slimport Gender		Slimport신호 → HDMI 변환
4. Quick Window Case	NFC Ant, WLC, Magnet	NFC, Wireless Charger 지원 Cover 열고 닫을 때 , LCD ON / OFF
5. Slim guard	NFC Ant, WLC	NFC, Wireless Charger 지원
6. Hard case	NFC Ant, WLC	NFC, Wireless Charger 지원

제6조(기능 및 성능)

1. 제품 주요기능

- 1) **Battery Indicator** : 잔여 배터리 용량에 따라 레벨을 표시
- 2) **착신음 선택/소거 기능** : 착신 멜로디(착신음 선택 또는 진동 전환 등을 통한 멜로디제거)
- 3) **SMS기능** : 단문 메시지를 수신 및 발신
- 4) **RSSI Display** : 수신감도를 나타냄
- 5) **단축 Dialing** : 99 개의 단축 다이얼
- 6) **DATA통신기능** : EVDO, LTE 망을 통한 인터넷접속
- 7) **Low Battery Alarm**
- 8) **진동기능**
- 9) **Menu선택기능**
- 10) **Last Number Redial** : 최근 통화한 번호를 메모리에서 지우지 않고 “통화” 버튼을 누르면 다시 다이얼링 가능
- 11) **Schedule Alarm기능** : 반복 이벤트 설정/Reminder기능
- 12) **Roaming표시** : 가입된 서비스지역을 벗어나 타 서비스 지역임을 표시(WCDMA, GSM)
- 13) **계산기 기능** : 기본적인 사칙연산 가능
- 14) **Mute 기능** : Screen상 Mute를 선택하여 통화 시 무음상태 유지/해제
- 15) **잠금 기능** : 전화 잠금, 전 원을 켤 때 잠금, 잠금 번호 변경, 전화 잠금 상태에서 비상전화, 선택 가능
- 16) **자기번호/ SW/HW 버전 확인 기능** : 설정 메뉴에서 자신의 폰 번호 확인, Software Version 확인
- 17) **Lock Code 설정/변경** : Lock Code 또는 Lock Pattern 입력으로 전화 잠금 해제
- 18) **Volume조절기능** : 전화벨/키톤/통화 음량 등을 조절
- 19) **시계기능** : 현재 시간을 기지국으로부터 수신한 시점부터 상태유지
- 20) **Backlight (LCD, 전면 키, Home Key)** :
 - ① 사용자가 LCD Backlight: 15초/30초/1분/2분/5분/10분/15분/화면 켜짐 유지로 설정가능
 - ② 전면 LED : ON/OFF, 모드 아이템별 설정/해제 가능
- 21) **전화 받기 기능** : 호 수신 시 통화 키를 누르면 통화가능
- 22) **Call Time Display기능** : 통화시간 표시
- 23) **Missed Call Alert 기능**
- 24) **착신 멜로디 Download 기능** : 사용자가 전화벨 등의 멜로디를 Down 입력
- 25) **PIM Phone Book 기능** : 전화번호부에 개수의 제한 없이 이름을 저장. 각 아이템마다 다수의 필드 저장. (휴대폰-휴대전화/직장/집/기본/기타, 그룹, 벨소리, 이벤트, 관계, 조직, 이메일, 메신저, 주소, 메모, 닉네임, 웹사이트 등)

- 26) Mobile-Sync : PC 혹은 phone에서 자료를 입력하고 download/upload 하는 기능
- 27) Caller ID Display : Call 수신 시 LCD를 통하여 Picture Caller ID 사진, Camera를 통해 입력된 화면 Display 및 사진촬영, Current Time, Missed Call Information, Alarm의 내용, Phone status 등을 Display 함
- 28) 28. CMX4.x MIDI Sound : 전화 착신 벨, 게임 Sounds를 지원하는 기능
- 29) MMS(Multimedia Messaging Service) : 사진,동영상포함 메일 등 지원 기능
- 30) Voice service : Voice Recognition System/ Voice Memo / Voice Recording
- 31) 전화번호 검색 시 알파벳 순으로 검색 기능
- 32) Internet Site bookmark 기능
- 33) Camcorder기능 : 동영상 저장/Play 기능 제공(UHD 30fps, 메모리잔량 동안 저장 가능)
- 34) 다양한 Multimedia contents(JPEG/PNG/QCELP/AAC/MPEG4) play 가능
- 35) SMS EMS 송/수신 기능:단문메시지를 편집하고 전송함
- 36) Inner charger : 배터리 충전회로 내장 및 충전 상태 표시 기능 제공
- 37) Provision service : 단말기 등록되기 전에 사업자가 제공 가능한 서비스를 사용자에게 보여주는 기능
- 38) 비행기모드 : 비행기 탑승 시 단말기에서 신호를 전파시키지 않도록 모드 지원
- 39) Bluetooth : Bluetooth 4.0 을 지원하며 지원되는 Profile은 headset/handsfree/file Transfer/
- 40) Dial-up/HID/A2DP/OPP/BIP/PBAP 을 지원 함
- 41) WLAN 기능 : IEEE 802.11 2.4G b/g/n/ac and 5G 11a/n/ac 지원 Audio, Video streaming 서비스 등 인터넷 망을 통한 멀티미디어 서비스 됨. E-mail 기능 지원 / 인터넷 콘텐츠 이용 가능
- 42) LTE 기능 : 3GPP 36.101 지원
- 43) PDN을 통한 인터넷 망 접속으로 다양한 멀티미디어 서비스 가능
- 44) Netcast 지원 / Skype 이용 가능
- 45) 음성촬영(Voice Shutter) : 음성 명령에 의해서 촬영이 되는 기능
- 46)HDR : 다중 노출 기술로 역광에서 보정된 사진을 촬영할 수 있게 하는 기능
- 47)파노라마 : 연속적으로 찍은 사진을 하나의 파노라마 사진으로 만드는 기능
- 48)자동 얼굴 인식 : 자동으로 얼굴을 인식하여, 해당 얼굴에 초점을 맞춰 주는 기능
- 49)Live Snapshot : 동영상 레코딩 중에 특정 장면을 사진으로 촬영 할 수 있는 기능
- 50)듀얼 레코딩 : 전/후면 동시 동영상 촬영하는 기능
- 51)스마트 스크린 : 화면을 보고 있을 때, 얼굴을 감지하여 화면을 꺼지지 않게 하는 기능
- 52) 스마트 비디오 : 비디오를 보고 있을 때, 얼굴을 감지하여 화면을 보지 않으면 재생을 멈추는 기능.
- 53)Audio Zooming : 주변 잡음을 감소시켜 특정 Target의 음성을 부각시키는 기능
- 54)톡톡 : double touch로 화면을 켜고, 끄는 기능
- 55)Q-Remote : 리모컨 기능 제공.

나. 규격 및 SPEC



LG Electronics



제7조(전기적 특성)

1. System 개요

항 목	Type / Spec.
1. Phone Type	DOP
2. 크기	146.31 x 74.64 x 9.15(mm)
3. 무게	약 151.4g(배터리 포함)
4. 적용 배터리	3000mAh (Typ),Li-ion
5. MSM Chipset	MSM8974, WTR1625L,WFR1620, PM8941, PM8841, WCD9320
6. Memory	24Gb(SDRAM DDR3) + 32GB eMMC
7. LCD	5.5", 1440x2560 Pixels, 16.7M Colors, TFT
8. Main Camera	13M OIS + 2.1M VT
9. Audio	Ø15x11x3.5, 1W FSD Speaker
10. Bluetooth	Bluetooth 4.0
11. WLAN	2.4GHz IEEE 802.11 b/g/n/ac , 5GHz IEEE 802.11 a/n/ac
12. LTE	3GPP 36.101

2. 사용 조건

항 목	Specification	단 위
사용 전압	3.8(Typ), 4.35 (Max)	V
동작 온도	-20 ~ +60	℃
저장 온도	-30 ~ +60	℃
동작 습도	65±5%	%

3. 무선 특성

1) LTE 송신부 전기적 특성

항목	Spec.	Min	Typ.	Max	단위
Maximum Output Power(ClassIII)	Power class 3 (Environmental Test)	20.3 (20.3)	23.0 (23.0)	25.7 (25.7)	dBm/ 10Mhz
Frequency Error	QPSK	-01.PPM -15hz			
EVM	QPSK 16QAM			17.5 12.5	%
Carrier leakage	Output Power > 0dBm			-24.2	dBc
In-band emissions for non allocated RB	QPSK, 12RB	0			dB
Spectrum Flatness (Range1 Ripple)	QPSK, 50RB			+5.4	dB
Occupied bandwidth	QPSK, 50RB			10	MHz
ACLR (Adjacent Channel Leakage Power Ratio)	E-UTRA ACLR1	29.2			dB
	UTRA ACLR1	32.2			dB
	UTRA ACLR2	35.2			dB
Transmitter Spurious emissions	9 kHz < f < 150 kHz			-36	dBm/1kHz
	150kHz < f < 30 MHz			-36	dBm/10kHz
	30MHz < f < 1GHz			-36	dBm/100kHz
	1GHz < f < 12.75GHz			-30	dBm/1MHz

※ 3GPP TS 36.521-1 .

-주파수 범위

S/K/L 공통 : B1 (1920~1940MHz/2110~2130MHz), B3(1710~1785MHz/1805 ~ 1880MHz),
B7(2500 ~ 2570MHz/2620 ~ 2690MHz), B17(704 ~ 716MHz/734 ~ 746MHz)

S/L 별도 : B5 (824~849MHz/869~894MHz)

K 별도 : B8 (864~869Mhz/824~819Mhz)

- 중간 주파수 : 없음

- 정격 최대 출력

S/K/L 공통 : B1 (23dbm), B17 (23.5dBm) , B3(24dbm)

S/L 별도 : B7 (24dBm), B5 (23dBm)

K 별도 : B7 (23.5dBm) B8 (24dBm)

(DL : QPSK, 50RB
UL : QPSK, 12RB)

2) LTE 수신부 전기적 특성

	Spec.	Min	Typ.	Max	
Sensitivity Level (DL : QPSK, 50RB UL : QPSK, 50RB)	Dual Receiver			SENS	dBm / 10MHz
	Single Receiver (exception in 3GPP)			SENS+3	dBm / 10MHz
Maximum input Level	Dual Receiver (DL : 64QAM, 50RB UL : QPSK, 12RB)			-25.7	dBm / 10MHz
ACS (Adjacent Channel Selectivity)	Dual Receiver (DL : QPSK, 50RB UL : QPSK, 12RB) $\Delta f = \pm 7.5\text{MHz}$, LTE BW 10MHz			33	dB
In-Band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f = \pm 7.5\text{MHz}$, LTE BW 10MHz $\Delta f = \pm 12.5\text{MHz}$, LTE BW 10MHz			SENS+6	dBm / 10MHz
Out of Band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f = \pm 15\text{MHz} \sim 85\text{MHz}$, CW			SENS+6	dBm / 10MHz
Narrow Band Blocking (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f = \pm 5.2125\text{MHz}$, CW			SENS+13	dBm / 10MHz
Intermodulation (DL : QPSK, 50RB UL : QPSK, 20RB)	Dual Receiver $\Delta f = \pm 10\text{MHz}$, LTE BW 10MHz			SENS+6	dBm / 10MHz
Spurious emissions (DL : QPSK, 0RB UL : QPSK, 0RB)	Dual Receiver $30\text{MHz} < f < 1\text{GHz}$			-57	dBm / 100kHz
	$1\text{GHz} < f < 12.75\text{GHz}$			-47	dBm / 1MHz

※ 3GPP TS 36.521-1 .

-SENS

B1 : -96.3 (DL : QPSK, 50RB UL : QPSK, 50RB)

B3 : -93.3 (DL : QPSK, 50RB UL : QPSK, 50RB)

B5 : -94.3 (DL : QPSK, 50RB UL : QPSK, 25RB) => S/L only

B8 : -94.3 (DL : QPSK, 50RB UL : QPSK, 25RB) => K only

B7 : -94.3 (DL : QPSK, 50RB UL : QPSK, 50RB)

B17 : -93.3 (DL : QPSK, 50RB UL : QPSK, 20RB)

3) WCDMA 송신부의 전기적 특성.

항 목	Spec.	Min	Typ.	Max	단 위
Maximum Output Power	Power class III	21.0	24	25.0	dBm/3.84MHz
Frequency Accuracy	-	-200	-	+200	Hz
Minimum Output Power	-	-	-	-50	dBm/3.84MHz
Occupied Bandwidth	-	-	-	4.6	MHz
Adjacent Channel Leakage Power Ratio	$\Delta f = \pm 5\text{MHz}$	-	-	-33	dBm/3.84MHz
	$\Delta f = \pm 10\text{MHz}$	-	-	-43	dBm/3.84MHz
Error Vector Magnitude(EVM)	-	-	-	17	%
Peak Code Domain Error	-	-	-	15	dBm/3.84MHz
Open Loop Output Power	Ior = -106.7dBm/3.84MHz	-46.7	-	-28.7	dBm/3.84MHz
	Ior = -65.7dBm/3.84MHz	-23.0	-	-5.0	dBm/3.84MHz
	Ior = -25.0dBm/3.84MHz	-0.0	-	14.0	dBm/3.84MHz
Transmit Off Power	-	-	-	-56	dBm/3.84MHz
Spectrum Emission Mask	$\Delta f = 2.5 \sim 3.5\text{MHz}$	$-35-15 * (\Delta f-2.5)$			dBc/30KHz
	$\Delta f = 3.5 \sim 7.5\text{MHz}$	$-35-1 * (\Delta f-3.5)$			dBc/1MHz
	$\Delta f = 7.5 \sim 8.5\text{MHz}$	$-39-10 * (\Delta f-7.5)$			dBc/1MHz
	$\Delta f = 8.5 \sim 12.5\text{MHz}$	-49			dBc/1MHz
Transmit Spurious Emission	9kHz ~ 150kHz			-36	dBm
	150kHz ~ 30MHz			-36	dBm
	30MHz ~ 1GHz			-36	dBm
	1GHz ~ 12.5GHz			-30	dBm

4) WCDMA 수신부의 전기적 특성.

항 목	Spec.	Min	Typ.	Max	단 위
Reference Sensitivity Level	BER < 0.001%	-	-	-106.7	dBm/3.84MHz
Maximum Input Level	Pin = -25dBm	-	-	0.1	BER(%)
Adjacent Channel Selectivity(ACS)	Jammer(Modulation) = -52dBm/3.84MHz $\Delta f = \pm 5\text{MHz}$ BER < 0.001%	-	-	-103	dBm/3.84MHz
In Band Blocking	Jammer(Modulation) = -56dBm/3.84MHz $\Delta f = \pm 10\text{MHz}$ BER < 0.001%	-	-	-103.7	dBm/3.84MHz
	Jammer(Modulation) = -44dBm/3.84MHz $\Delta f = \pm 15\text{MHz}$ BER < 0.001%	-	-	-103.7	dBm/3.84MHz
Inter-modulation Desensitization	Jammer1(CW)= -46dBm Jammer2(Modulation) = -46dBm/3.84MHz $\Delta f = \pm 10\text{MHz}$ BER < 0.001%	-	-	-103.7	dBm/3.84MHz
	Jammer1= -46dBm(CW) Jammer2(Modulation) = -46dBm/3.84MHz $\Delta f = \pm 20\text{MHz}$ BER < 0.001%	-	-	-103.7	dBm/3.84MHz
Spurious Emissions	30kHz ~ 1GHz	-	-	-57	dBm/30kHz
	1GHz ~ 1.92GHz	-	-	-47	dBm/1MHz
	1.92GHz ~ 1.98GHz	-	-	-60	dBm/3.84MHz
	1.98GHz ~ 2.11GHz	-	-	-47	dBm/3.84MHz
	2.11GHz ~ 2.17GHz	-	-	-60	dBm/3.84MHz
	2.17GHz ~ 12.75GHz	-	-	-47	dBm/1MHz

- 주파수 범위 : B1(1922.8~1977.2MHz/2112.8~2167.2MHz),
B2(1852.4~1907.6MHz/1932.4~1987.6MHz)
B5(826.4~846.6MHz/871.4~891.6MHz)

- 중간 주파수 : 없음

-정격 최대 출력

B1 : (24dBm) , B2 /B5 : (23dBm)

5) GSM 송수신부의 전기적 특성.

항 목	Spec.
Phase Error	Rms : 5° Peak : 20 °
Frequency Error	GSM : 0.1 ppm DCS/PCS : 0.1 ppm
EMC(Radiated Spurious Emission Disturbance)	GSM/DCS : < -28dBm
Transmitter Output power and Burst Timing	GSM : 5dBm – 33dBm ± 3dB DCS/PCS : 0dBm – 30dBm ± 3dB
Burst Timing	<3.69us
Spectrum due to modulation out to less than 1800kHz offset	200kHz : -36dBm 600kHz : -51dBm/-56dBm
Spectrum due to modulation out to larger than 1800kHz offset to the edge of the transmit band	GSM : 1800-3000kHz : < -63dBc(-46dBm) 3000kHz-6000kHz : < -65dBc(-46dBm) 6000kHz < : < -71dBc(-46dBm) DCS : 1800-3000kHz : < -65dBc(-51dBm) 6000kHz < : < -73dBc(-51dBm)
Spectrum due to switching transient	400kHz :-19dBm/-22dBm(5/0), -23dBm 600kHz :-21dBm/-24dBm(5/0), -26dBm
Reference Sensitivity – TCH/FS	Class II(RBER) : -105dBm(2.439%)
Usable receiver input level range	0.012(-15 - -40dBm)
Intermodulation rejection – Speech channels	± 800kHz, ± 1600kHz : -98dBm/-96dBm (2.439%)
AM Suppression -GSM : -31dBm - DCS : -29dBm	-98dBm/-96dBm (2.439%)
Timing Advance	± 0.5T

- 주파수 범위 : GSM850(824.2~848.8MHz/869.2~893.8MHz),
GSM900(880~915MHz/925~960MHz),
DCS(1710~1785MHz/1805~1880MHz),
PCS (1850~1910MHz/1830~1990MHz)
- 국부발진 주파수 범위 : 1688 ~ 1736 MHz
- 중간 주파수 : 없음
- 정격 최대 출력 : GSM850:(32.5dBm), GSM900:(32.5dBm), DCS/PCS:(29.5dBm)

6) HSDPA 송수신부의 전기적 특성.

	항 목	Spec.		Min	Typ.	Max	단 위
송 신	5.2.A Maximum Output Power with HS-DPCCH	$1/15 \leq \beta_c/\beta_d \leq 12/15$ $13/15 \leq \beta_c/\beta_d \leq 15/8$ $15/7 \leq \beta_c/\beta_d \leq 15/0$		21 20 19	23 22.5 22	25 25 25	dBm/3.84MHz
	5.7A HS-DPCCH	MAX / 0dBm	Step 0	0 dBm +/- 1 dB			dB
			Step 1 – Step 0	6 dB +/- 2.3 dB			dB
			Step 2 – Step 3	1 dB +/- 0.6 dB			dB
			Step 5 – Step 4	0 dB +/- 0.6 dB			dB
			Step 6 – Step 7	5 dB +/- 2.3 dB			dB
	5.9A Spectrum Emission Mask with HS-DPCCH	$\Delta f = 2.5 \sim 3.5\text{MHz}$		$-35-15 * (\Delta f-2.5)$			dBc/30KHz
		$\Delta f = 3.5 \sim 7.5\text{MHz}$		$-35-1 * (\Delta f-3.5)$			dBc/1MHz
		$\Delta f = 7.5 \sim 8.5\text{MHz}$		$-39 \ 10 * (\Delta f-7.5)$			dBc/1MHz
		$\Delta f = 8.5 \sim 12.5\text{MHz}$		-49			dBc/1MHz
	5.10A Adjacent Channel Leakage Power Ratio with HS-DPCCH	$\Delta f = \pm 5\text{MHz}$		-	-	-33	dBm/3.84MHz
		$\Delta f = \pm 10\text{MHz}$		-	-	-43	dBm/3.84MHz
	5.13.1A Error Vector Magnitude (EVM) with HS-DPCCH	MAX Power -20 dBm		-	-	17.5	%
수 신	6.3A Maximum Input Level for HS-PDSCH Reception	Pin = -25dBm		-	-	10	BLER(%)

7) GPS 전기적 특성

항 목	Spec.
Sensitivity Test	Success rate ≥ 50% @ -154 dBm

※ 상기 Spec. : QSPR GNSS Tracking Sensitivity 측정 항목

- ※ F400L/S/K - CDMA 1X 미 지원으로, IS-916 규격 제외됨.
- ※ WCDMA Roaming Band : 기타 항목 - 3GPP TS 34.171에 준함.

8) Bluetooth 수신부 전기적 특성

항 목	Spec.		Min	Typ.	Max	단 위
Sensitivity (Single Slot Packet)	DH1 PRBS9	BER @ 0.1%		-	-70	dBm
Sensitivity (Multi Slot Packet)	DH5 PRBS9	BER @ 0.1%		-	-70	dBm
C/I Performance	DH1 PRBS9	Co-Channel -60dBm@ BER0.1%		-	11	dB
		C/I @1MHz -60dBm@ BER 0.1%		-	0	
		C/I @2MHz -60dBm@ BER 0.1%		-	-30	
		C/I @≥3MHz -67dBm@ BER0.1%		-	-40	
		C/I @Image -67dBm@ BER 0.1%		-	-9	
		C/I @1MHz Image -67dBm@ BER 0.1%		-	-20	
Blocking Performance	DH1 PRBS9	-10dBm @ 30M ~2 GHz		-	-67	dBm
		-27dBm @ 2 ~ 2.4 GHz				
		-27dBm @ 2.5 ~ 3 GHz				
		-10dBm @ 3 ~ 12.75 GHz				
Intermodulation Performance	DH1 PRBS9	$\Delta f = +5\text{MHz}$ (CW) $\Delta f = +10\text{MHz}$ (BT) BER @ 0.1%		-	-64	dBm
		$\Delta f = -5\text{MHz}$ (CW) $\Delta f = -10\text{MHz}$ (BT) BER @ 0.1%		-	-64	dBm
Maximum Input Level	DH1 PRBS9	BER @ 0.1%	-20	-		dBm

9) Bluetooth 송신부 전기적 특성(Continued)

항 목	Spec.		Min	Typ.	Max	단 위
Output Power	DH5 PRBS9	Nominal Pwr	-6	-	+20	dBm
		Peak Pwr	-	-	+23	dBm
Power Density	DH5 PRBS9	@100kHz BW		-	+20	dBm
Frequency Range	DH5 PRBS9	-30dBm @100kHz BW	2400	-	2483	MHz
Tx Output Spectrum (20dB BW)	DH5 PRBS9	$\Delta f = f_H - f_L$		-	1	MHz
Tx Output Spectrum (Adjacent Output Power)	DH5 PRBS9	@M-N = 2		-	-20	dBm
		@M-N >= 3		-	-40	dBm
Modulation Index	DH5 11110000	f_{1_AVG}	140	-	175	kHz
	DH5 10101010	f_{2_MAX}	115	-		kHz
		f_{2_AVG}/f_{1_AVG}	0.8	-		
Initial Carrier Freq. Tolerance	DH5 PRBS9		-75	-	+75	kHz
Carrier Frequency Drift	DH1 10101010	Freq. Drift	-25	-	+25	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH3 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
	DH5 10101010	Freq. Drift	-40	-	+40	kHz
		Drift Rate	-20	-	+20	kHz/50us
Out of Band Spurious Emission	DH1 PRBS9	@ 30MHz ~ 12.75GHz	20			dBc

★ 음영으로 표시된 항목은 현재 측정할 수 없는 항목임

10) 지상파 DMB 전기적 특성

항목	Spec.	Min	Typ.	Max	단위
Sensitivity	BER 10 ⁻⁴ 이하 @Viterbi Decoder Output			-95	dBm
Maximum Input Power	BER 10 ⁻⁴ 이하 @Viterbi Decoder Output	-10			dBm
Adjacent Channel Selectivity	BER 10 ⁻⁴ 이하 @Viterbi Decoder Output 조건 만족시의 인접채널 신호와의 레벨차이	30			dB
Far-off Selectivity	BER 10 ⁻⁴ 이하 @Viterbi Decoder Output 조건 만족시의 ±5MHz 떨어진 신호와의 레벨차이	40			dB
Acquisition time after Synchronization loss	10초 이상 신호를 차단한 후에 ±500Hz 벗어난 신호인가 시 동기를 잡는데 까지 걸리는 시간			3	sec
Performance in a Rayleigh Channel	A. Urban(@25Km/s) 조건에서의 감도 (BER 10 ⁻⁴ 이하@Viterbi Decoder Output)			-75	dBm
	B. Rural(@120Km/s) 조건에서의 감도 (BER 10 ⁻⁴ 이하@Viterbi Decoder Output)			-75	dBm
	C. SFN(@60Km/s) 조건에서의 감도 (BER 10 ⁻⁴ 이하@Viterbi Decoder Output)			-75	dBm
무선감도	BER 10 ⁻⁴ 이하@Viterbi Decoder Output			TBD	dBm

11) WLAN 802.11b 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	$\leq 20\text{dBm}$ under (Europe), $\leq 30\text{dBm}$ under (USA)
Frequency Tolerance	within ± 25 PPM
Chip clock Frequency Tolerance	within ± 25 PPM
Spectrum Mask	≤ -30 @ $f_c - 22\text{MHz} < f < f_c - 11\text{MHz}$ and $f_c + 11\text{MHz} < f < f_c + 22\text{MHz}$ ≤ -50 @ $f < f_c - 22\text{MHz}$ and $f > f_c + 22\text{MHz}$
Power ramp on/off time	$\leq 2\mu\text{s}$
Carrier Suppression	$\leq -15\text{dB}$
Modulation Accuracy (Peak EVM)	$\leq 35\%$
Spurious Emissions	$< -36\text{ dBm}$ @ 30MHz ~ 1GHz $< -30\text{ dBm}$ above @ 1GHz ~ 12.75GHz $< -47\text{ dBm}$ @ 1.8GHz ~ 1.9GHz $< -47\text{ dBm}$ @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	$\leq -76\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Max input Sensitivity	$\geq -10\text{dBm}$ (1Mbps,2Mbps,5.5Mbps,11Mbps) @ FER $\leq 8\%$
Rx Adjacent Channel Rejection	$\geq 35\text{dB}$ @FER $\leq 8\%$, interference input signal -70dBm @ $f_c \pm 25\text{MHz}$ (11Mbps)

12) WLAN 802.11g 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	2400 MHz ~ 2483.5 MHz (CH1~CH13)
Tx Power Level	≤ 20dBm under (Europe), ≤ 30dBm under (USA)
Frequency Tolerance	within ±25 PPM
Chip clock Frequency Tolerance	within ±25 PPM
Spectrum Mask	≤ -20 @ ±11MHz offset (9Mhz ~ 11MHz) ≤ -28 @ ±20MHz offset (11MHz ~ 20Mhz) ≤ -40 @ ±30MHz offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	≤ -5dB@6Mbps, ≤ -8dB@9Mbps, ≤ -10dB@12Mbps, ≤ -13dB@18Mbps, ≤ -16dB@24Mbps, ≤ -19dB@36Mbps, ≤ -22dB@48Mbps, ≤ -25dB@54Mbps
Spurious Emissions	< -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6Mbps, -81dBm@9Mbps, -79dBm@12Mbps -77dBm@18Mbps, -74dBm@24Mbps, -70dBm@36Mbps -66dBm@48Mbps, -65dBm@54Mbps
Rx Max input Sensitivity	≥ -20dBm(6,9,12,18,24,36,48,54Mbps) @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

13-1) WLAN 802.11a 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 20 PPM
Chip clock Frequency Tolerance	within ± 20 PPM
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9MHz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20MHz) ≤ -40 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6\text{Mbps}$, $\leq -8\text{dB}@9\text{Mbps}$, $\leq -10\text{dB}@12\text{Mbps}$, $\leq -13\text{dB}@18\text{Mbps}$, $\leq -16\text{dB}@24\text{Mbps}$, $\leq -19\text{dB}@36\text{Mbps}$, $\leq -22\text{dB}@48\text{Mbps}$, $\leq -25\text{dB}@54\text{Mbps}$
Spurious Emissions	$< -36\text{ dBm @ } 30\text{ MHz to } 47\text{ MHz}$ $< -54\text{ dBm @ } 47\text{ MHz to } 74\text{ MHz}$ $< -36\text{ dBm @ } 74\text{ MHz to } 87,5\text{ MHz}$ $< -54\text{ dBm @ } 87,5\text{ MHz to } 118\text{ MHz}$ $< -36\text{ dBm @ } 118\text{ MHz to } 174\text{ MHz}$ $< -54\text{ dBm @ } 174\text{ MHz to } 230\text{ MHz}$ $< -36\text{ dBm @ } 230\text{ MHz to } 470\text{ MHz}$ $< -54\text{ dBm @ } 470\text{ MHz to } 862\text{ MHz}$ $< -36\text{ dBm @ } 862\text{ MHz to } 1\text{ GHz}$ $< -30\text{ dBm @ } 1\text{ GHz to } 5.15\text{ GHz}$ $< -30\text{ dBm @ } 5.35\text{ GHz to } 5.47\text{ GHz}$ $< -30\text{ dBm @ } 5.725\text{ GHz to } 26.5\text{ GHz}$
Rx Min input Sensitivity	PER $\leq 10\%$ $-82\text{dBm}@6\text{Mbps}$, $-81\text{dBm}@9\text{Mbps}$, $-79\text{dBm}@12\text{Mbps}$ $-77\text{dBm}@18\text{Mbps}$, $-74\text{dBm}@24\text{Mbps}$, $-70\text{dBm}@36\text{Mbps}$ $-66\text{dBm}@48\text{Mbps}$, $-65\text{dBm}@54\text{Mbps}$
Rx Max input Sensitivity	$\geq -30\text{dBm}(6,9,12,18,24,36,48,54\text{Mbps})$ @ PER $\leq 10\%$

13-2) WLAN 802.11a 송수신부의 전기적 특성(Conducted)

항목	Spec.
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6Mbps, ACR ≥ 15dB@9Mbps, ACR ≥ 13dB@12Mbps, ACR ≥ 11dB@18Mbps, ACR ≥ 8dB@24Mbps, ACR ≥ 4dB@36Mbps ACR ≥ 0dB@48Mbps, ACR ≥ -1dB@54Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

14-1) WLAN 802.11n(HT20) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	2400 ~ 2483.5 MHz (CH1 ~ 13) 5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 10\text{mW/MHz}$ (CH1 ~ 13) $\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5170 ~ 5815 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5170 ~ 5815 MHz)
Spectrum Mask	≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -28\text{dB}@65\text{Mbps}$
Spurious Emissions	[2400 MHz ~ 2483.5 MHz (CH1 ~ 13)] < -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz [5170 MHz ~ 5815 MHz (CH36 ~ 161)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz

14-2) WLAN 802.11n(HT20) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps
Rx Max input Sensitivity	≥ -20dBm @ PER ≤ 10%(2400 ~ 2483.5 MHz) ≥ -30dBm @ PER ≤ 10%(5170 ~ 5815 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

15-1) WLAN 802.11n(HT40) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Chip clock Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Spectrum Mask	≤ -20 @ $\pm 21\text{MHz}$ offset (19Mhz ~ 21MHz) ≤ -28 @ $\pm 40\text{MHz}$ offset (21MHz ~ 40Mhz) ≤ -45 @ $\pm 60\text{MHz}$ offset (40MHz ~ 60Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -28\text{dB}@65\text{Mbps}$
Spurious Emissions	[5170 MHz ~ 5815 MHz (CH36 ~ 161)] $< -36\text{ dBm @ } 30\text{ MHz to } 47\text{ MHz}$ $< -54\text{ dBm @ } 47\text{ MHz to } 74\text{ MHz}$ $< -36\text{ dBm @ } 74\text{ MHz to } 87,5\text{ MHz}$ $< -54\text{ dBm @ } 87,5\text{ MHz to } 118\text{ MHz}$ $< -36\text{ dBm @ } 118\text{ MHz to } 174\text{ MHz}$ $< -54\text{ dBm @ } 174\text{ MHz to } 230\text{ MHz}$ $< -36\text{ dBm @ } 230\text{ MHz to } 470\text{ MHz}$ $< -54\text{ dBm @ } 470\text{ MHz to } 862\text{ MHz}$ $< -36\text{ dBm @ } 862\text{ MHz to } 1\text{ GHz}$ $< -30\text{ dBm @ } 1\text{ GHz to } 5.15\text{ GHz}$ $< -30\text{ dBm @ } 5.35\text{ GHz to } 5.47\text{ GHz}$ $< -30\text{ dBm @ } 5.725\text{ GHz to } 26.5\text{ GHz}$

15-2) WLAN 802.11n(HT40) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Rx Min input Sensitivity	PER ≤ 10% -79dBm@6.5Mbps, -76dBm@13Mbps, -74dBm@19.5Mbps -71dBm@26Mbps, -67dBm@39Mbps, -63dBm@52Mbps -62dBm@58.5Mbps, -61dBm@65Mbps
Rx Max input Sensitivity	≥ -30dBm @ PER ≤ 10%(5170 ~ 5815 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

16-1) WLAN 802.11ac(VHT20) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	2400 ~ 2483.5 MHz (CH1 ~ 13) 5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 10\text{mW/MHz}$ (CH1 ~ 13) $\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5170 ~ 5815 MHz)
Chip clock Frequency Tolerance	within ± 25 PPM(2400 ~ 2483.5 MHz) within ± 20 PPM(5170 ~ 5815 MHz)
Spectrum Mask	2.4G (2400 ~ 2483.5 MHz) ≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -45 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz) 5G (5170 ~ 5710 MHz) ≤ -20 @ $\pm 11\text{MHz}$ offset (9Mhz ~ 11MHz) ≤ -28 @ $\pm 20\text{MHz}$ offset (11MHz ~ 20Mhz) ≤ -40 @ $\pm 30\text{MHz}$ offset (20MHz ~ 30Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@6.5\text{Mbps}$, $\leq -10\text{dB}@13\text{Mbps}$, $\leq -13\text{dB}@19.5\text{Mbps}$, $\leq -16\text{dB}@26\text{Mbps}$, $\leq -19\text{dB}@39\text{Mbps}$, $\leq -22\text{dB}@52\text{Mbps}$, $\leq -25\text{dB}@58.5\text{Mbps}$, $\leq -27\text{dB}@65\text{Mbps}$, $\leq -30\text{dB}@78\text{Mbps}$
Spurious Emissions	[2400 MHz ~ 2483.5 MHz (CH1 ~ 13)] < -36 dBm @ 30MHz ~ 1GHz < -30 dBm above @ 1GHz ~ 12.75GHz < -47 dBm @ 1.8GHz ~ 1.9GHz < -47 dBm @ 5.15GHz ~ 5.3GHz

16-2) WLAN 802.11ac(VHT20) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Spurious Emissions	[5170 MHz ~ 5815 MHz (CH36 ~ 161)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz
Rx Min input Sensitivity	PER ≤ 10% -82dBm@6.5Mbps, -79dBm@13Mbps, -77dBm@19.5Mbps -74dBm@26Mbps, -70dBm@39Mbps, -66dBm@52Mbps -65dBm@58.5Mbps, -64dBm@65Mbps, -59dBm@78Mbps
Rx Max input Sensitivity	≥ -30dBm @ PER ≤ 10%
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@6.5Mbps, ACR ≥ 13dB@13Mbps, ACR ≥ 11dB@19.5Mbps, ACR ≥ 8dB@26Mbps, ACR ≥ 4dB@39Mbps, ACR ≥ 0dB@52Mbps ACR ≥ -1dB@58.5Mbps, ACR ≥ -2dB@65Mbps, ACR ≥ -7dB@78Mbps ※ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

17-1) WLAN 802.11ac(VHT40) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Chip clock Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Spectrum Mask	≤ -20 @ $\pm 21\text{MHz}$ offset (19Mhz ~ 21MHz) ≤ -28 @ $\pm 40\text{MHz}$ offset (21MHz ~ 40Mhz) ≤ -40 @ $\pm 60\text{MHz}$ offset (40MHz ~ 60Mhz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@13.5\text{Mbps}$, $\leq -10\text{dB}@27\text{Mbps}$, $\leq -13\text{dB}@40.5\text{Mbps}$, $\leq -16\text{dB}@54\text{Mbps}$, $\leq -19\text{dB}@81\text{Mbps}$, $\leq -22\text{dB}@108\text{Mbps}$, $\leq -25\text{dB}@121.5\text{Mbps}$, $\leq -27\text{dB}@135\text{Mbps}$, $\leq -30\text{dB}@162\text{Mbps}$, $\leq -32\text{dB}@180\text{Mbps}$
Spurious Emissions	[5170 MHz ~ 5815 MHz (CH36 ~ 161)] < -36 dBm @ 30 MHz to 47 MHz < -54 dBm @ 47 MHz to 74 MHz < -36 dBm @ 74 MHz to 87,5 MHz < -54 dBm @ 87,5 MHz to 118 MHz < -36 dBm @ 118 MHz to 174 MHz < -54 dBm @ 174 MHz to 230 MHz < -36 dBm @ 230 MHz to 470 MHz < -54 dBm @ 470 MHz to 862 MHz < -36 dBm @ 862 MHz to 1 GHz < -30 dBm @ 1 GHz to 5.15 GHz < -30 dBm @ 5.35 GHz to 5.47 GHz < -30 dBm @ 5.725 GHz to 26.5 GHz

17-2) WLAN 802.11ac(VHT40) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Rx Min input Sensitivity	PER ≤ 10% -79dBm@13.5Mbps, -76dBm@27Mbps, -74dBm@40.5Mbps -71dBm@54Mbps, -67dBm@81Mbps, -63dBm@108Mbps -62dBm@121.5Mbps, -61dBm@135Mbps, -56dBm@162Mbps, -54dBm@180Mbps
Rx Max input Sensitivity	≥ -30dBm @ PER ≤ 10%(5170 ~ 5815 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@13.5Mbps, ACR ≥ 13dB@27Mbps, ACR ≥ 11dB@40.5Mbps, ACR ≥ 8dB@54Mbps, ACR ≥ 4dB@81Mbps, ACR ≥ 0dB@108Mbps ACR ≥ -1dB@121.5Mbps, ACR ≥ -2dB@135Mbps ACR ≥ -7dB@162Mbps, ACR ≥ -9dB@180Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

18-1) WLAN 802.11ac(VHT80) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Transmit Frequency	5170 ~ 5815 MHz (CH36 ~ 64, CH100 ~ 124, CH149 ~ 161)
Tx Power Level	$\leq 2.5\text{mW/MHz}$ (CH36 ~ 48), $\leq 10\text{mW/MHz}$ (CH52 ~ 161)
Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Chip clock Frequency Tolerance	within ± 20 PPM(5170 ~ 5815 MHz)
Spectrum Mask	≤ -20 @ $\pm 41\text{MHz}$ offset (39MHz ~ 41MHz) ≤ -28 @ $\pm 80\text{MHz}$ offset (41MHz ~ 80MHz) ≤ -40 @ $\pm 120\text{MHz}$ offset (80MHz ~ 120MHz)
Transmitter constellation error (rms EVM)	$\leq -5\text{dB}@29.3\text{Mbps}$, $\leq -10\text{dB}@58.5\text{Mbps}$, $\leq -13\text{dB}@87.8\text{Mbps}$, $\leq -16\text{dB}@117\text{Mbps}$, $\leq -19\text{dB}@175.5\text{Mbps}$, $\leq -22\text{dB}@234\text{Mbps}$, $\leq -25\text{dB}@263.3\text{Mbps}$, $\leq -27\text{dB}@292.5\text{Mbps}$, $\leq -30\text{dB}@351\text{Mbps}$, $\leq -32\text{dB}@390\text{Mbps}$
Spurious Emissions	[5170 MHz ~ 5815 MHz (CH36 ~ 161)] $< -36\text{ dBm @ } 30\text{ MHz to } 47\text{ MHz}$ $< -54\text{ dBm @ } 47\text{ MHz to } 74\text{ MHz}$ $< -36\text{ dBm @ } 74\text{ MHz to } 87,5\text{ MHz}$ $< -54\text{ dBm @ } 87,5\text{ MHz to } 118\text{ MHz}$ $< -36\text{ dBm @ } 118\text{ MHz to } 174\text{ MHz}$ $< -54\text{ dBm @ } 174\text{ MHz to } 230\text{ MHz}$ $< -36\text{ dBm @ } 230\text{ MHz to } 470\text{ MHz}$ $< -54\text{ dBm @ } 470\text{ MHz to } 862\text{ MHz}$ $< -36\text{ dBm @ } 862\text{ MHz to } 1\text{ GHz}$ $< -30\text{ dBm @ } 1\text{ GHz to } 5.15\text{ GHz}$ $< -30\text{ dBm @ } 5.35\text{ GHz to } 5.47\text{ GHz}$ $< -30\text{ dBm @ } 5.725\text{ GHz to } 26.5\text{ GHz}$

18-2) WLAN 802.11ac(VHT80) 송수신부의 전기적 특성(Conducted)

항목	Spec.
Rx Min input Sensitivity	PER ≤ 10% -76dBm@29.3Mbps, -73dBm@58.5Mbps, -71dBm@87.8Mbps -68dBm@117Mbps, -64dBm@175.5Mbps, -60dBm@234Mbps -59dBm@263.3Mbps, -58dBm@292.5Mbps, -53dBm@351Mbps, -51dBm@390Mbps
Rx Max input Sensitivity	≥ -30dBm @ PER ≤ 10%(5170 ~ 5815 MHz)
Rx Adjacent Channel Rejection	PER ≤ 10%, ACR ≥ 16dB@29.3Mbps, ACR ≥ 13dB@58.5Mbps, ACR ≥ 11dB@87.8Mbps, ACR ≥ 8dB@117Mbps, ACR ≥ 4dB@175.5Mbps, ACR ≥ 0dB@234Mbps ACR ≥ -1dB@263.3Mbps, ACR ≥ -2dB@292.5Mbps ACR ≥ -7dB@351Mbps, ACR ≥ -9dB@390Mbps ※ ACR shall be measured by setting the desired signal's strength 3 dB above the rate-dependent sensitivity specified in min input sensitivity

제8조 부가기능

1. 소모전류

◆ RF 소모 전류 측정 시 Network Mode는 각 Band Only, Backlight Off

항 목	Specification		
	VOLTE	WCDMA	GSM
1. Sleep Mode	8mA 이하	8mA 이하	8mA 이하
2. Sleep : Ear jack 연결 시	8mA 이하	8mA 이하	8mA 이하
3. 대기전류 (Sleep & Idle 평균)	8.5mA 이하@2.56s	8.5mA 이하 @ DRX 7	8.5mA 이하 @ PP5
4. BT Connected 대기	11.5mA 이하 (대기전류 대비 3mA 이하)	11.5mA 이하 (대기전류 대비 3mA 이하)	11.5mA 이하 (대기전류 대비 3mA 이하)
5. Talk Mode	-Max Pwr시 850mA이하@ Avg -Tx 10dBm 350mA이하@ Avg	-Max Pwr 시 650mA이하@ Avg - Tx 10dBm 200 mA이하@ Avg	GSM LVL 5 350mA이하@ Avg
6. No SVC Mode	200mA 이하	200mA 이하	200mA 이하
7. Power Off Mode (전원검사 공정시 별도 spec적용)	500uA 이하 (1mA 이하)	500uA 이하 (1mA 이하)	500uA 이하 (1mA 이하)
8. LTE	Standby Mode (2.56s, Average Current)		8.5mA 이하
	LTE Cat4 @ Tx 10dBm (LCD OFF)		350 mA 이하

2. RSSI

F400S

- Network Mode는 각 Band Only상태에서 측정함.
- RF 소모 전류 측정 시 LCD Off 상태에서 측정함.

bar	Specification			단위
	LTE	WCDMA	GSM	
4→3	-100~-106	-97~-105	-91~-97	Antenna BAR
3→2	-106~-112	-101~-109	-95~-101	
2→1	-112~-118	-104~-112	-99~-105	
1→0	-117~-123	-106~-114	-103~-109	

- ◆ RSSI 측정 시 Power 변동 후 10초 이상의 공백 후 측정해야 함.(RSSI의 Stability를 확보하기 위한 시간임)
- ◆ WCDMA의 경우, Uplink Parameters→Maximum Uplink Transmit Power Level을 24dBm으로 설정 후 측정
- ◆ GSM의 경우,Cell Parameters→MS Tx Power Max CCH을 DCS,PCS 는 0 / EGSM,GSM850은 5로 설정후 측정

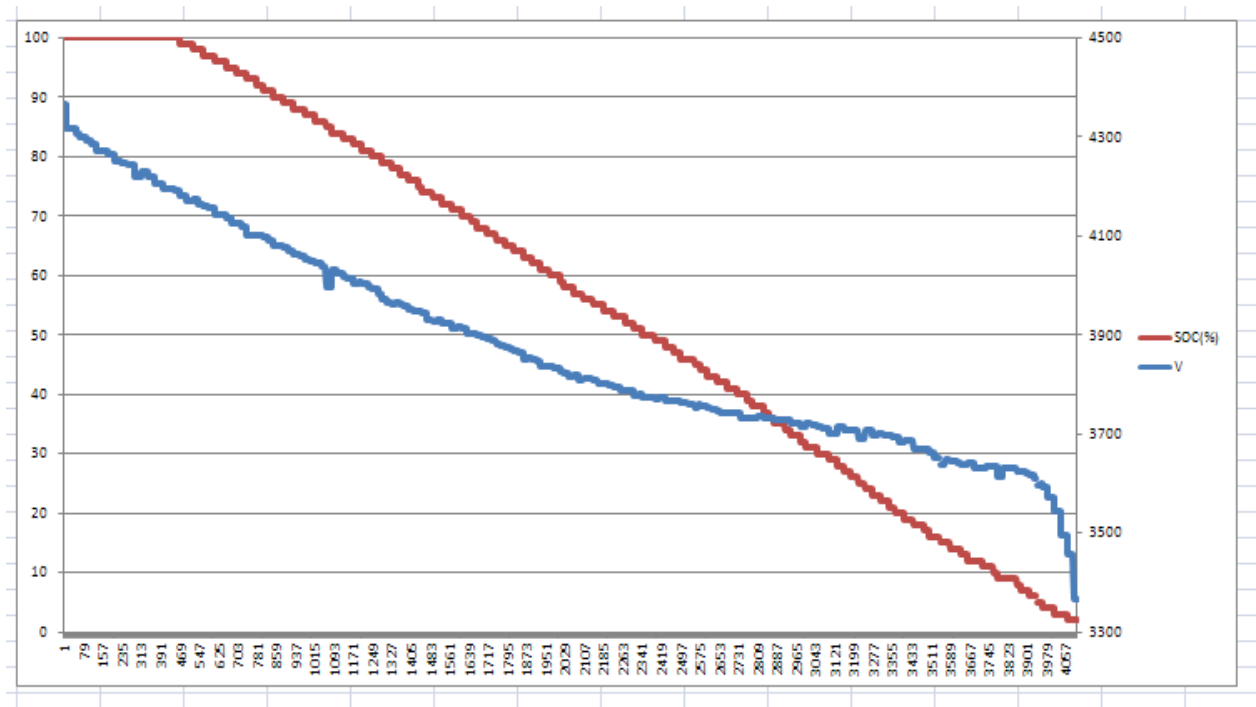
F400L/K

- Network Mode는 각 Band Only상태에서 측정함.
- RF 소모 전류 측정 시 LCD Off 상태에서 측정함.

bar	Specification			단위
	LTE	WCDMA	GSM	
5→4	-107~-113	-93~-101	-87~-93	Antenna BAR
4→3	-111~-117	-97~-105	-91~-97	
3→2	-115~-121	-101~-109	-95~-101	
2→1	-119~-125	-104~-112	-99~-105	
1→0	-122~-128	-106~-114	-103~-109	

- ◆ RSSI 측정 시 Power 변동 후 10초 이상의 공백 후 측정해야 함.(RSSI의 Stability를 확보하기 위한 시간임)
- ◆ WCDMA의 경우, Uplink Parameters→Maximum Uplink Transmit Power Level을 24dBm으로 설정 후 측정
- ◆ GSM의 경우,Cell Parameters→MS Tx Power Max CCH을 DCS,PCS 는 0 / EGSM,GSM850은 5로 설정후 측정

3. Battery Bar 방전 표시



Battery Bar	Specification	Battery Bar	Specification
Bar 20(Full)	98%이상	Bar 9 -> Bar 8	43% -> 42%
Bar 20 -> Bar 19	98% -> 97%	Bar 8 -> Bar 7	38% -> 37%
Bar 19 -> Bar 18	93% -> 92%	Bar 7 -> Bar 6	33% -> 32%
Bar 18 -> Bar 17	88% -> 87%	Bar 6 -> Bar 5	28% -> 27%
Bar 17 -> Bar 16	83% -> 82%	Bar 5 -> Bar 4	23% -> 22%
Bar 16 -> Bar 15	78% -> 77%	Bar 4 -> Bar 3	16% -> 15%
Bar 15 -> Bar 14	73% -> 72%	Bar 3 -> Bar 2	13% -> 12%
Bar 14 -> Bar 13	68% -> 67%	Bar 2 -> Bar 1	8% -> 7%
Bar 13 -> Bar 12	63% -> 62%	Bar 1 -> Bar 0	3% -> 2%
Bar 12 -> Bar 11	58% -> 57%	Power off	2% -> 1%
Bar 11 -> Bar 10	53% -> 52%	Low battery pop-up	15% , 5%, 1%
Bar 10 -> Bar 9	48% -> 47%		

4. Battery 표준 사용 시간

구분	Specification	측정 조건
대기 시간 (LTE)	350시간 이상 8.5mA 이하	2.56s @ BATT 용량 : 3000mAh
대기 시간 (WCDMA)	350시간 이상 8.5mA 이하	2.56s @ BATT 용량 : 3000mAh
대기 시간 (GSM)	350시간 이상 8.5mA 이하	EGSM PCL5 @ BATT 용량 : 3000mAh
통화 시간 (WCDMA)	900분 이상 200mA 이하	10dBm @ BATT 용량 : 3000mAh (AMR Mode시)
통화 시간 (GSM)	500분 이상 350mA 이하	EGSM, PCL5 @ BATT 용량 : 3000mAh
통화 시간 (VT)	250분 이상 700mA 이하	4G , Speaker Mode, Max volume @ BATT 용량 : 3000mAh
통화시간 (LTE)	510분 이상 350mA 이하	LTE Cat4 @ TX 10dBm & LCD OFF
충전 시간	210분 이하	LCD OFF
무선 충전시간	400분 이하	LCD OFF

5. MP3 사용시간

구 분	Specification	측정 조건
Speaker	20시간 이상	기준 곡 : ELT_Mode.mp3 -92dBm, Half Rate, Backlight Off, Volume Max.
Ear Phone	55시간 이상	

7. 8대 기본 품질

구분		Specification	Target	측정 조건
DOU (신규) (Day Of Usage)		1.0	1.0	LTE Profile Ver4.0.5 기준
발열 (ΔT) *	Game	$\Delta 20^{\circ}\text{C}$	$\Delta 20^{\circ}\text{C}$	With TA, Max 밝기
		$\Delta 18^{\circ}\text{C}$	$\Delta 18^{\circ}\text{C}$	W/O TA, Max 밝기
	WIFI Streaming	$\Delta 17^{\circ}\text{C}$	$\Delta 17^{\circ}\text{C}$	Real streaming (main Band)
	LTE	$\Delta 15^{\circ}\text{C}$	$\Delta 15^{\circ}\text{C}$	LTE TX 15dBm
	WCDMA	$\Delta 18^{\circ}\text{C}$	$\Delta 18^{\circ}\text{C}$	WCDMA TX 23dBm
Handset 수신 음량 (Acoustic)		$-10 \pm 3 \text{ dB}$	$-10 \pm 3 \text{ dB}$	Torso type 3.3 거치 (Hot spot 기준)
Earphone 수신 음량 (Ear Acoustic)		$-10 \pm 3 \text{ dB}$	$-10 \pm 3 \text{ dB}$	Torso type 3.3 거치
LCD 색 좌표		Wx = 0.3013 Wy = 0.3182	Wx = 0.3013 ± 0.025 Wy = 0.3182 ± 0.025	White 이미지 밝기 Max
LCD 밝기		Typ 400 nit Typ 140 nit	Typ 400 nit Typ 140 nit	UI Max UI Default
야외 시인성		TBD	TBD	기 출시된 타사 대표 모델 동등 이상
Camera Shutter lag		50ms ↓ (ZSL)	50ms ↓ (ZSL)	High Tier 500 Lux 기준

* ΔT = 발열 측정 부위 (MAX) – 주변온도
측정 조건 : 별도 언급 없을 시, LCD Off 상태 측정

제9조 시험 측정방법 및 규정

1. 전기적 특성 측정방법
: 제품성능 시험업무세척 LG(45)-B-4602-51 표준 시험방법에 준한다.
2. 제품 신뢰성 시험
: 제품 신뢰성 시험업무세척 LG(45)-B-4602-54 표준 시험방법에 준한다.
3. U/I검증 업무세척
: QE U/I검증 업무세척 LG(45)-B-4602-53 표준 시험방법에 준한다.
4. Field Test 업무세척
: QE Field Test 업무세척 LG(45)-B-4602-52 표준 시험방법에 준한다.

제10조 신기능 추가에 따른 시험 방법 및 SPEC.정의

1. S/W적인 신기능은 사용설명서에 언급된 방법에 준한다.

LTE DebugScreen



- 1) PLMN : MCC (mobile country code) MNC (mobile network code)
Ex) LG U+: 450 05
- 2) SRV Status: 현재 단말의 서비스 Status
Ex) No service. / Limited Service
- 3) Band / BW : 단말의 Band 및 BandWidth
Ex) B1 / 5M
- 4) EARFCN (DL/UL) : DL/UL Channel 정보 :
- 5) RSRP / TX: Serving Cell RSRP / Tx Power
- 6) RSRQ / SINR : Serving Cell RSRQ/RSRP
- 7) RSSI/Tx : RSSI / Tx power dBm기준
- 8) EMM Cause : LTE EPS Network으로 부터 받은 reject cause.
Ex) LTE_NAS_ILLEGAL_ME / LTE_NAS_NW_FAILURE ...
- 9) EMM Status
Ex) EMM_DEREGISTERED / EMM_REGISTERED
- 10) ECM Status
Ex) EMM_IDLE_STATE / EMM_CONNECTED_STATE
- 11) Cell_ID(PCI) : 현재 cell 의 정보
- 12) IMSI
- 13) GUTI
- 14) IP : 할당 받은 IP 주소

WCDMA DebugScreen(1/2)



DLCH : 단말이 잡고 있는 Downlink WCDMA channels

ULCH : 단말이 잡고 있는 Uplink WCDMA channels

PLMN : MCC (mobile country code) MNC (mobile network code)

Ex) China Unicom: 460 01

Service Domain : 현재 단말이 지원 가능한 서비스 도메인

Ex) CS (Circuit switched), PS (Packet switched), CSDPS

NOM (Network Operation Mode) : 망 동작 모드

Ex) 1 (Combined 로 등록 및 서비스), 2 (CS, PS 각각 등록 및 서비스)

LAC (Location Area Code)/RAC (Routing Area Code): CS/PS용 지역 코드

Cell : 현재 cell 의 정보:

Ex) XXX(MSC)-XX(RNC)-XXX(NODE_B)-XX(sector)-X(fa)-X(reserved)

MM (Mobility Management) Cause, state/substate/service 상태:

MM 도메인에서 망으로부터 받은 reject cause, MM state, MM substate, Service 상태

Ex) 0x0/IDLE/NORMAL/AVAIL : CS Domain으로 정상 등록 후 Idle 상태

Substate

MM_NULL_SUBSTATE - 'NULL'

MM_NO_IMSI - 'NO IMSI'

MM_PLMN_SEARCH - 'PLMN SRCH '

MM_LIMITED_SERVICE - 'LIMITED'

MM_ATTEMPTING_TO_UPDATE - 'ATTEMP UP'

MM_LOCATION_UPDATE_NEEDED - 'LU NEED'

MM_NO_CELL_AVAILABLE - 'NO CELL'

MM_PLMN_SEARCH_NORMAL_SERVICE - 'PLMN SRCH N'

MM_NORMAL_SERVICE - 'NORMAL'

GMM (GPRS MM) Cause, state/substate/ service 상태:

GMM 도메인에서 망으로부터 받은 reject cause, MM state, MM substate, service 상태

Ex) 0x0/REG/NORMAL/AVAIL : PS Domain으로 정상 REG 상태

WCDMA DebugScreen(2/2)



WL1 : Layer 1 state 정보

: IDLE , FS, ACQ, BCH, PCH, FACH, DCH, DEACTIVE,

DRX (Discontinuous Reception cycle length) : 단말기가 수신한 DRX cycle length 계수에 의해 계산된 DRX cycle length 값

Ex) DRX cycle length가 6인 경우 : $2^6 = 640\text{ms}$

RRC (Radio Resource Control): RRC state 정보

DISCONN : RRC_STATE_DISCONNECTED

CONNECTING : RRC_STATE_CONNECTING

CELL FACH : RRC_STATE_CELL_FACH

CELL DCH : RRC_STATE_CELL_DCH

CELL PCH : RRC_STATE_CELL_PCH

URA PCH : RRC_STATE_URA_PCH

RSSI/Tx : RSSI / Tx power dBm기준

ASET: Active SET의 PSC 정보 (최대 3개까지 display)

RSCP: Active SET의 RSCP (Received Signal Channel Power) dBm

EcIo: Active SET의 Energy / Interference 정보 dB

NASET: Neighbor SET의 PSC 정보 (최대 3개까지 display)

NRSCP : Neighbor SET의 RSCP (Received Signal Channel Power) dBm

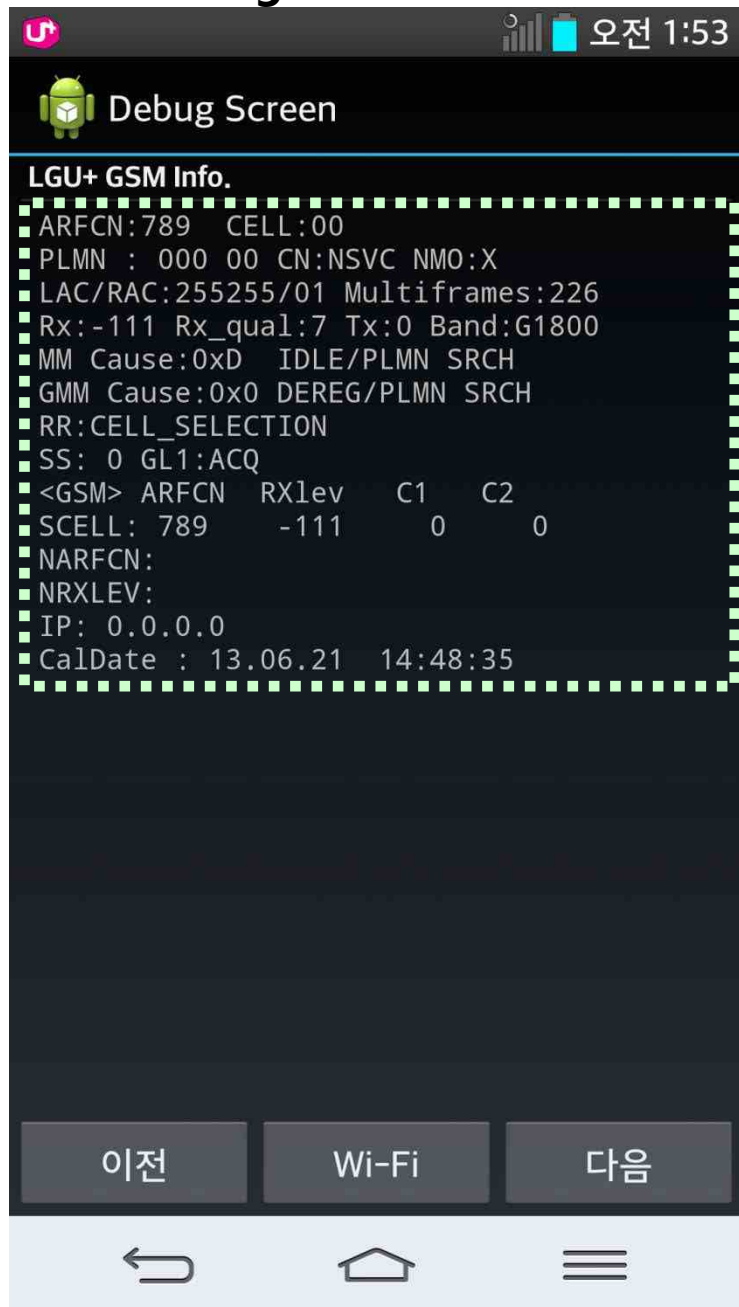
NEcIo : Neighbor SET의 Energy / Interference 정보 dB

BLER : Block Error Rate

TMSI : 위치등록 결과 망에서 받은 TMSI 정보

PTMSI : 위치등록 결과 망에서 받은 PTMSI 정보

GSM DebugScreen



ARFCN : 단말이 잡고 있는 GSM channel

CELL : 단말이 잡고 있는 GSM Cell Identifier

PLMN :: MCC MNC

Ex) China Unicom: 460 01

CN : 현재 단말이 지원 가능한 서비스 도메인

Ex) CS (Circuit switched), PS (Packet switched), CSPA

NMO : 망 동작 모드

Ex) 1 (Combined 로 등록 및 서비스), 2 (CS, PS 각각 등록 및 서비스)

LAC/RAC: CS/PS용 지역 코드

Multiframe : sleep에서 깨어나는 주기

Rx : Rx Power

RX_qual : BER 에 준한 Rx quality level 정보

Tx : Tx Power

Band: 현재 단말이 획득하고 있는 망 band 종류 Ex) IMT

MM Cause, state/substate: MM 도메인에서 망으로부터 받은 reject cause, MM state, MM substate 정보

Ex) 0x0 IDLE/NORMAL

GMM Cause, state/substate: GMM 도메인에서 망으로부터 받은 reject cause, MM state, MM substate 정보

Ex) 0x0 REG/NORMAL

RR: RR state 정보

SS : 현재 Service status 정보

Ex) 2 (In service state)

GL1 : Layer 1 state 정보

Ex) ACQ

<GSM> ARFCN Rxlev C1 C2

SCell: 현재 단말이 획득하고 있는 망 정보

(C1/C2 = Cell selection을 위한 param.)

NARFCN: 현재 단말 주변 cell channel 정보

NRXLEV : 현재 단말 주변 Rx power

다. 제품 분해 방법

Battery cover Remove

1. 분해 방법



손톱 홈을 이용하여
Cover battery를 분해

Cover Battery Remove

1. 분해 순서

A → B → C → D → E)



Battery Remove

1. 분해 순서



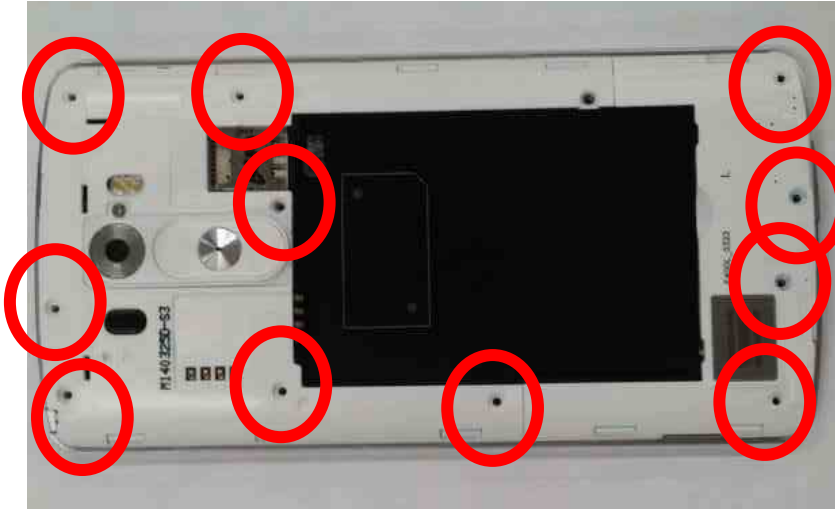
손톱홈을 이용하여 Battery 분리



Rear Cover Remove

1. 분해 방법

붉은 원형 안 Screw 12ea 를 분리 한다



Rear 양쪽을 잡아 위쪽으로 올린다.



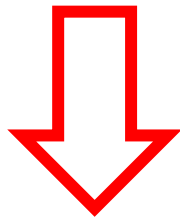
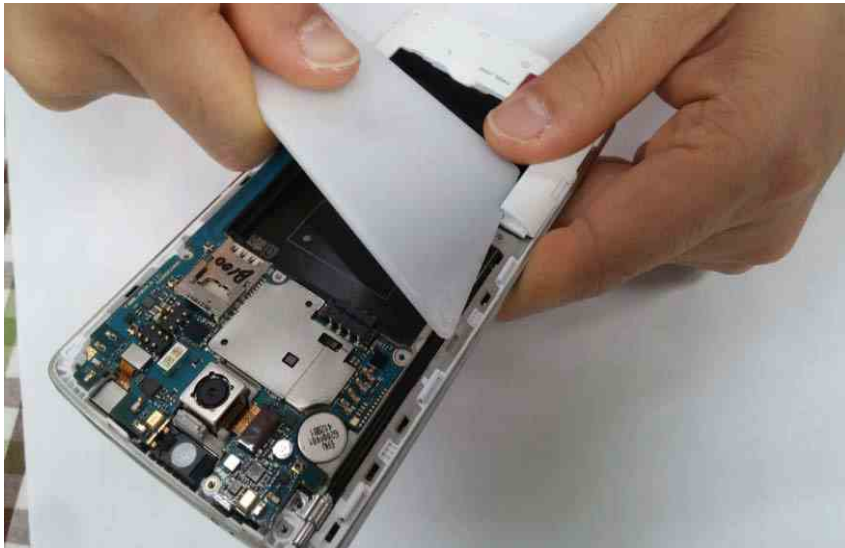
Rear Cover 를 분리 한다



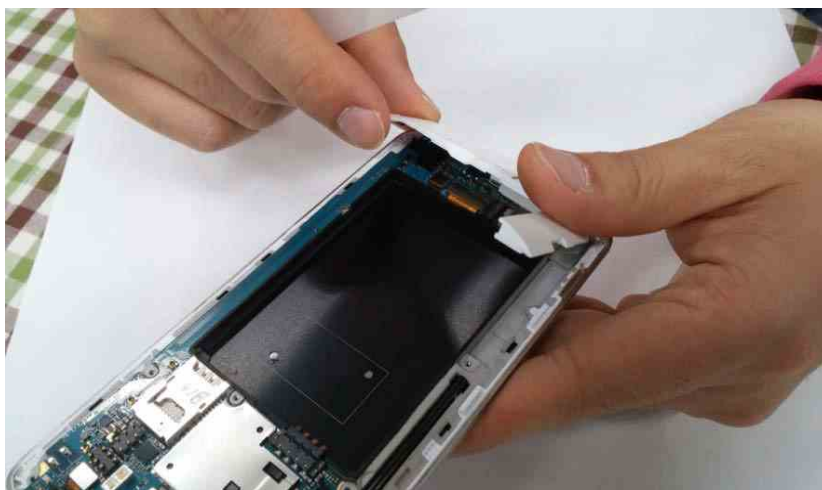
Main Antenna Remove

1. 분해 방법

헤라를 이용하여 MainAntenna를 들어올린다.



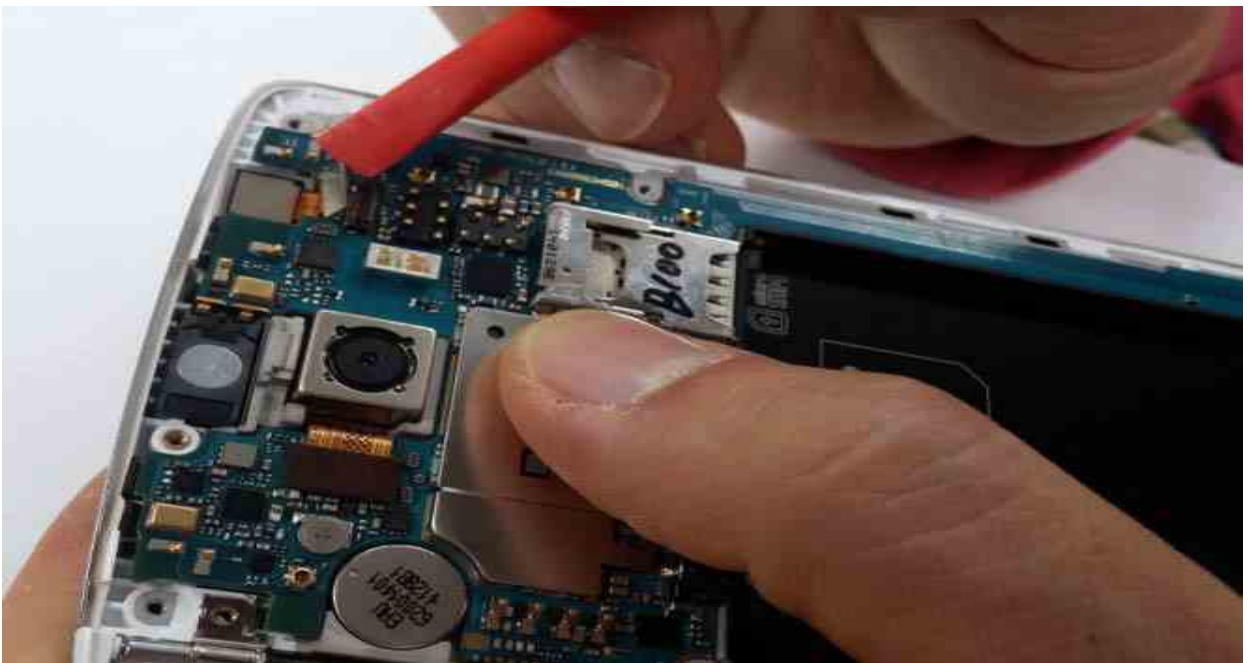
이후 손으로 MainAntenna 를 분리한다.



PCB Remove

1. 분해 방법

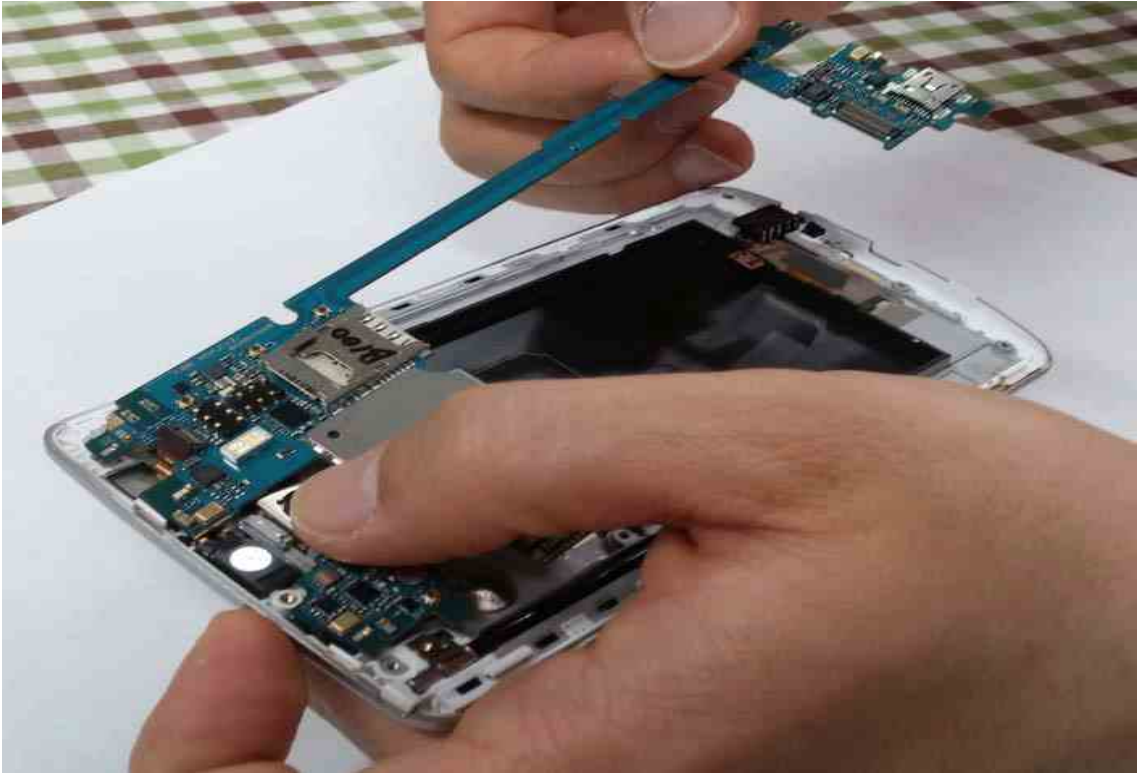
헤라를 이용하여 Connector 를 분리 후(4개소)
PCB를 분리 한다(PCB 분해 참조)



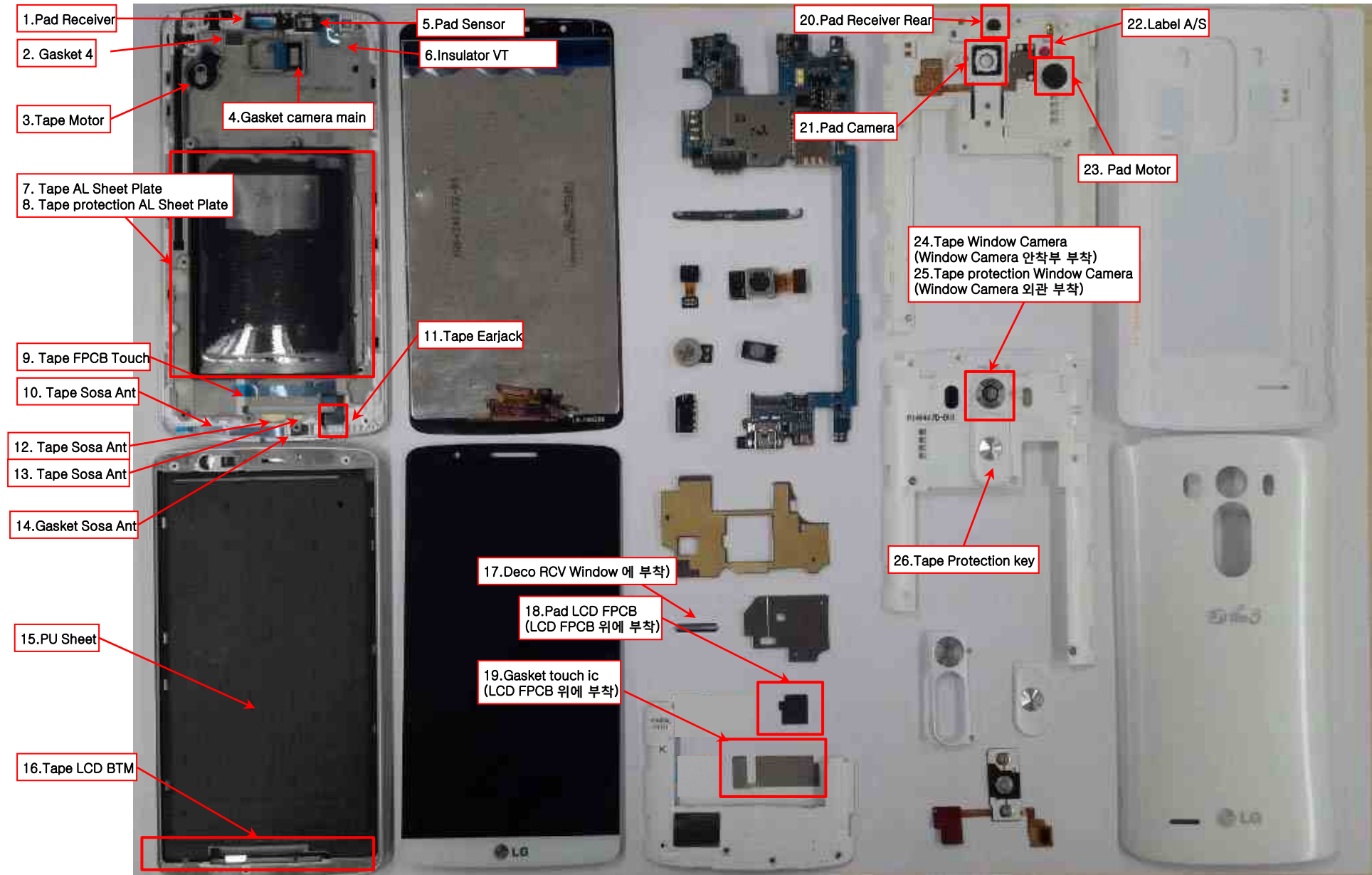
(참조) Main PCB 분해방법

1. 분해 방법

4. PCB를 들어 올린다



3. LG-F400 SVC Assembly (제품분해 전체도 및 부자재 위치)



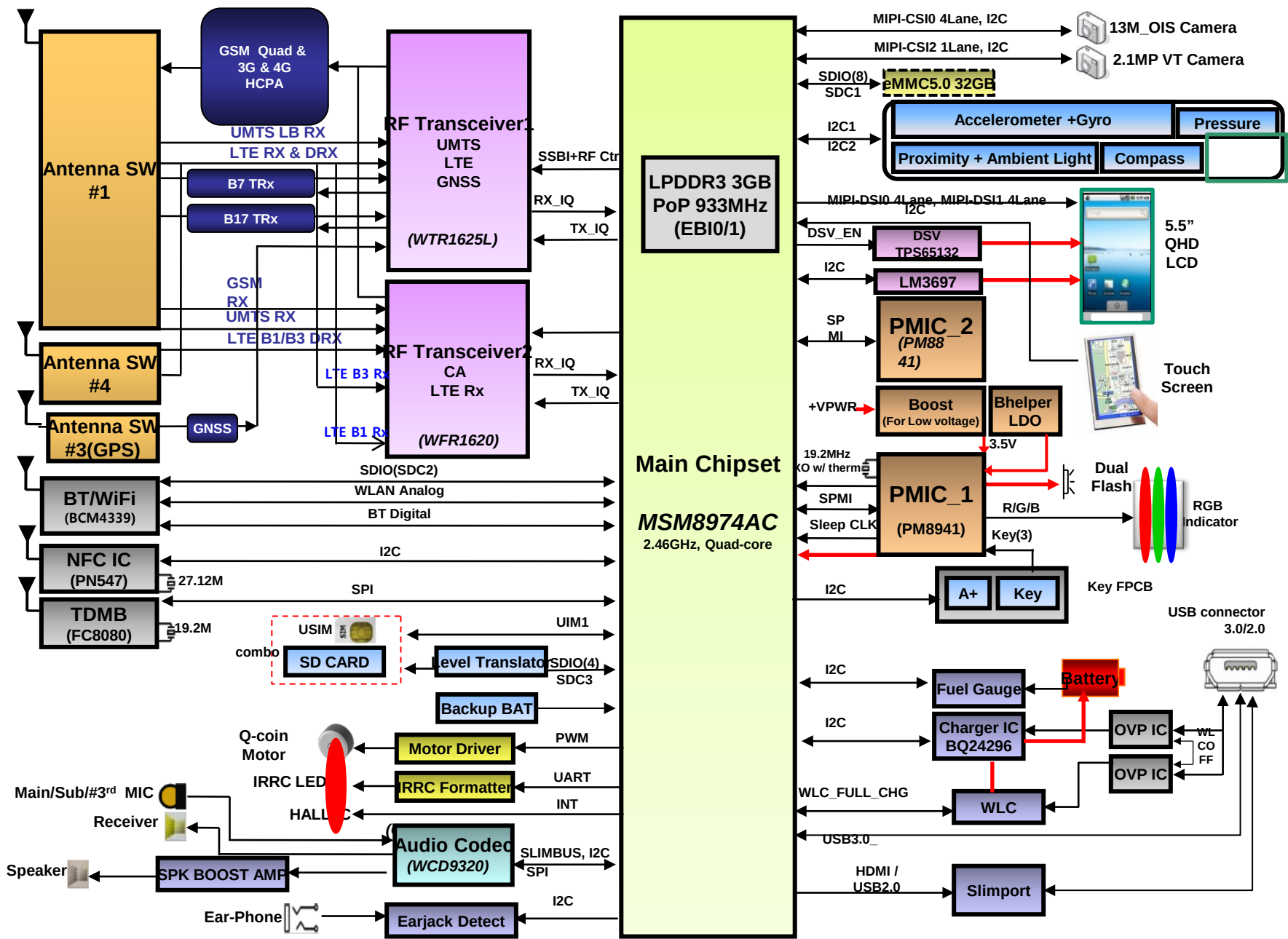
라. 블록도



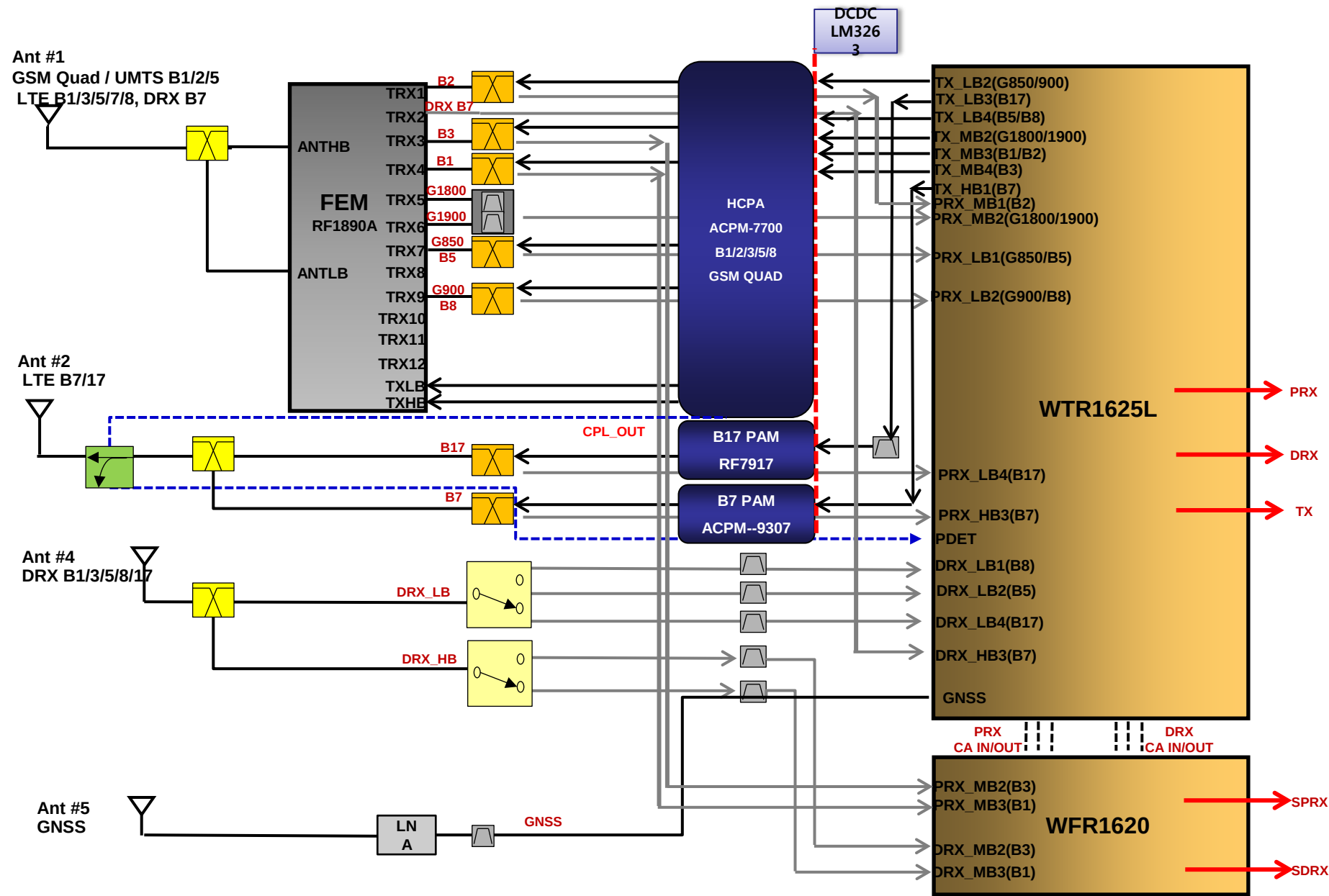
LG Electronics



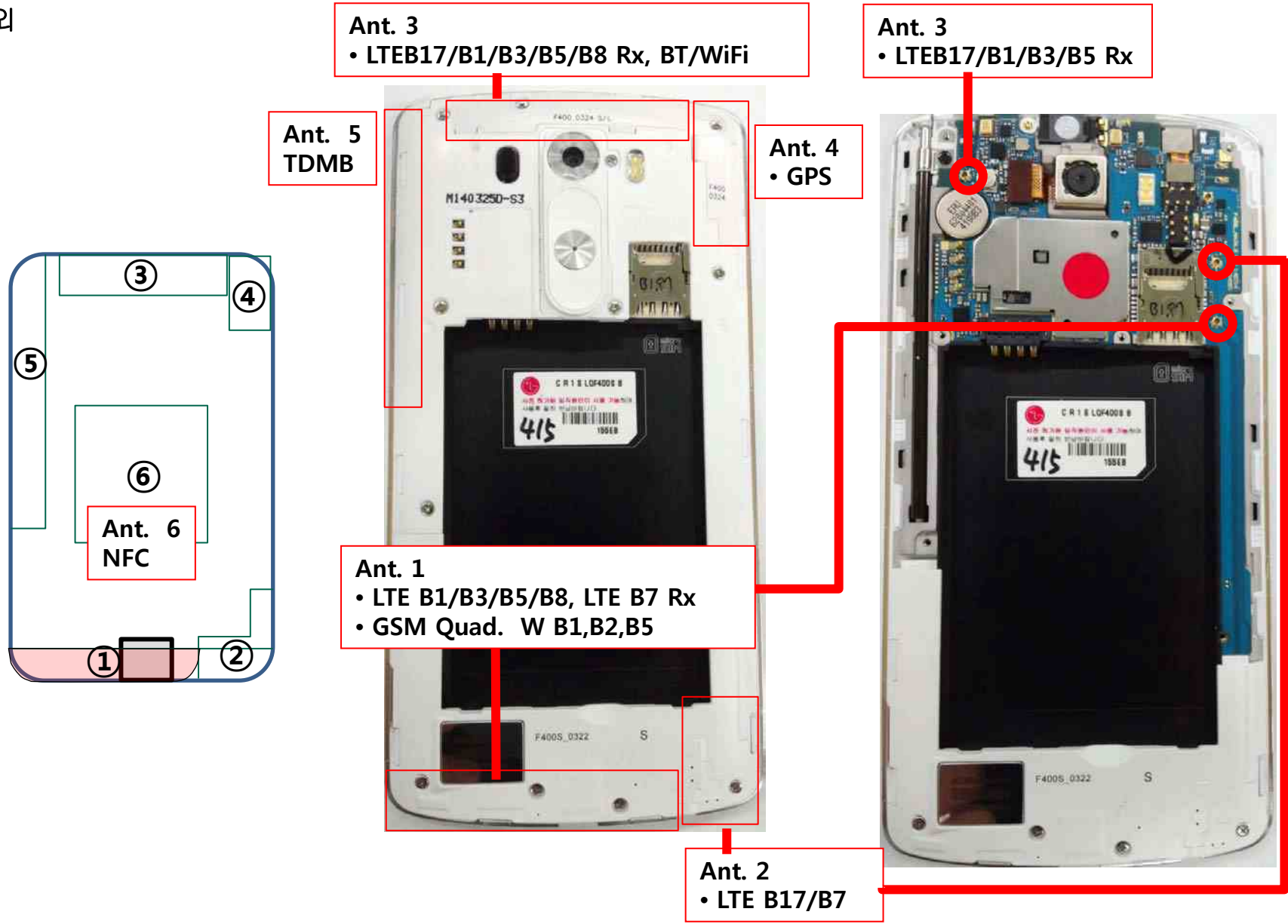
1. Total Block Diagram



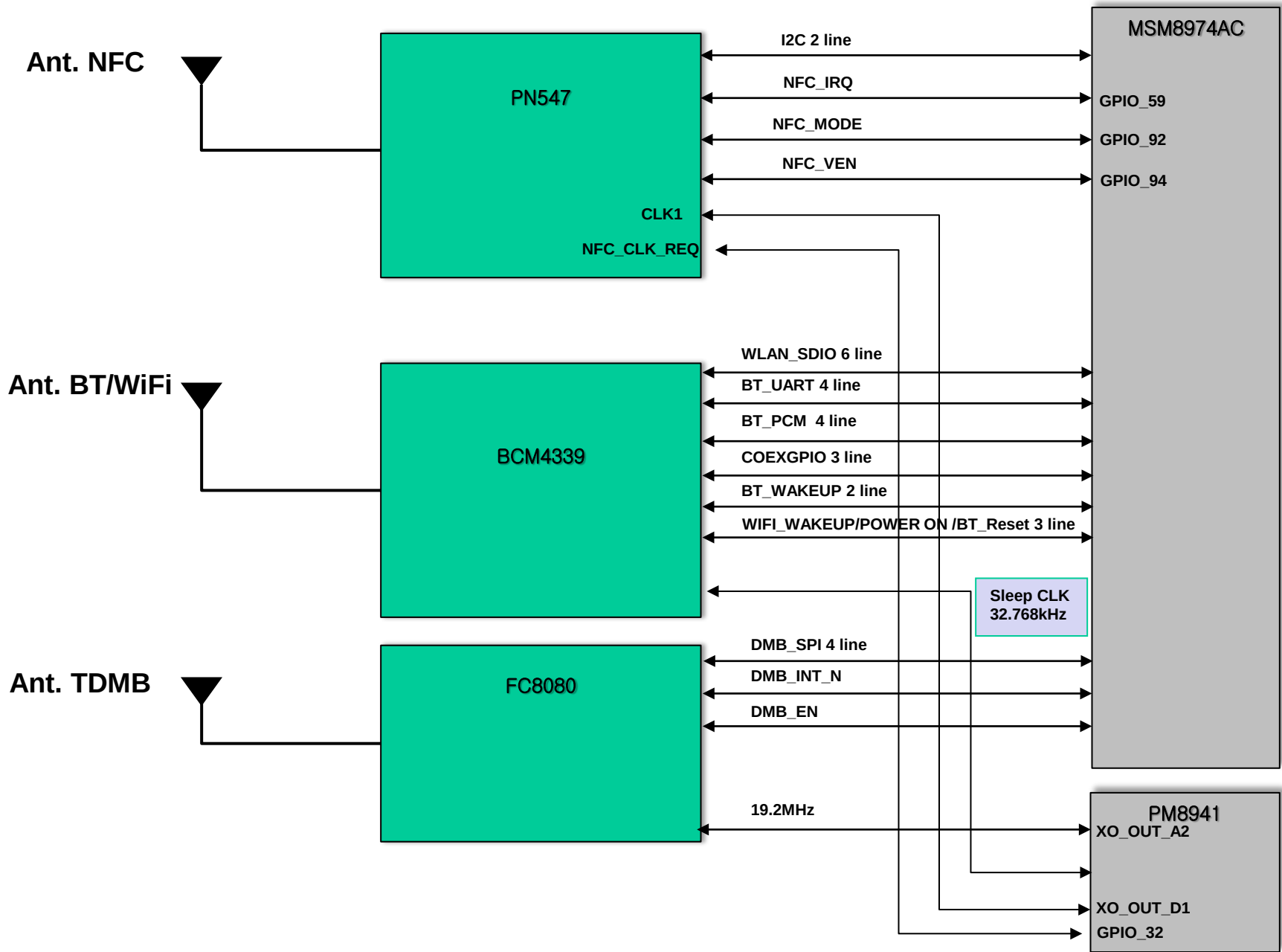
2. RF-Front Block-diagram



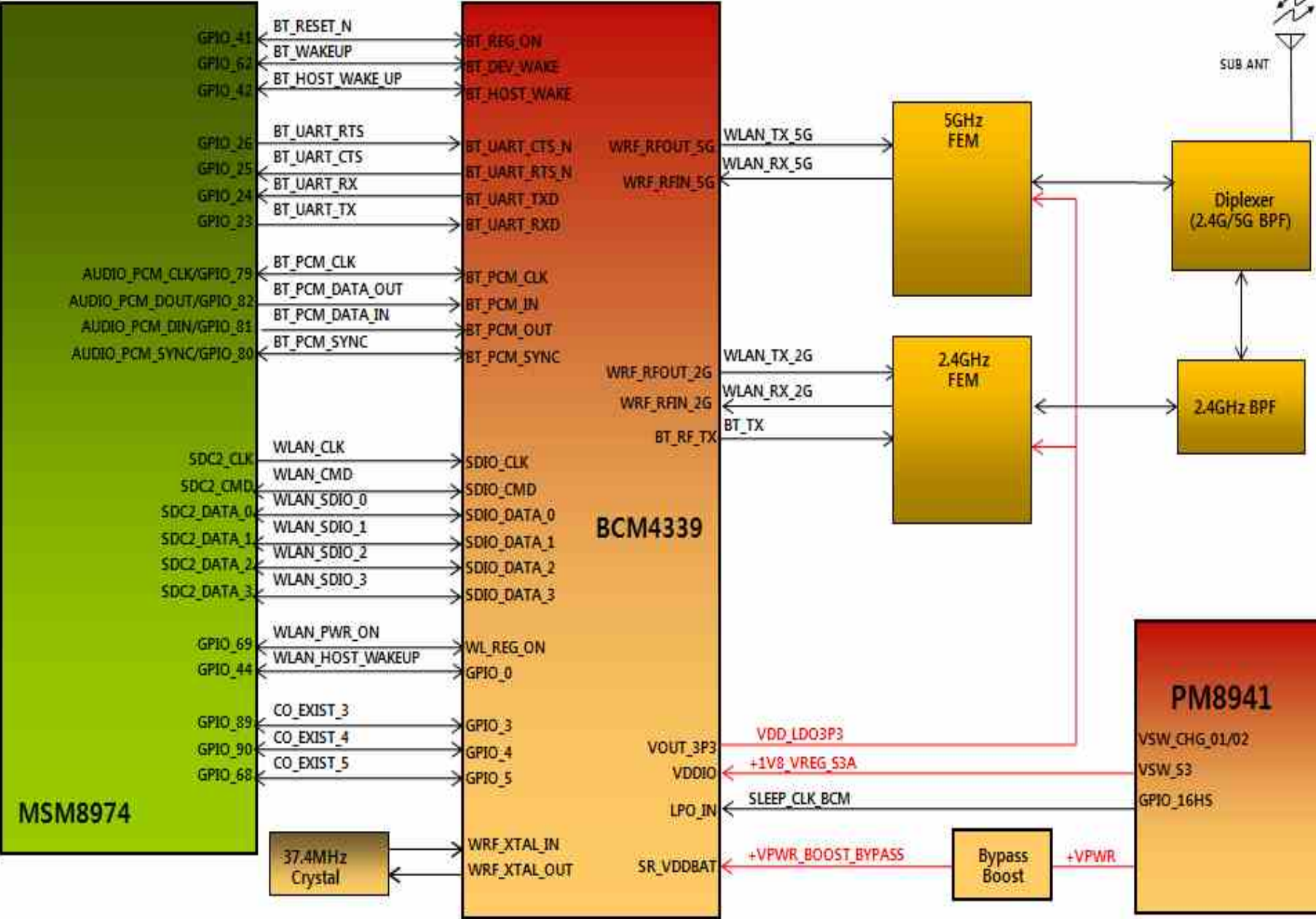
3. System HW Block Diagram ; Antenna 배치
S/L : B8 제외
K : B5 제외



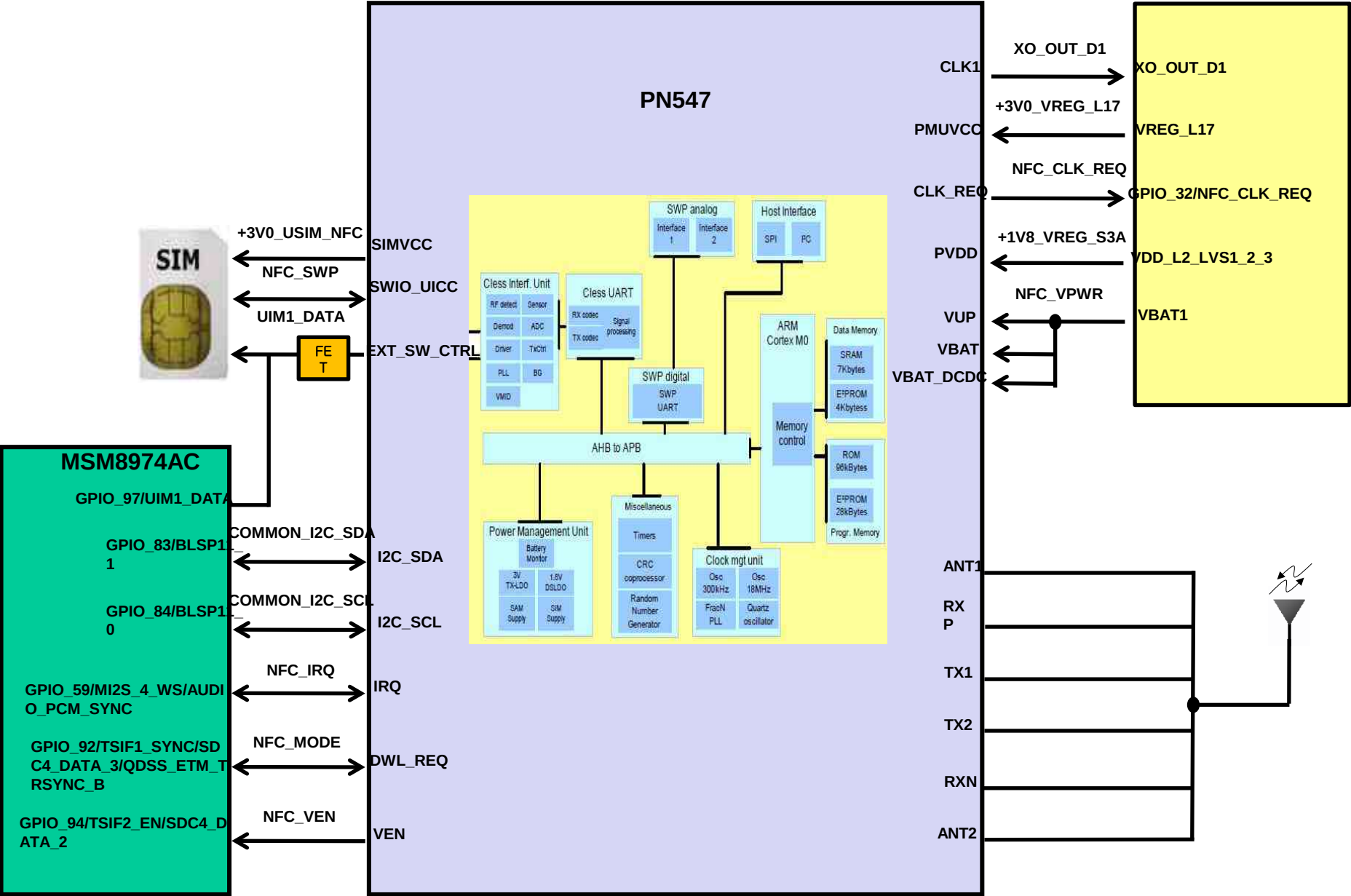
4. Connectivity Block-diagram



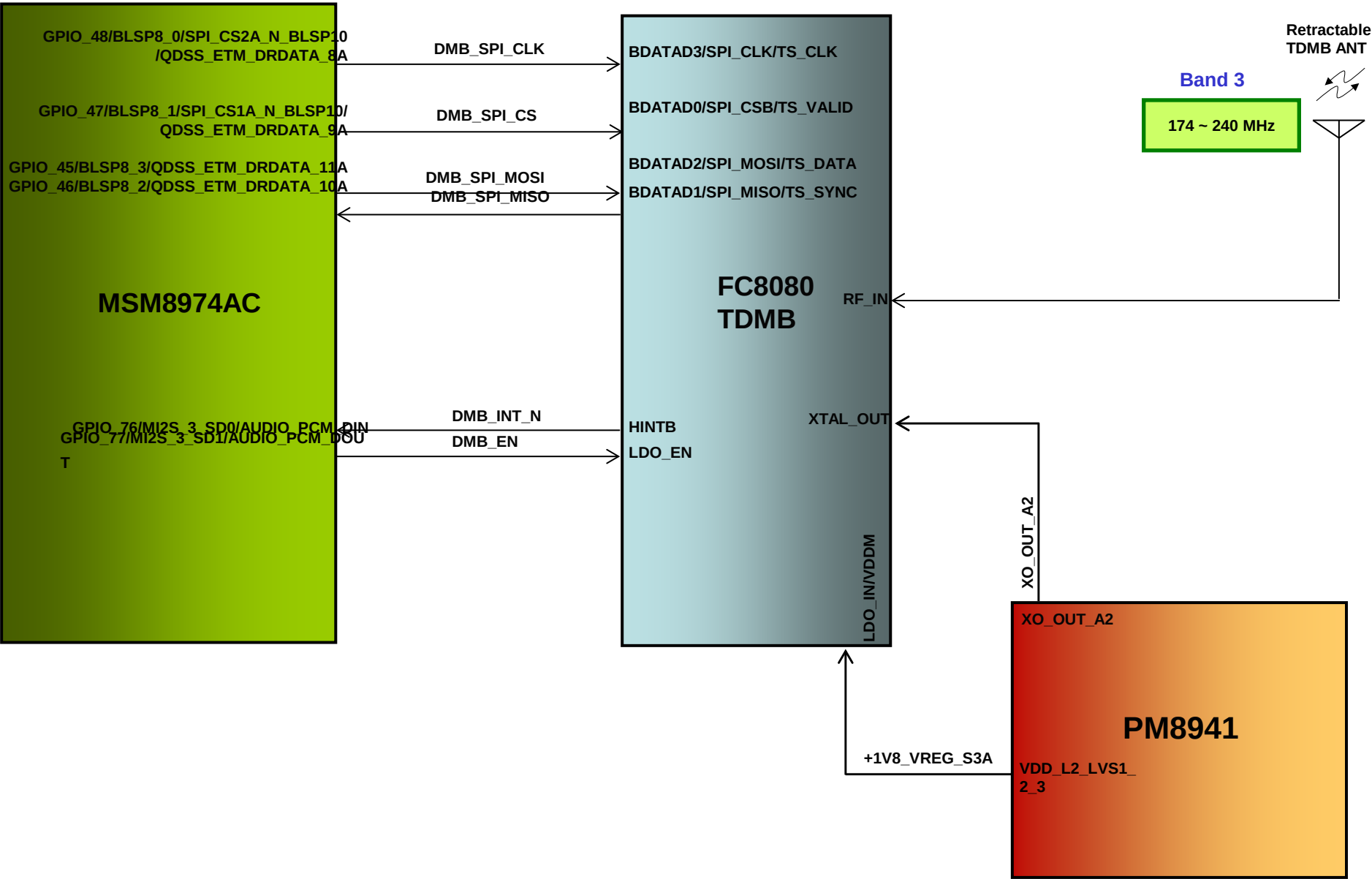
5. Connectivity Block-diagram (BT/Wi-Fi)



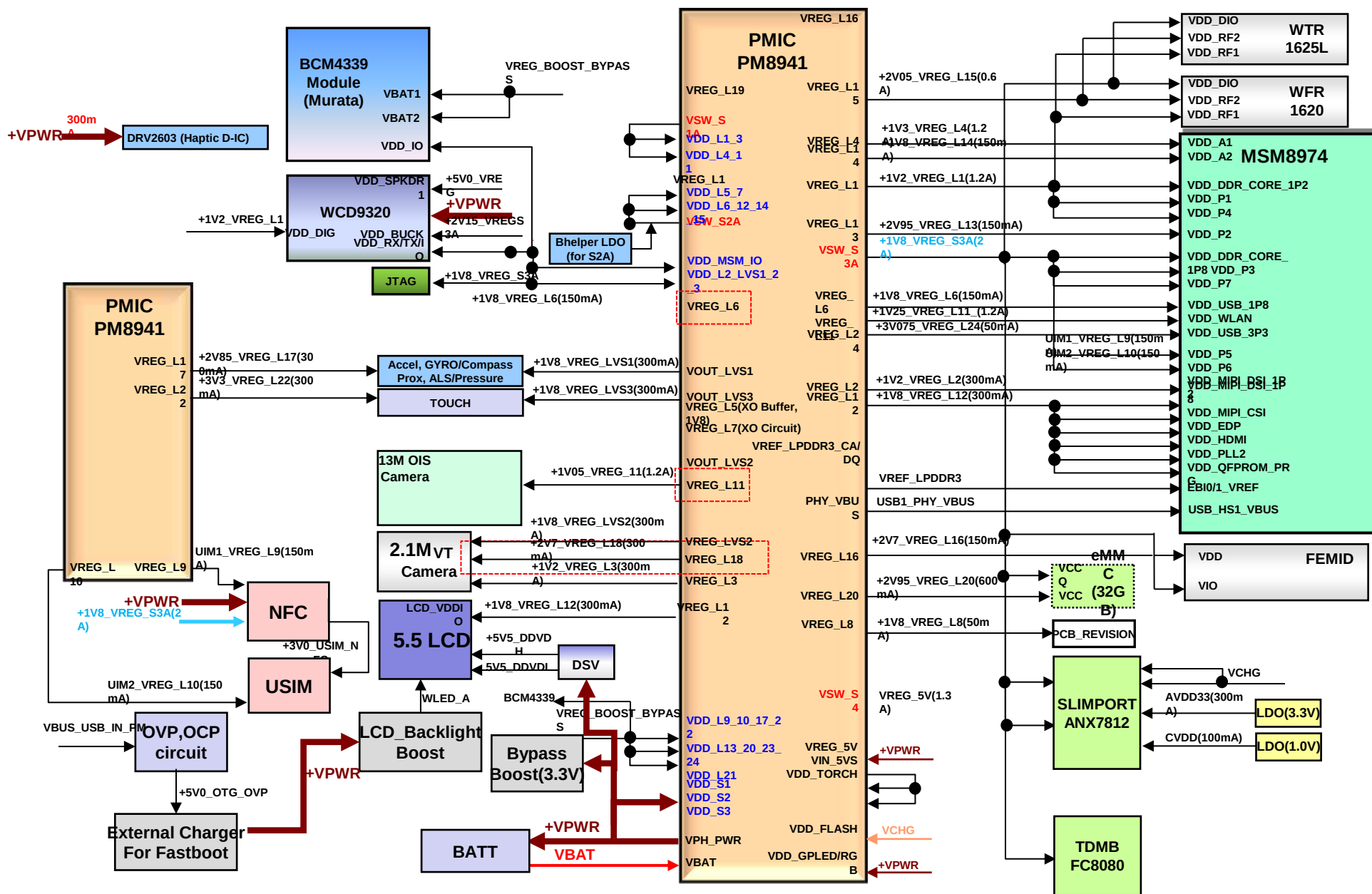
6. Connectivity Block-diagram (NFC)



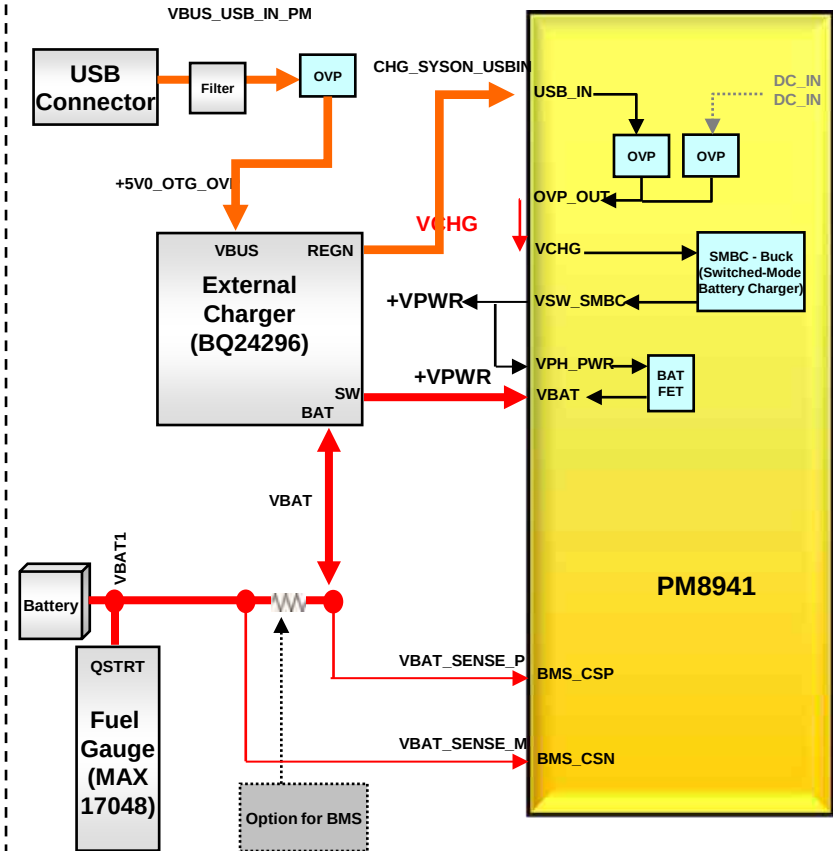
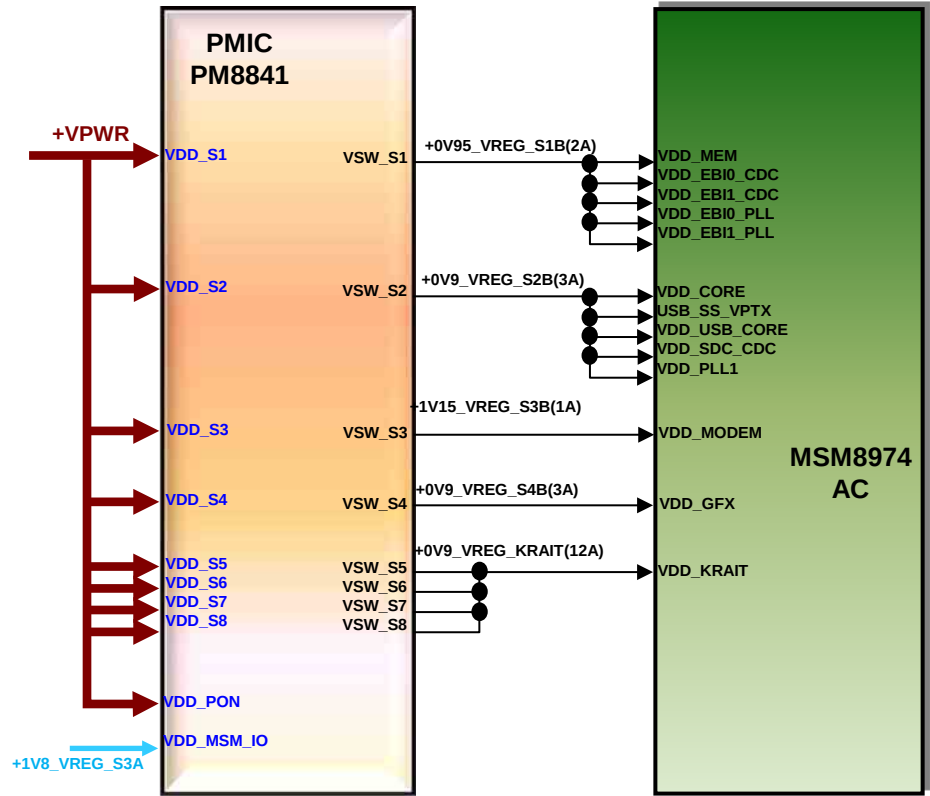
7. Connectivity Block-diagram (TDMB)



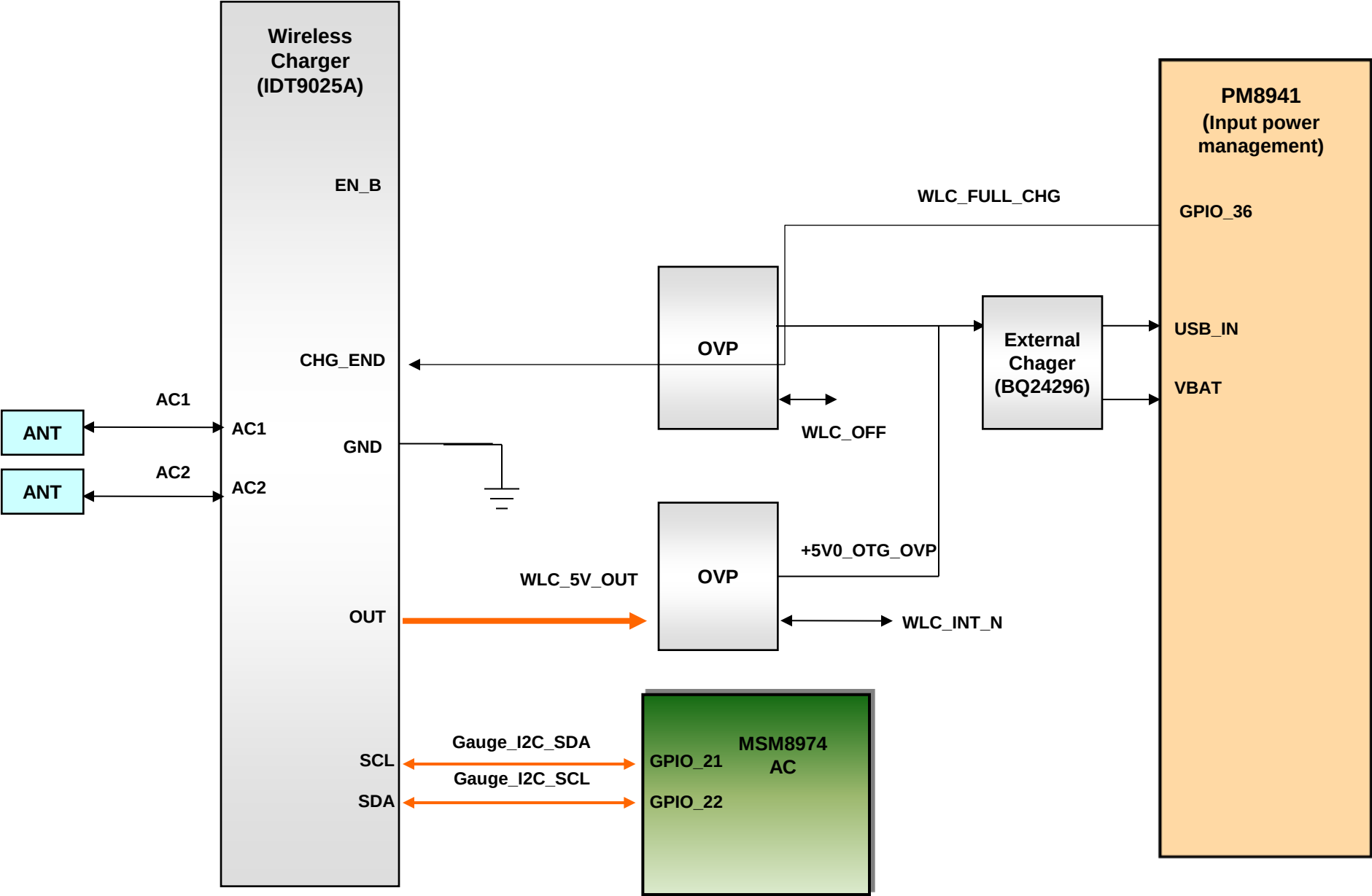
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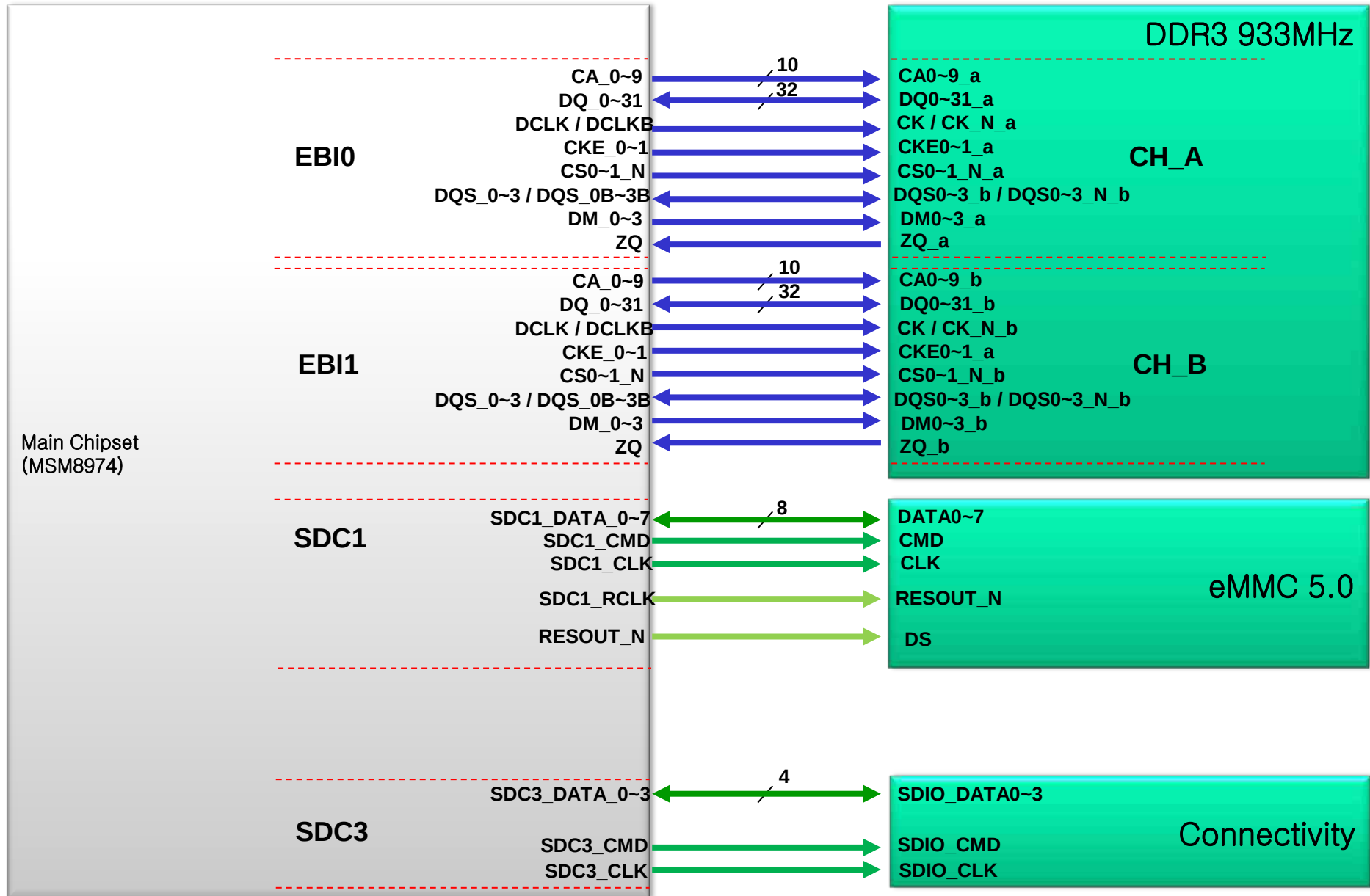
9. PM8841 SMPS Power



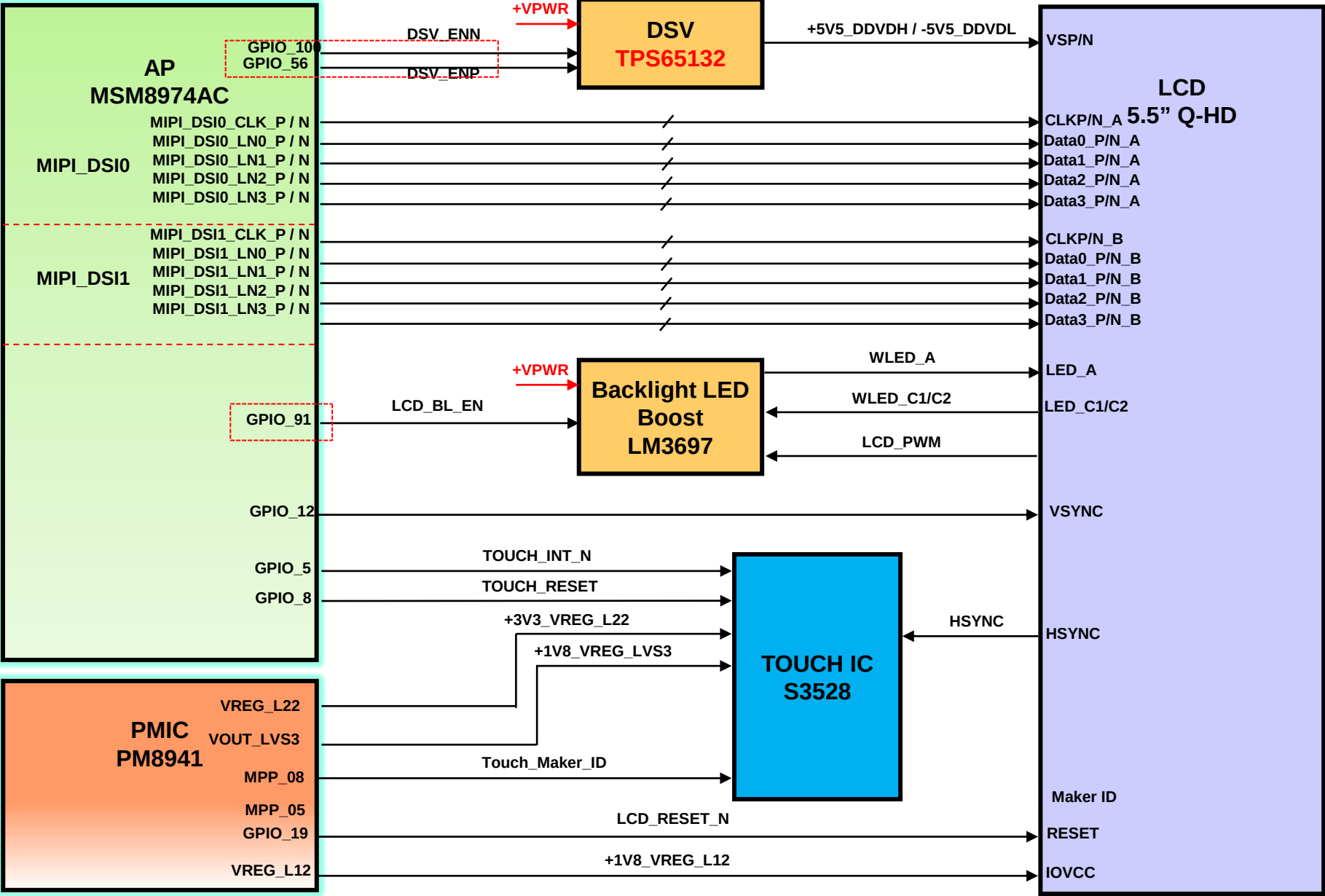
10. Wireless charger



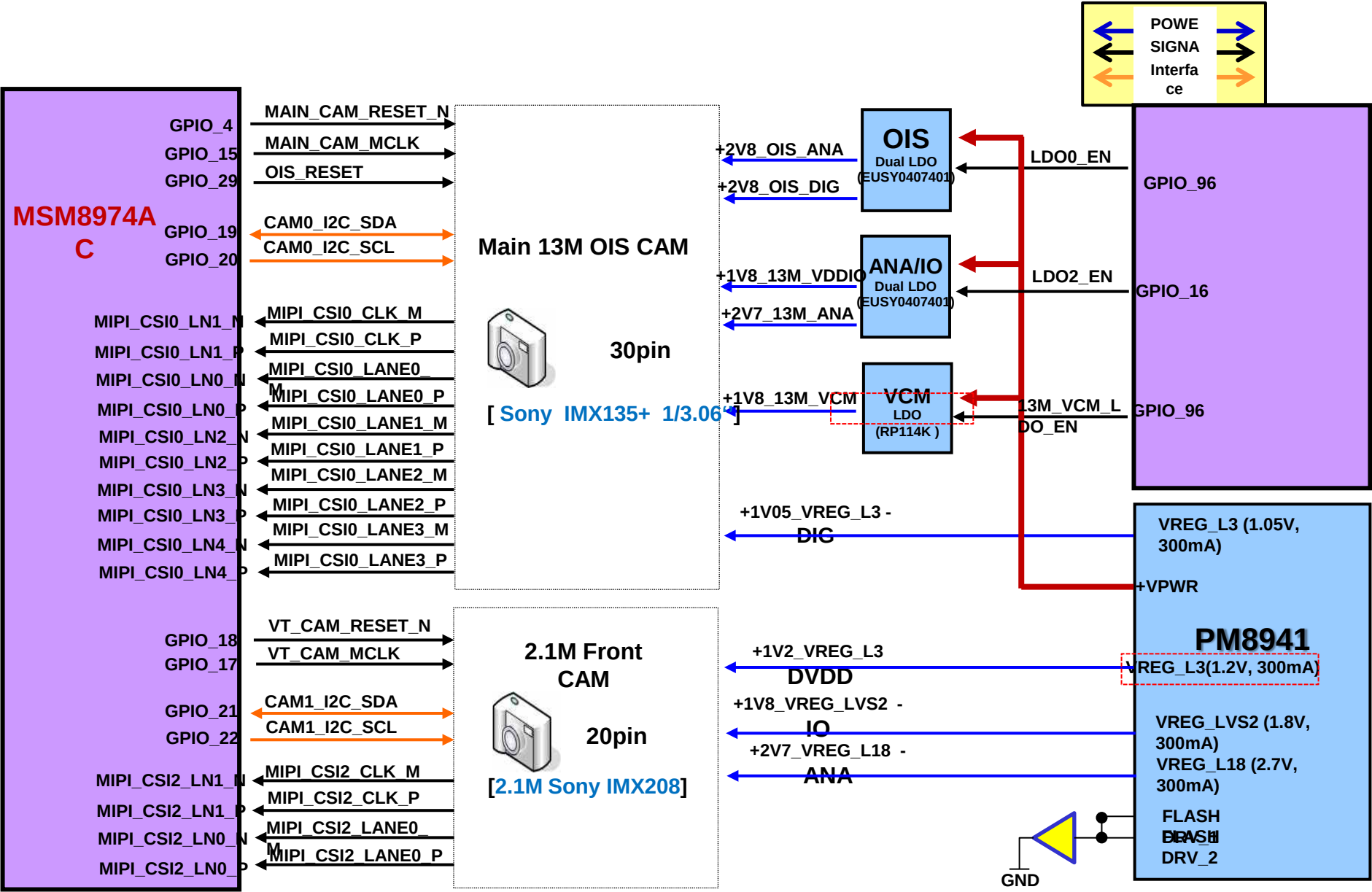
11. Memory I/F



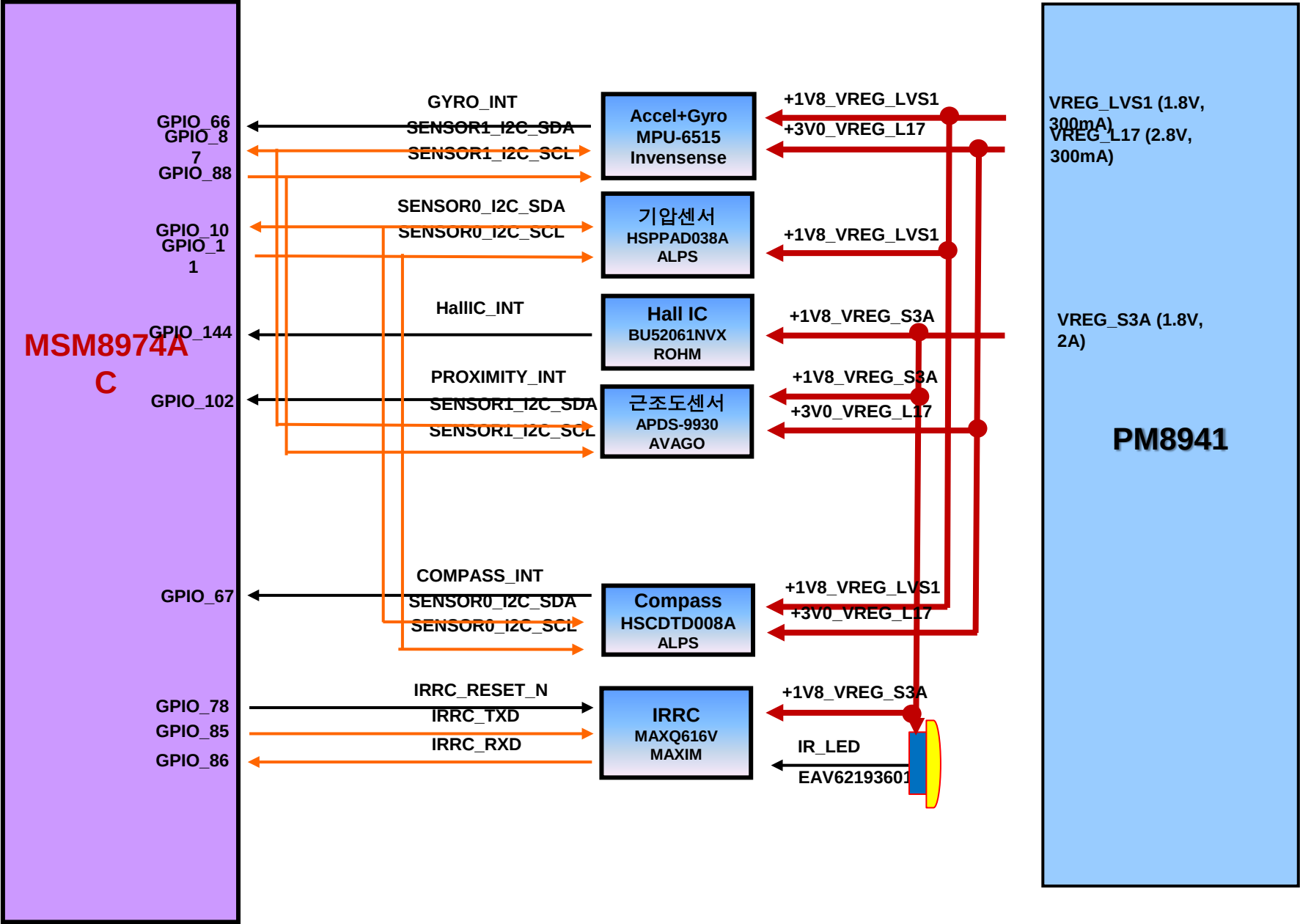
12. Main Display I/F



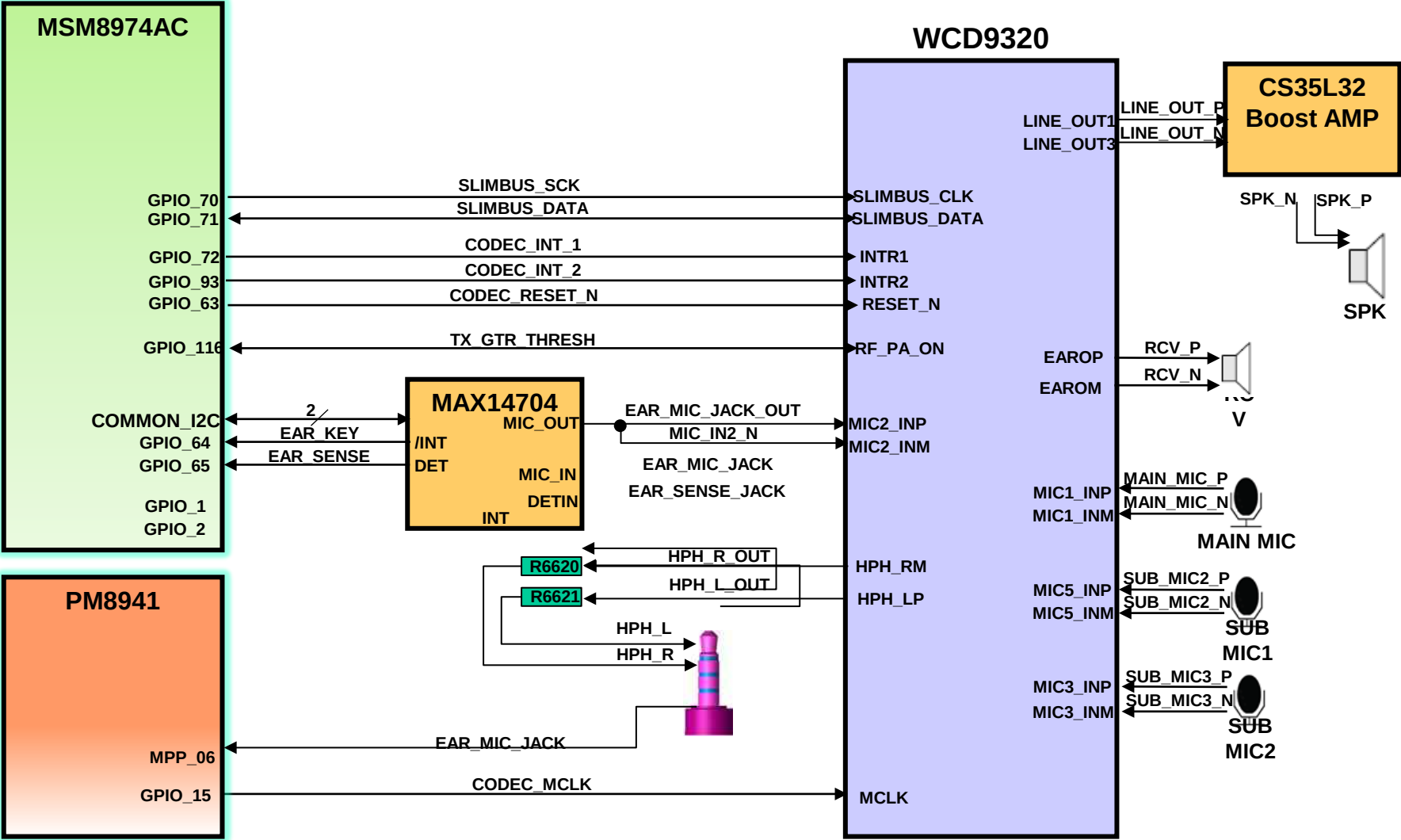
13. 13M OIS Camera, 2.1M VT Camera



14. Sensor



15. Audio Codec I/F



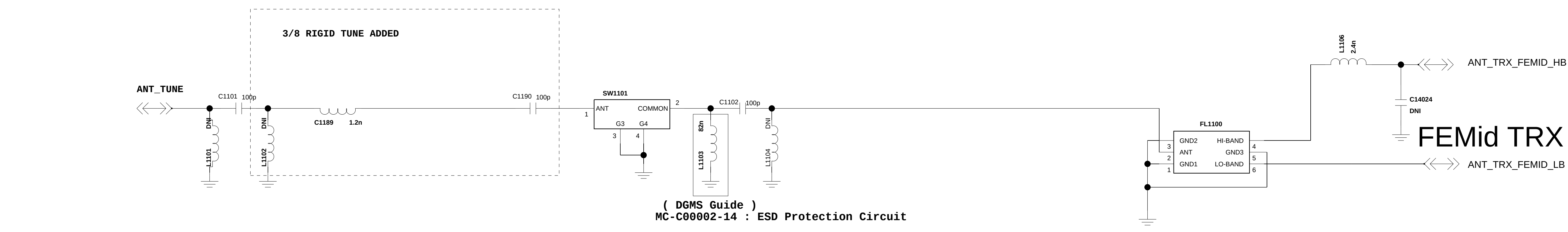
마. 회로도 및 부품 배치도



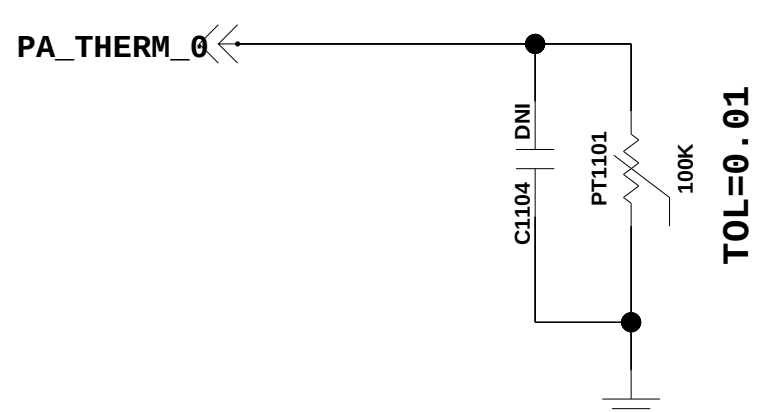
<1-3-1-20_ANT_CA_KR_2>

ANT1 (LTE 1, 3, 5, 7, 8, 17 / WCMDA 1, 2, 5 / MIMO 7/ GSM)

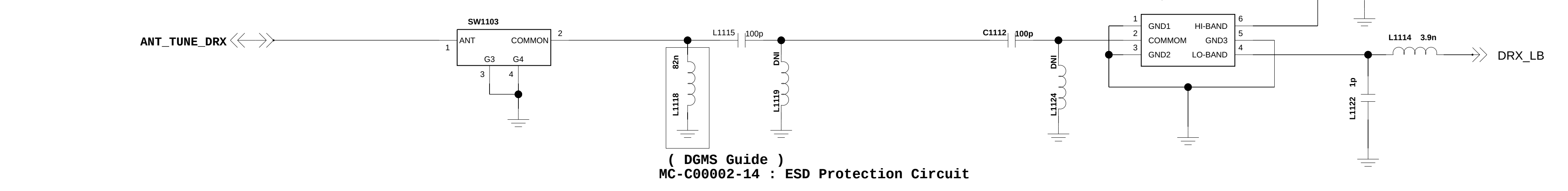
Rev_0.1



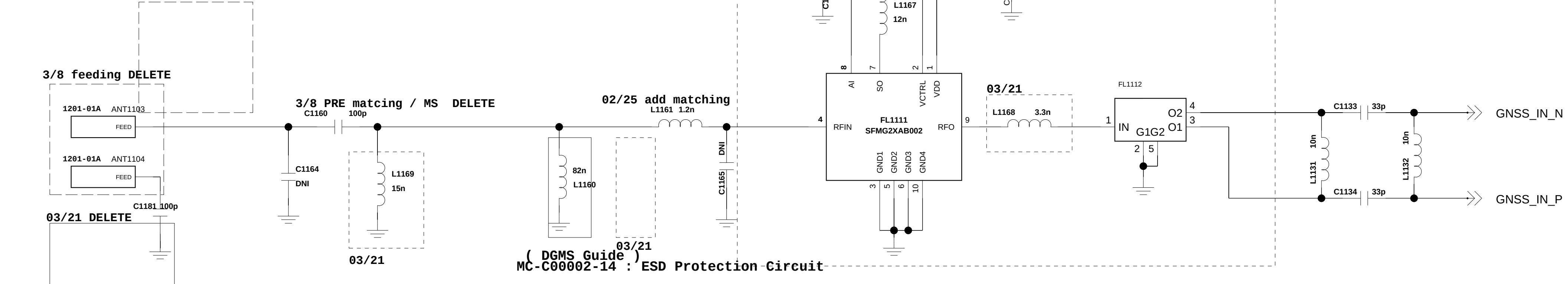
PA THERM_0



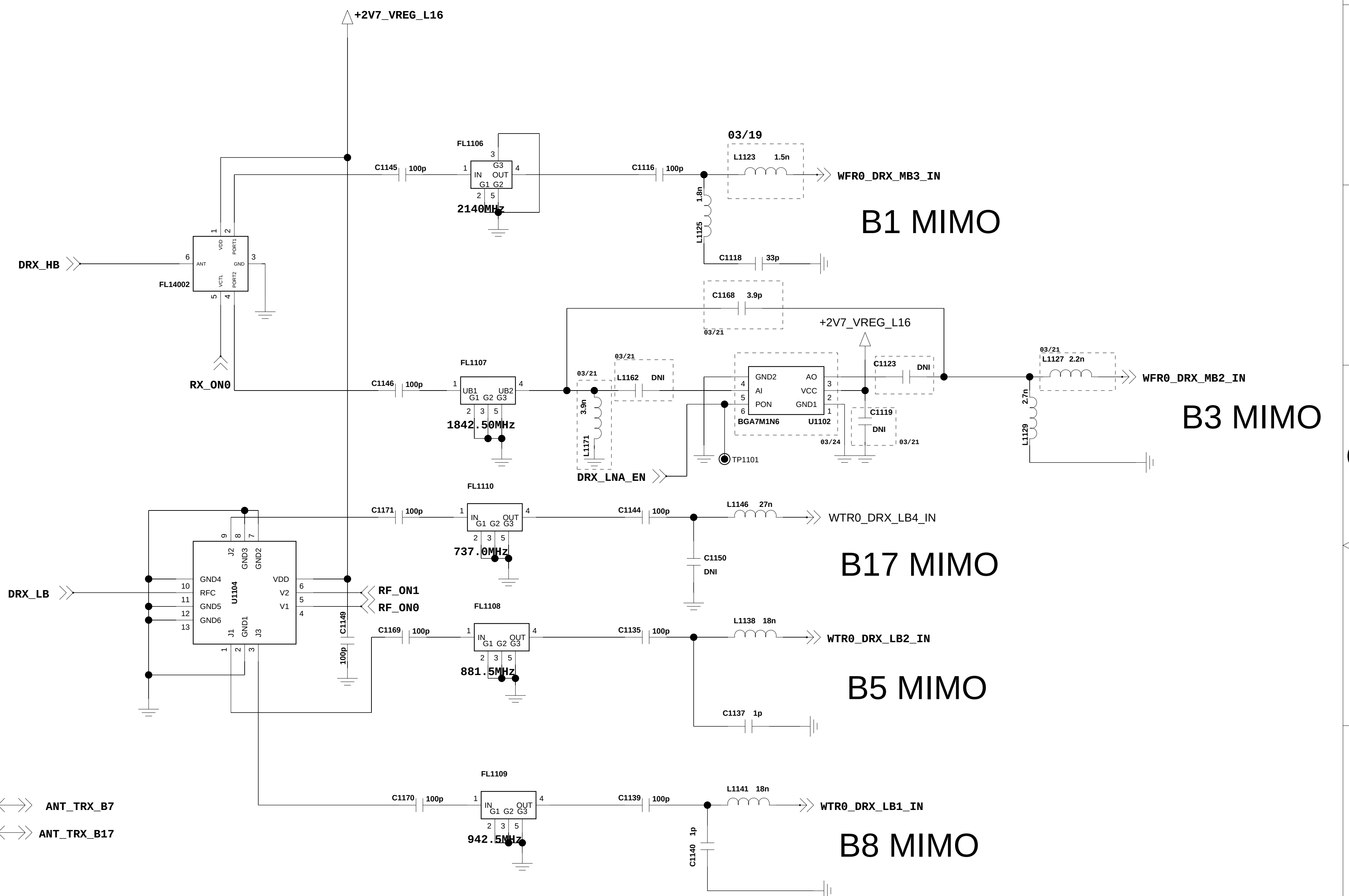
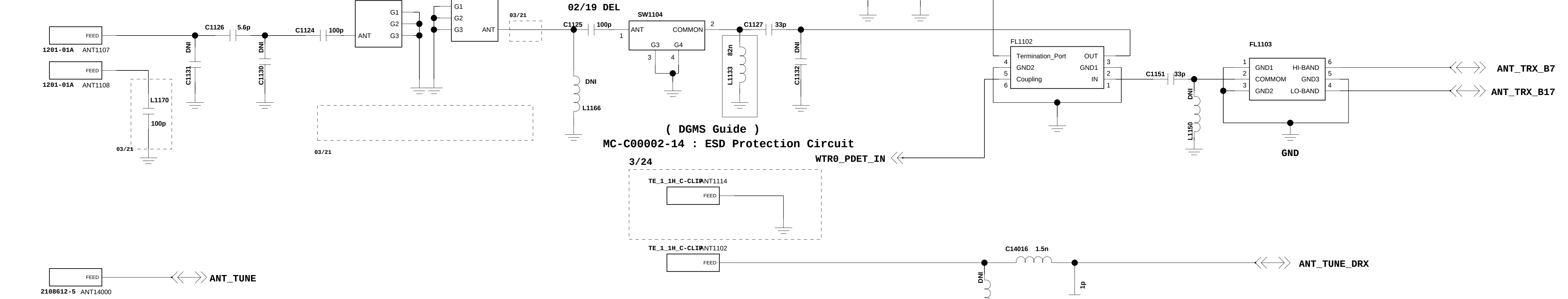
ANT3 (MIMO B1, 3, 5, 8, 17)



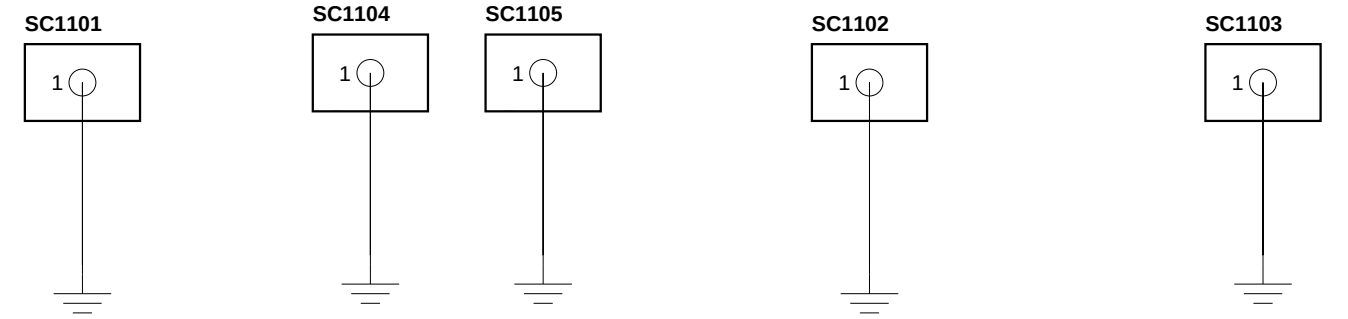
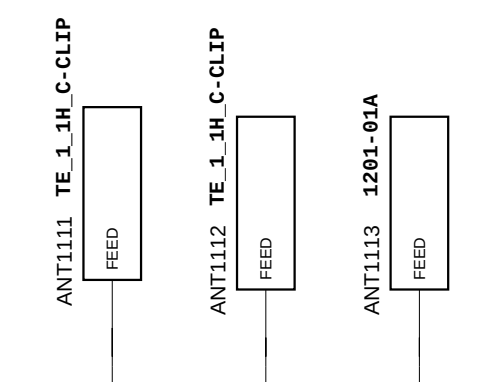
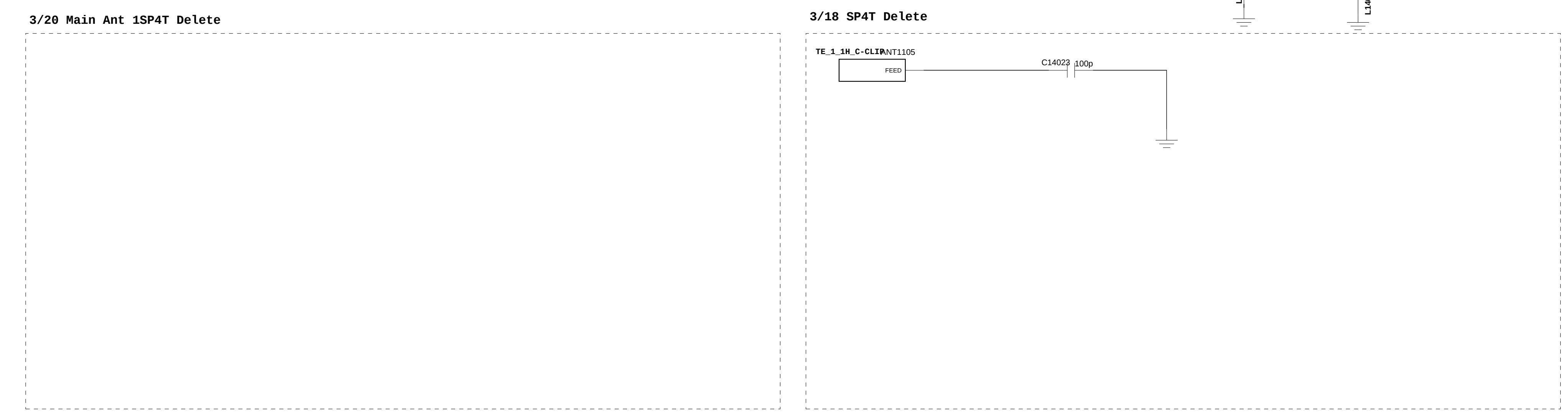
ANT5 (GNSS)



ANT2 (B7, 17)

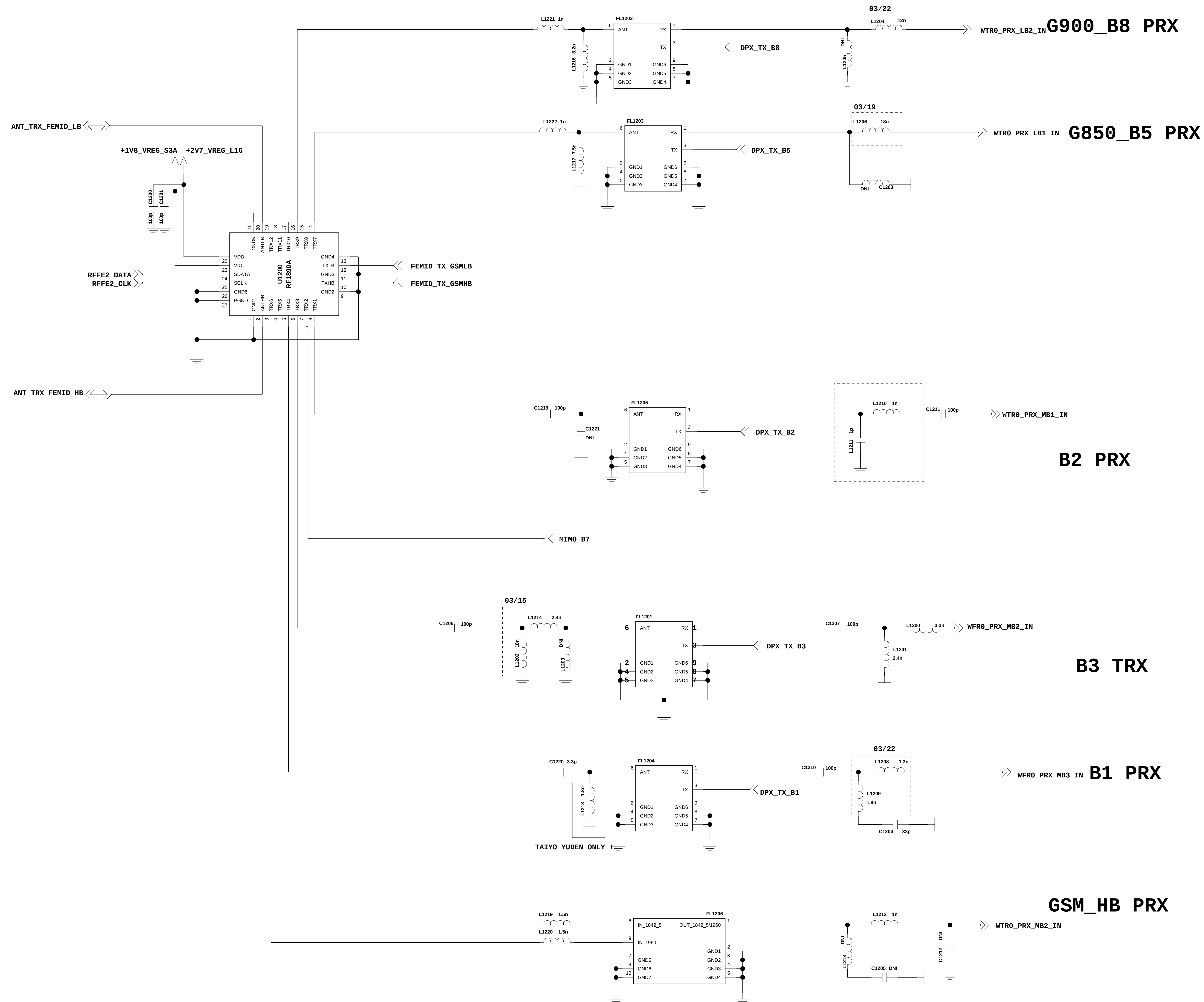


< SUS plate >

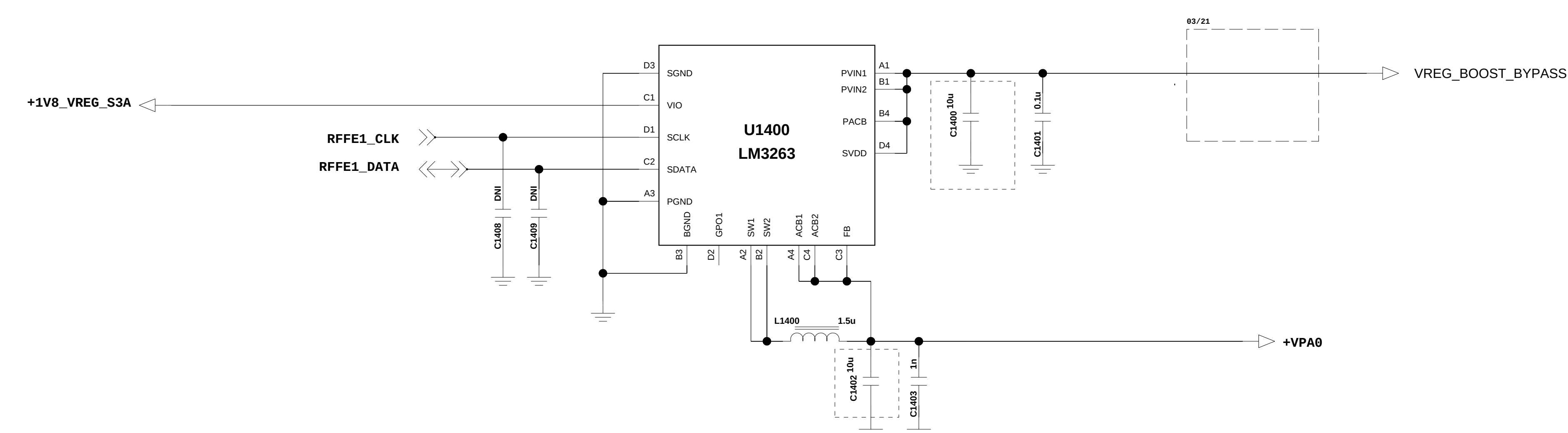


LGE Internal Use Only

Rev_0.1

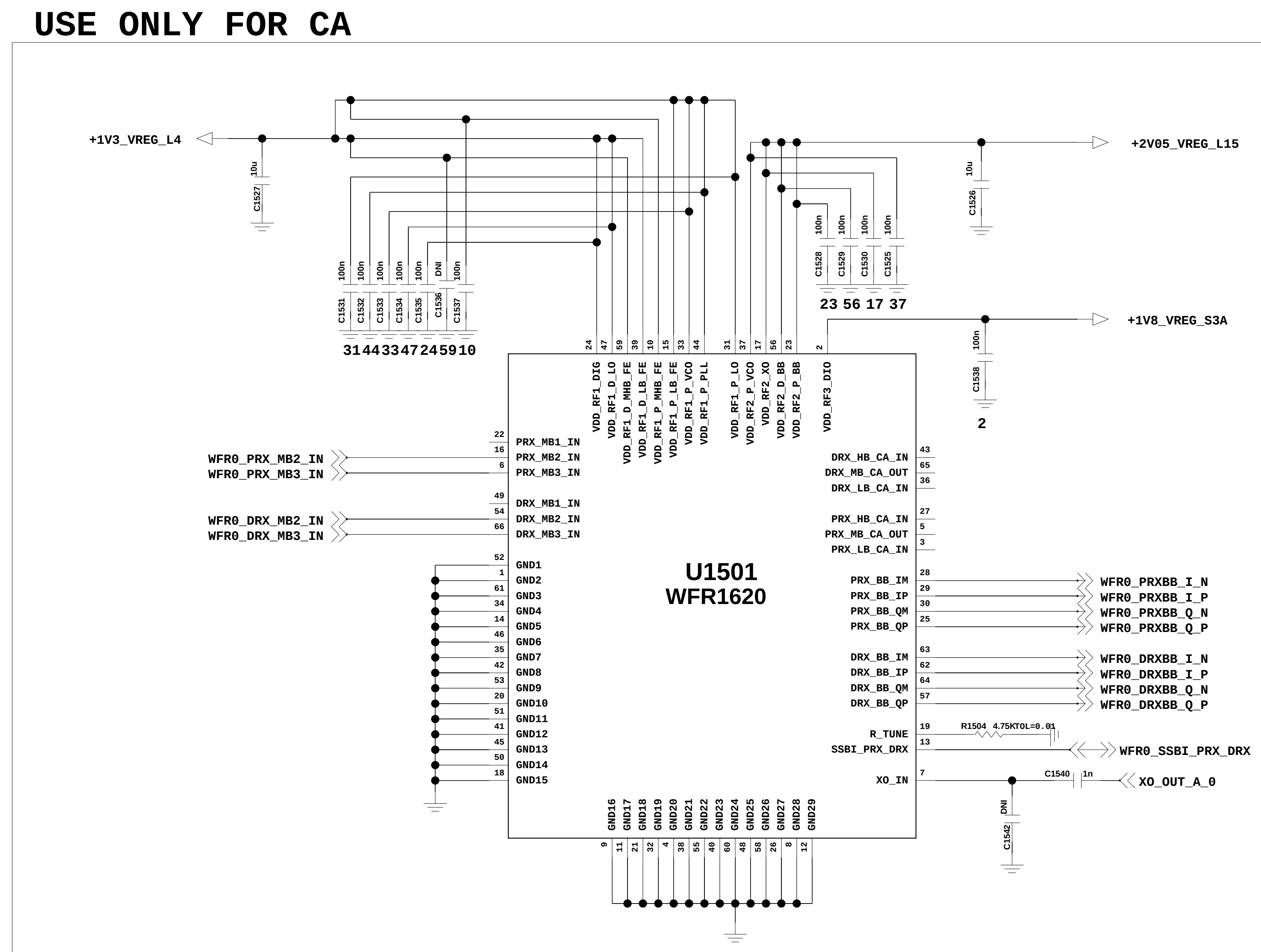


< 1-3-4-2_PT_APT_LM3263 > Rev_0.4



Rev_0.4

Rev_0.4



L

K



1

H



F

E

D

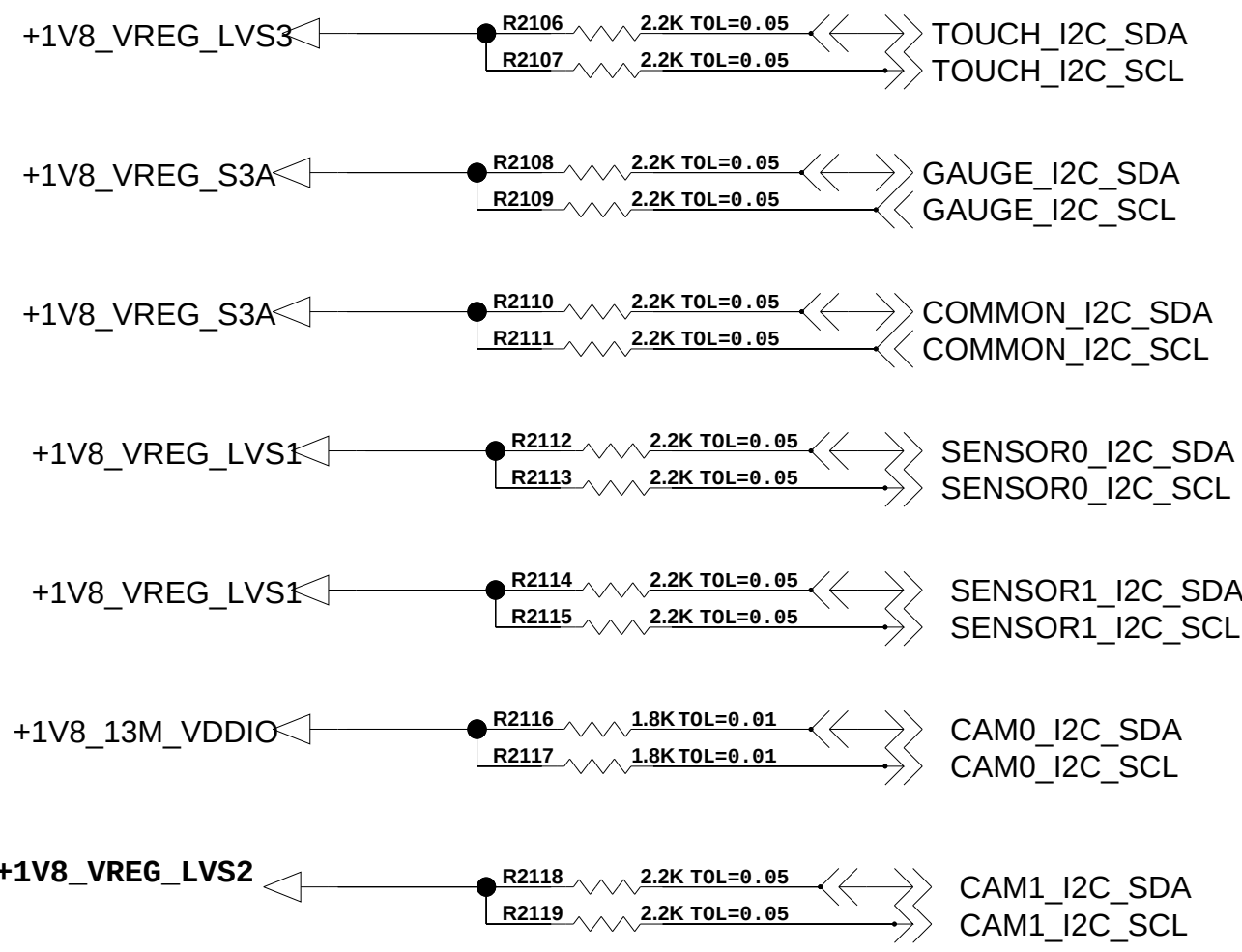
C

B

A

<2-1-2-2-1_MSM8974AC_Control_GPIO>

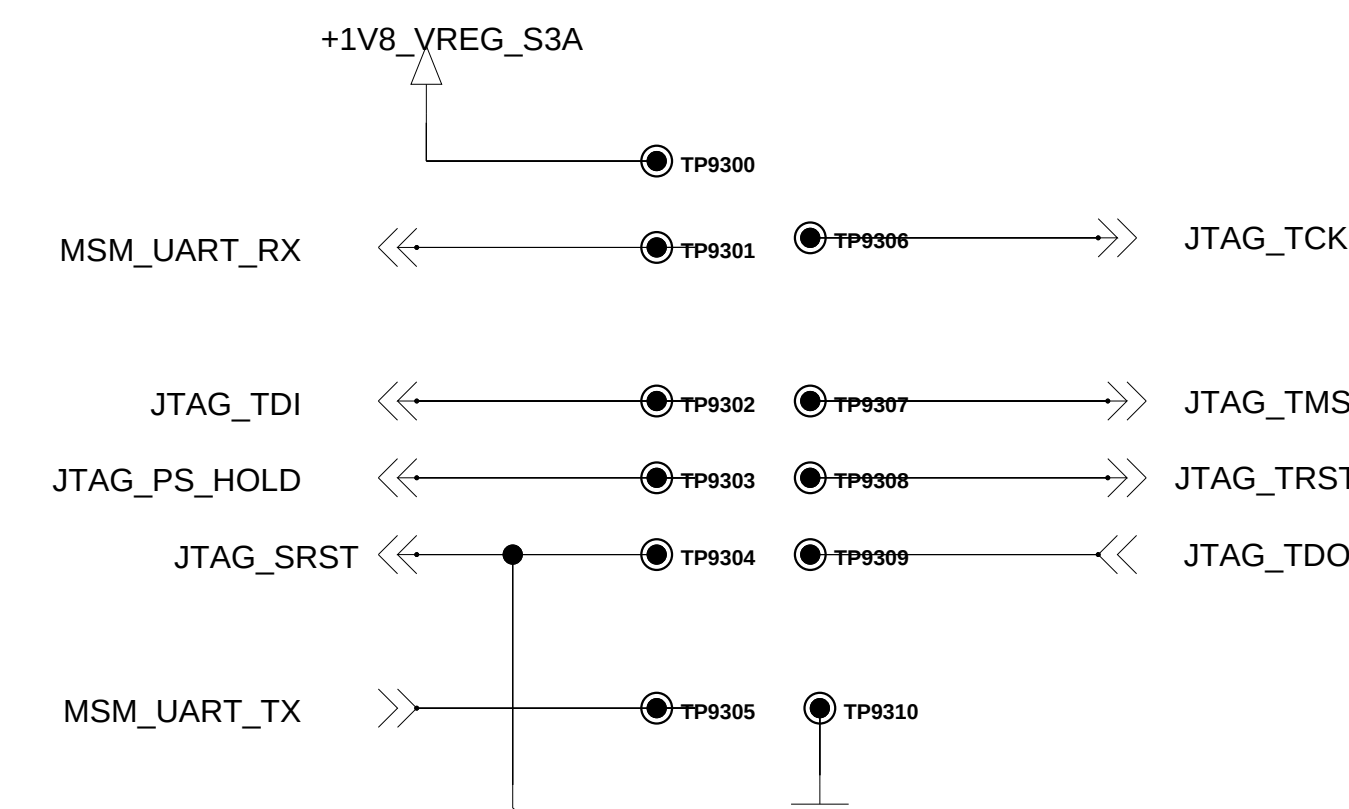
I2C Pull-Up



U2100 MSM8974AC

Place Close to MSM (Max 10mm)
eMMC v4.5 : NC / eMMC v5.0 : RCLK
if eMMC v5.0 use, SD_C1_RCLK need to route to eMMC

< 9-3-1_JTAG > Rev_0.4



Schematic1

2-1-2-2-1_MSM8974AB_Data123.Dwg

LGE Internal Use Only

16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1

<2-1-2-2-3_MSM8974AC_POWER>

CCDS CARD Information	
Release Date	2013.08.26
Based on Reference Schematic	Rev.M
89-NA437-41_MSM8974-MSM8974AB_BASEBAND_REFERENCE_SCHEMATIC_REV.M	

NOTE 2:QMC Checklist Rev.C
1 μ F 1 μ F cap each close to the following pin or pin group: (BD38, BC37), BF28

NOTE 1:QMC Reference Schematic Rev.L
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

NOTE 7:QMC Checklist Rev.C
3 μ F 1 μ F cap, one cap each close to pin F2, P4, K4

NOTE 9:QMC Checklist Rev.K
VSENSE_KRAIT_0P9, REMOTE_GND_SNS
VSENSE_KRAIT_0P9 and REMOTE_GND_SNS
need to be routed differentially

NOTE 8:QMC Reference Schematic Rev.L
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

NOTE 3:QMC Chekclist Rev.C
If HSIC is used, connect VDD_P4 to 1.2 V (VREG_L1_1P2)
If GPIO144/145 is used, connect VDD_P4 to 1.80 V(VREG_S3A_1P8)

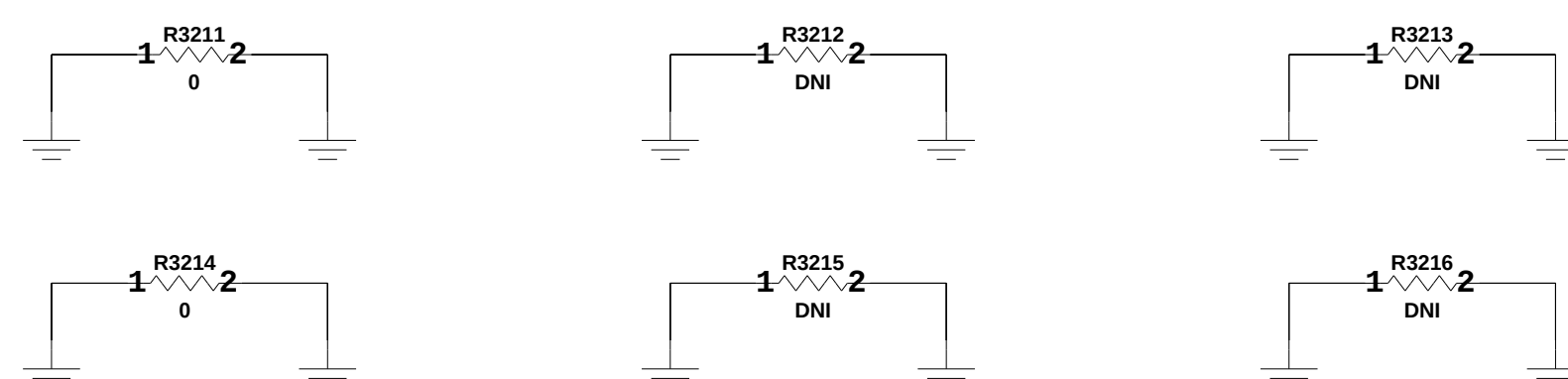
NOTE 4:QMC Checklist Rev.C
VREG_S1B_0P95
Place a 0 Ω resistors between VREG_S1B_0P95 and the VDD_EBI CDC and VDD_EBI PLL pins are to star connects to VDD_MX rather than just short to a plane.
The resistor forces a separate netname and a separate routing and does not allow shorting to the plane.
We recommend to keep the resistors in the design during the routing stage to achieve the desired isolation. Once routing is complete customer can remove the resistor and simply short the isolated net back to the main plane at the location of the resistor.

NOTE 10:QMC Reference Schematic Rev.K
PLACE S2B, S4B, S5B, S6B, S7B, S8B
OUTPUT CAPACITORS CLOSE TO MSM

U2100 MSM8974AC

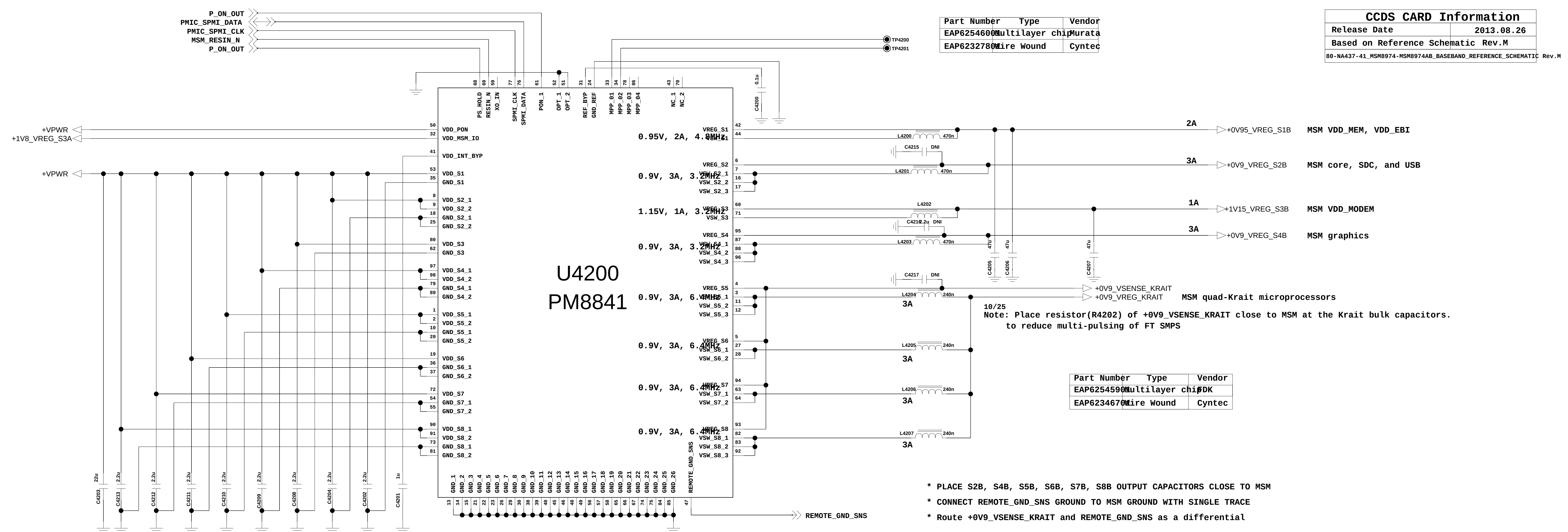
LG Electronics	
Schematic1	
Rev A2	PDM NUMBER
2-1-2-2-3_MSM8974AB	Power

<LPDDR3_POP_SK Hynix_933MHz>



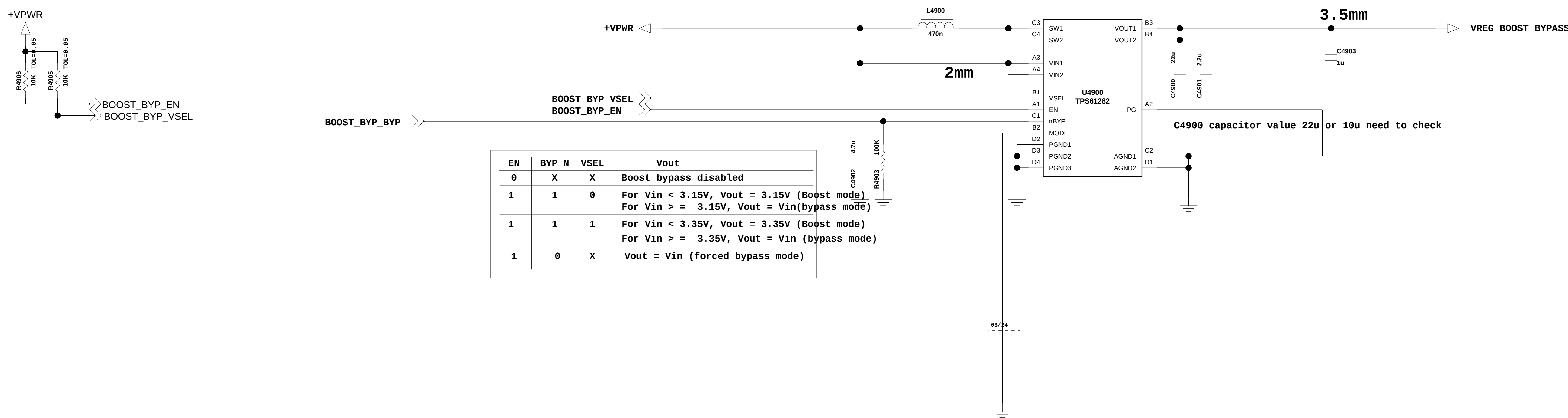
03/15
It should be Closed to

<4-2-1_PMIC_PM8841>

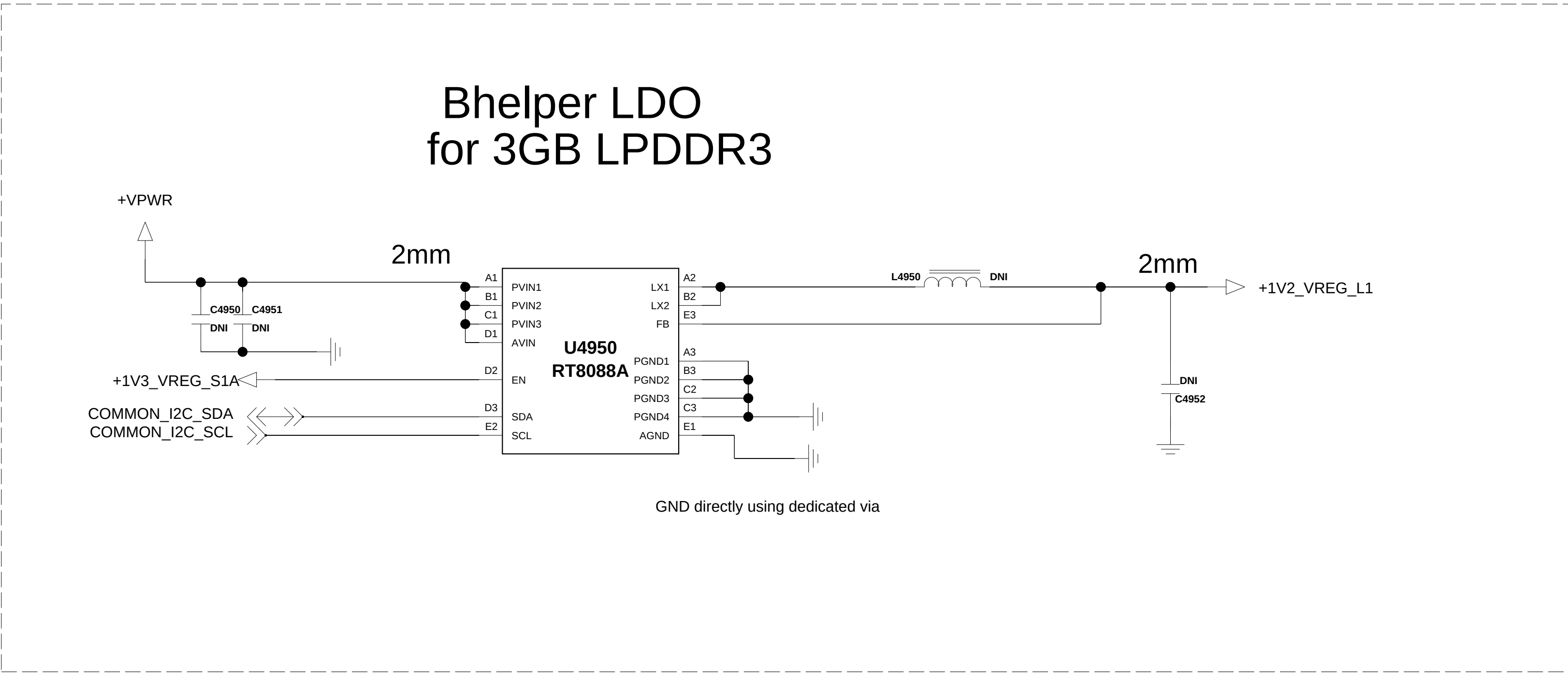


<4-9-1-4-3_Bypass_Booster_Single_TPS61282>

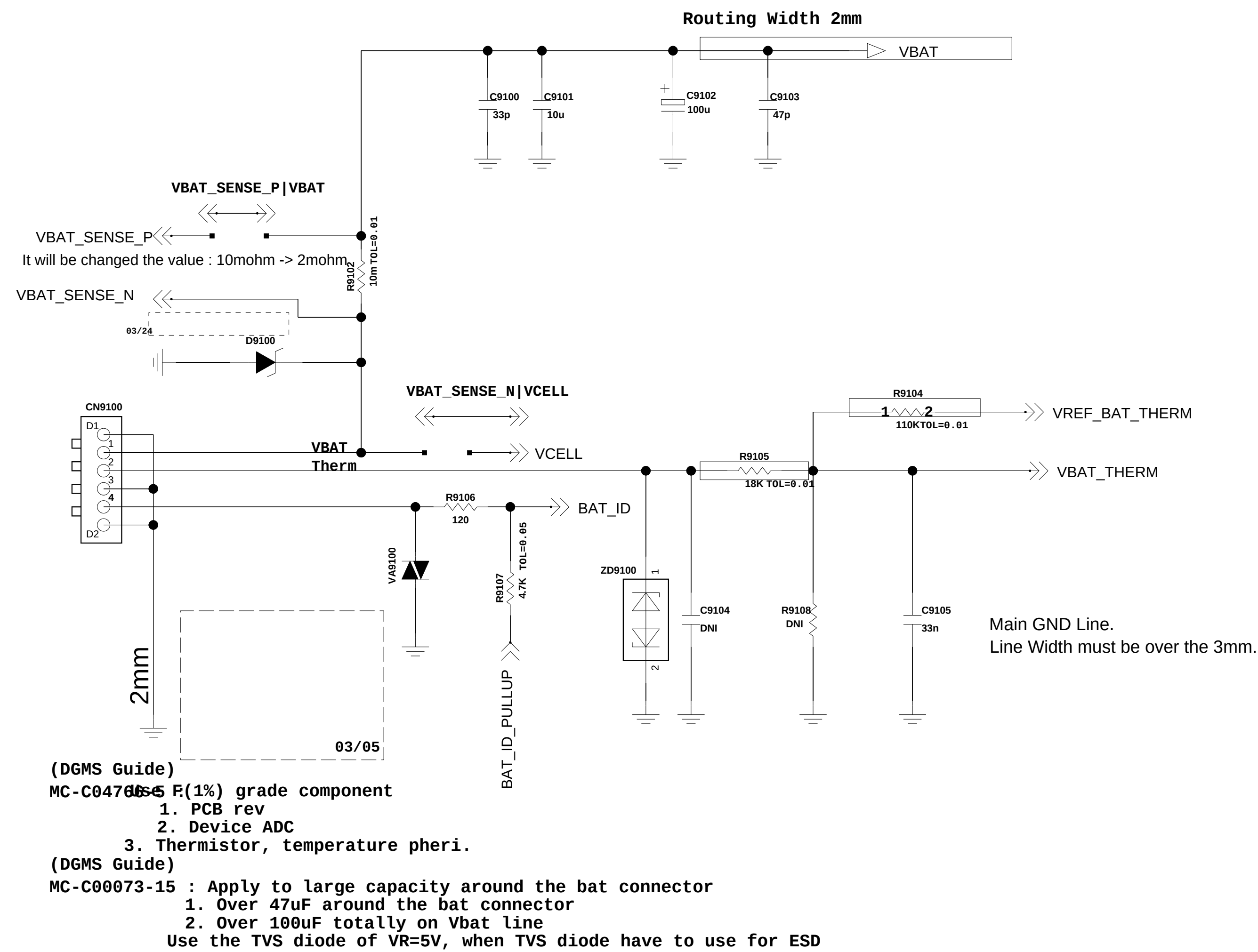
H Volt LDO Bypass boost



03/15 DNI

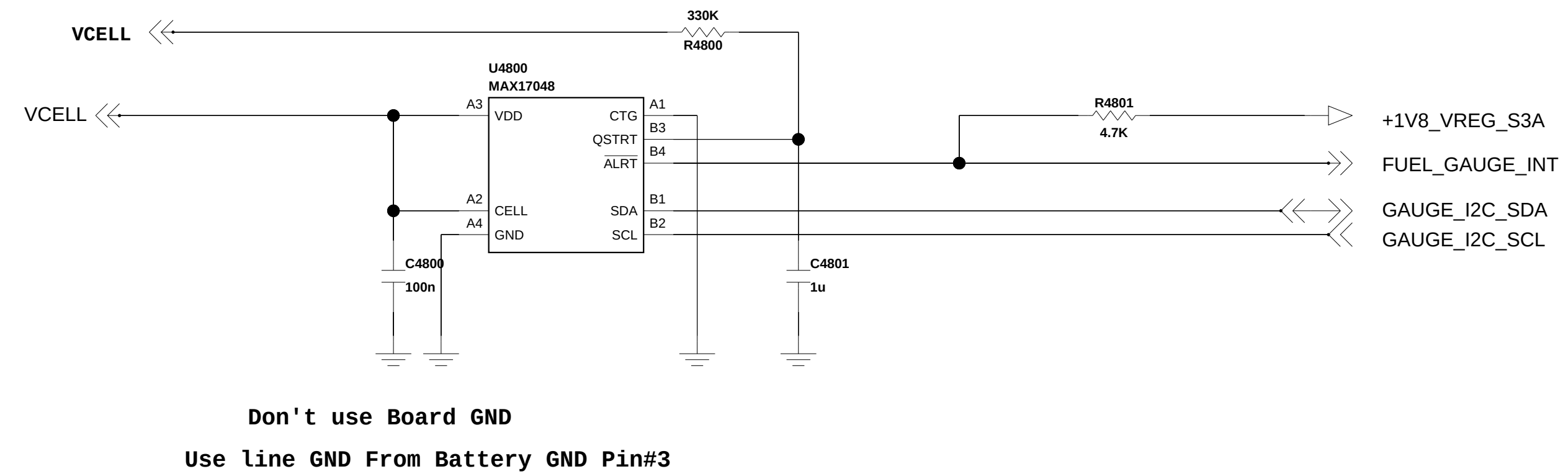


<9-1-2_Battery_CNT_4>REV.0.5



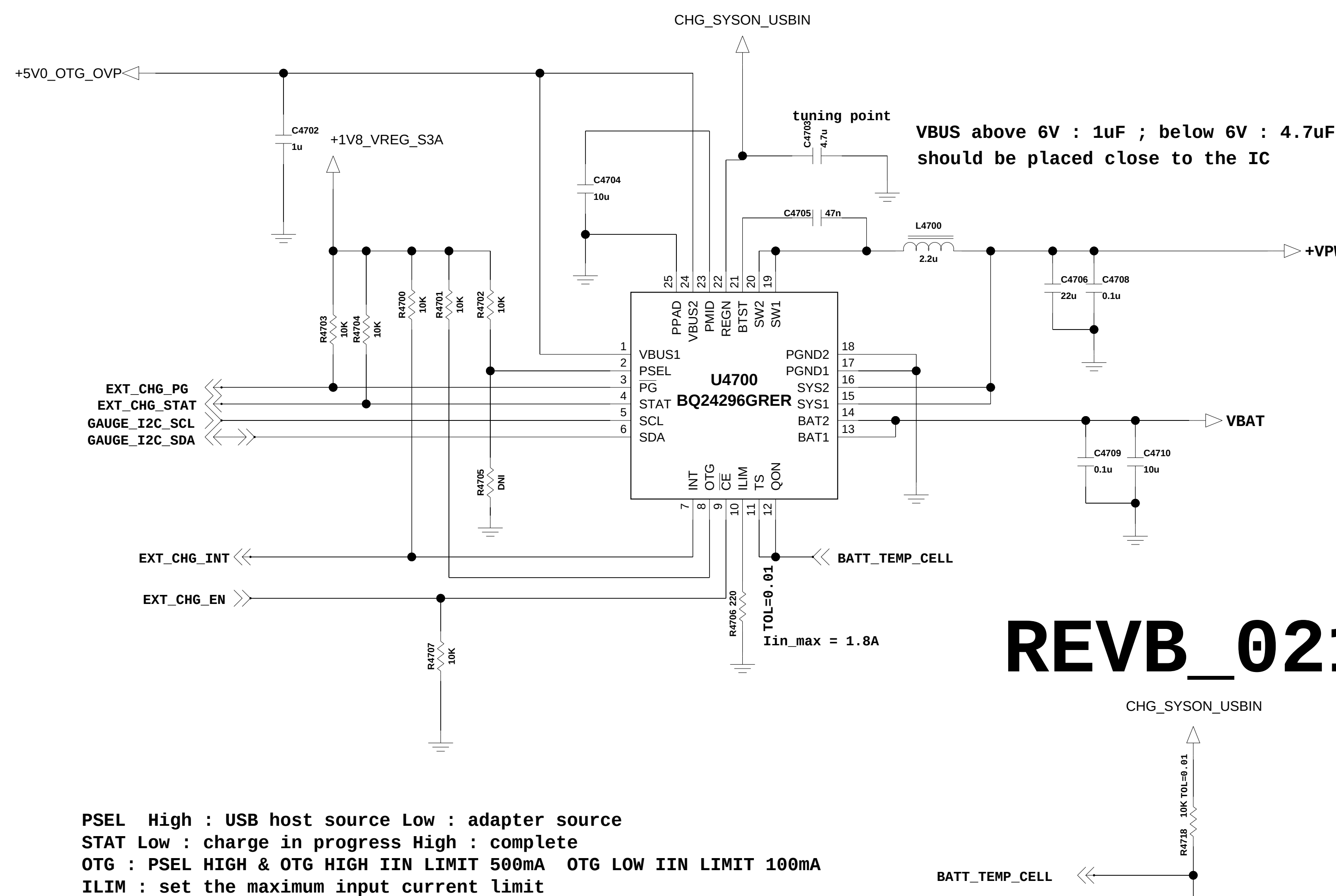
<4-8-1_Fuel_Gauge_MAX17048>Rev_0.3

Only use line VBAT1 from Battery Connector.
CLOSE TO Battery CONNECTOR



Single charger IC

<4-7-2-4_Charger_Switching_BQ24192>



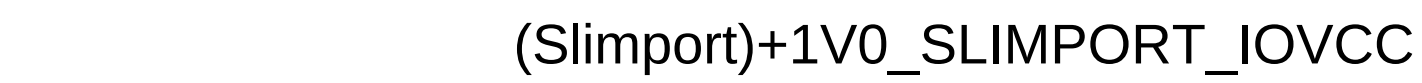
REVB_0213

REVB_0213

NOTE1: QMC Checklist Rev.C
A 47 kΩ resistor and 1μF shunt RC filter is required on the USB_IN lines of the PMIC for noise rejection and for charger removal detection.
--> not included (VBUS_USB_IN connected to slimport)

Rev_0.3

RP114K331D-TRB U7600 **0.3mm**



Note :
V1 and V2 are in the range of 0.6 to 1.2V
The difference between V1 and V2 is within $\pm 100\text{mV}$

(DGMS Guide)
MC-C06026-2 : If the AP is from Qualcomm
the TP on HDMI_CEC is required
for SW Test in production

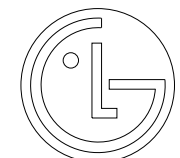
<4-7-3-1_Wireless_Charger_MAX14670EWL_T

The schematic diagram illustrates the WLC control circuit. It features two MAX14670EWL+T comparators, U4750 and U4751. U4750's non-inverting input (IN1) is connected to the OTG_EN signal through a 47K resistor (R4752) and a 1uF capacitor (C4750). Its inverting input (IN2) is connected to a reference voltage through a 200K resistor (R4752) and a 1uF capacitor (C4750). The output of U4750 (OUT1) is connected to the WLC_OFF signal. U4751's non-inverting input (IN1) is connected to the WLC_OFF signal through a 1uF capacitor (C4756). Its inverting input (IN2) is connected to a reference voltage through a 1uF capacitor (C4754) and a 1uF capacitor (C4753). The output of U4751 (OUT1) is connected to the WLC_CHG_INT signal. The circuit also includes a USB input (VBUS_USB_IN_PM) and a 5V0_OTG_OVP signal.

Drawing Date

Rev

Schematic1



LG Electronics

LGE Internal Use Only

L



J



H

C

F



C

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E

D

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12

11

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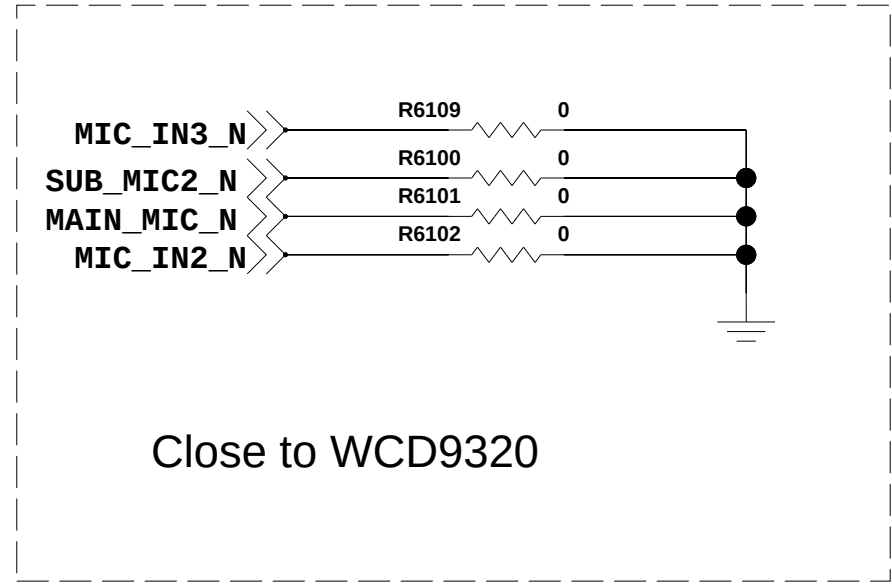
E

D

C

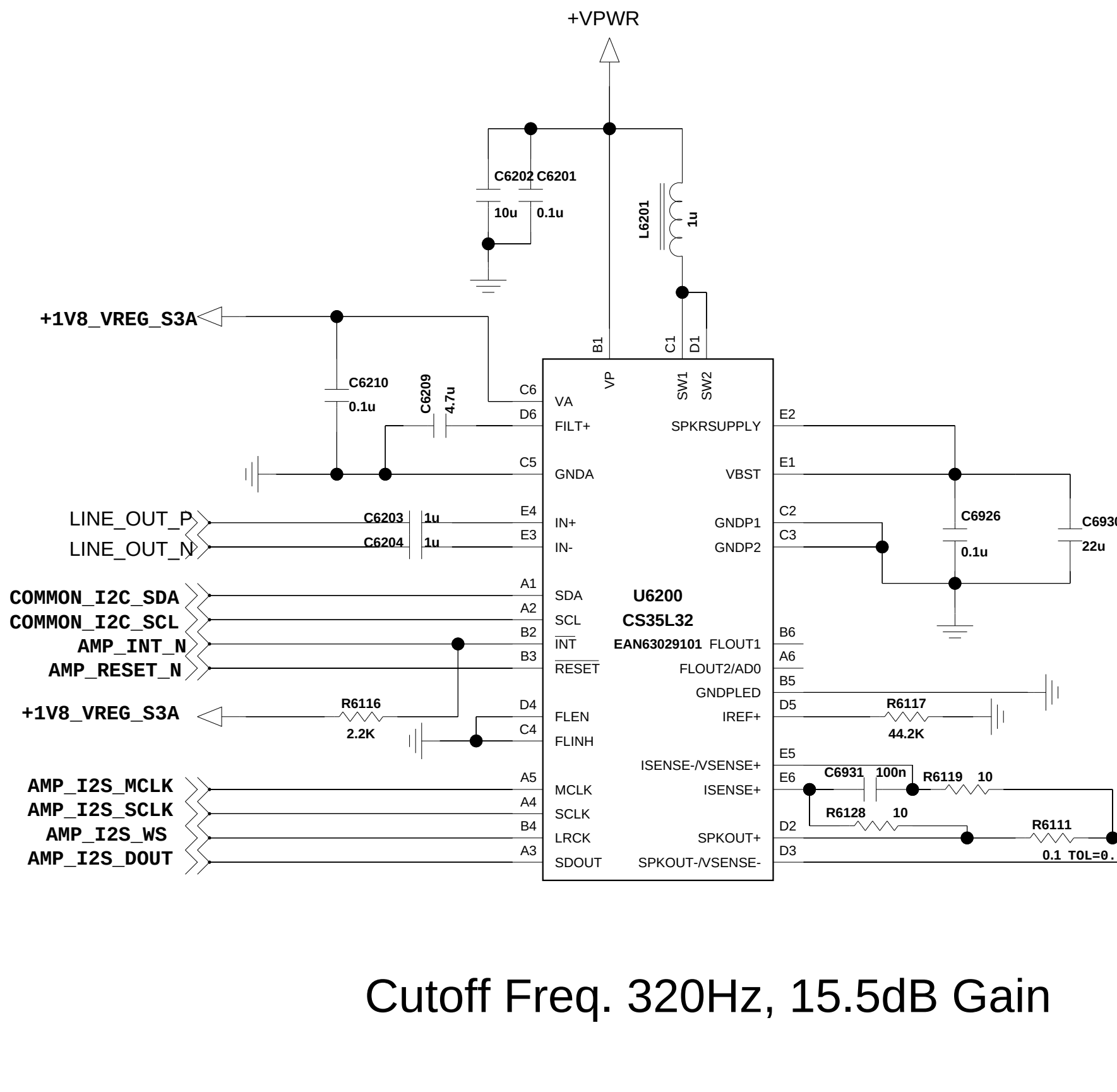
B

A



Close to WCD9320

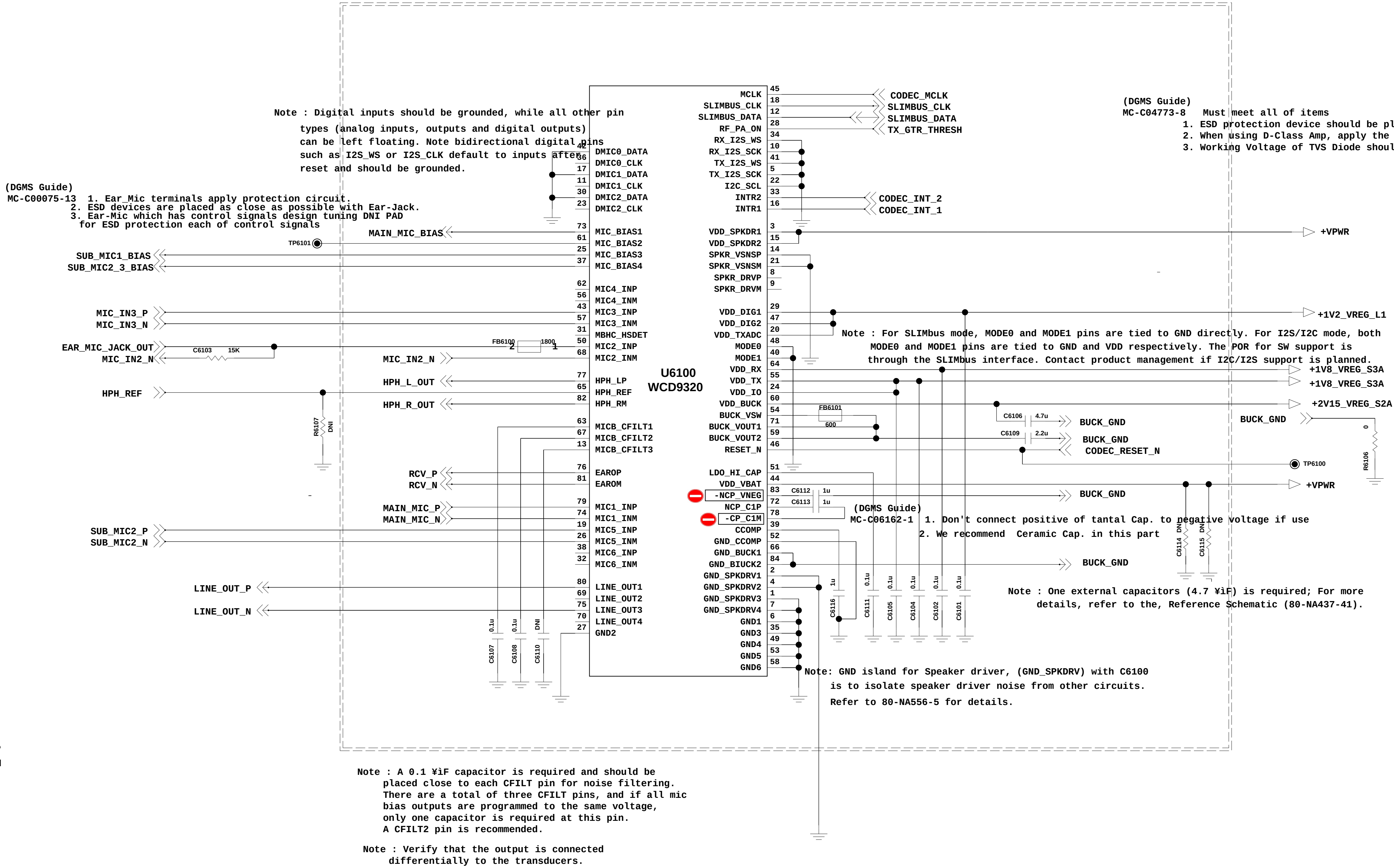
SPK BOOST AMP



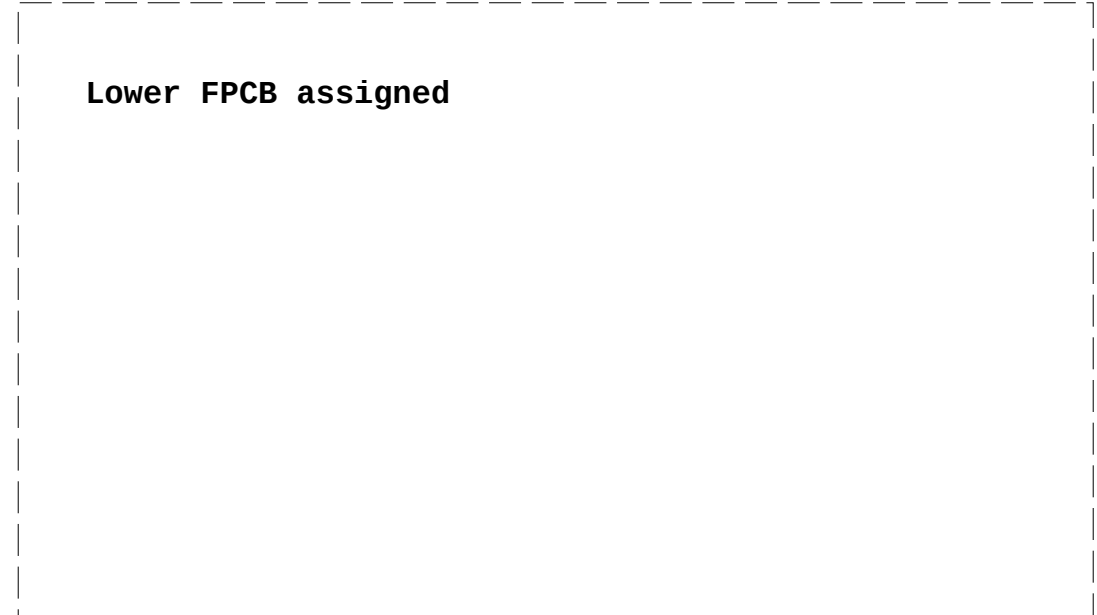
Cutoff Freq. 320Hz, 15.5dB Gain

<6-1-1-5_WCD9320>

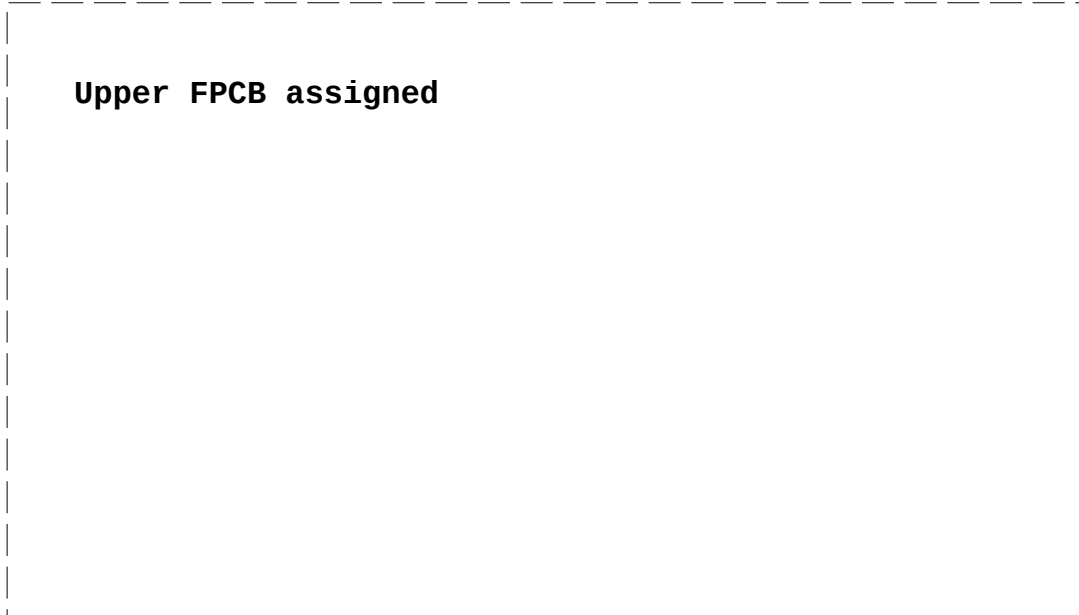
REV_0.3



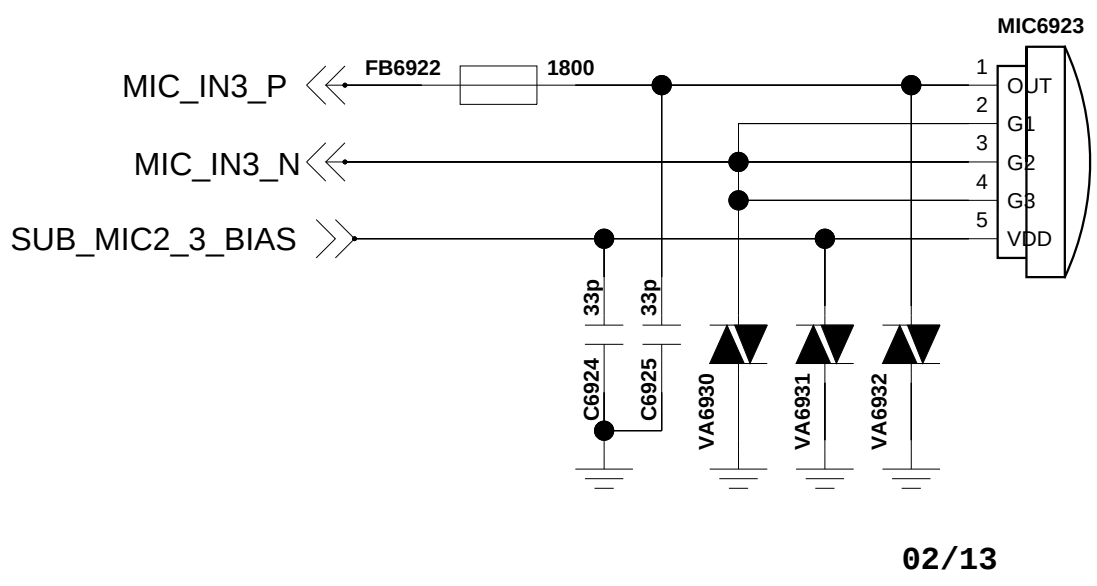
Main MIC



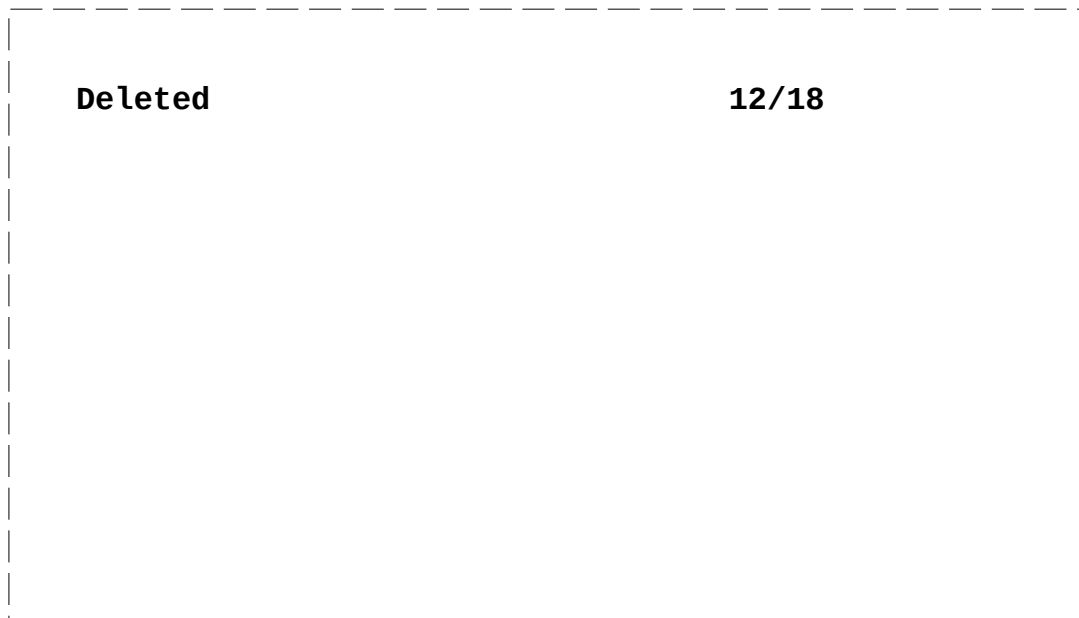
Sub_MIC1



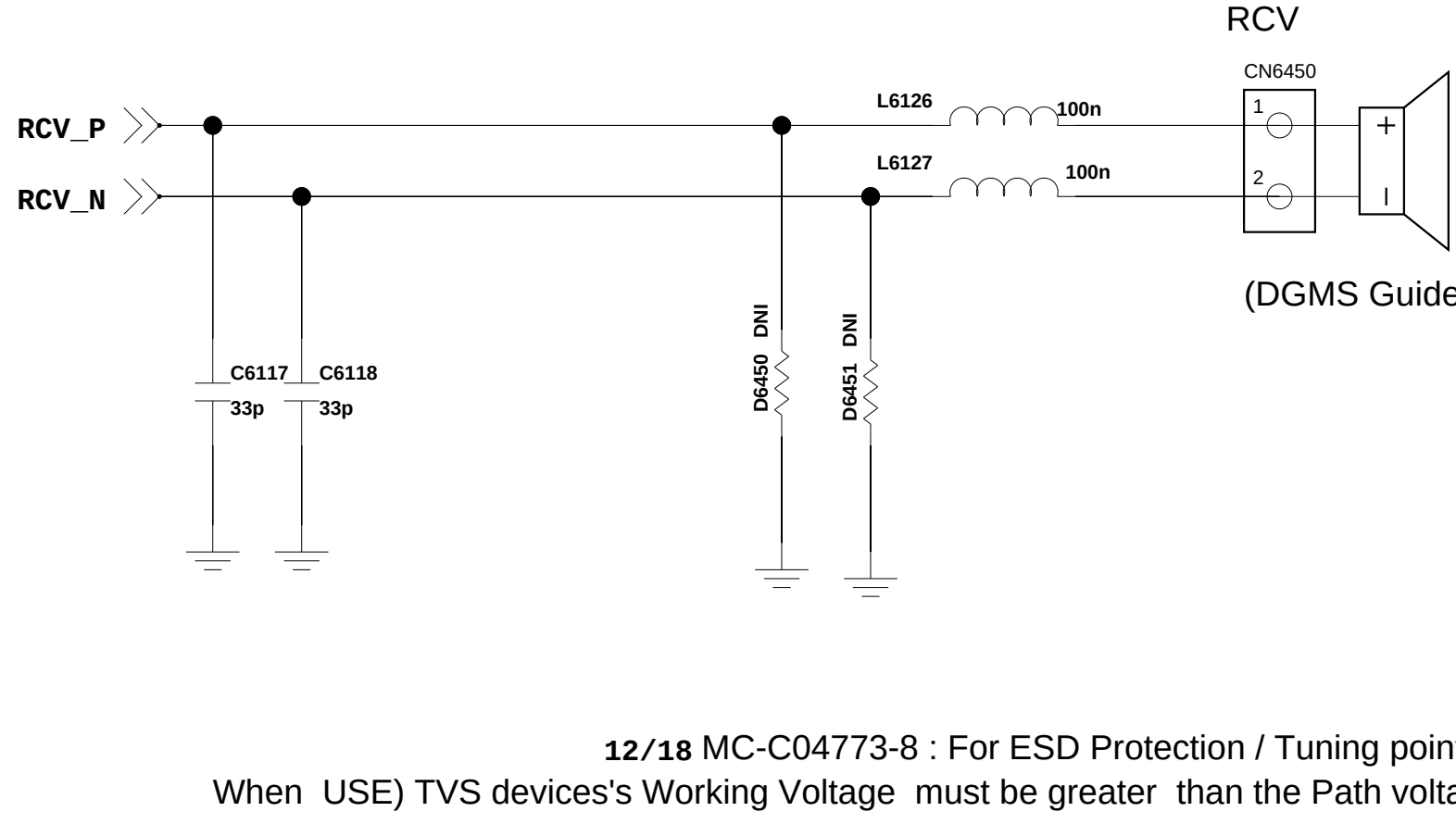
Sub_MIC2



Sub_MIC3

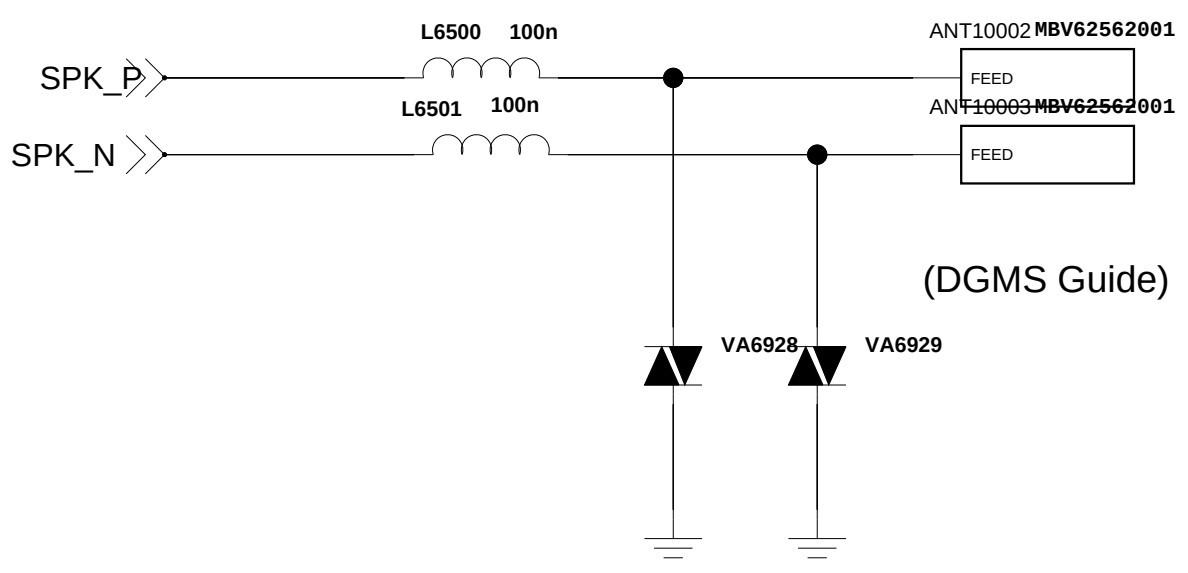


< 6-8-2_Receiver >



< 6-7-1_Speaker >

Rev_0.4



LGE Internal Use Only

16

15

14

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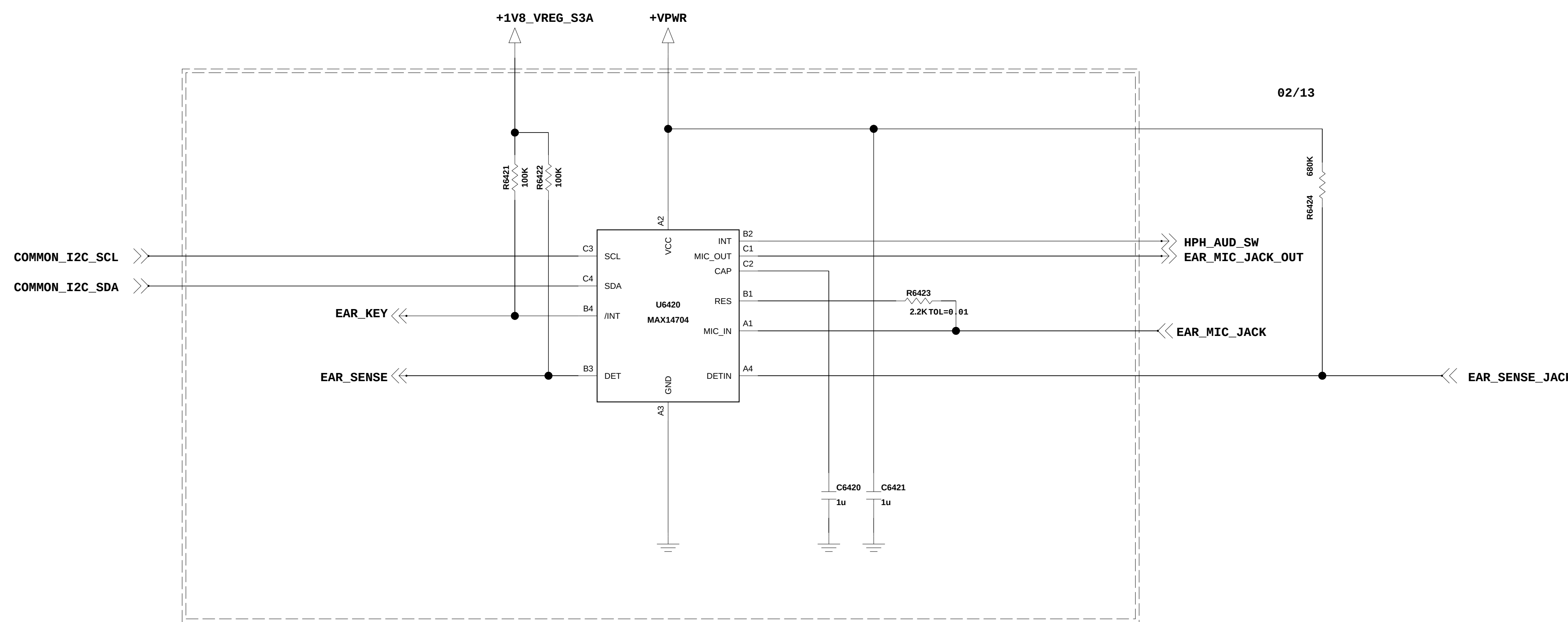
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3

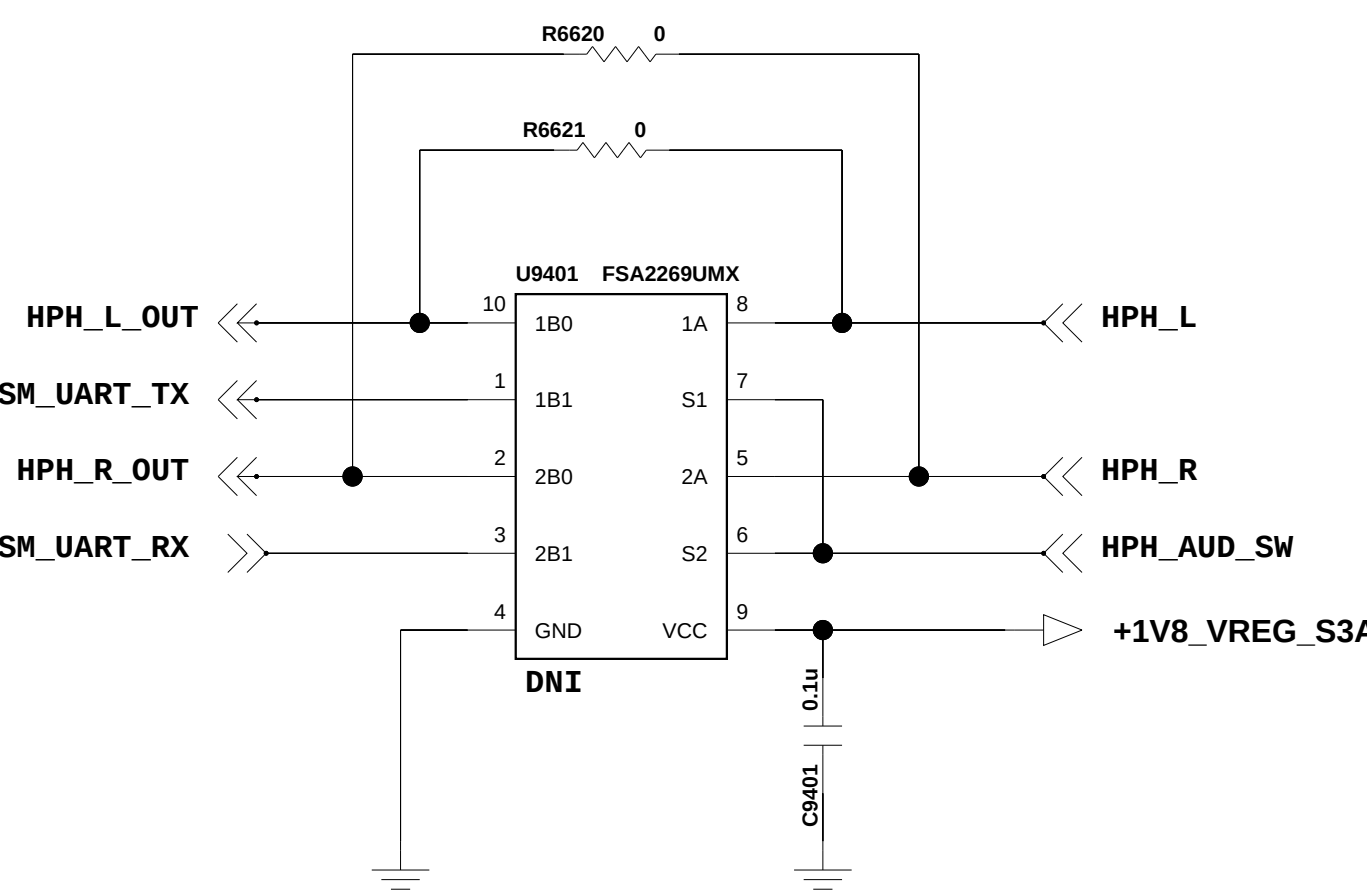
2

1

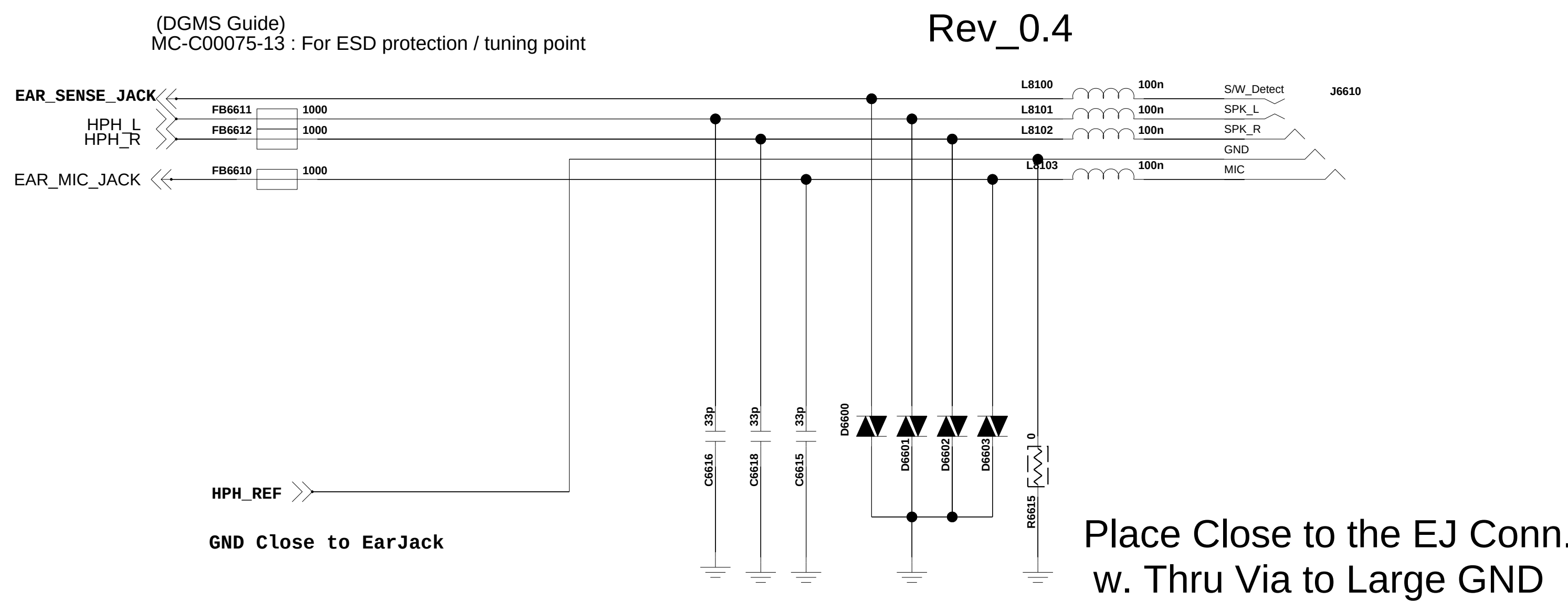
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<Earjack_Debug_Port >



< 6-6-1_Earjack >



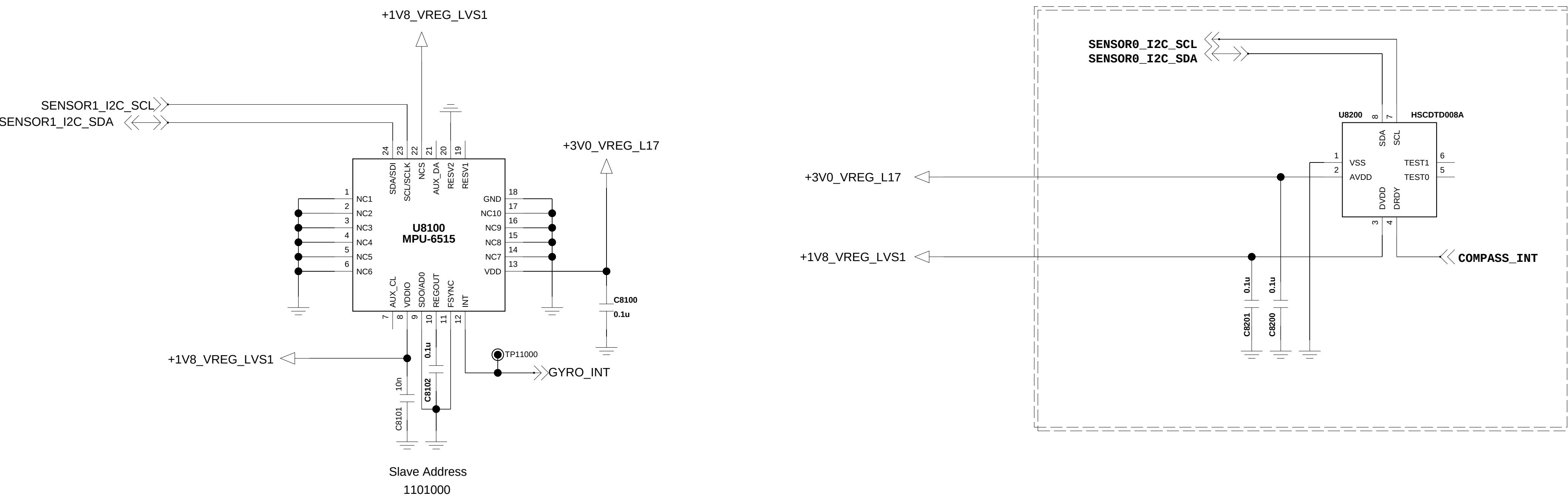
SENSOR/MOTOR/IRRRC/SIM/SD/LED

I2C	Devices
Sensor0	Compass, Temp+Humid+Pressure
Sensor1	Accel+Gyro, Pressure, UV, Prox+RGB+IR Gesture

<8-1-3-3_Accel_Gyro_MPU-6515 >

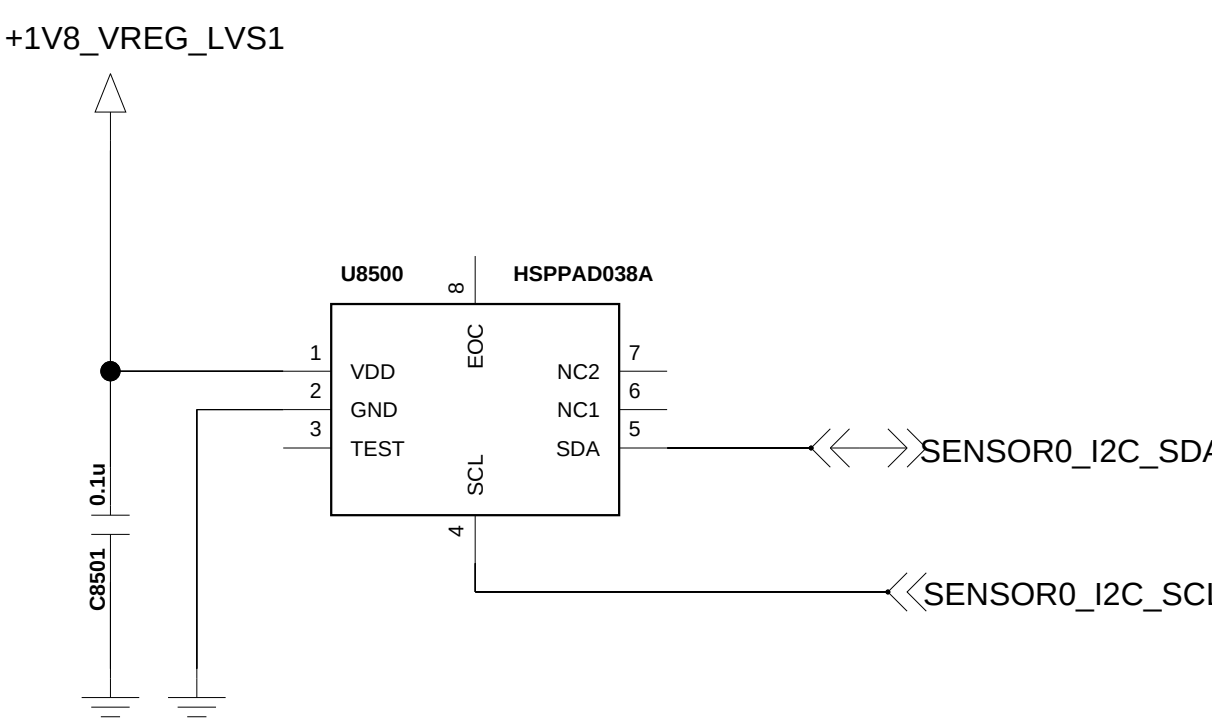
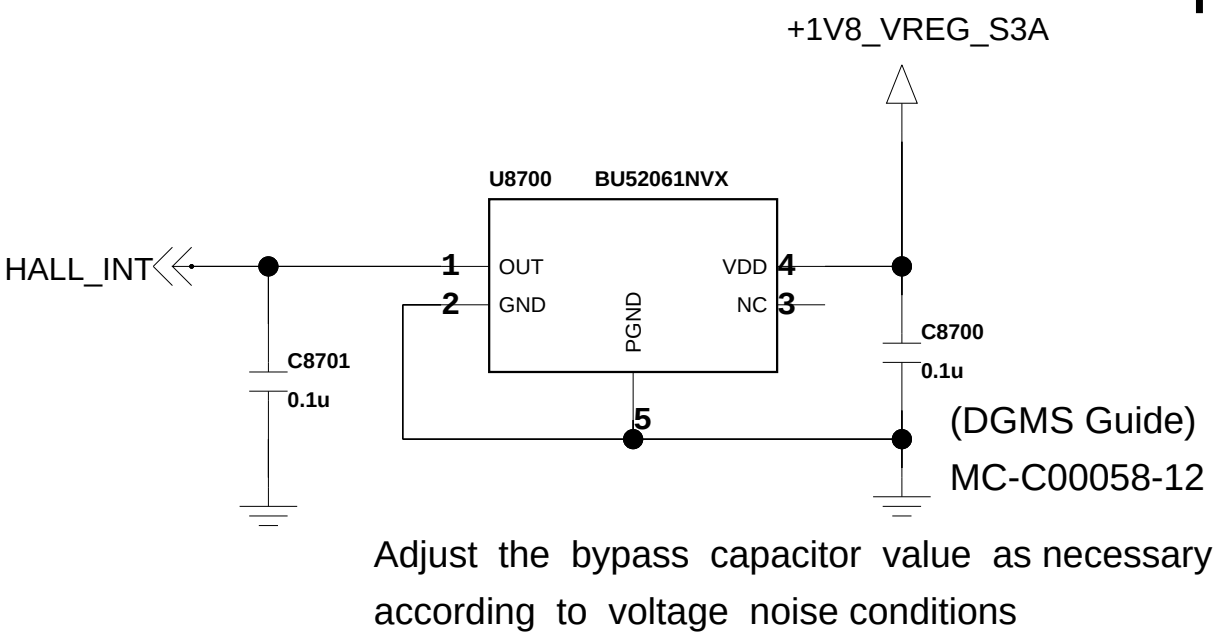
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Rev_0.3

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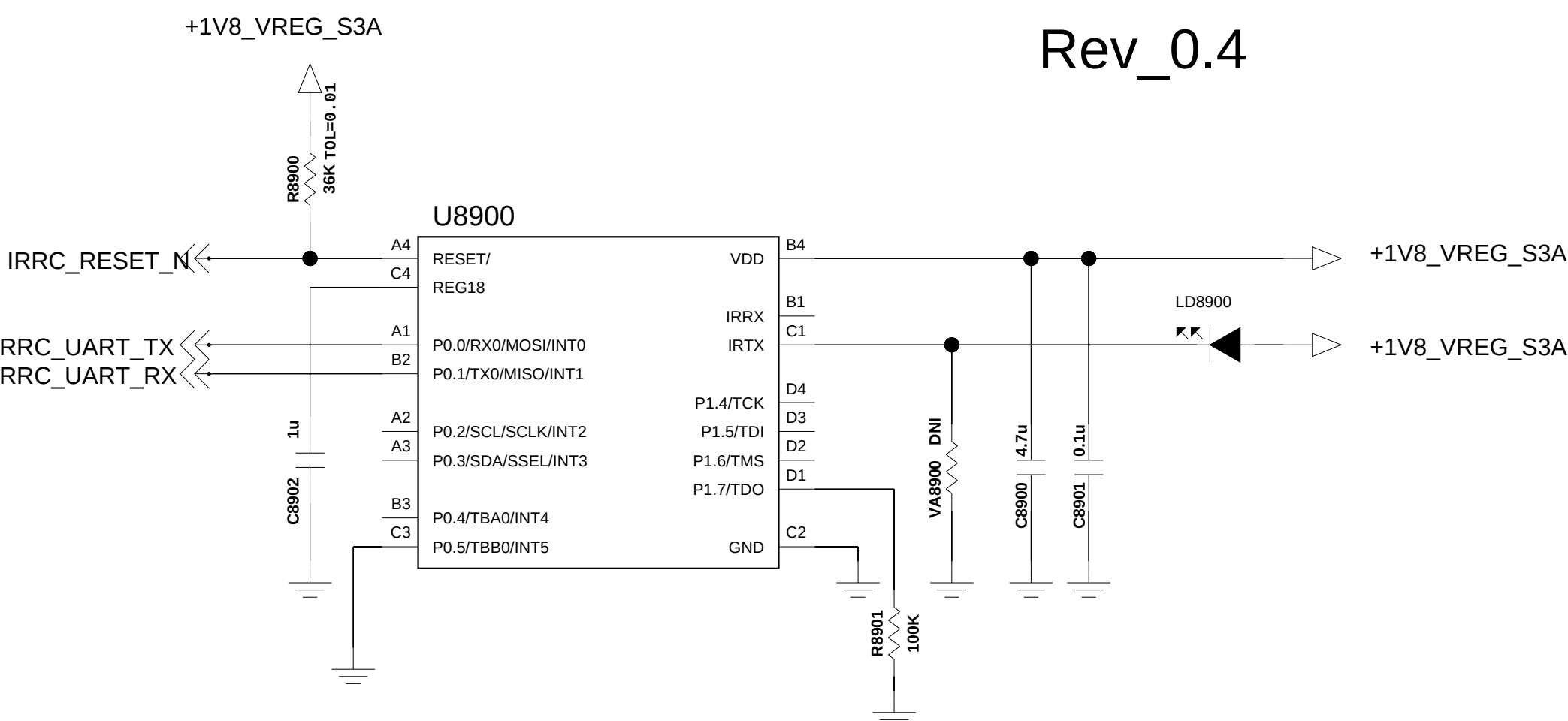
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Rev_0.3



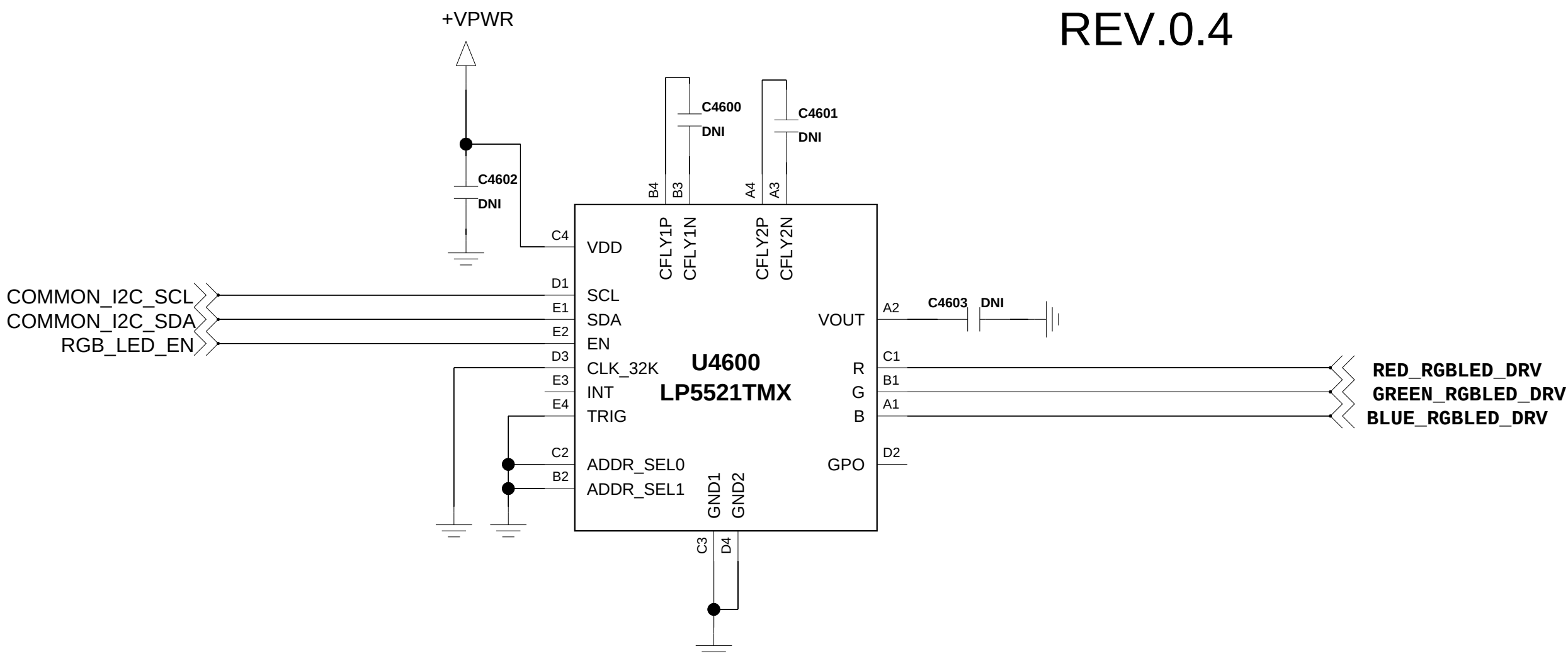
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Rev_0.4



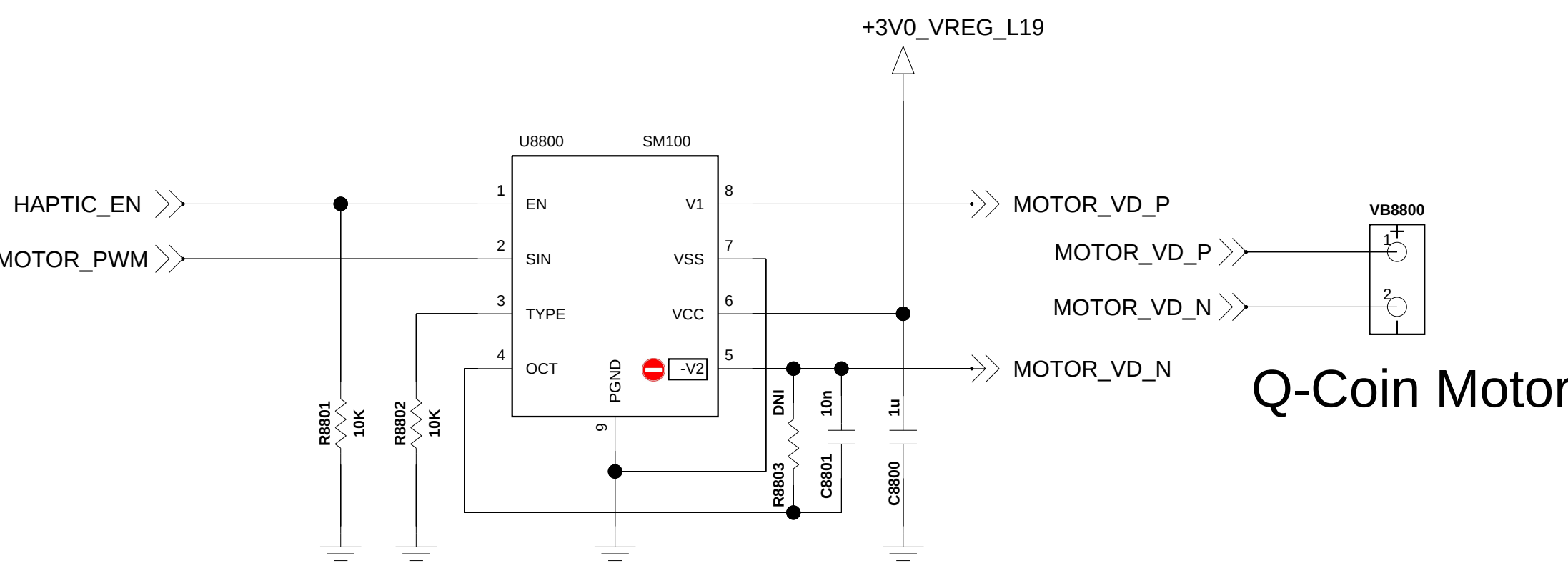
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REV.0.4



< 8-8-1-1_Haptic_SM100 >

Rev_0.3

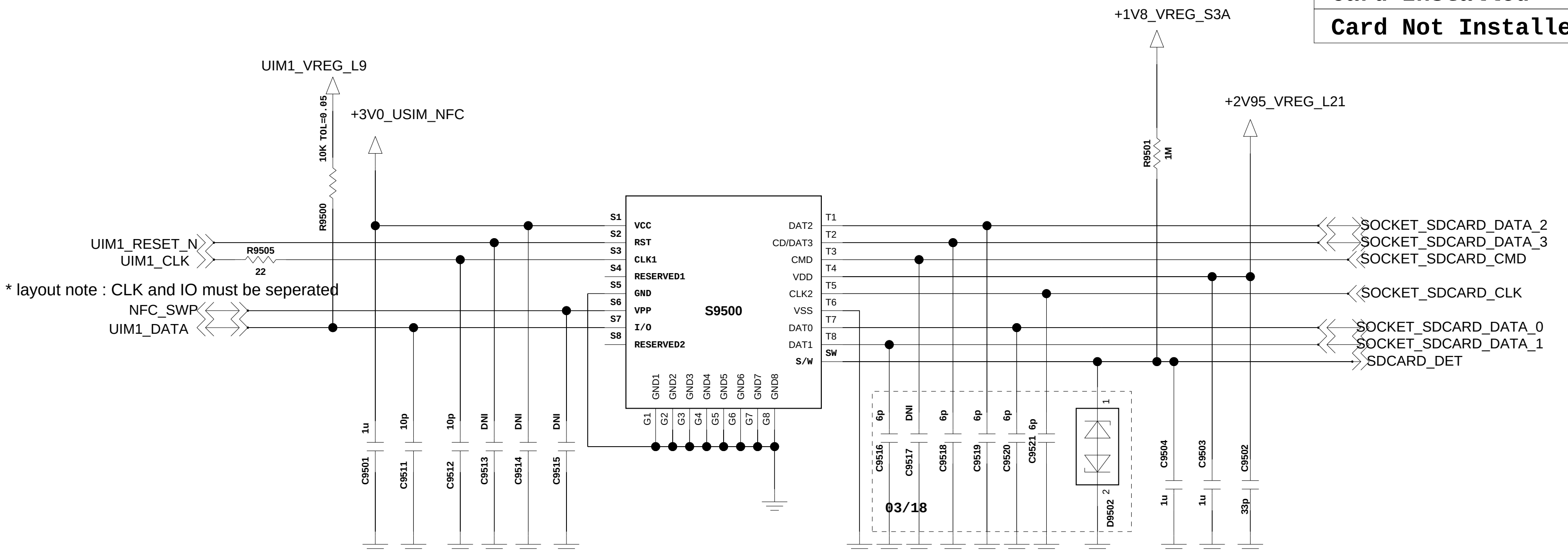
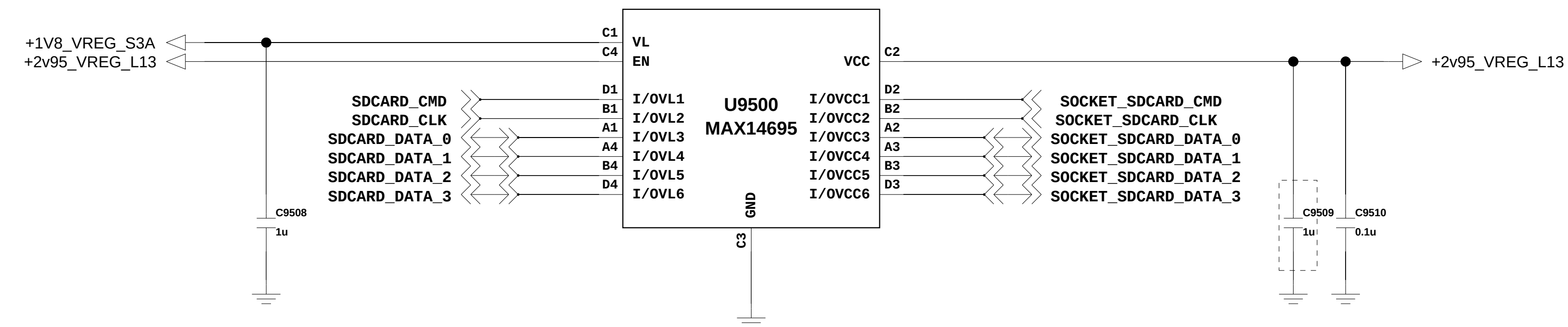


<LEVEL TRANSLATOR>

< 9-5-3_Micro_SD_Combo >

Rev_0.3

Card Detect	SW	VSS(3V)	CARD_DET
Card Installed			GND
Card Not Installed			1.8V



A

A

A

A

A

A

L

K



1



G



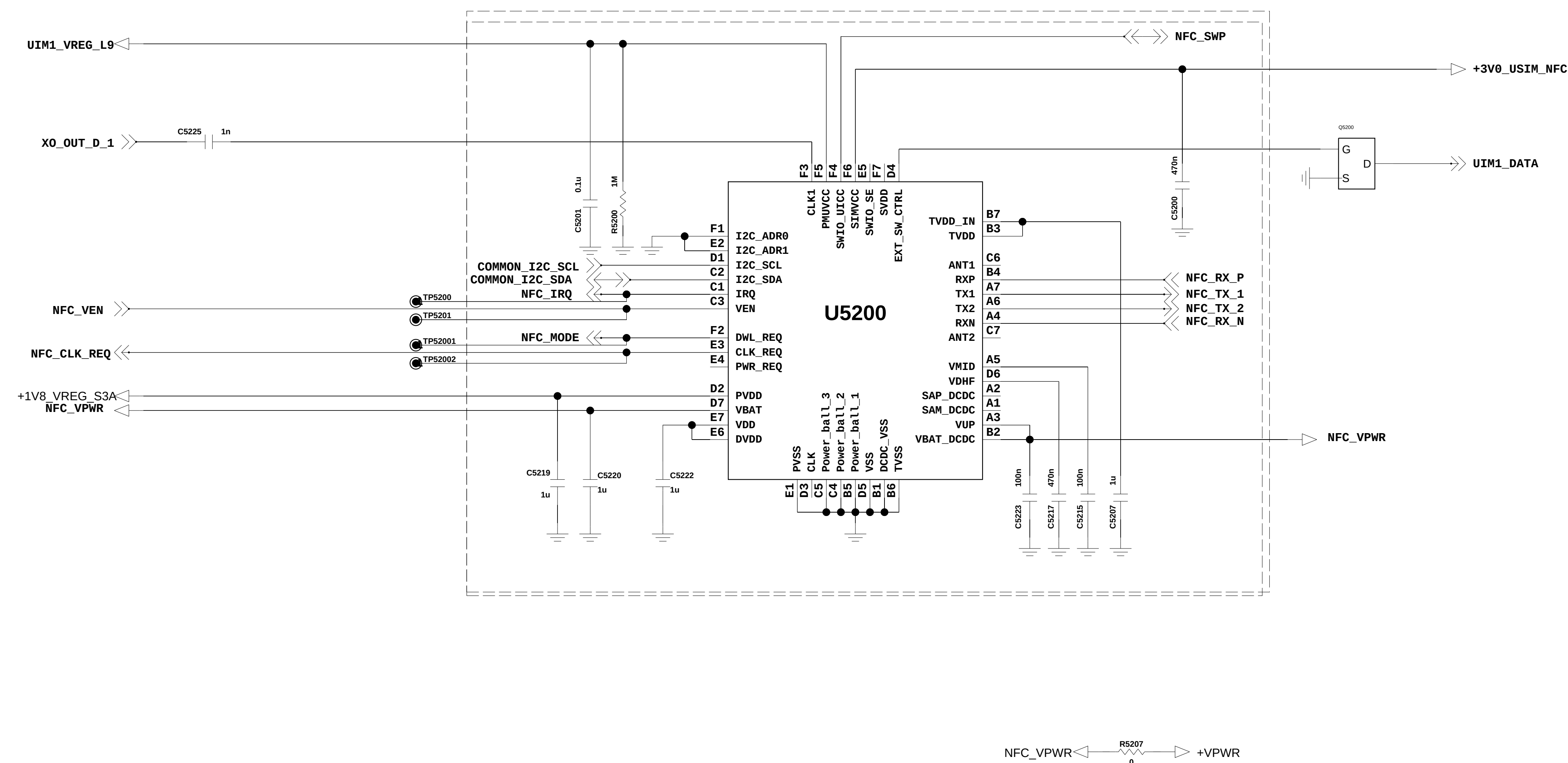
C



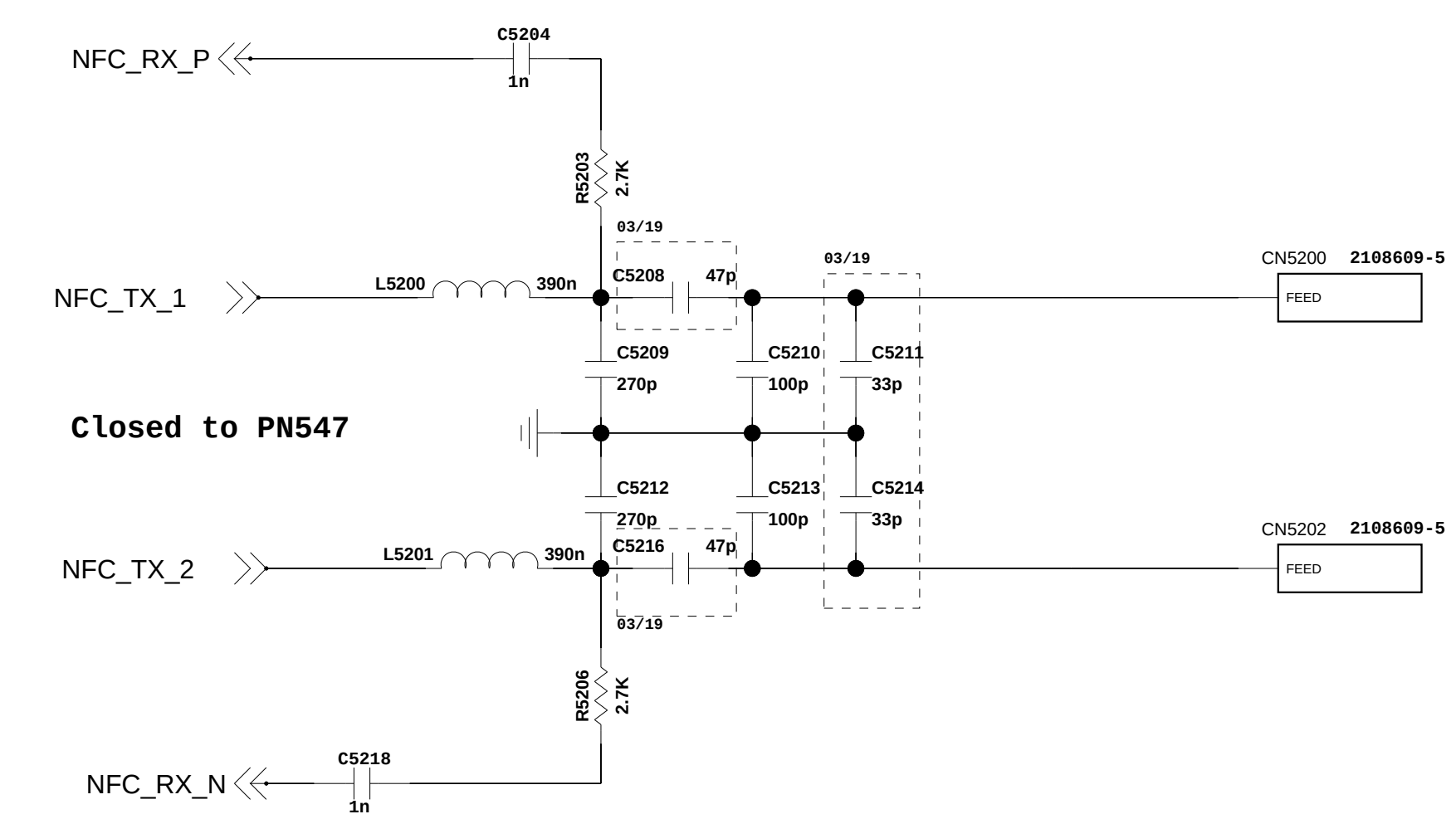
D



<5-2-1-4_NFC_PN547> Rev_0.5

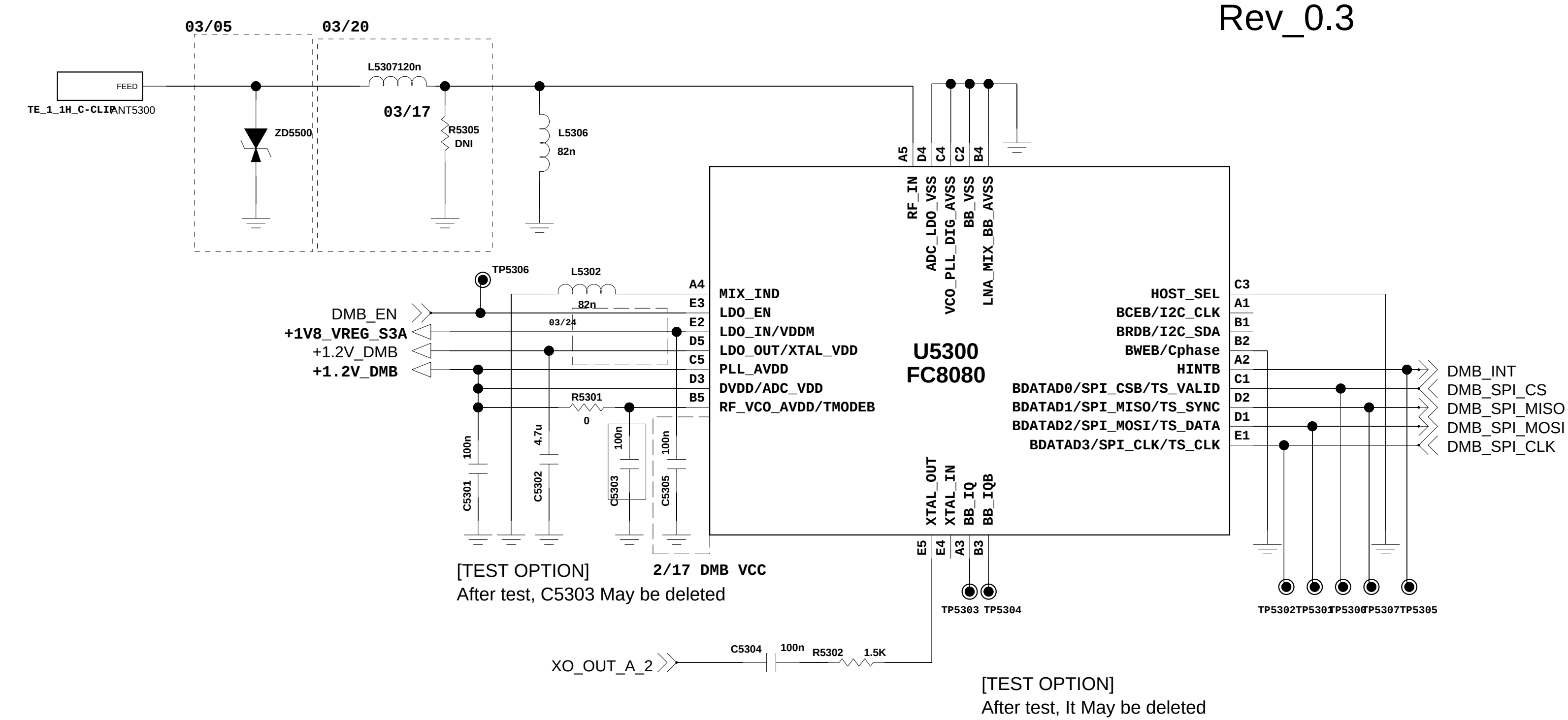


NFC Antenna



< 5-3-1-5_TDMB_FC8080 >

Rev_0.3



L

K

J



H

G

F

E

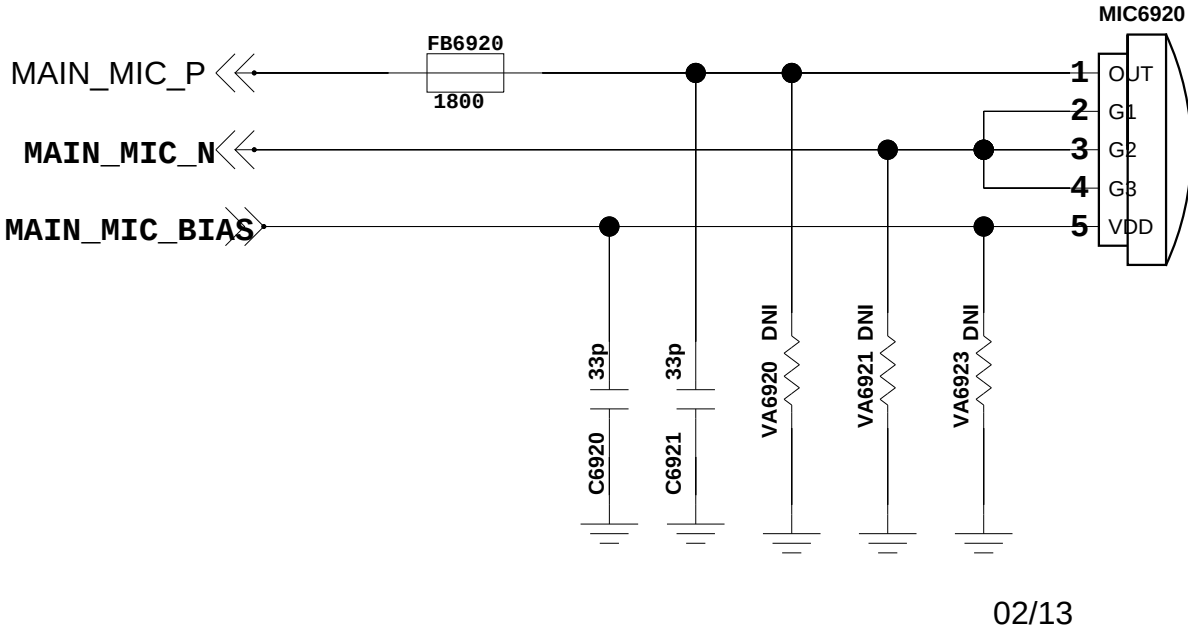
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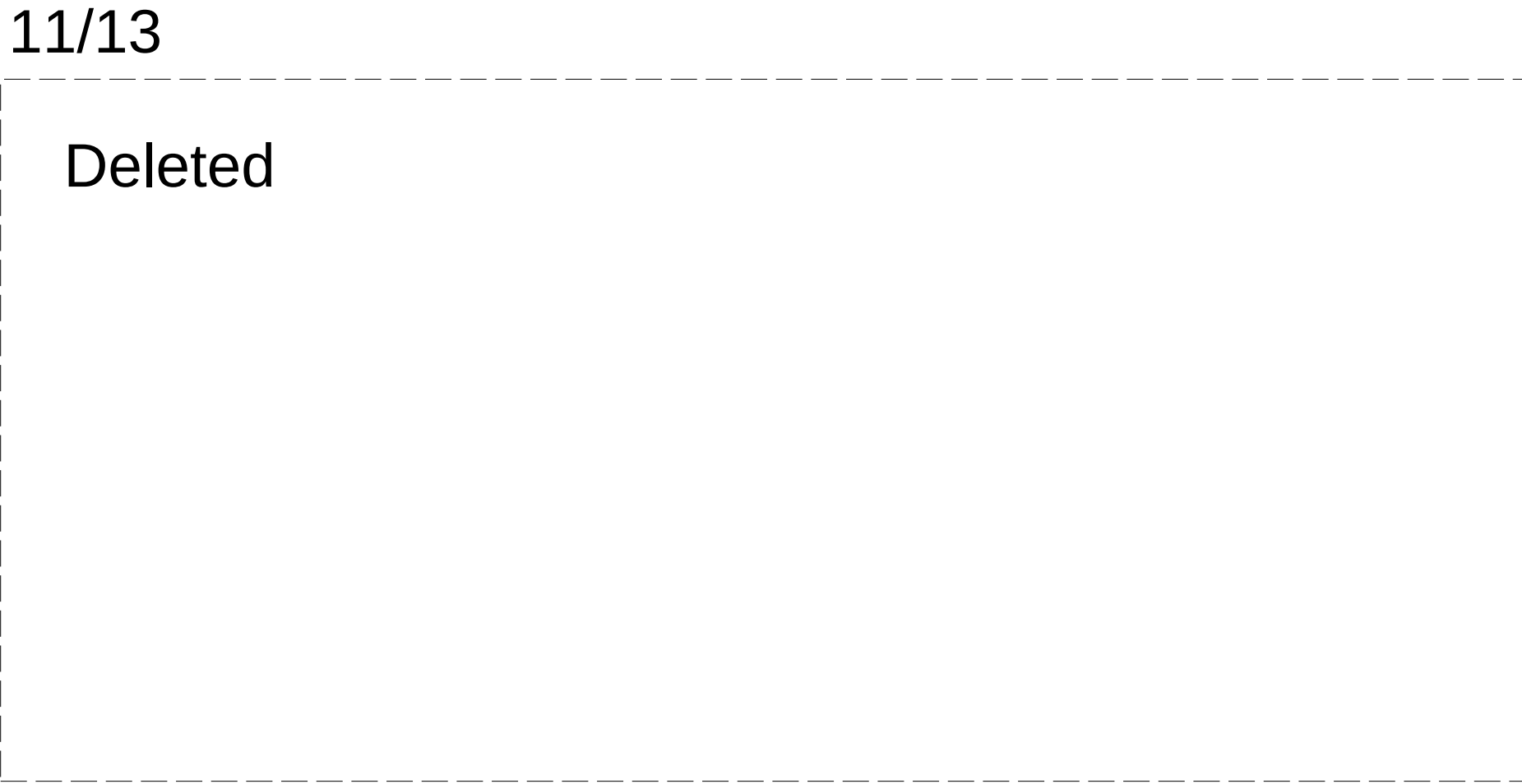
B



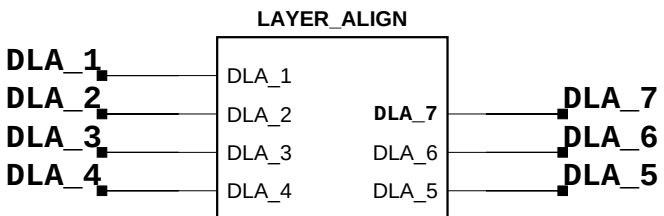
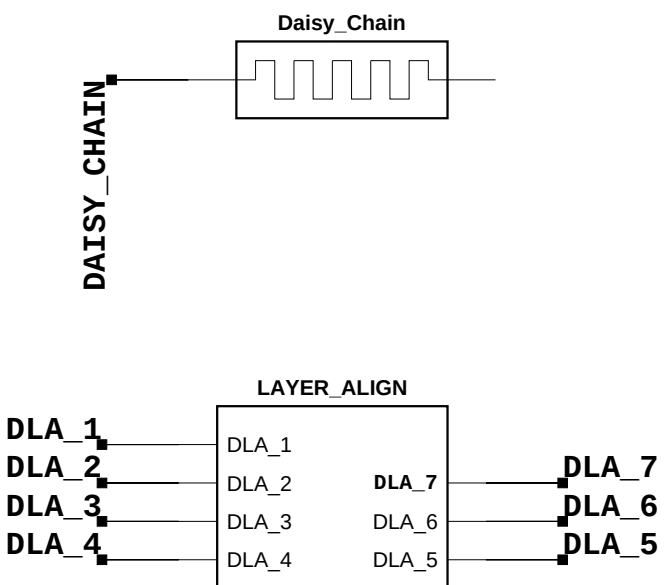
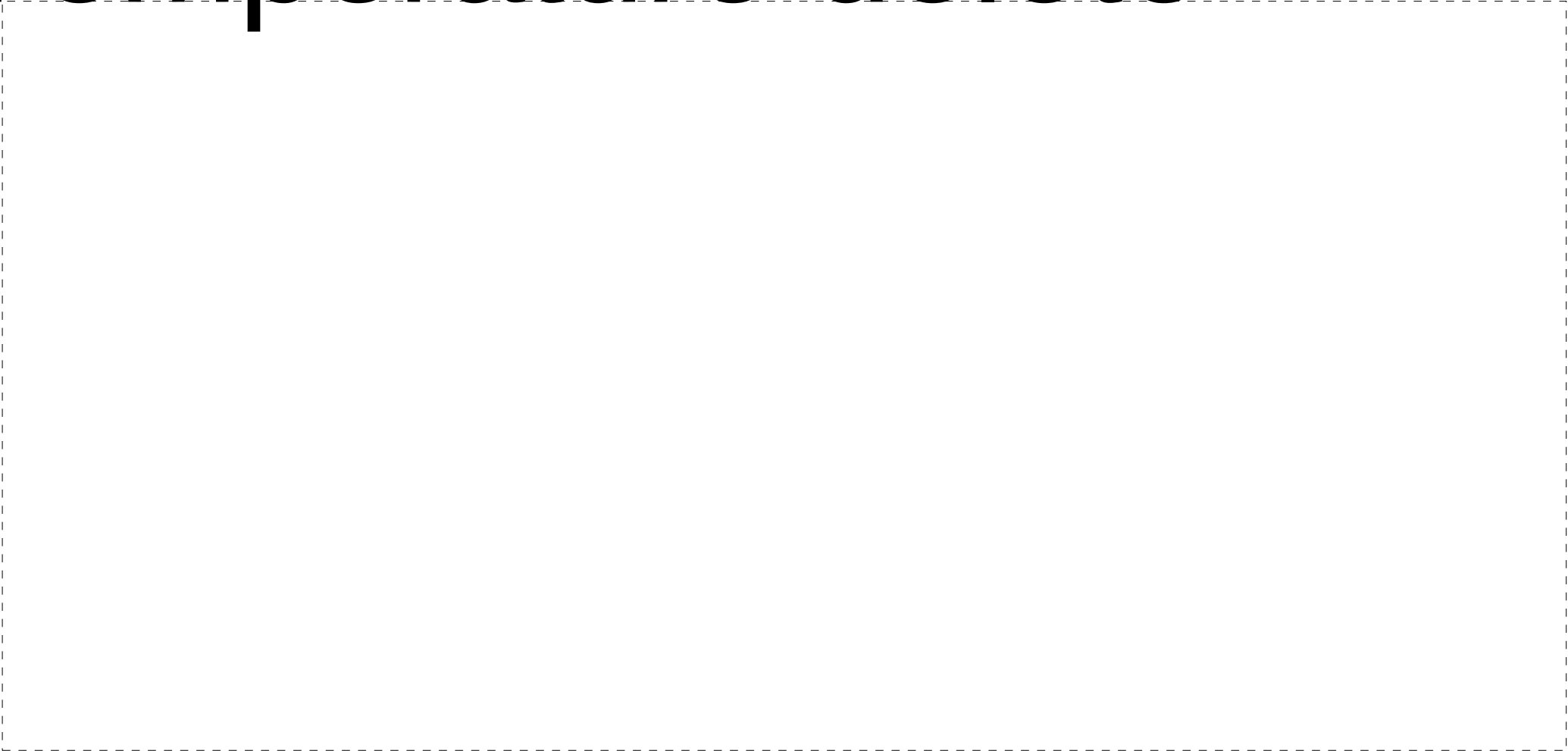
From previously Lower FPCB to Main PCB (11/13 now)

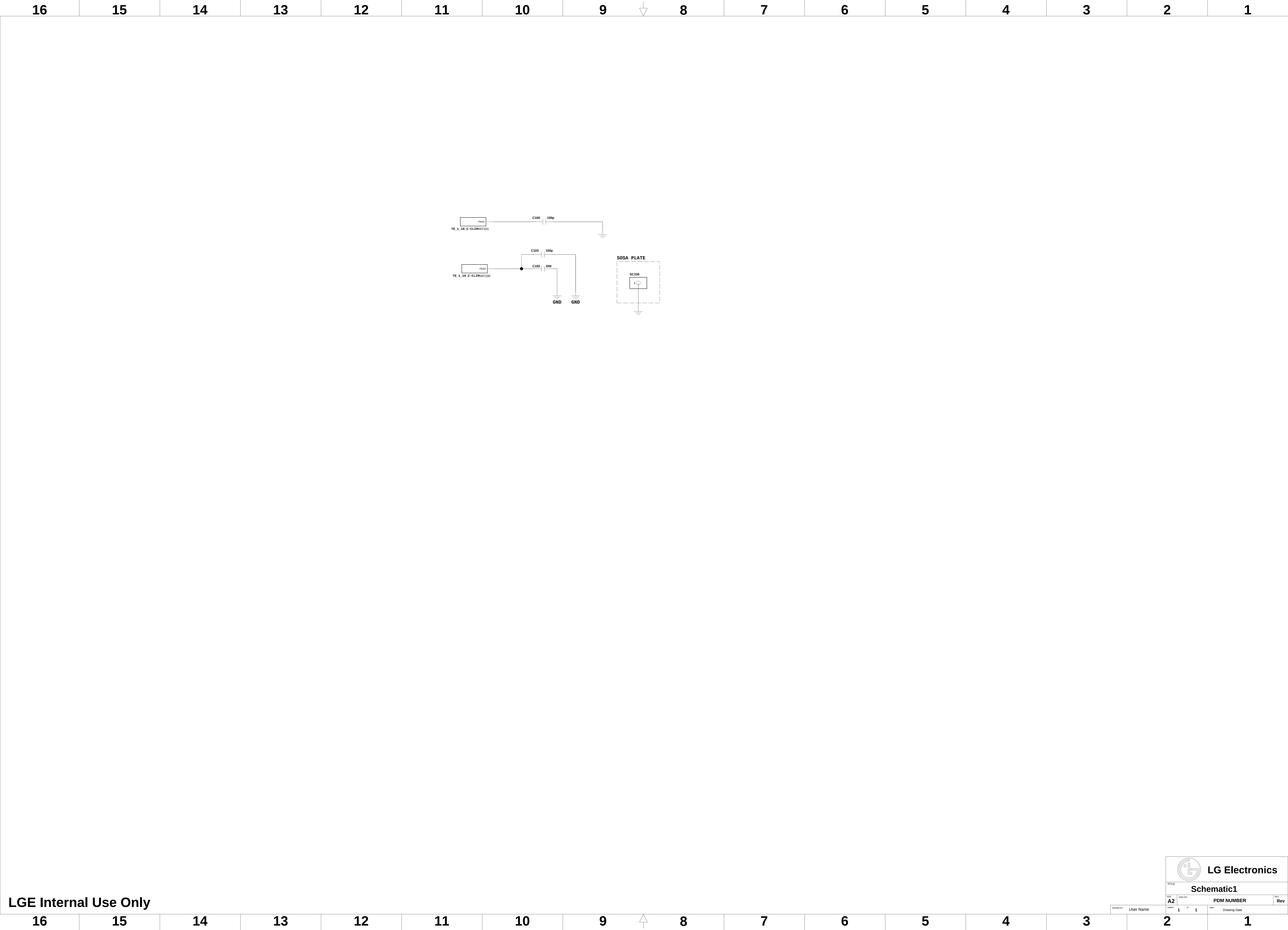


RGB LED

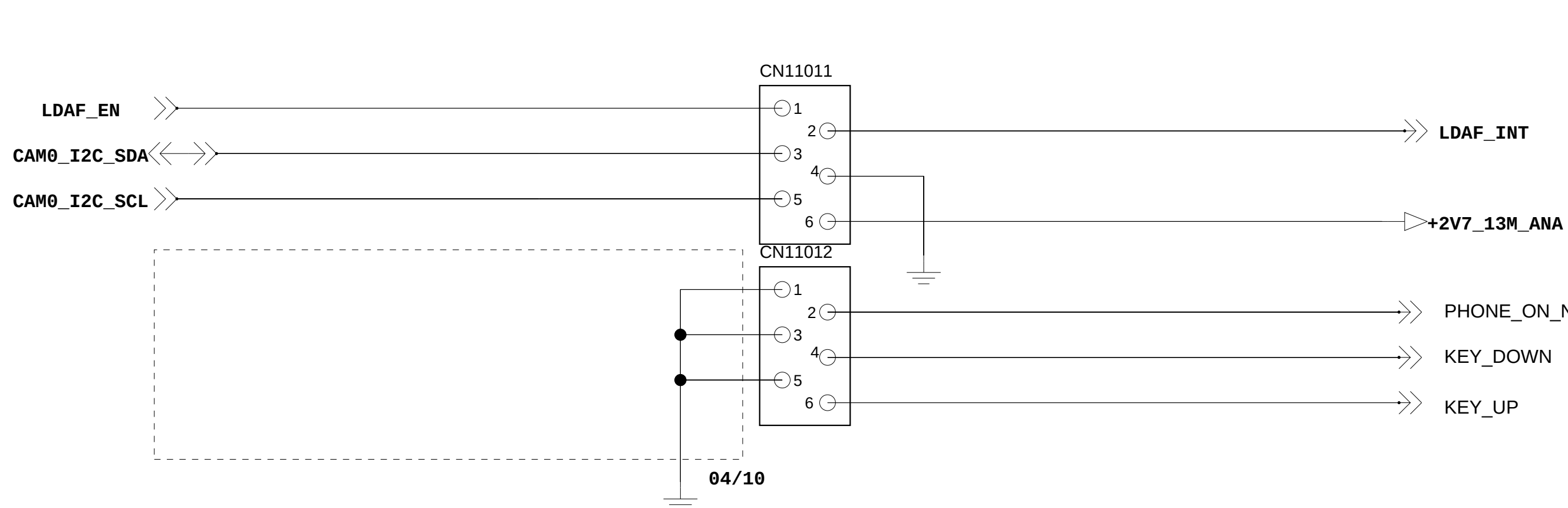


Temperature delete

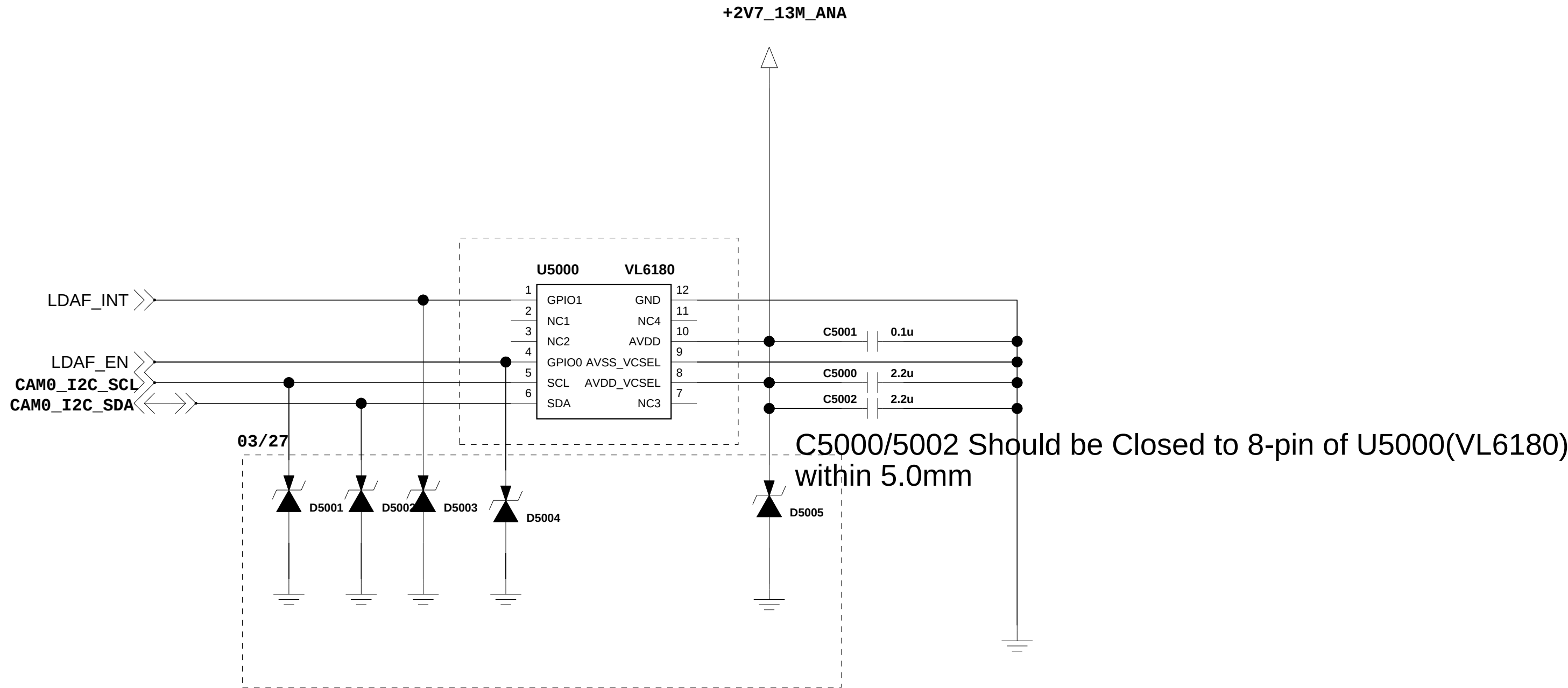




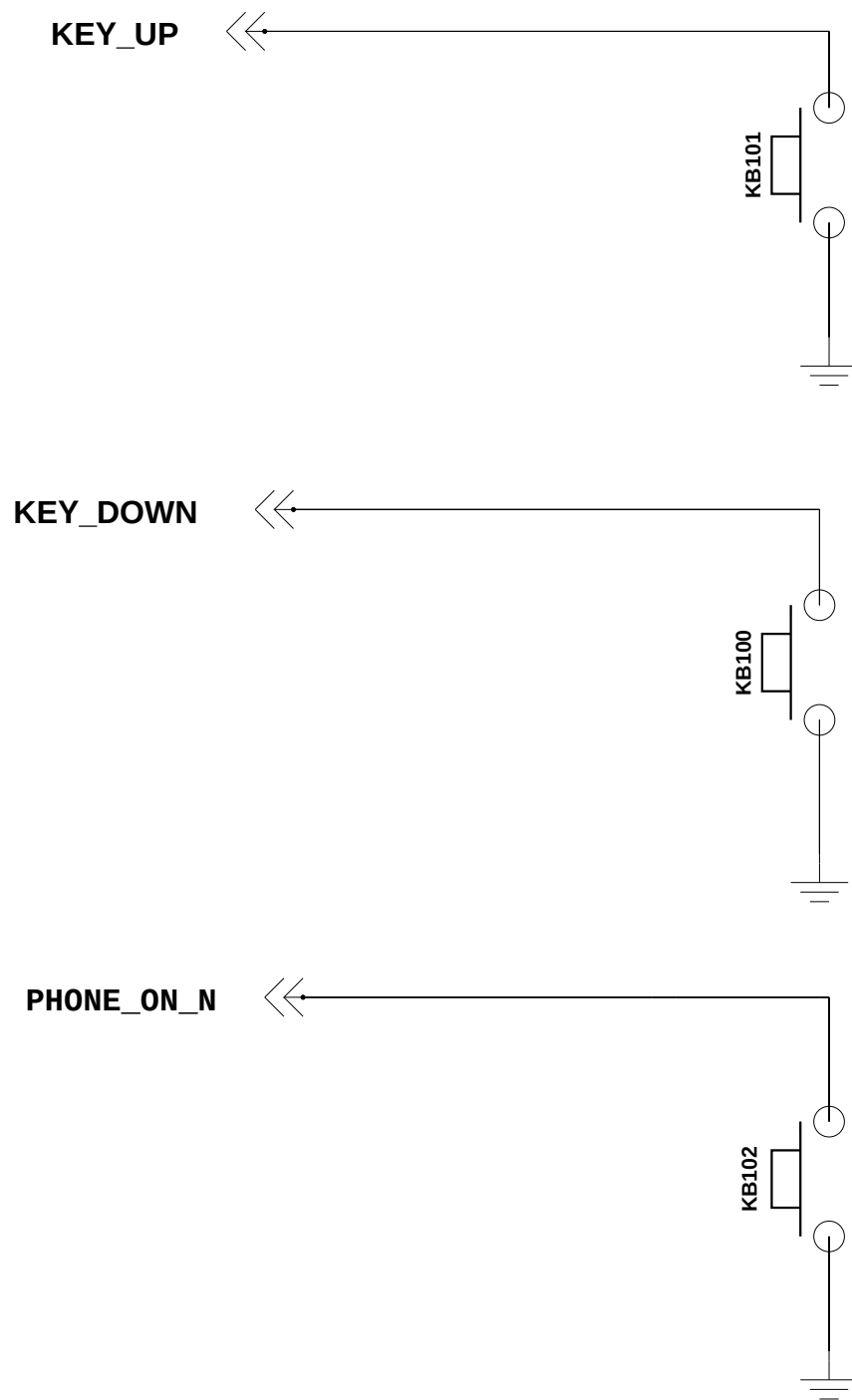
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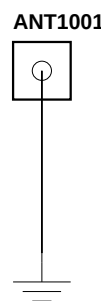
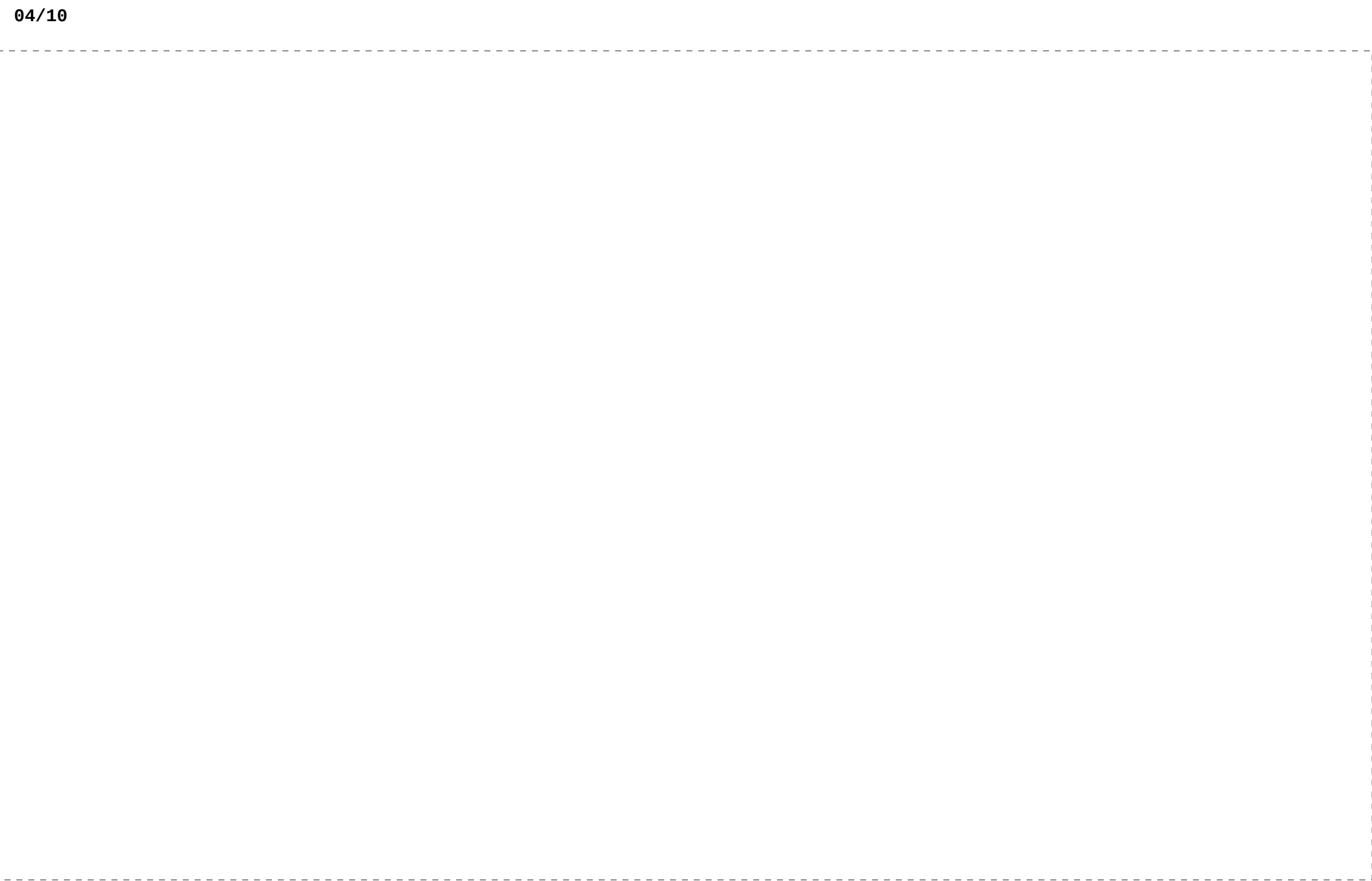
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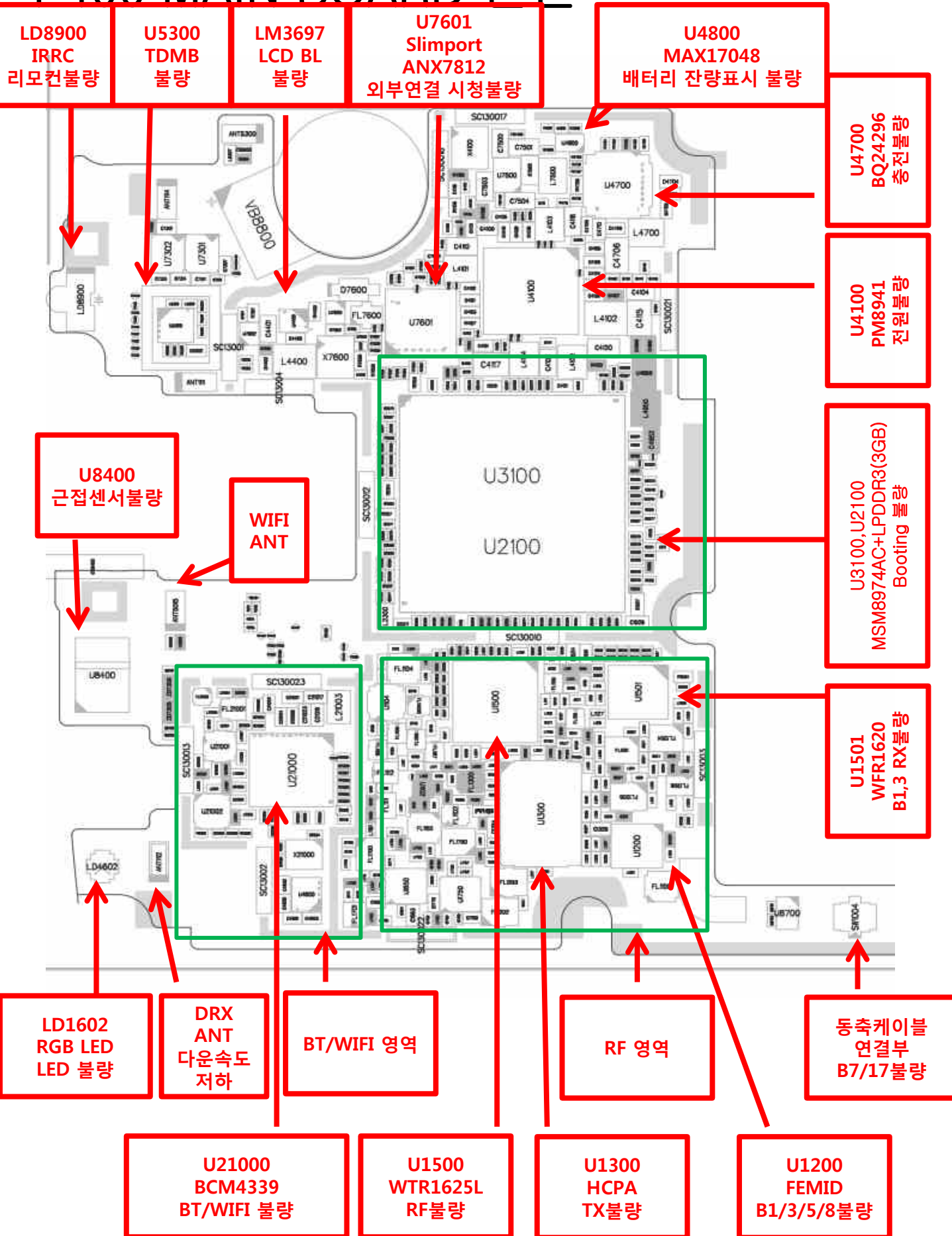
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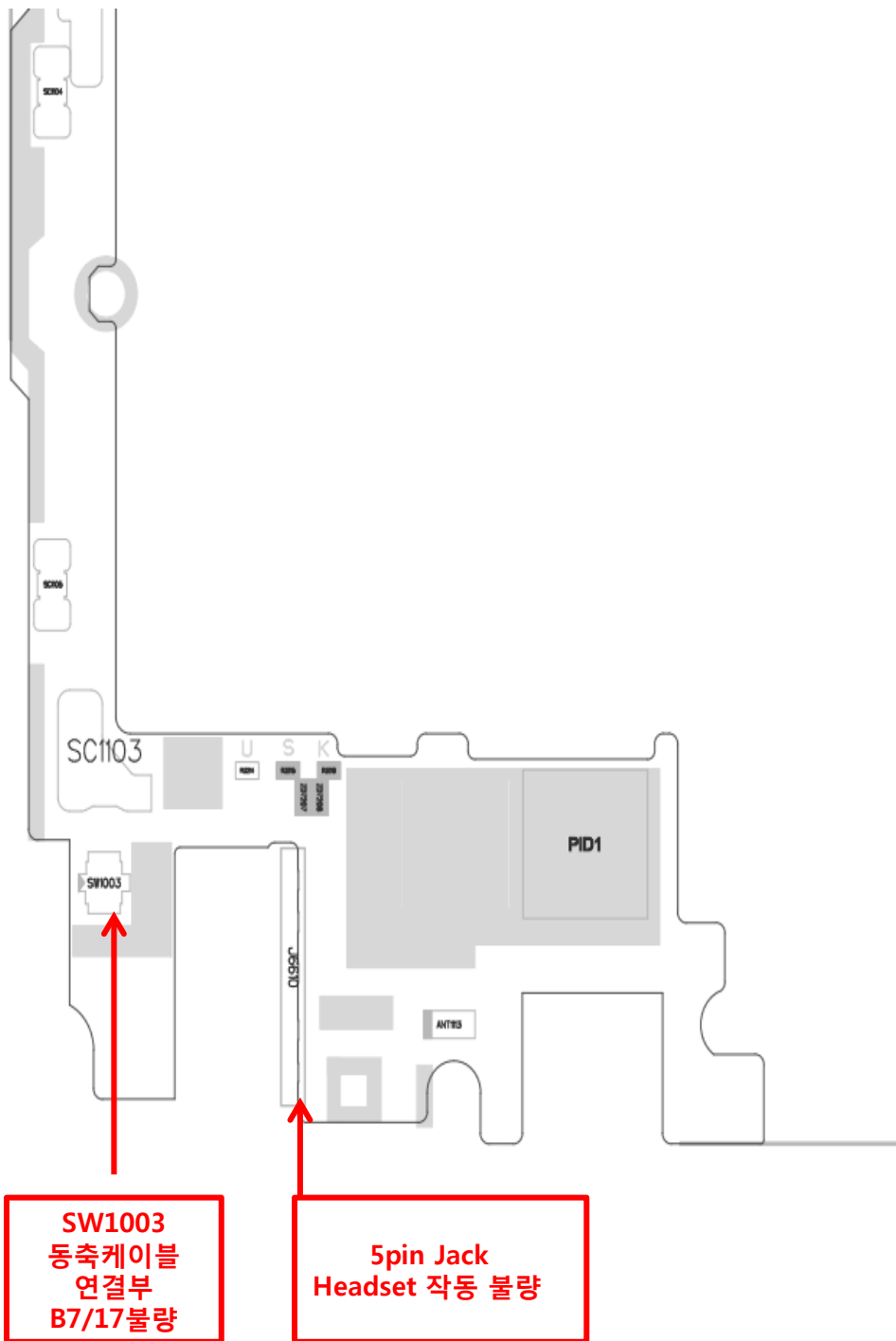
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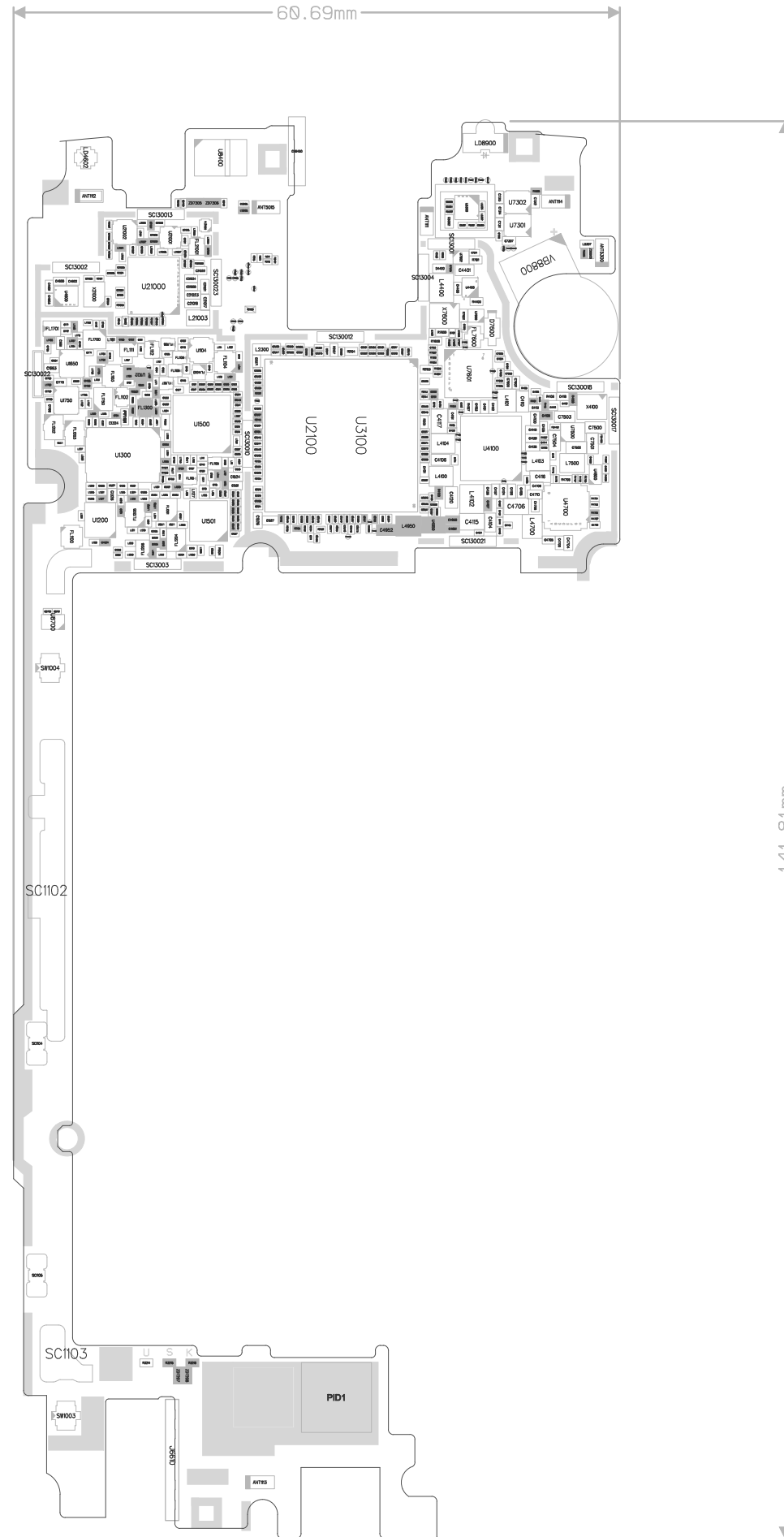


F400 MAIN BOARD 전면

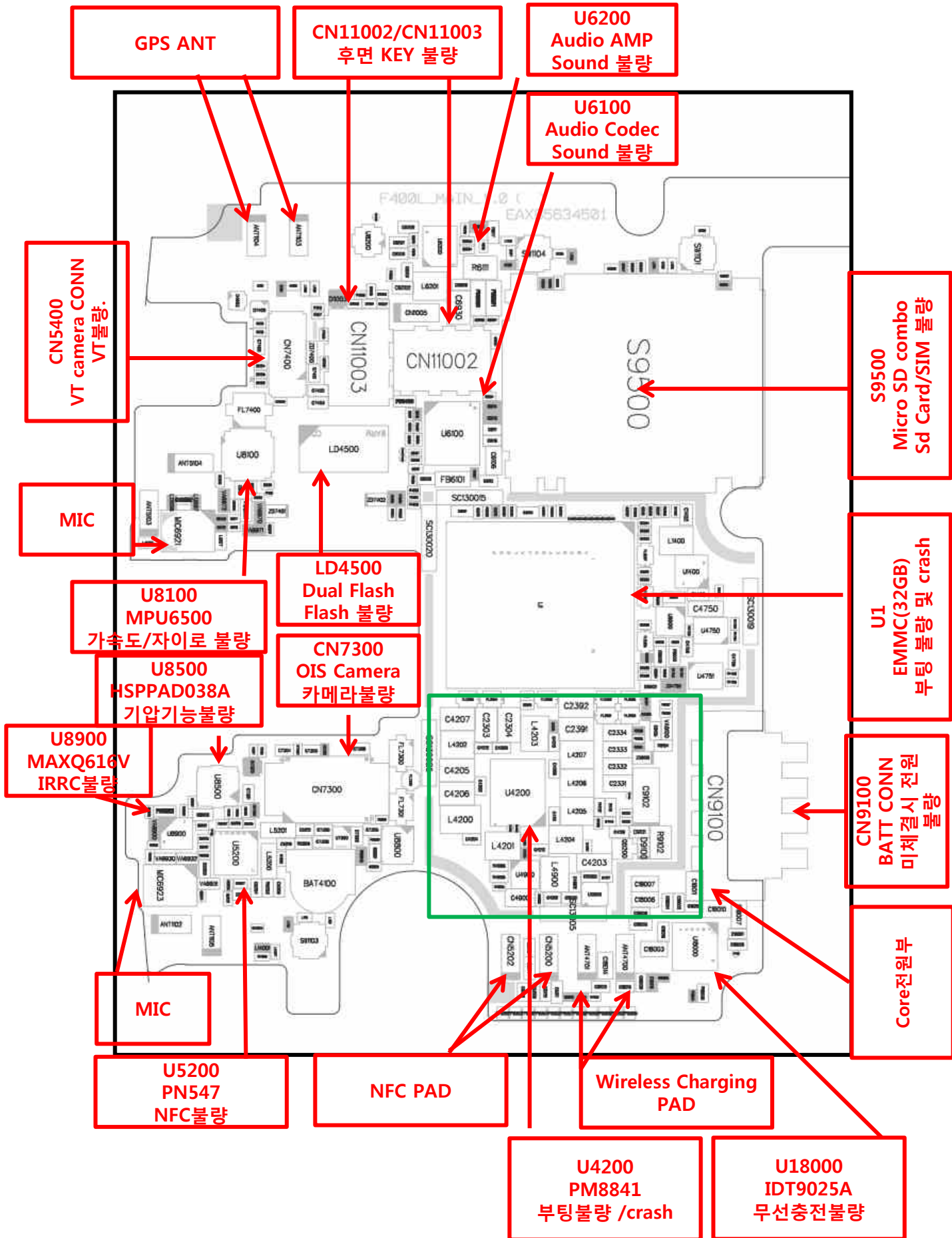


F400 MAIN BOARD 전면

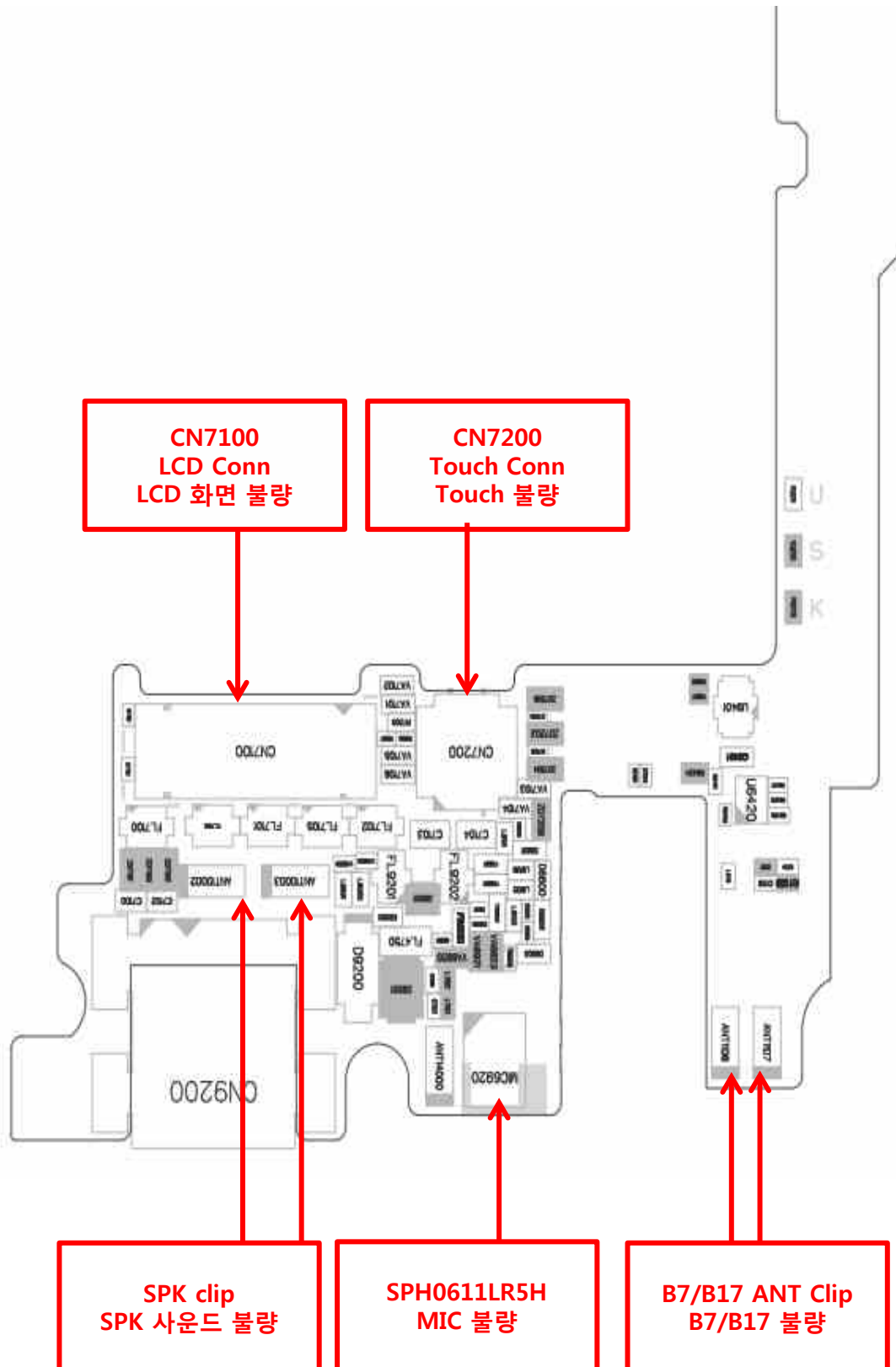


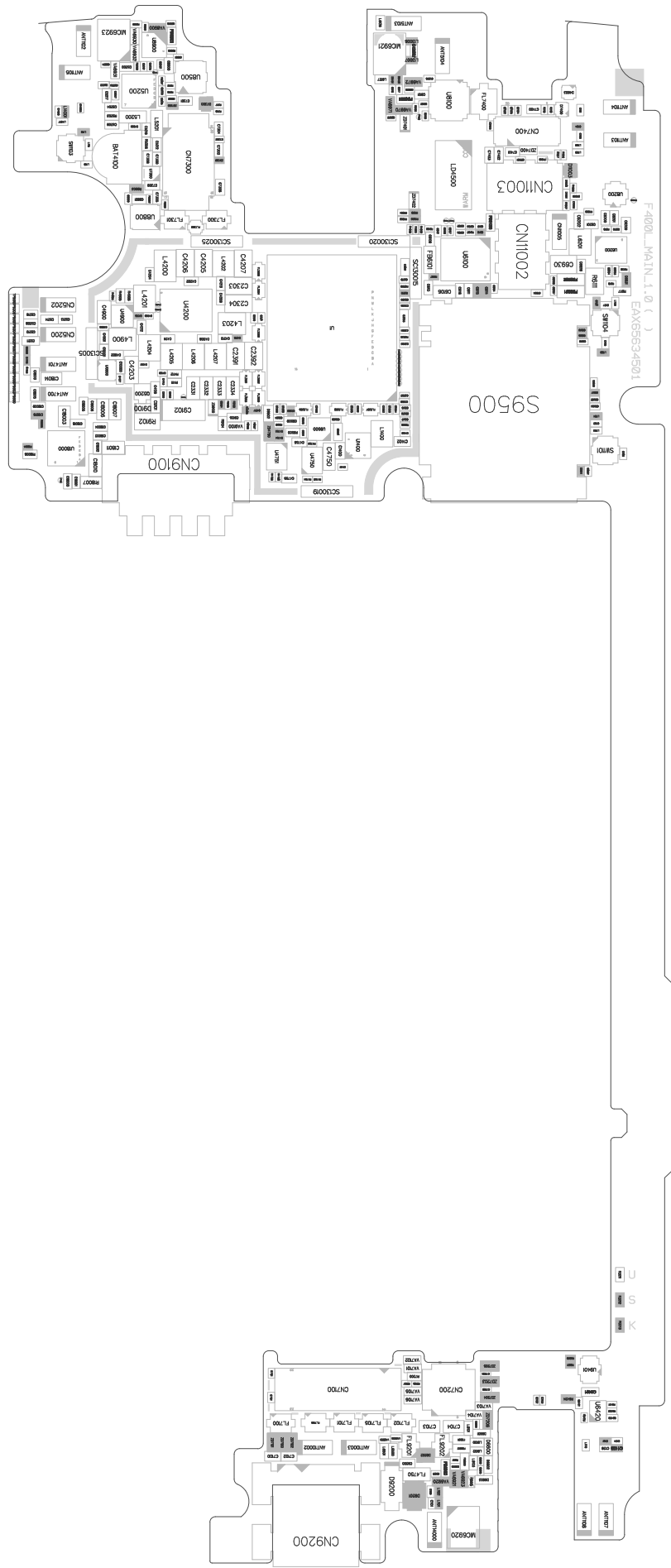


F400 MAIN BOARD 후면



F400 MAIN BOARD 후면





F400
SOSA
PCB

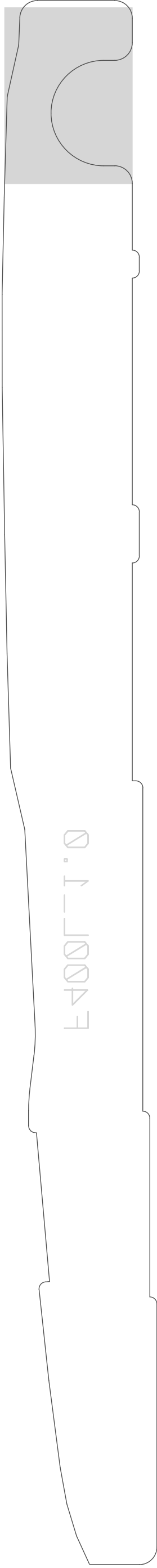
SOSA TOP

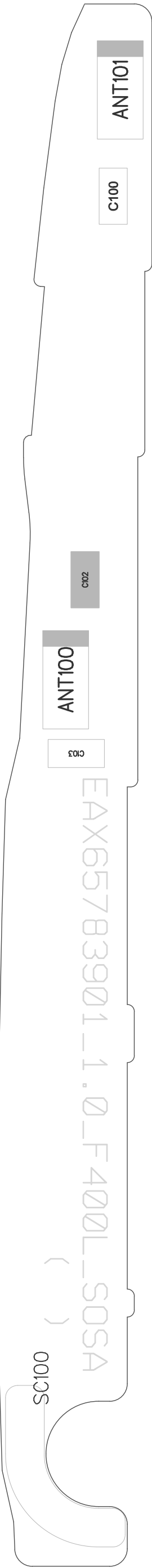


SOSA BOT



ANT clip
무선 성능 불량

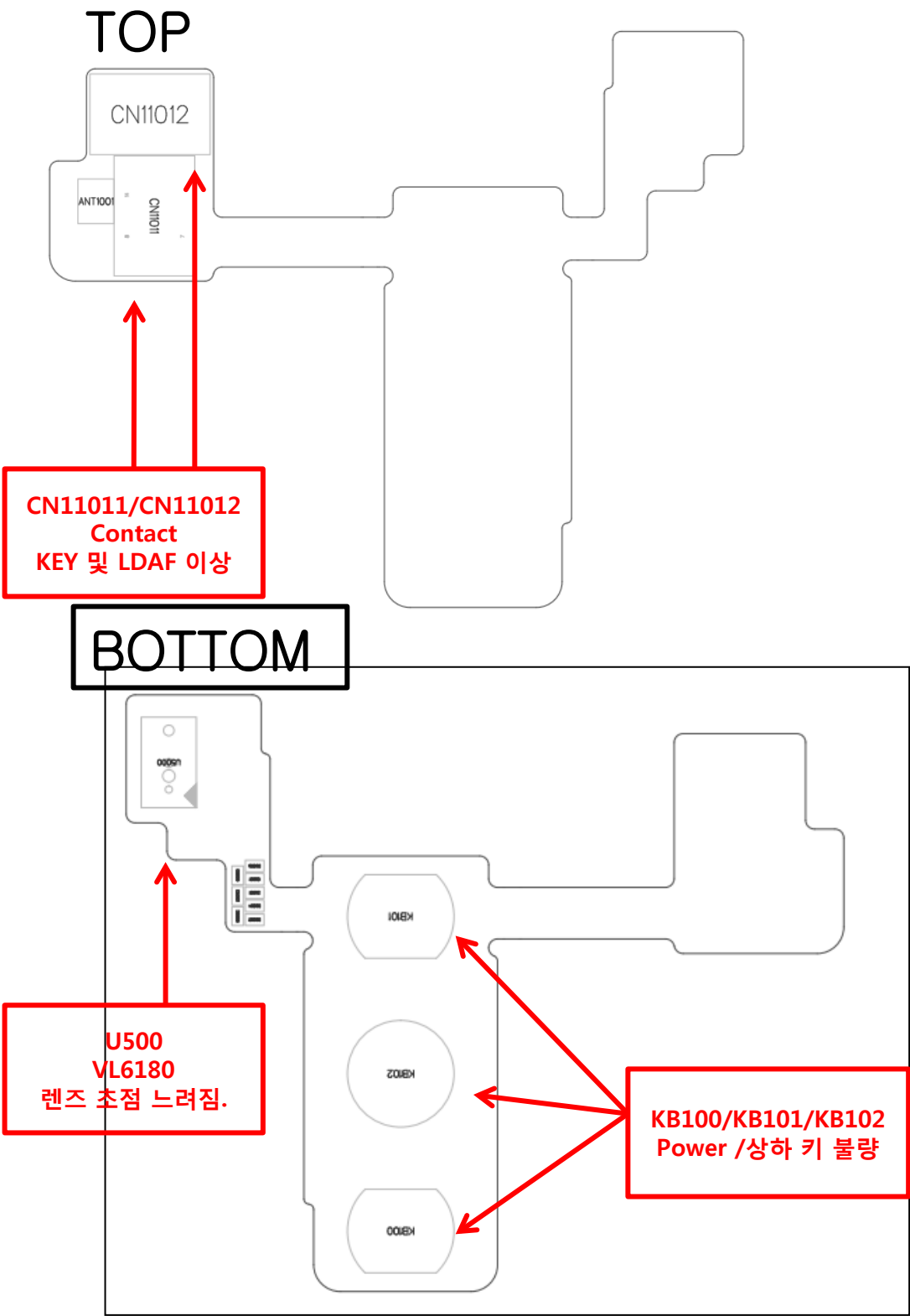


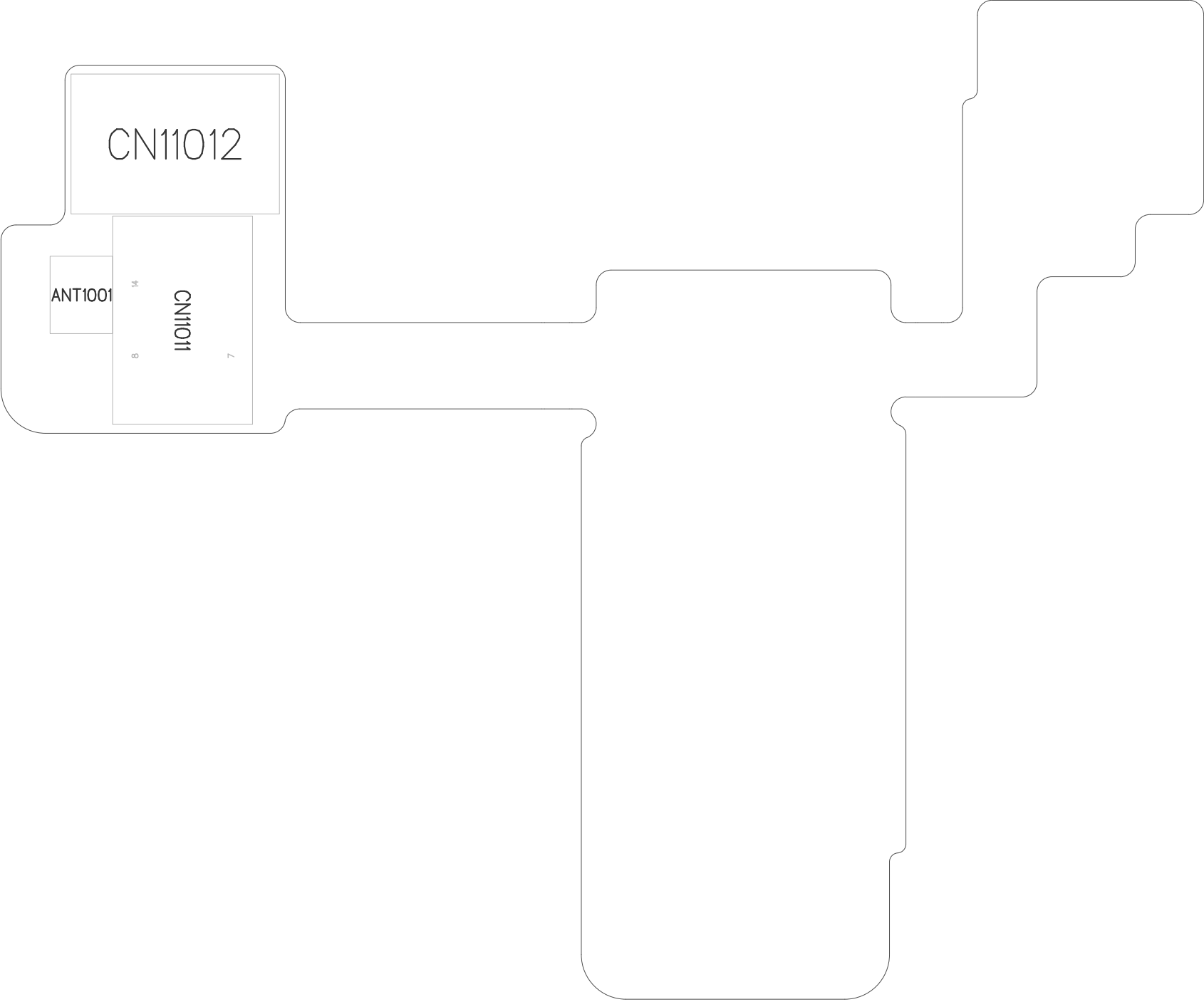


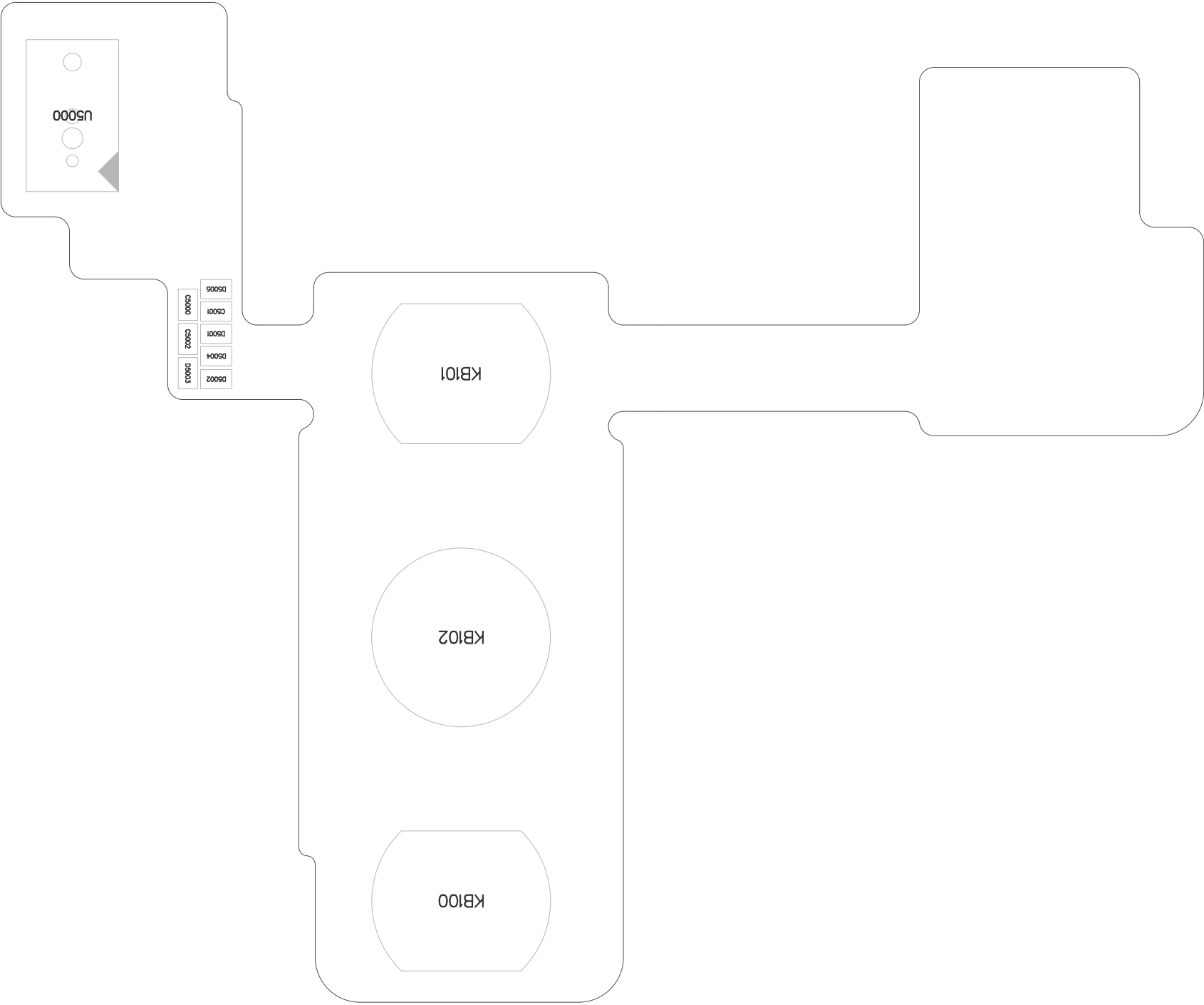
F400

Power_Key

FPCB



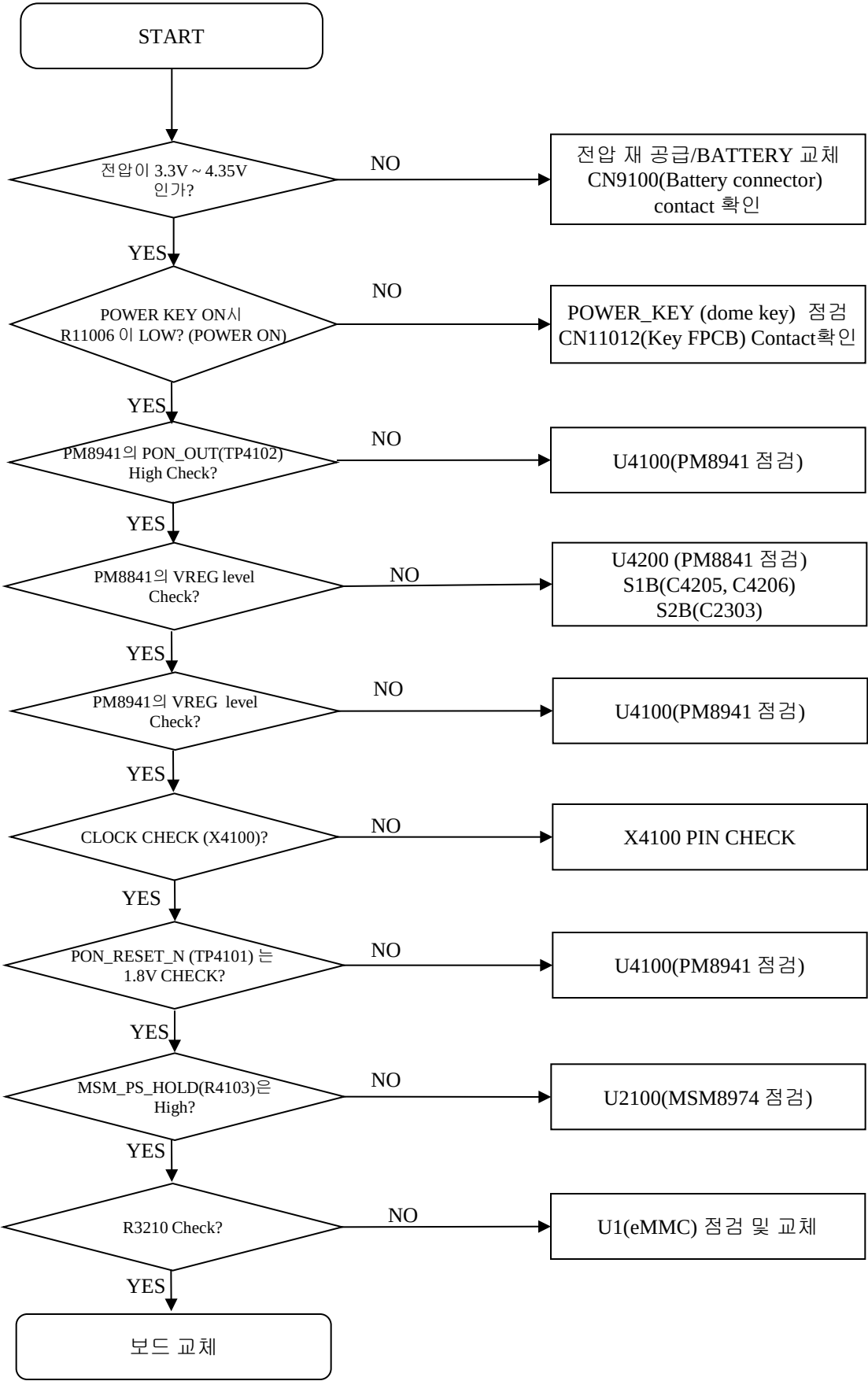




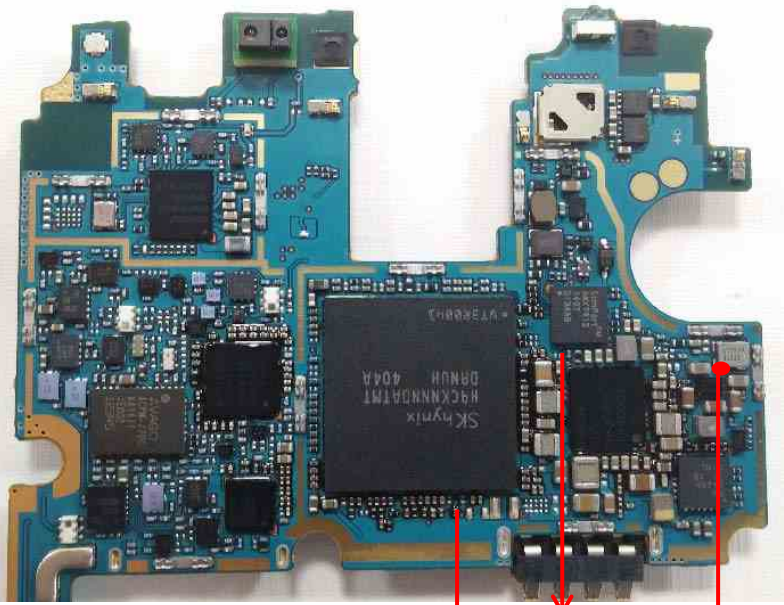
바. 주요기능 수리방법



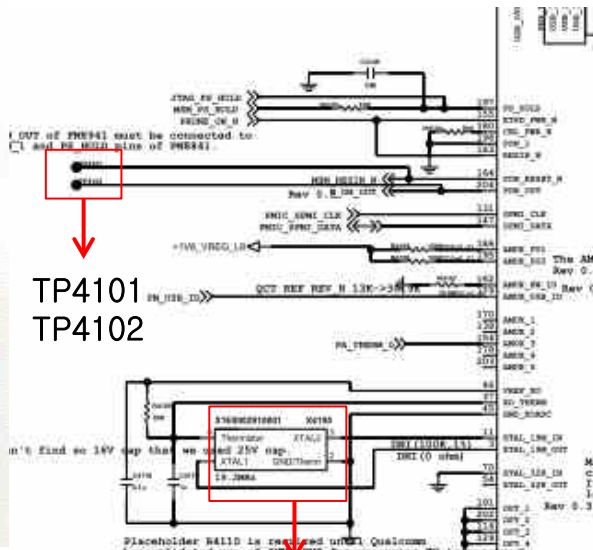
1. POWER



TOP

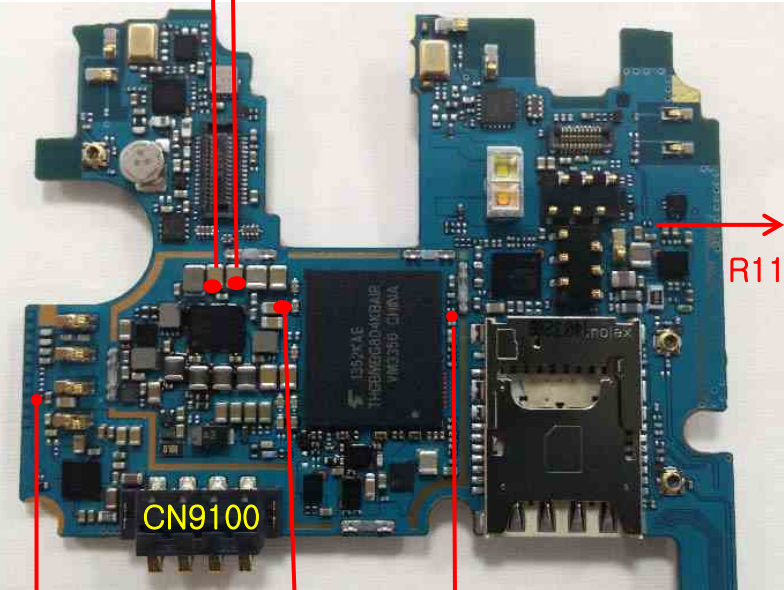


TP4102
TP4101
X4100
좌측하단

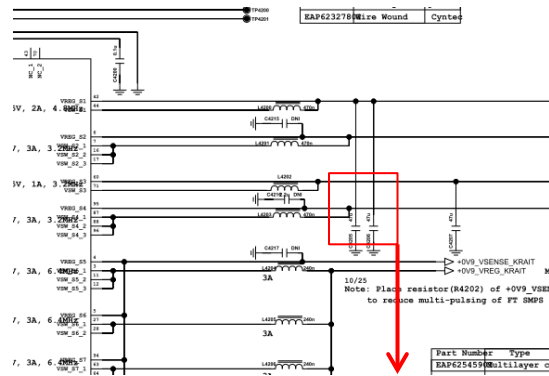


X4100

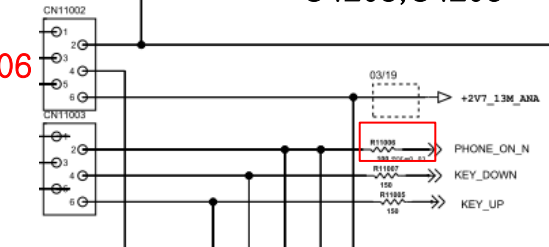
Bottom



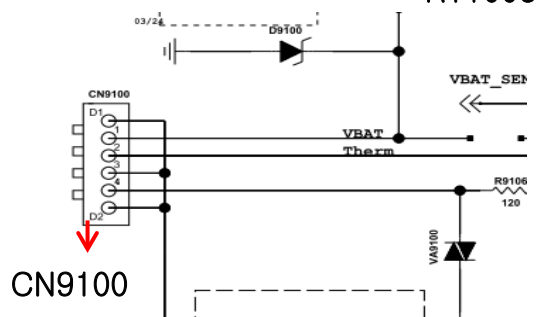
R4103
C2303
C2303



C4205,C4206

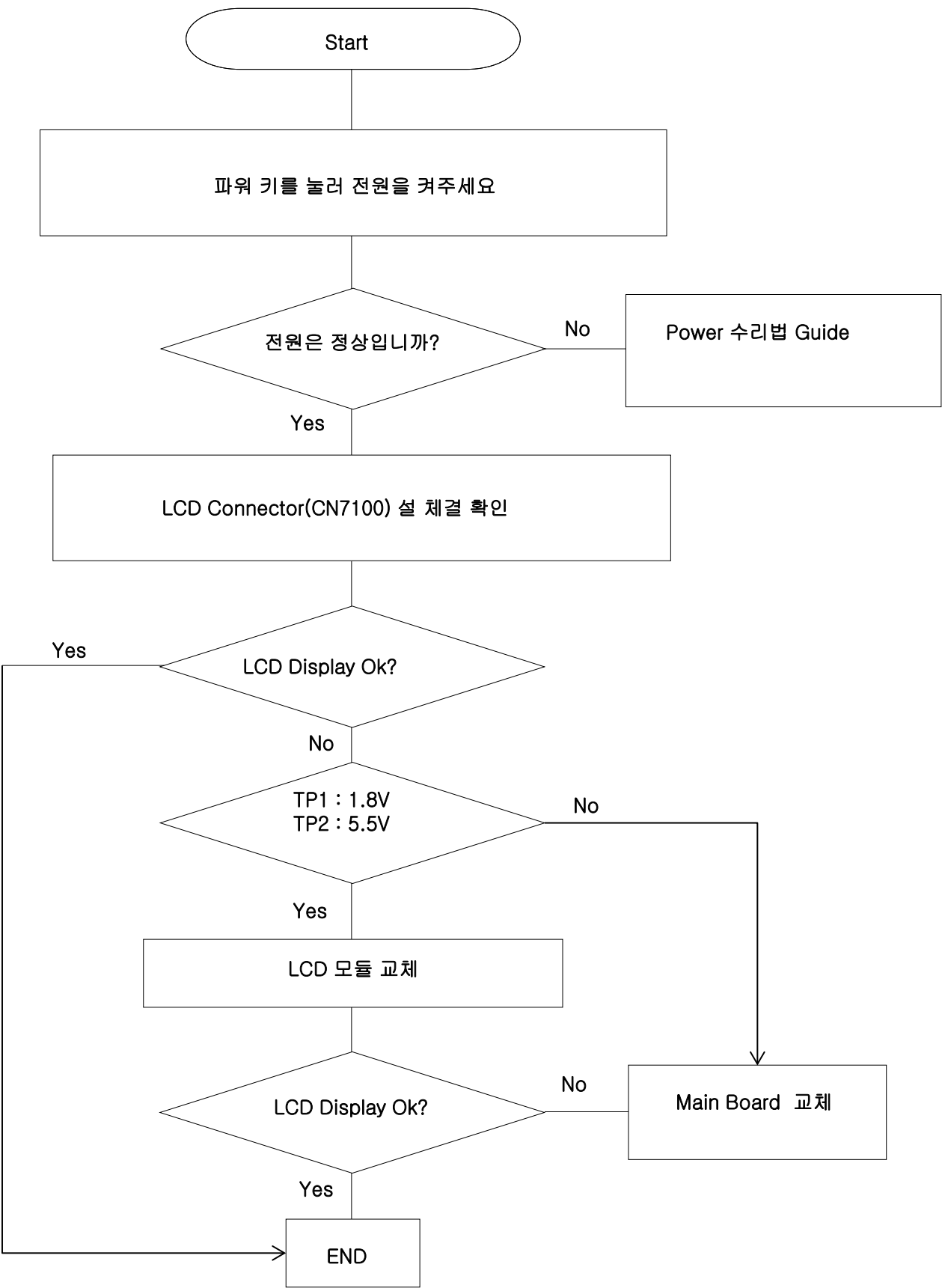


R11006

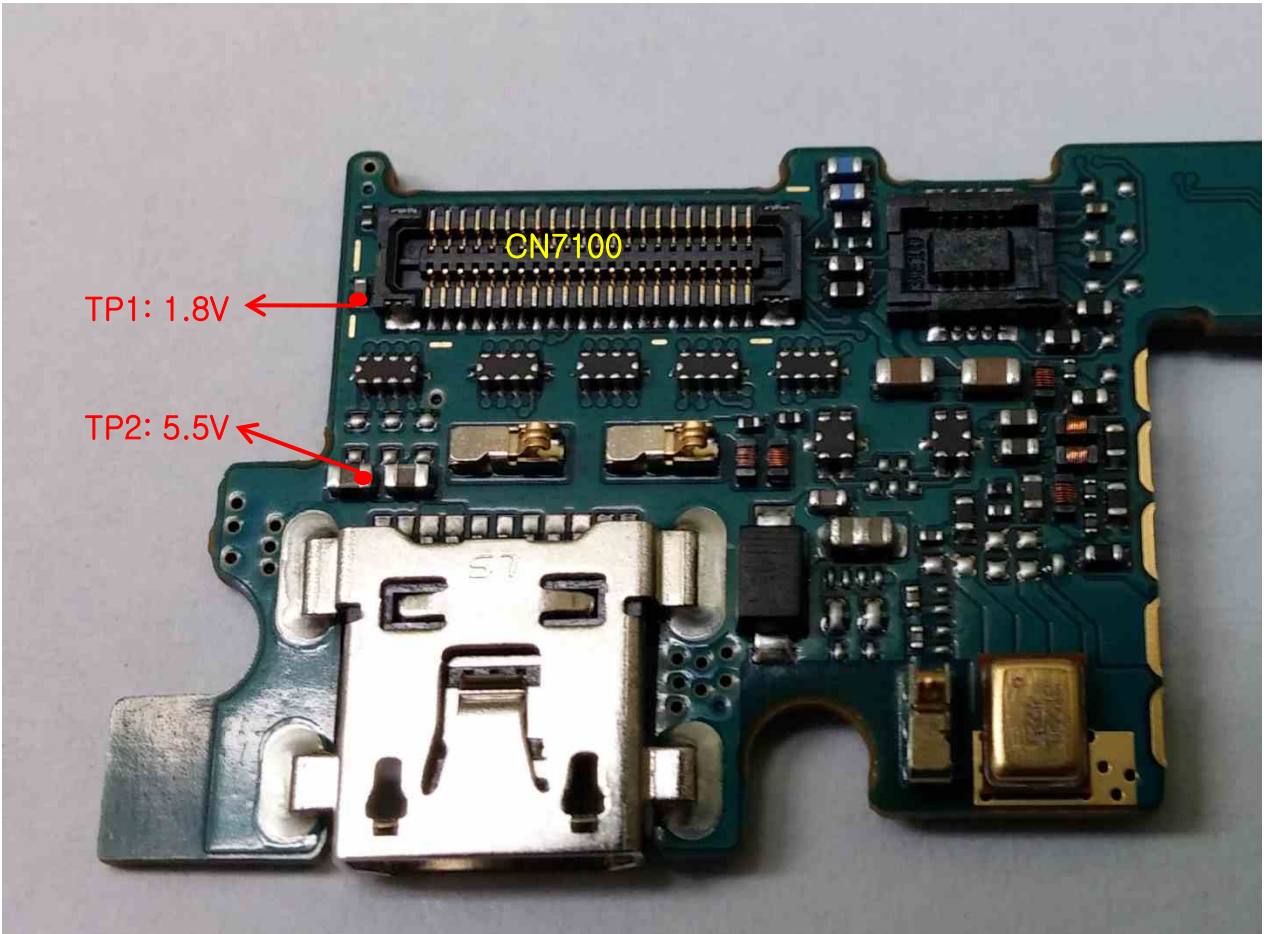
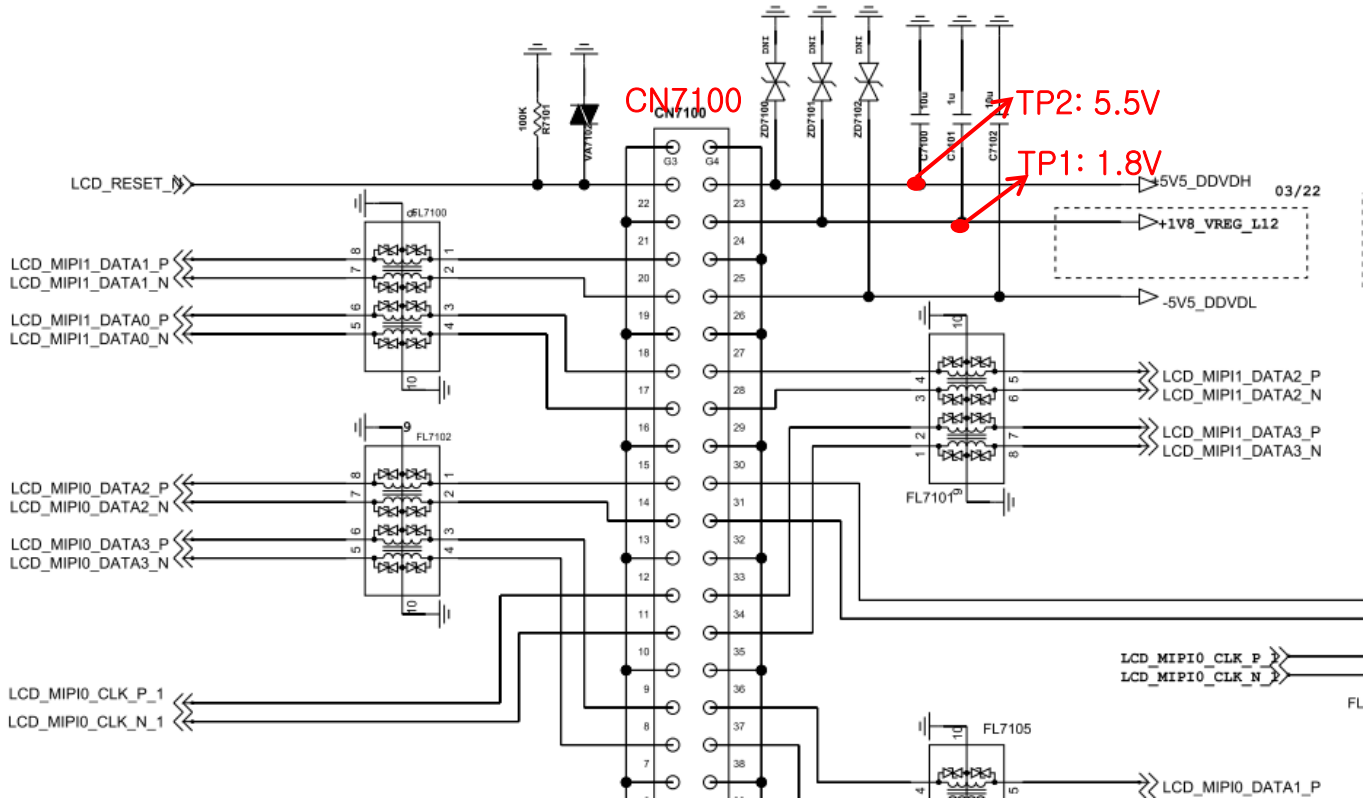


CN9100

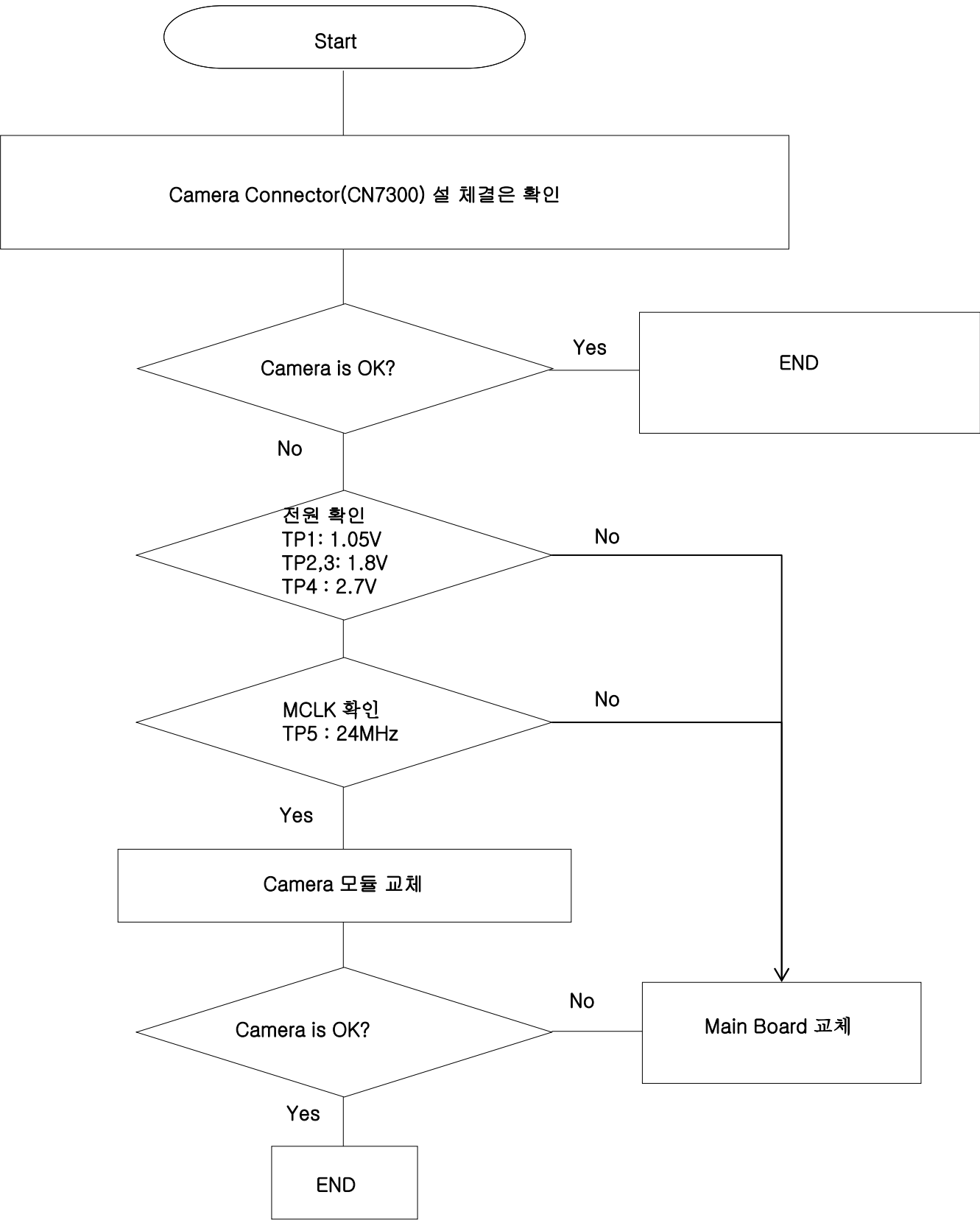
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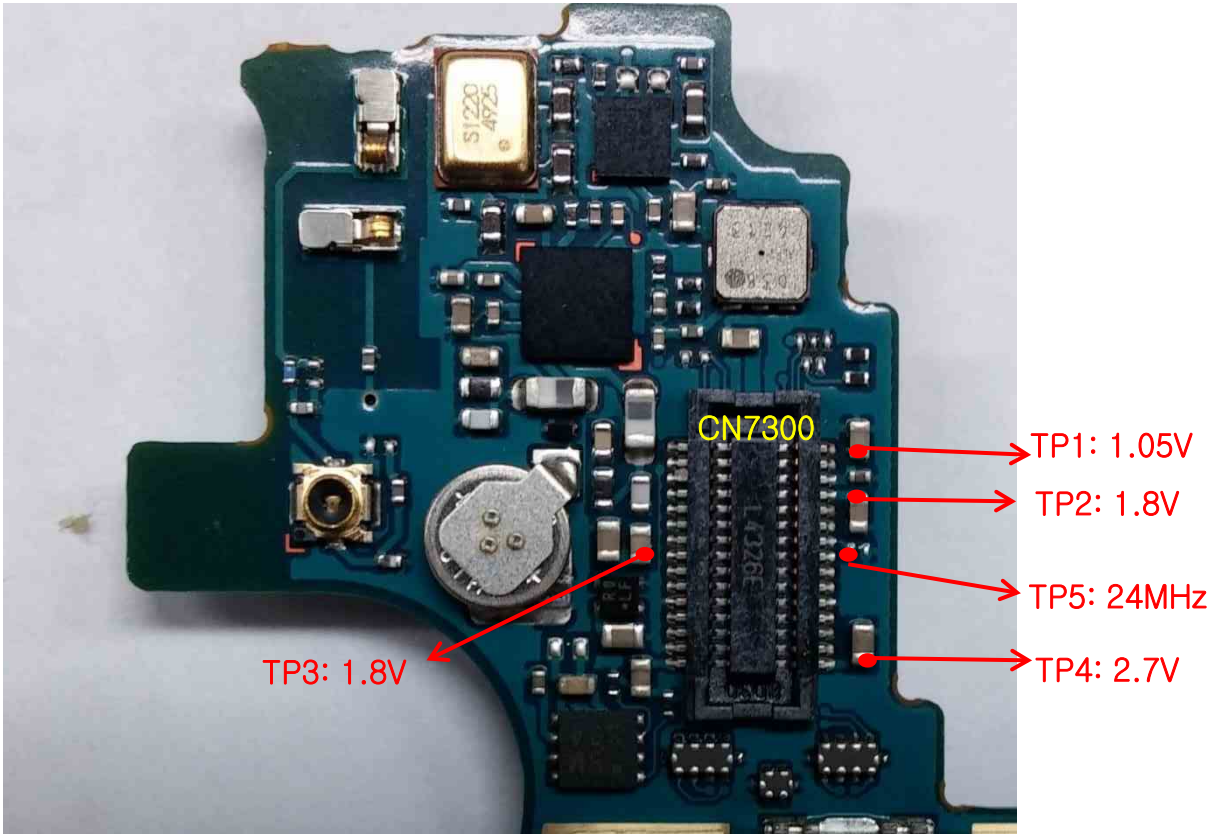
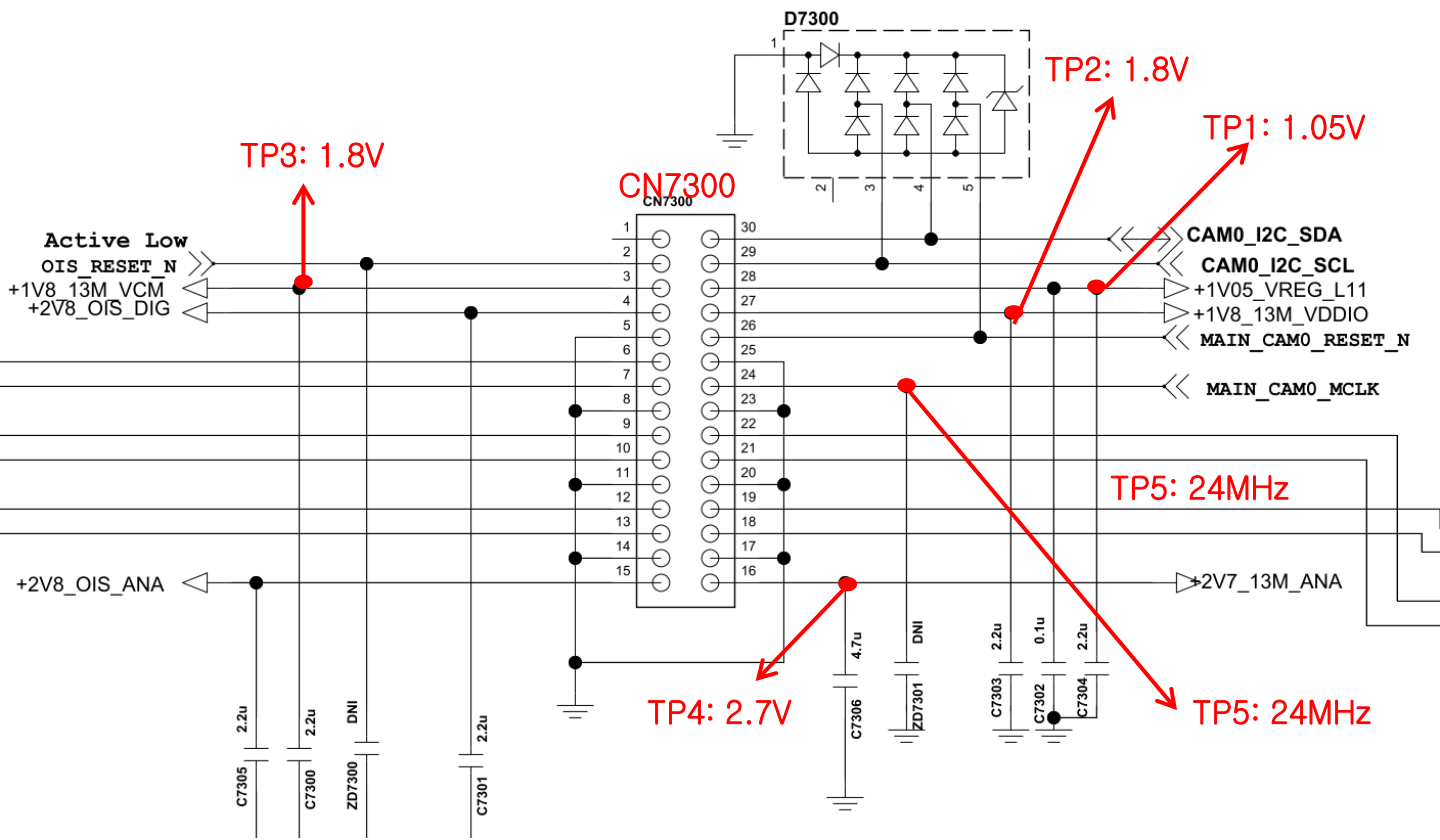
2. Main Display



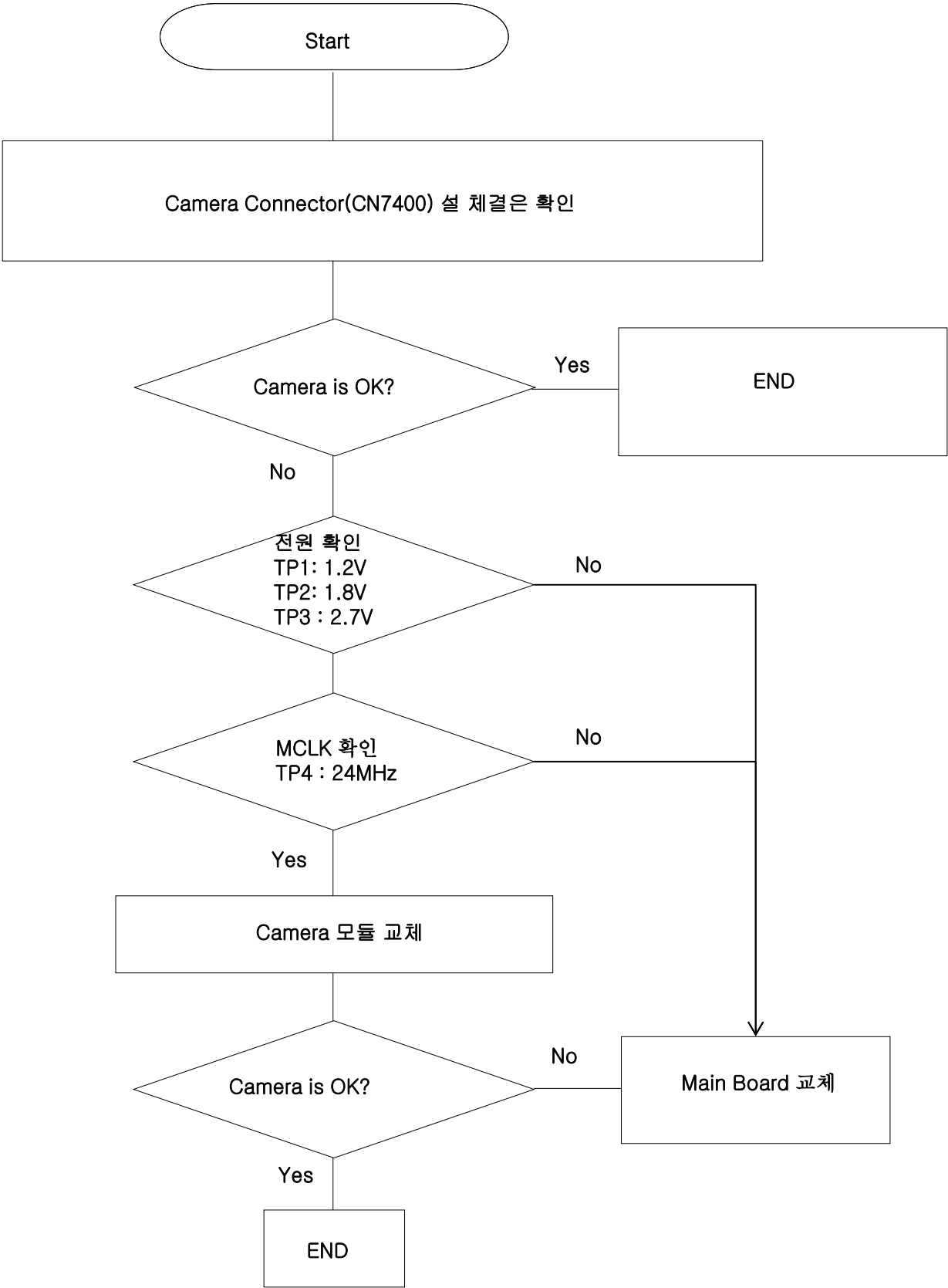
3. 13M Camera



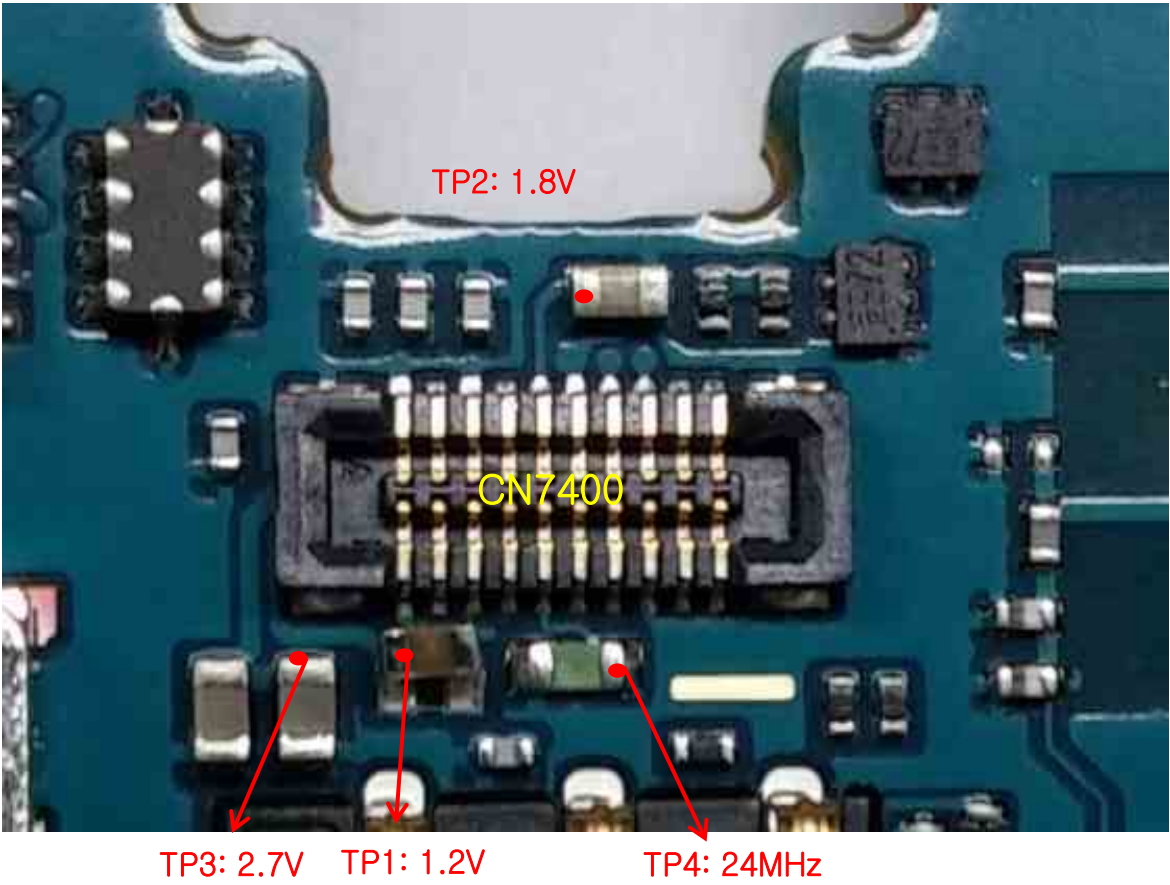
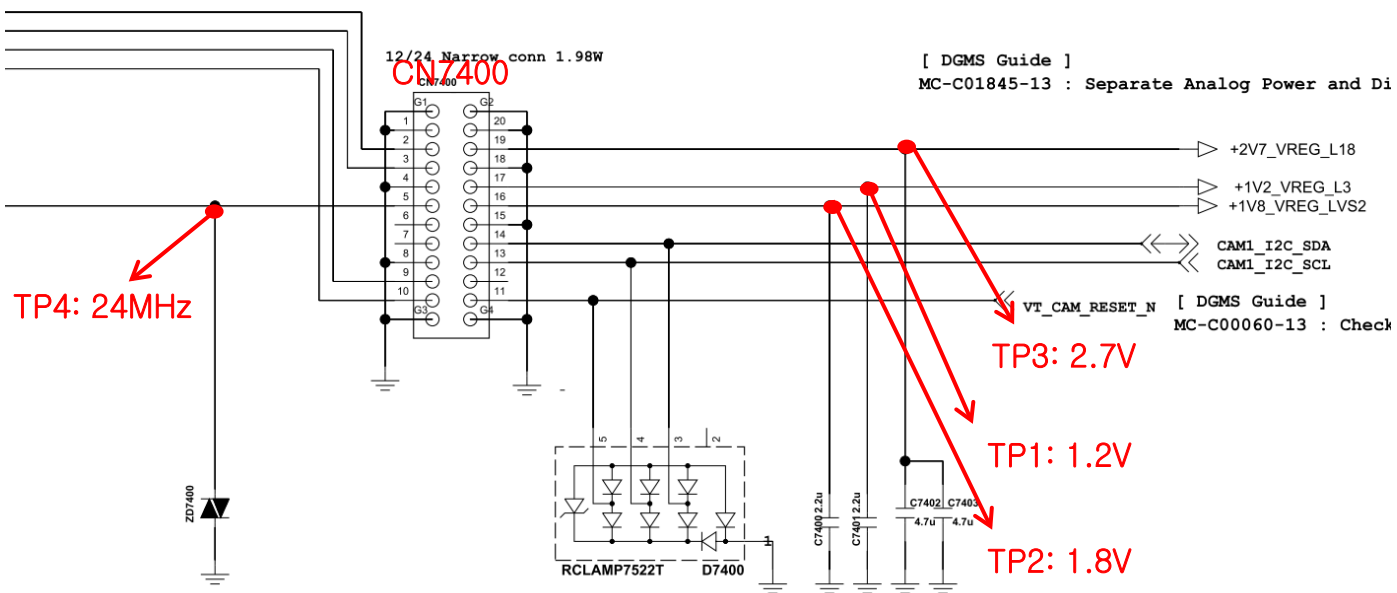
3. 13M Camera



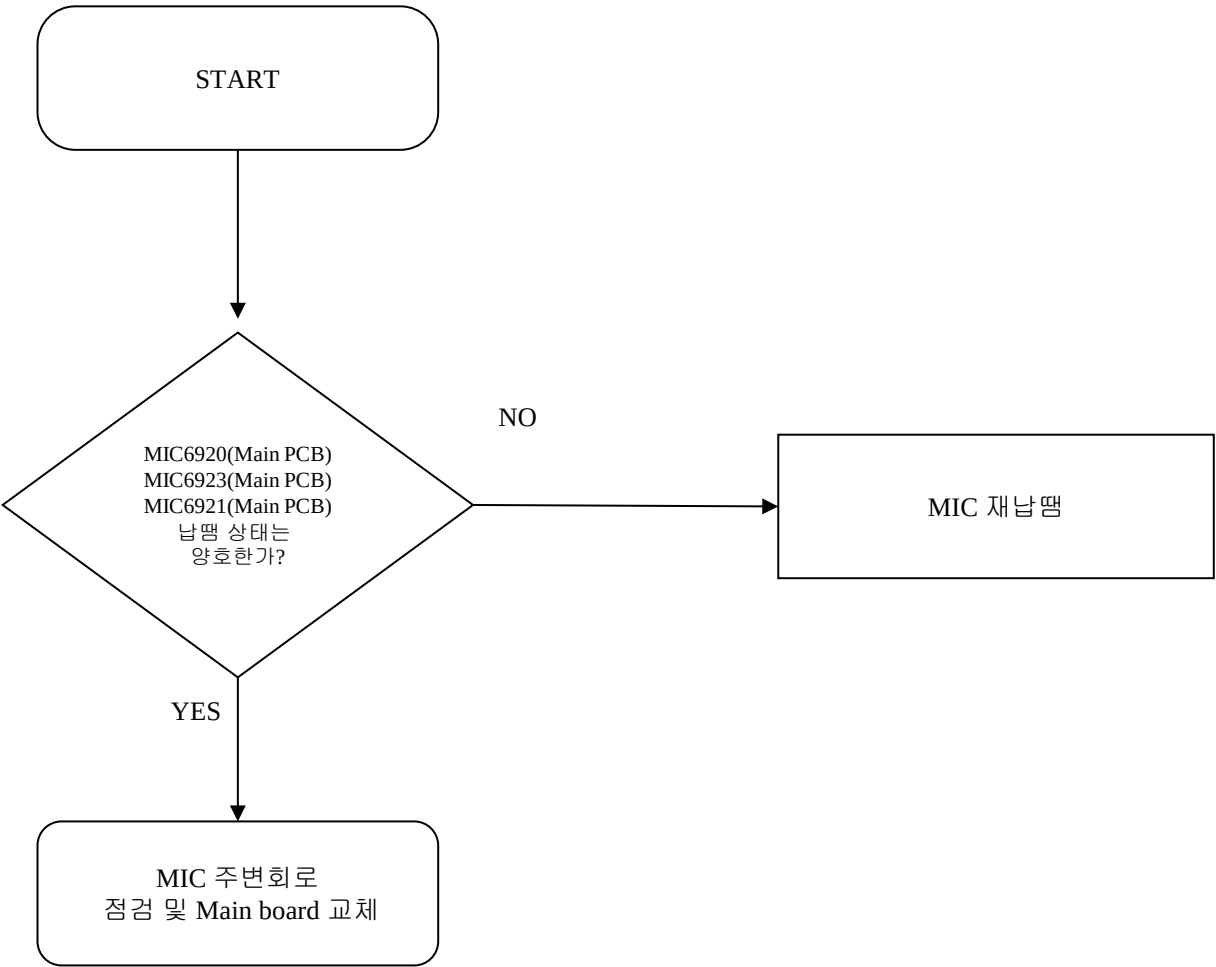
4. 2.1M Camera



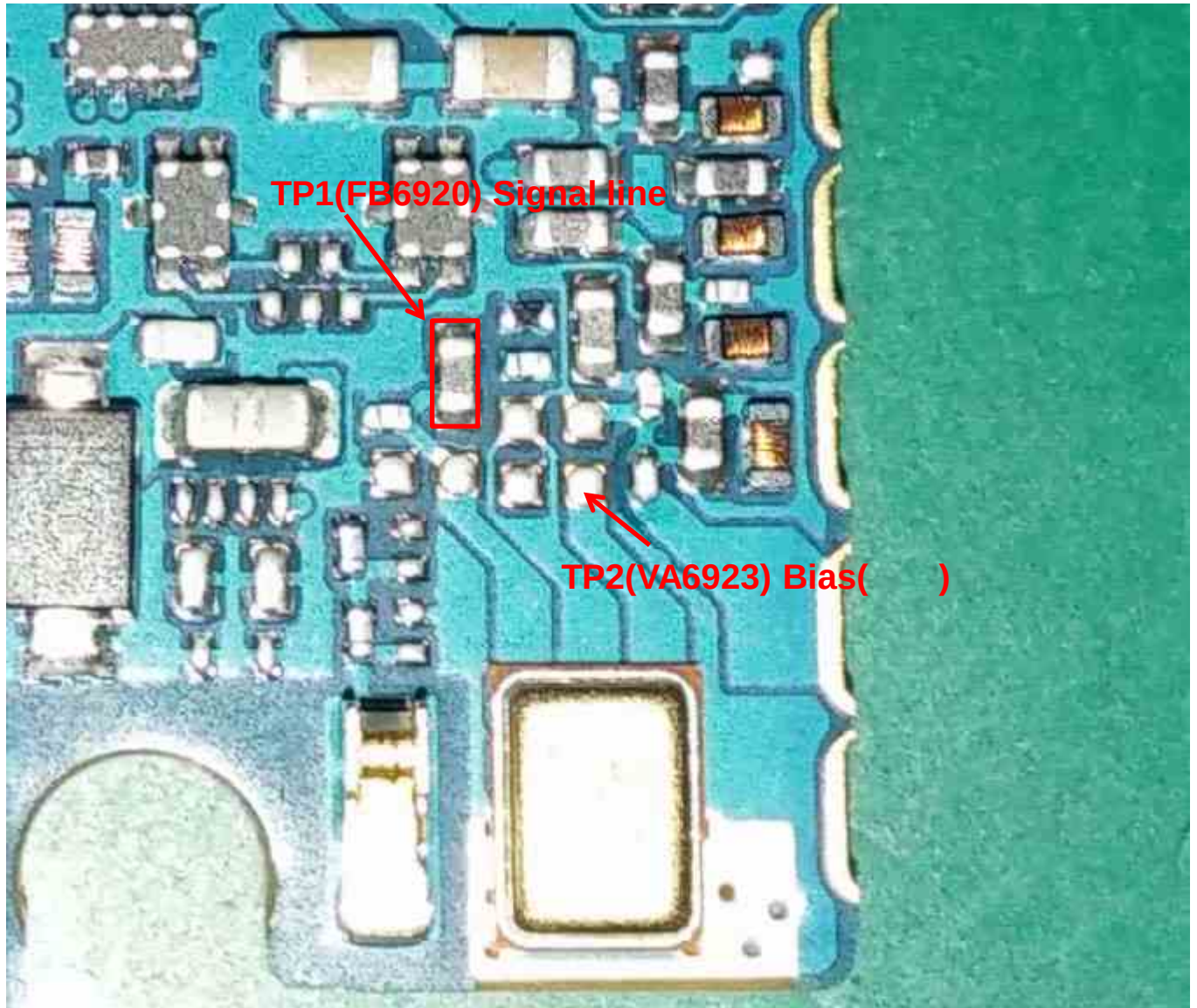
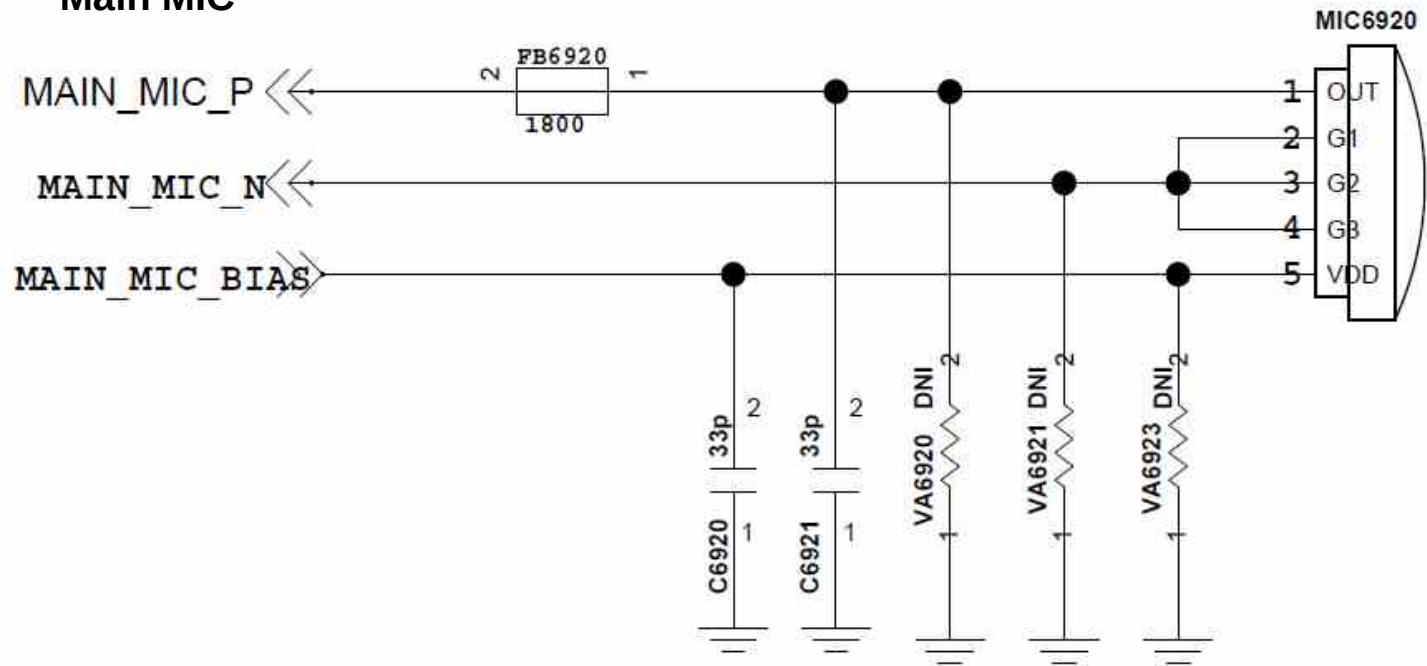
4. 2.1M Camera



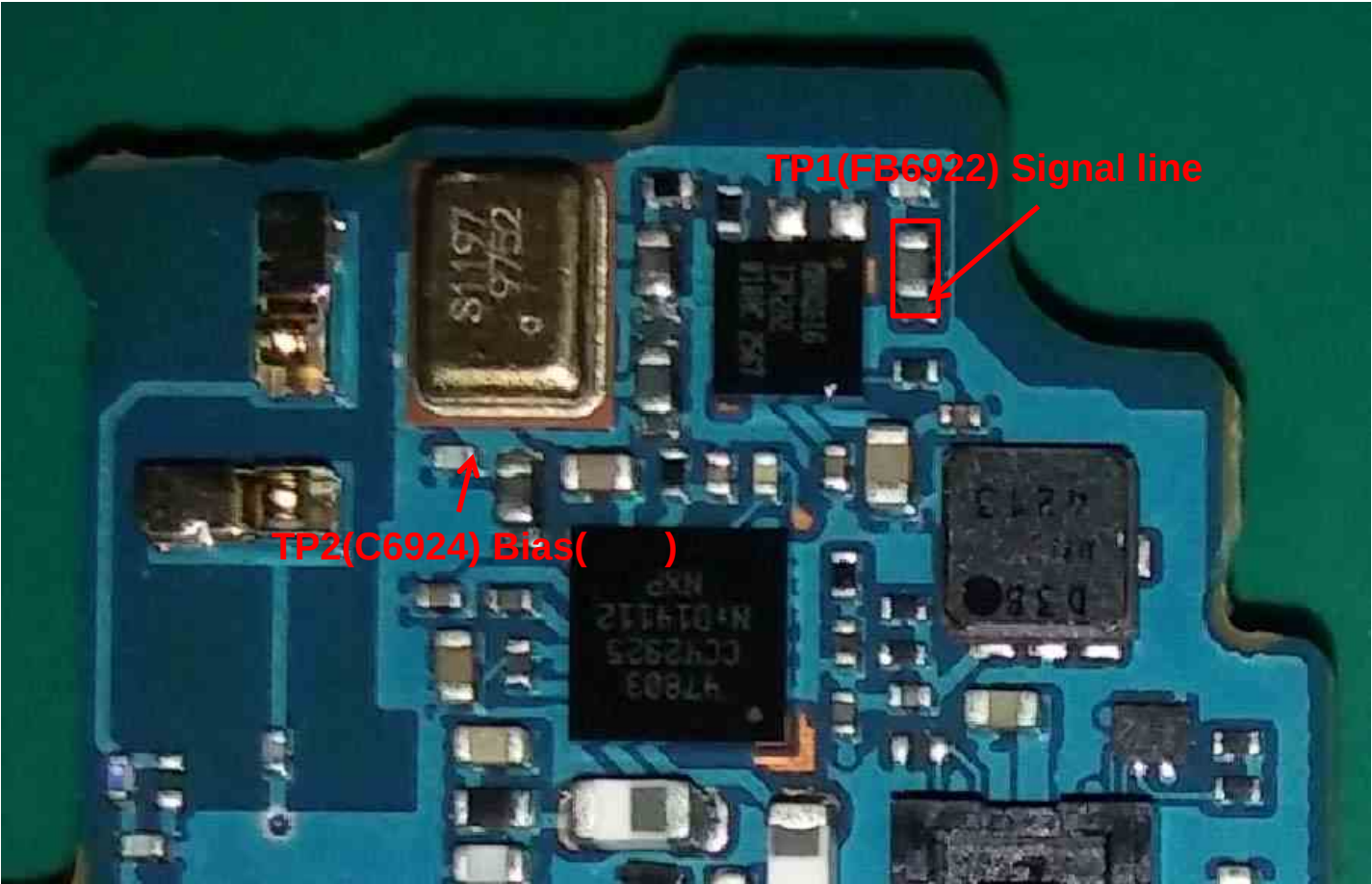
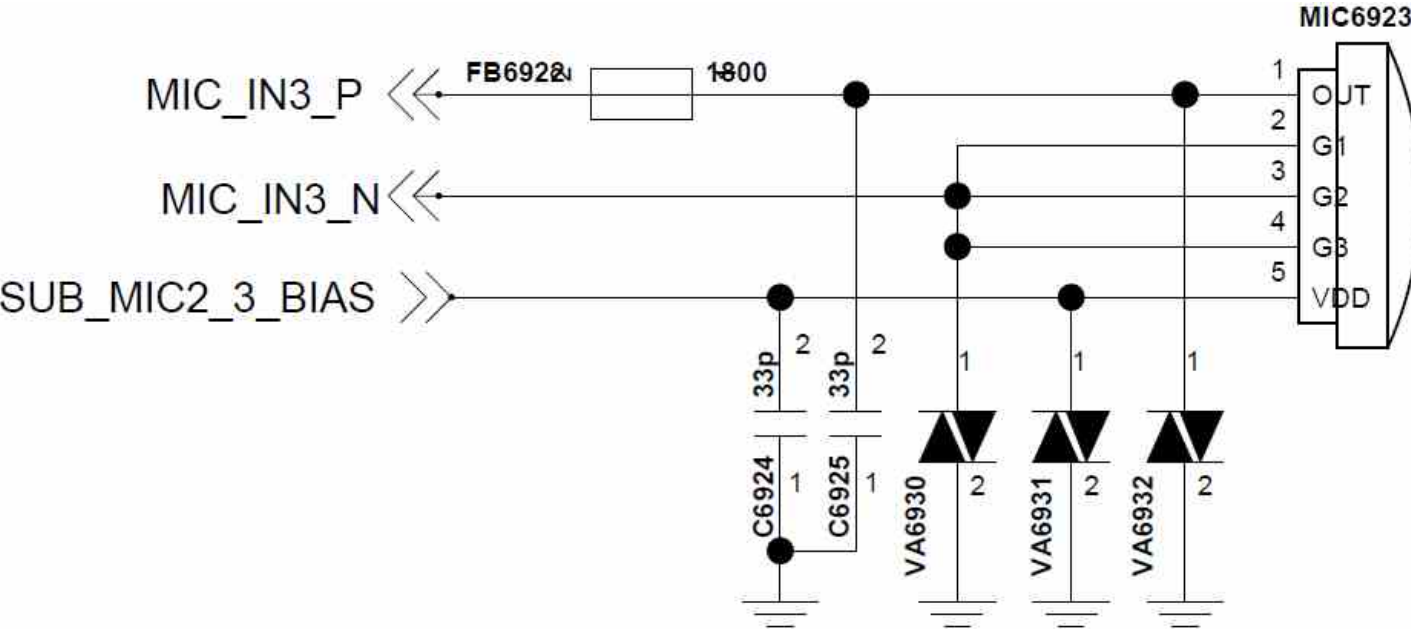
5. Mic



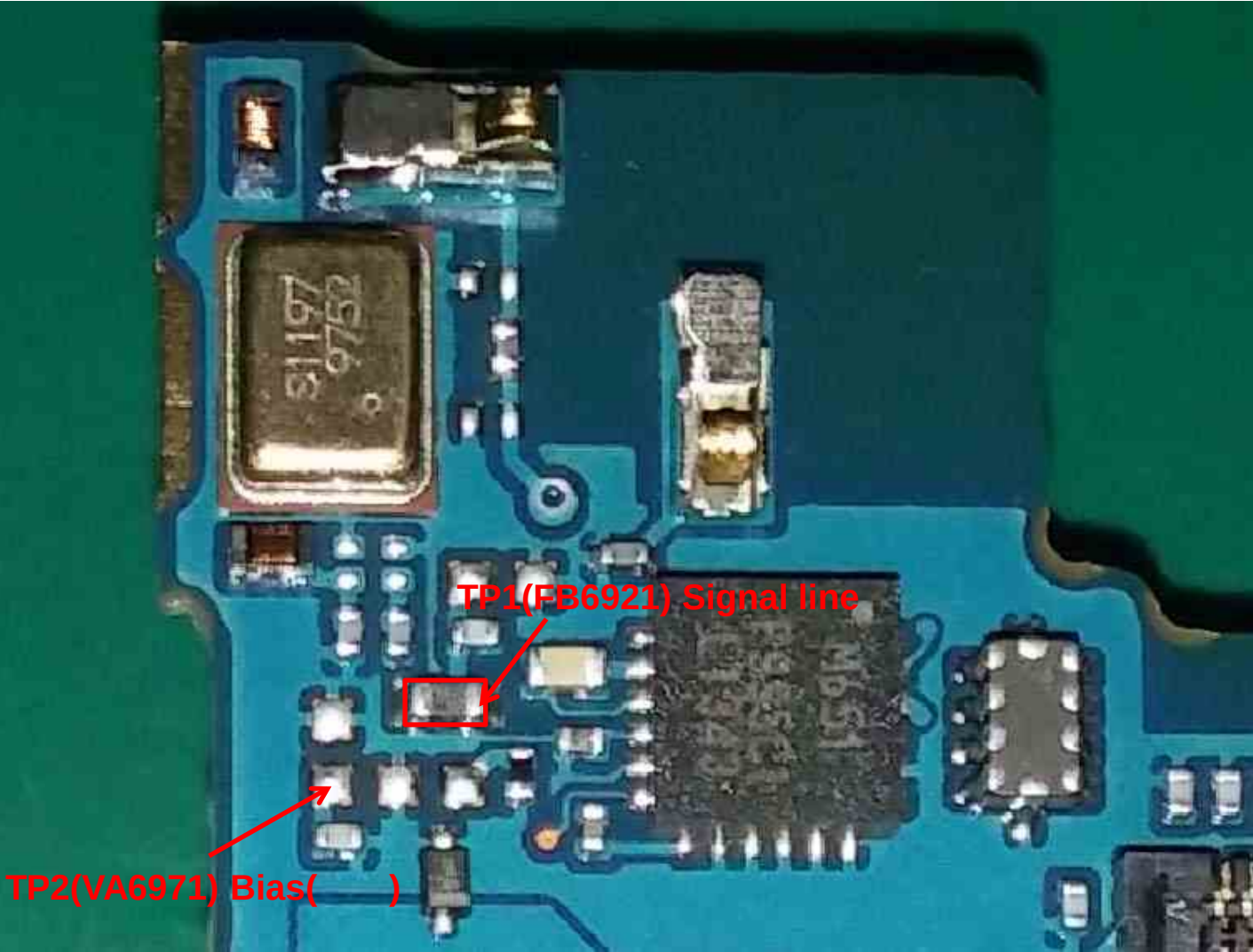
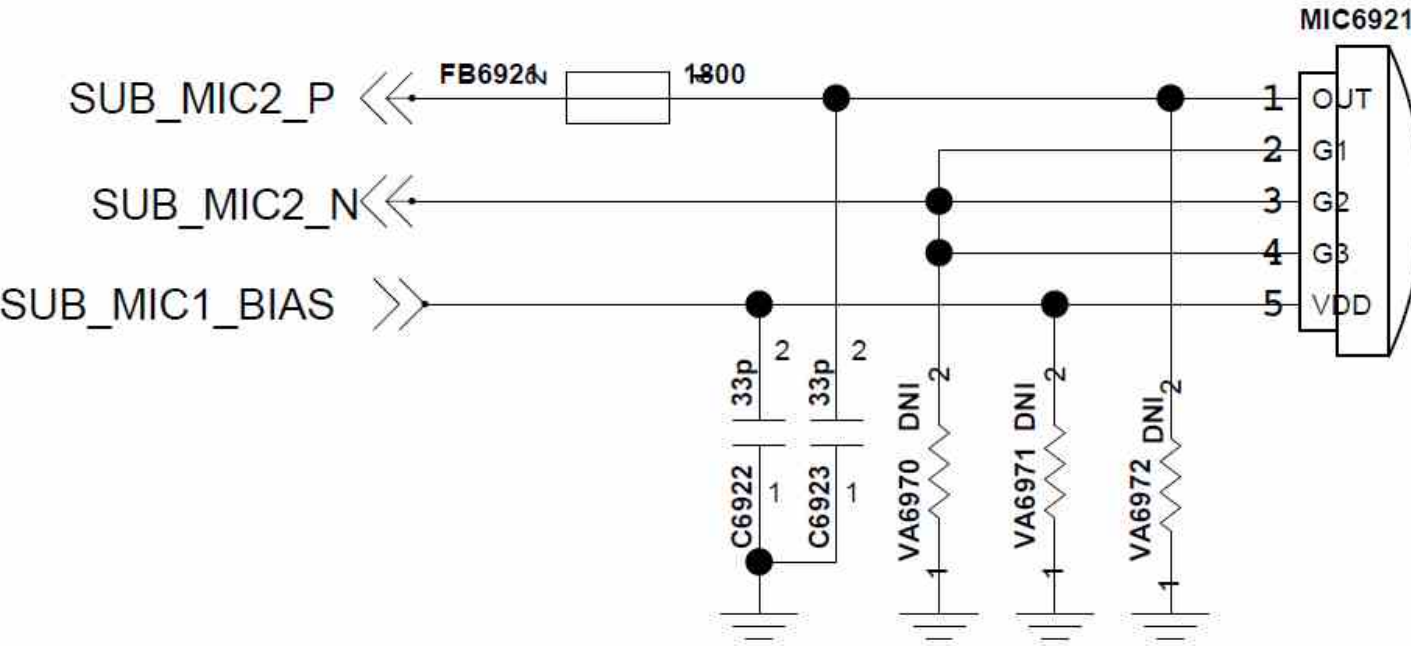
Main MIC



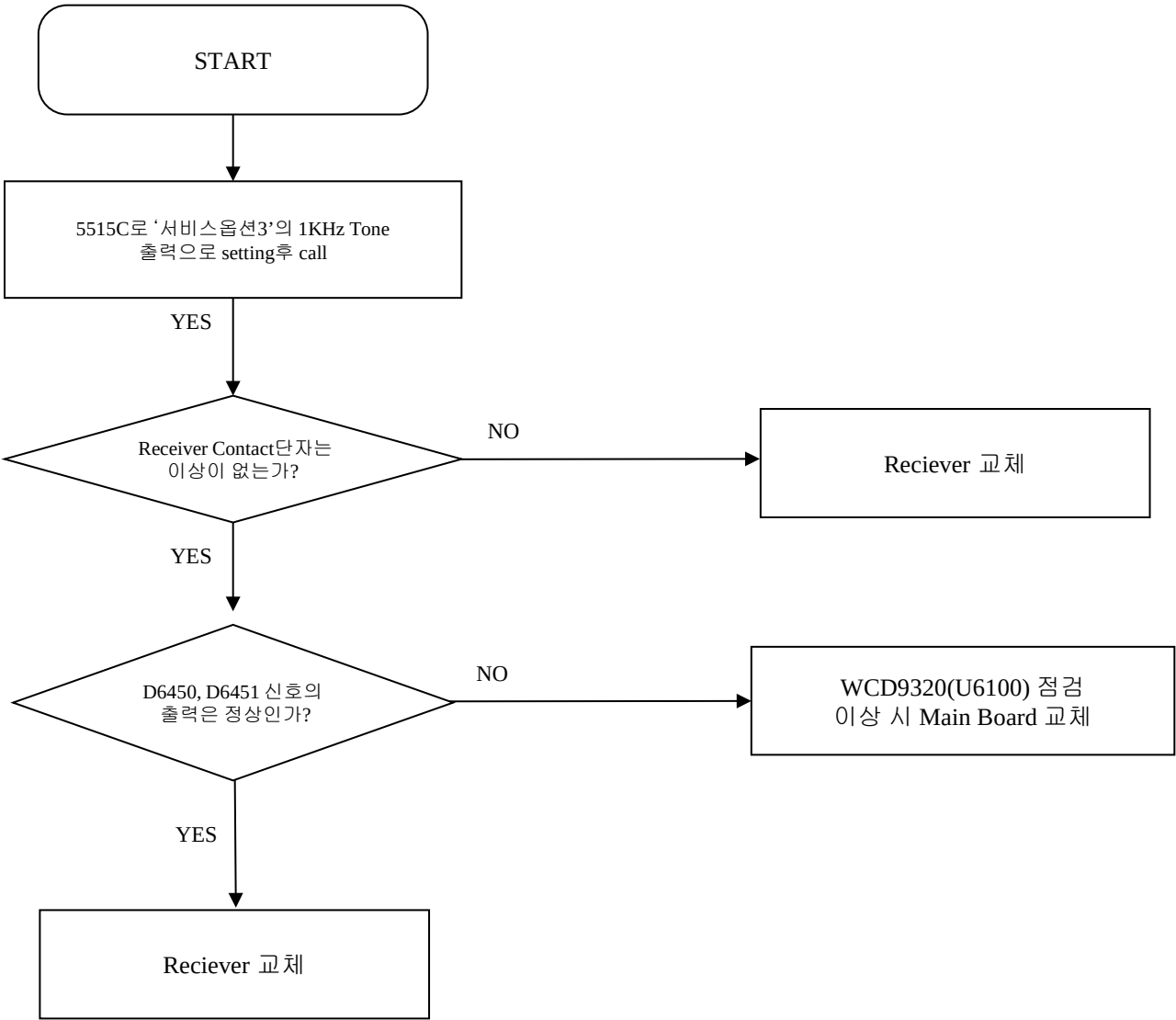
Sub MIC2



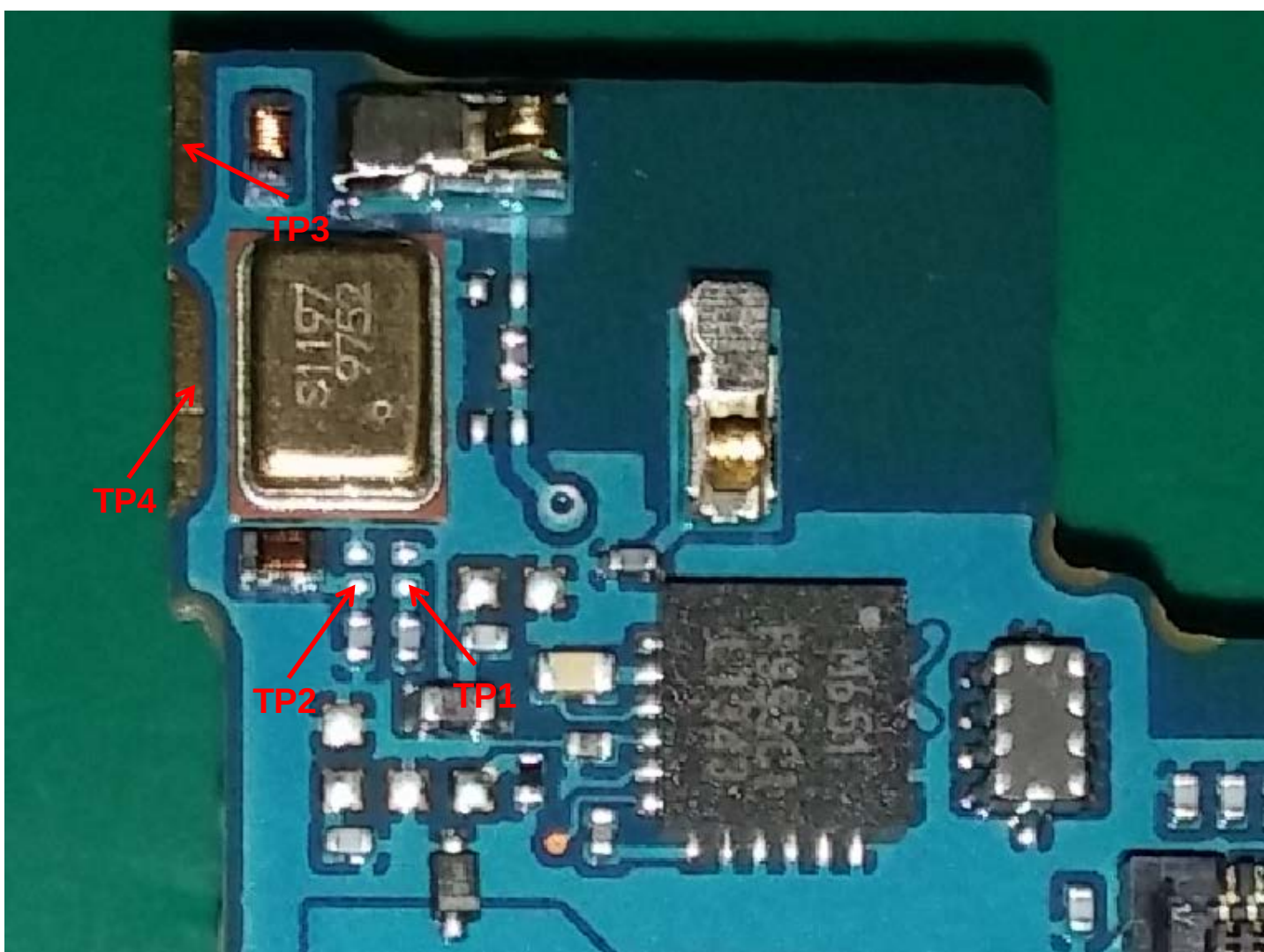
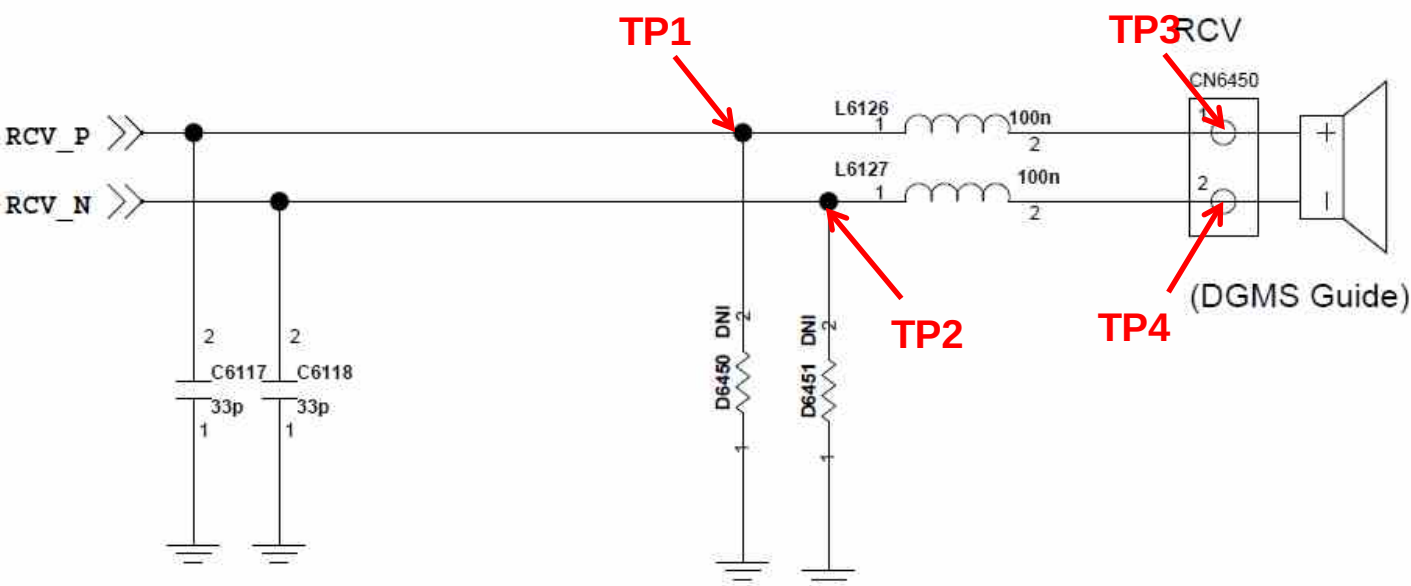
Sub MIC1



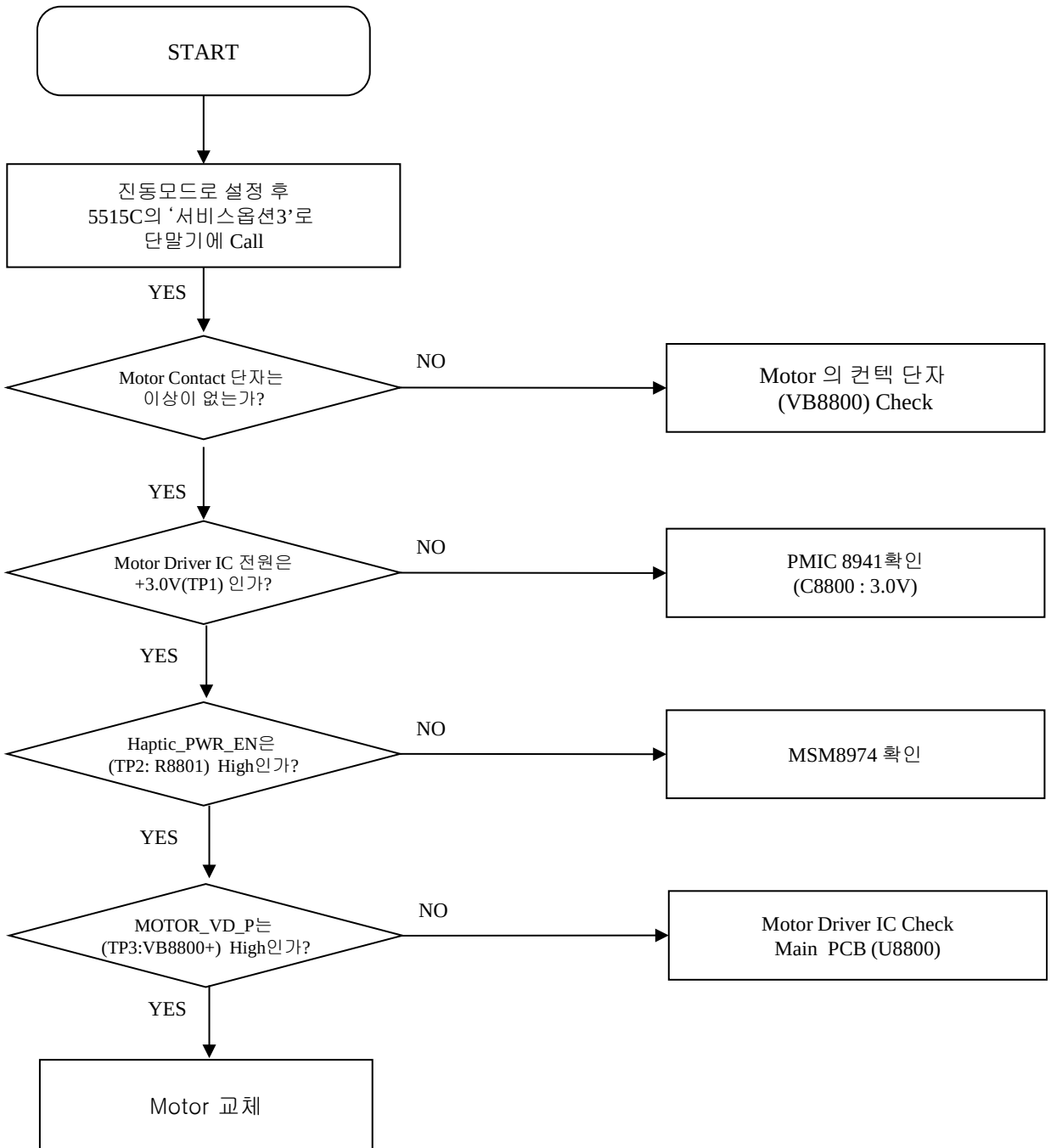
6.



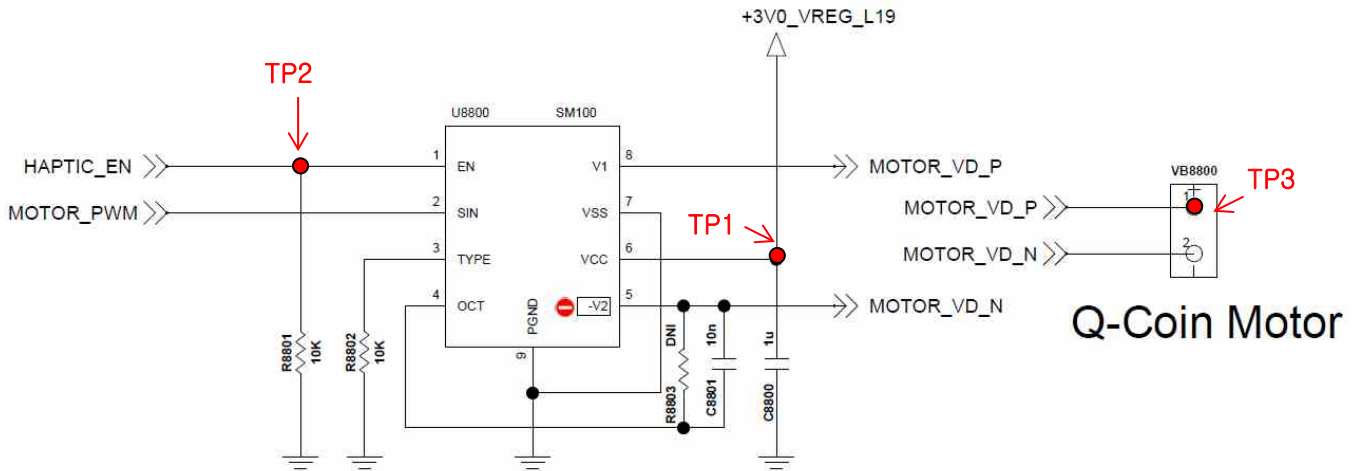
Receiver



7.

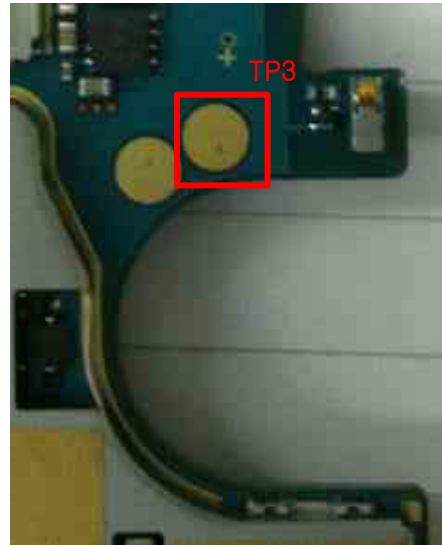
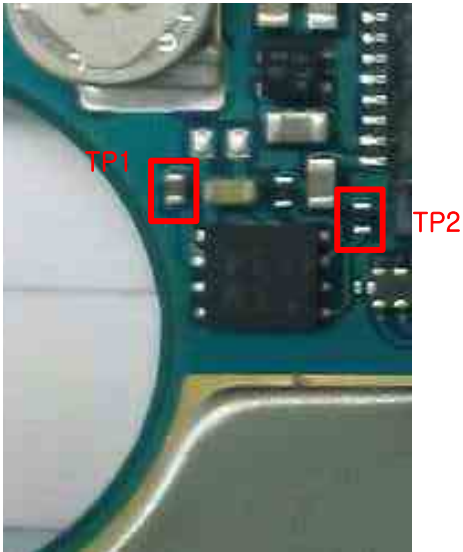


MOTOR

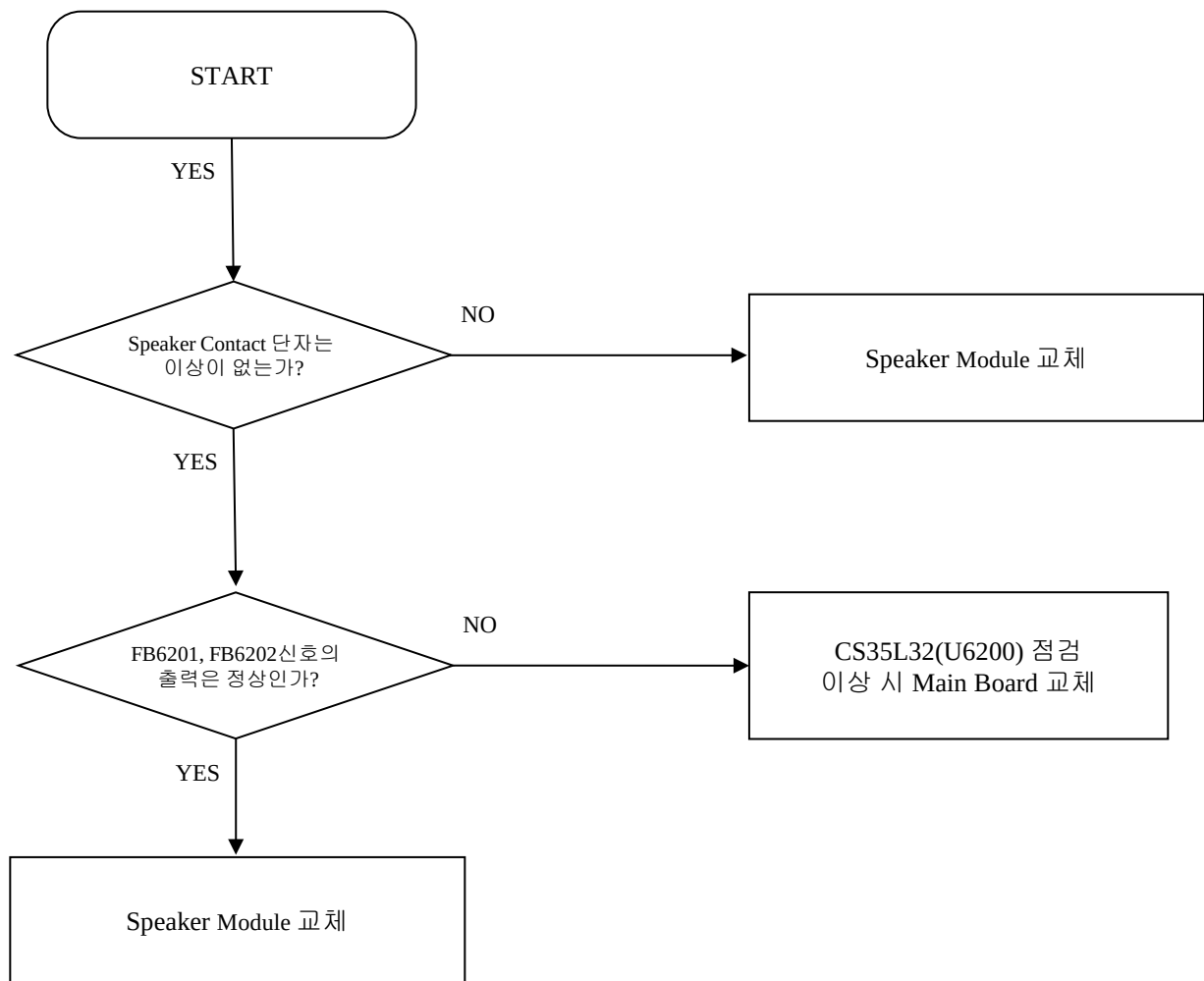


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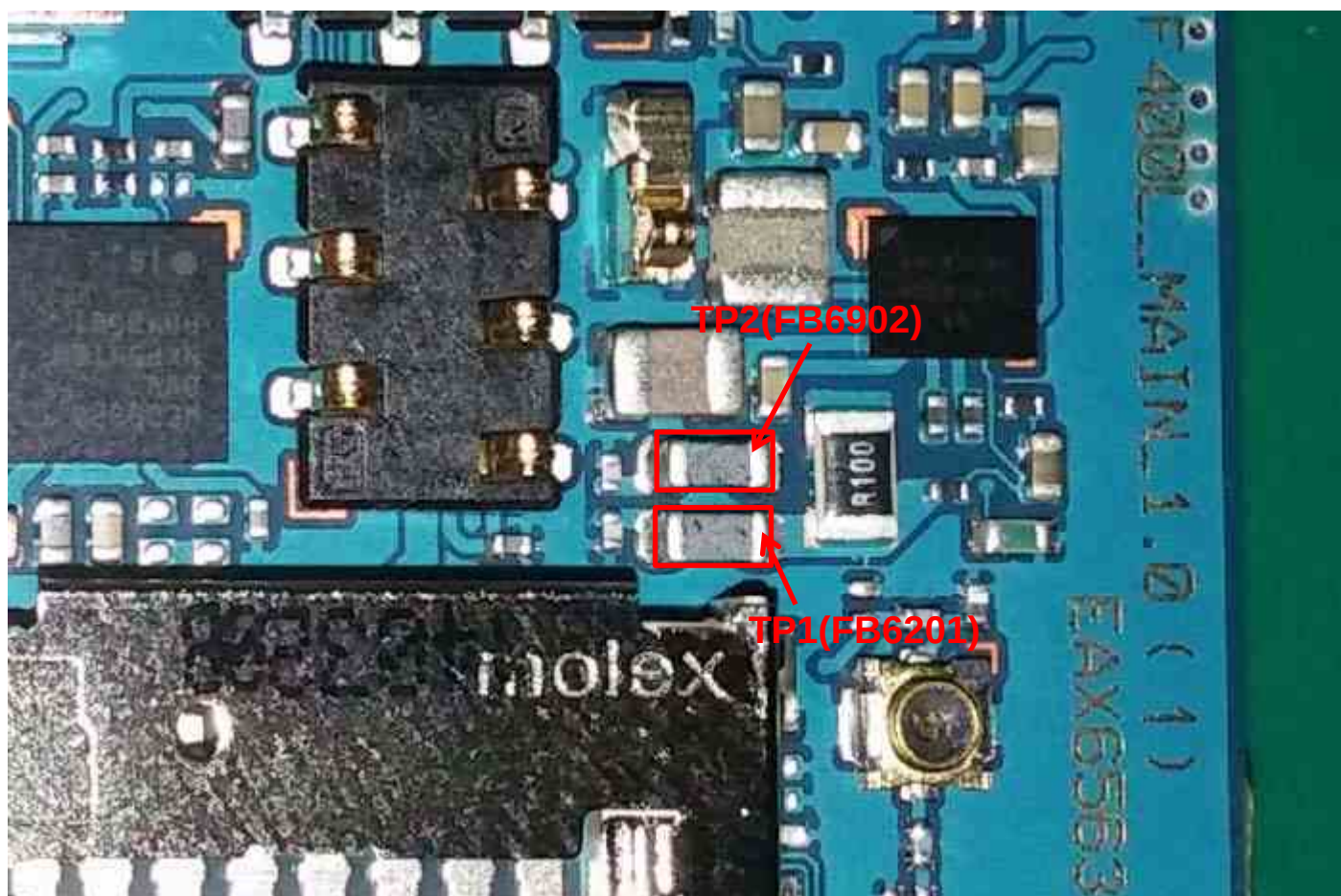
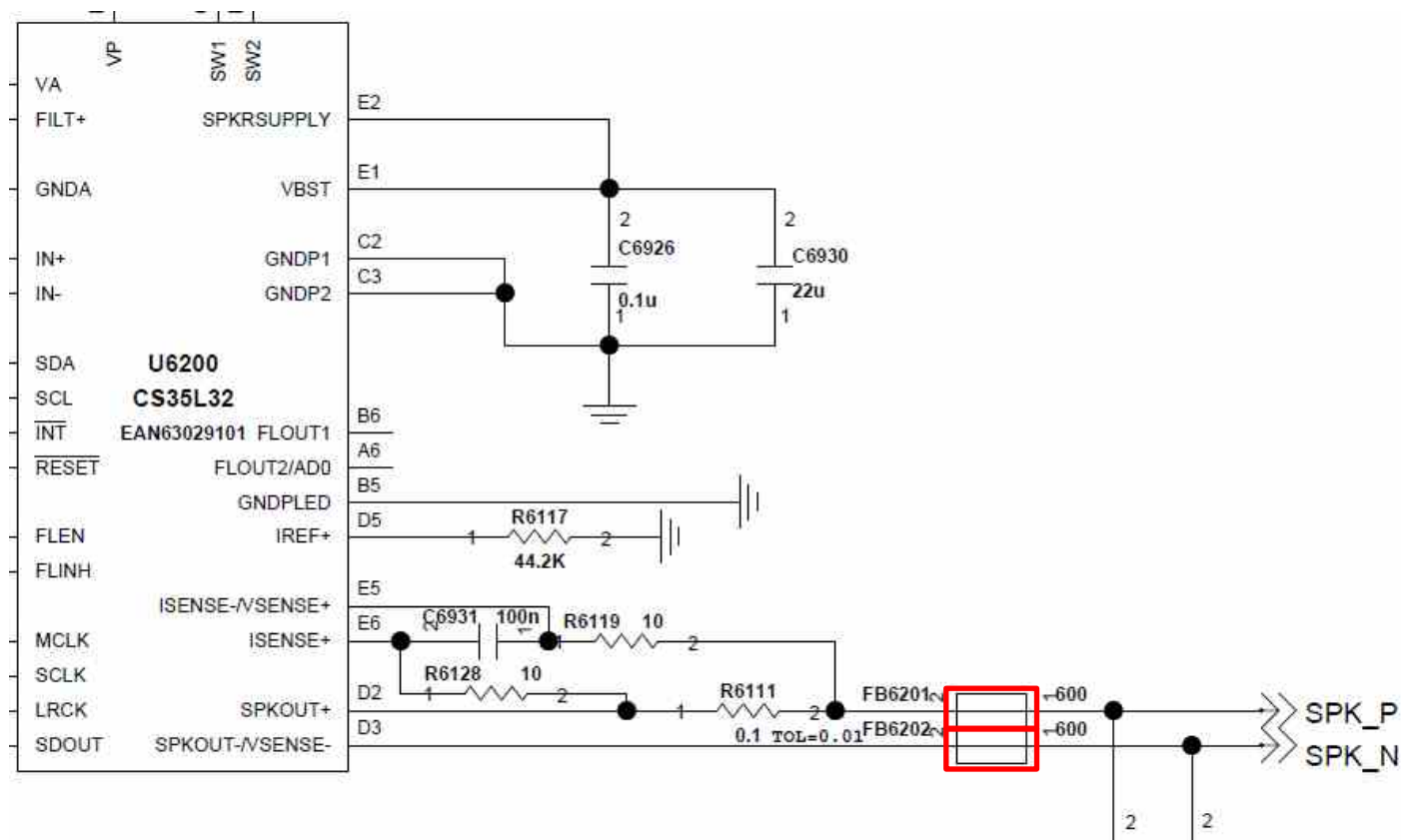
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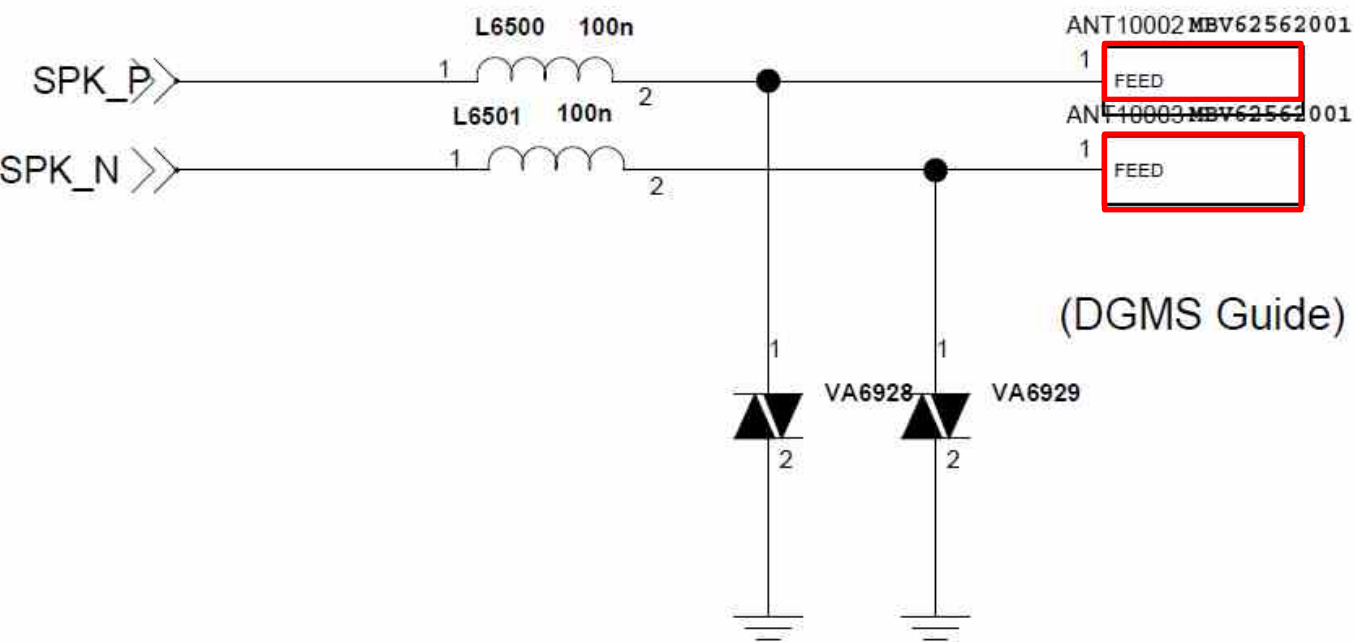
8. MP3 Sound



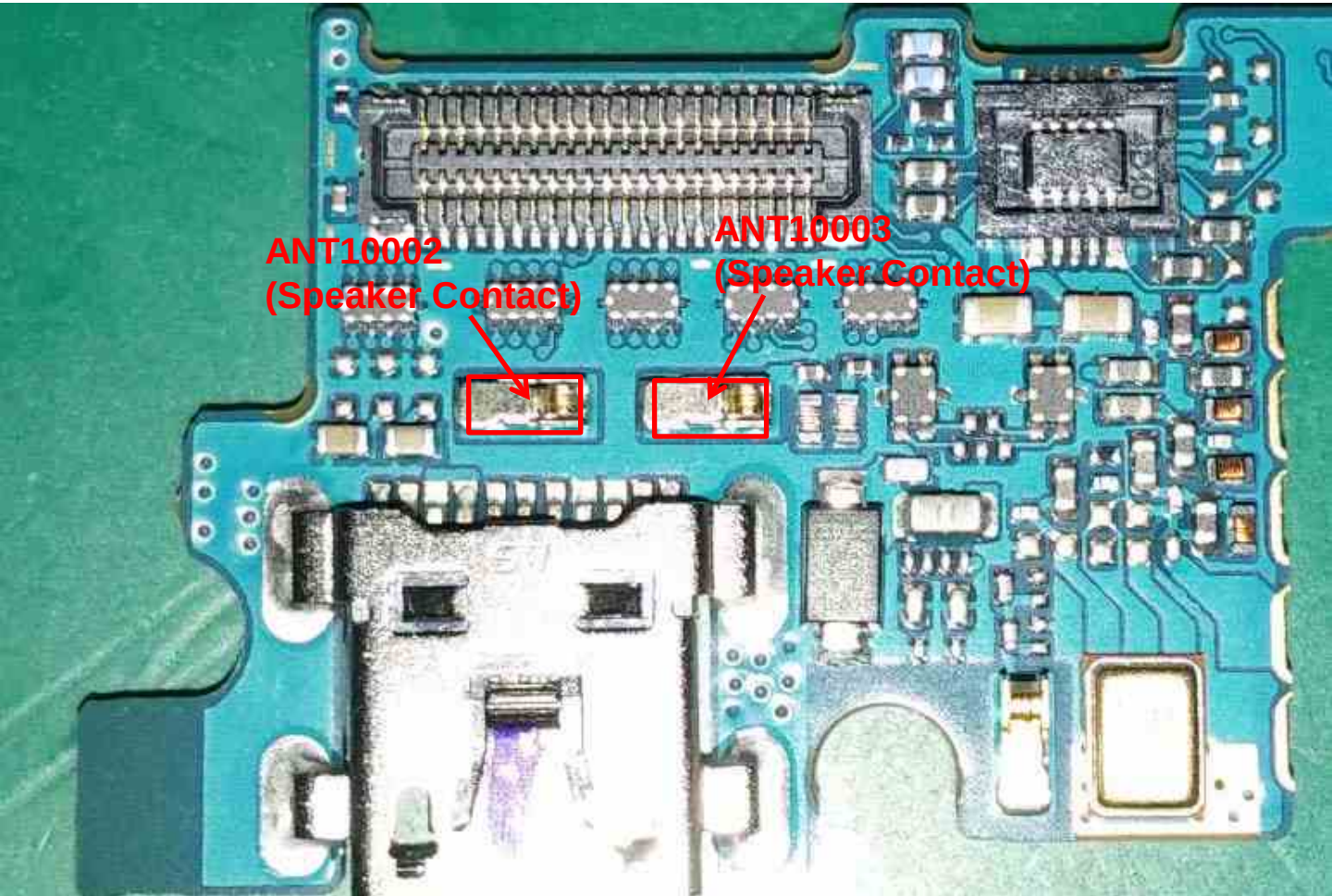
C35L32



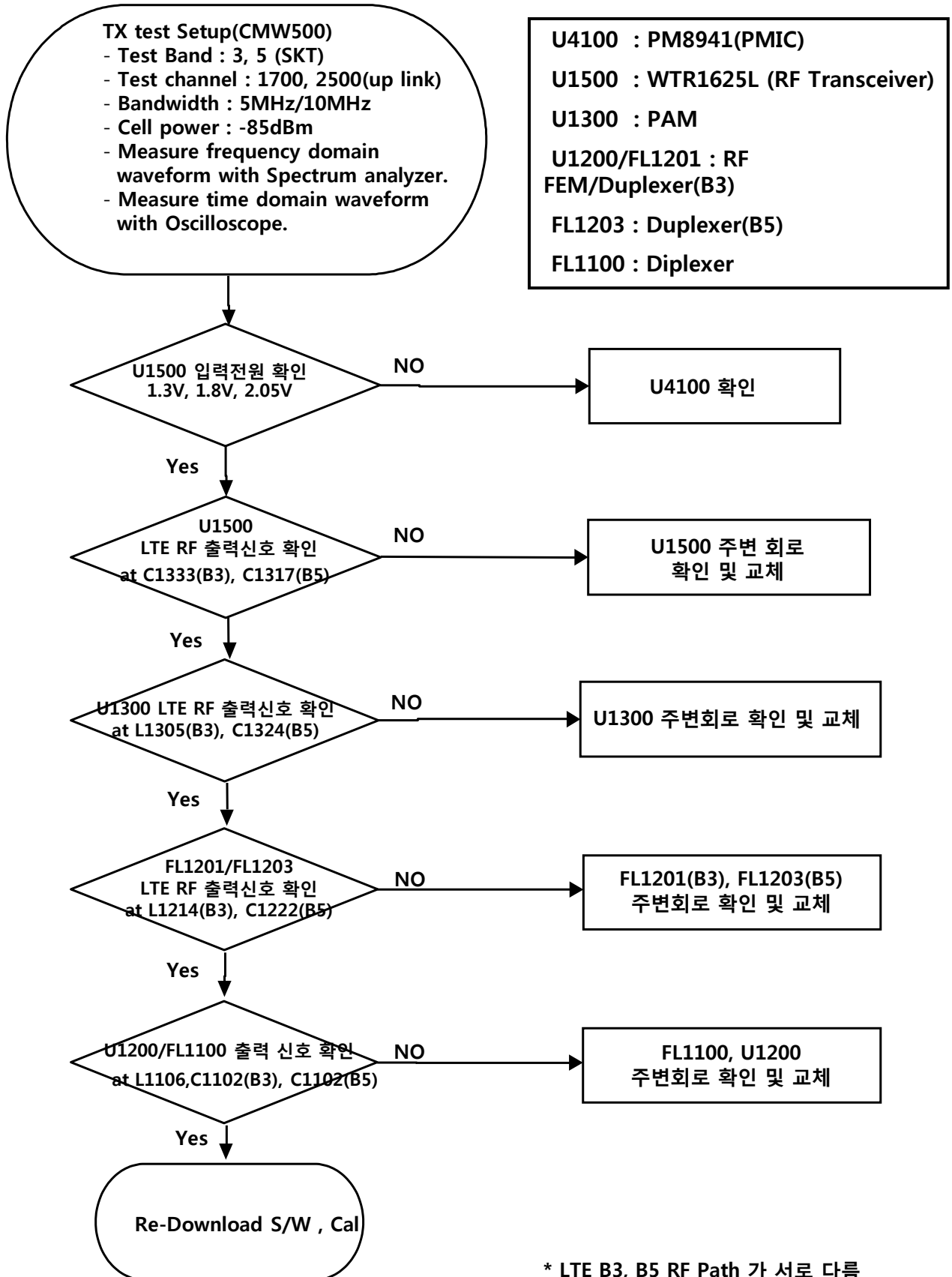
Speaker



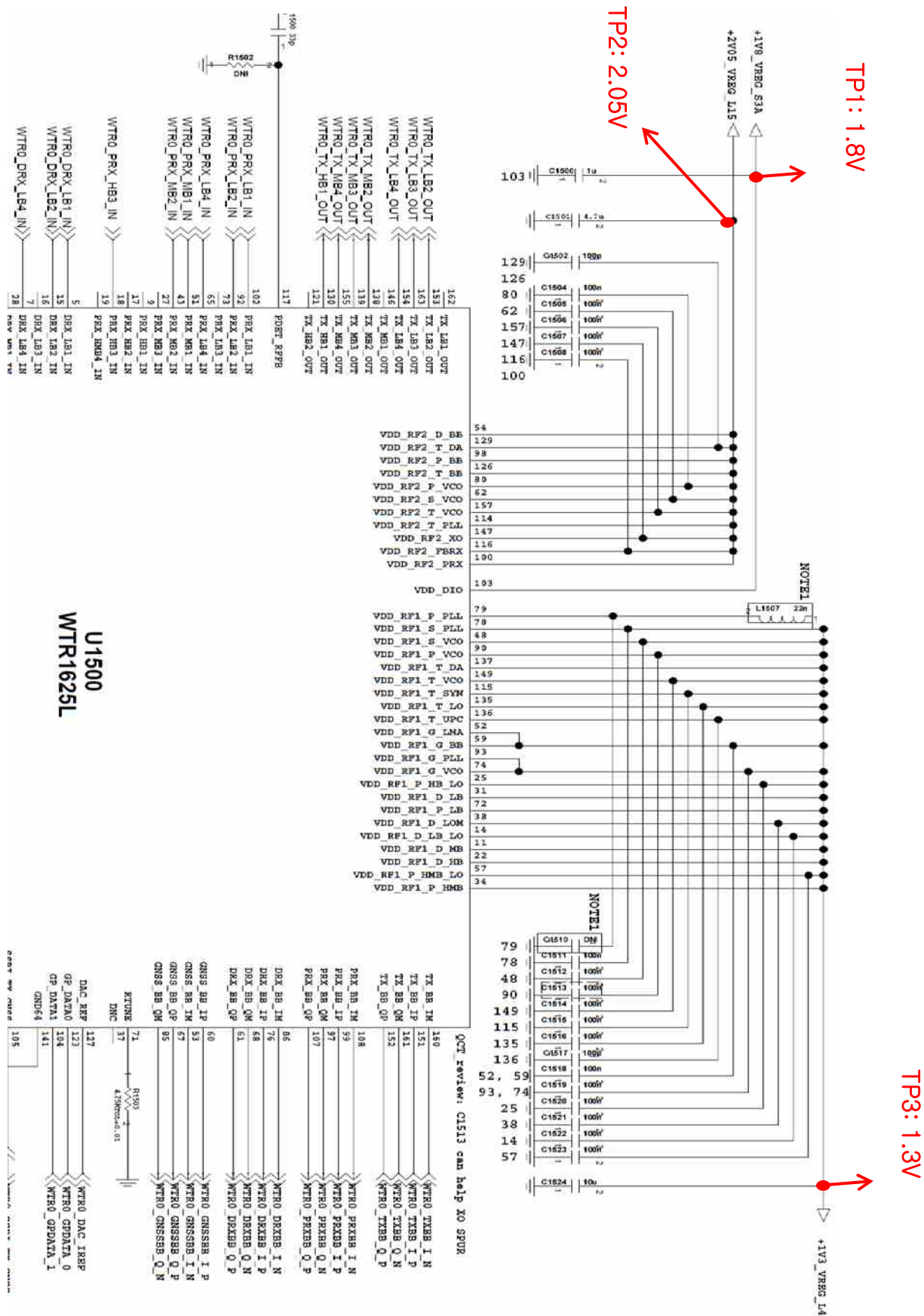
(DGMS Guide)



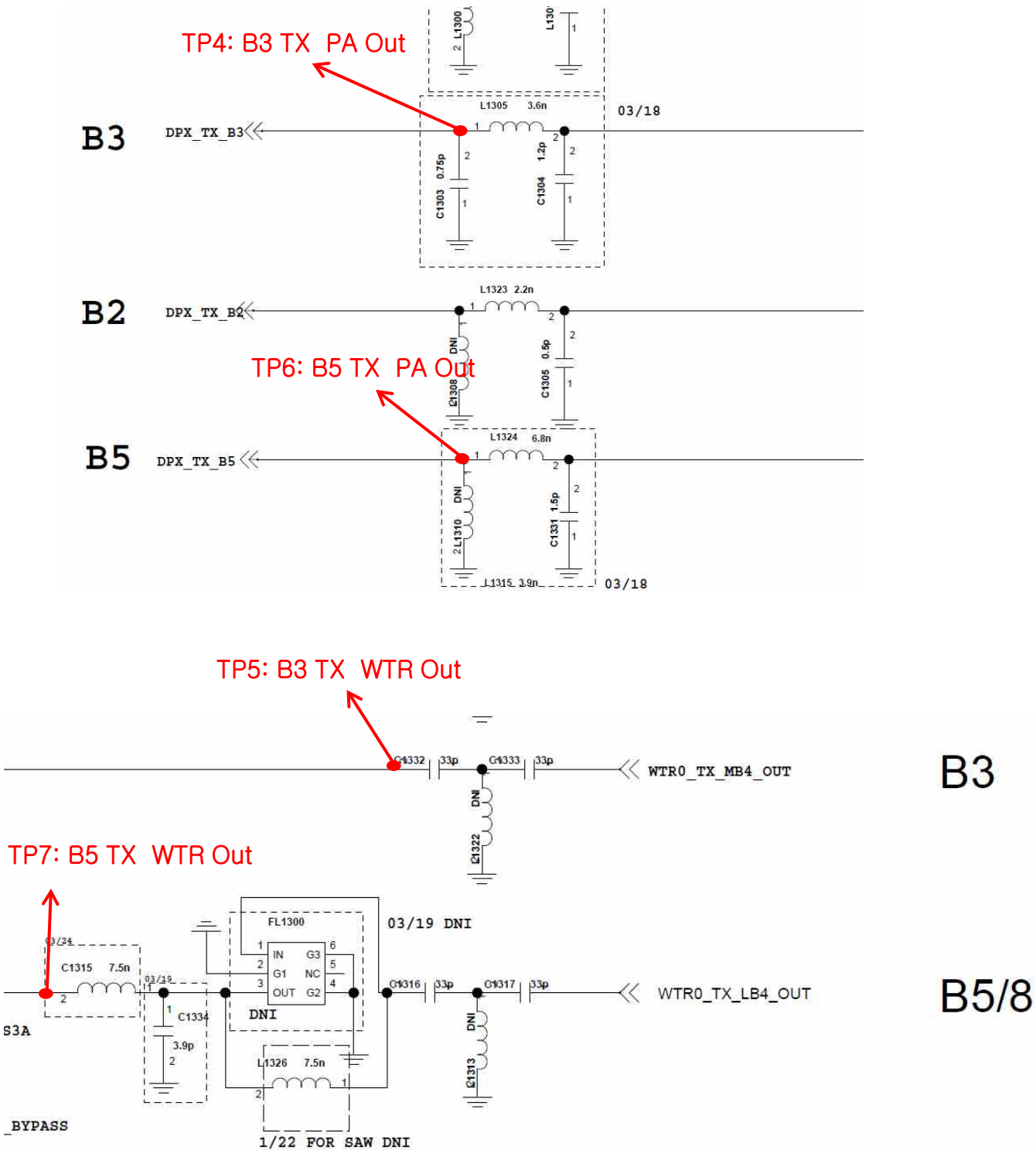
9-1. TX POWER가 나오지 않을 때 (LTE Tx, SKT)



9-1. TX POWER가 나오지 않을 때 (LTE Tx, SKT)

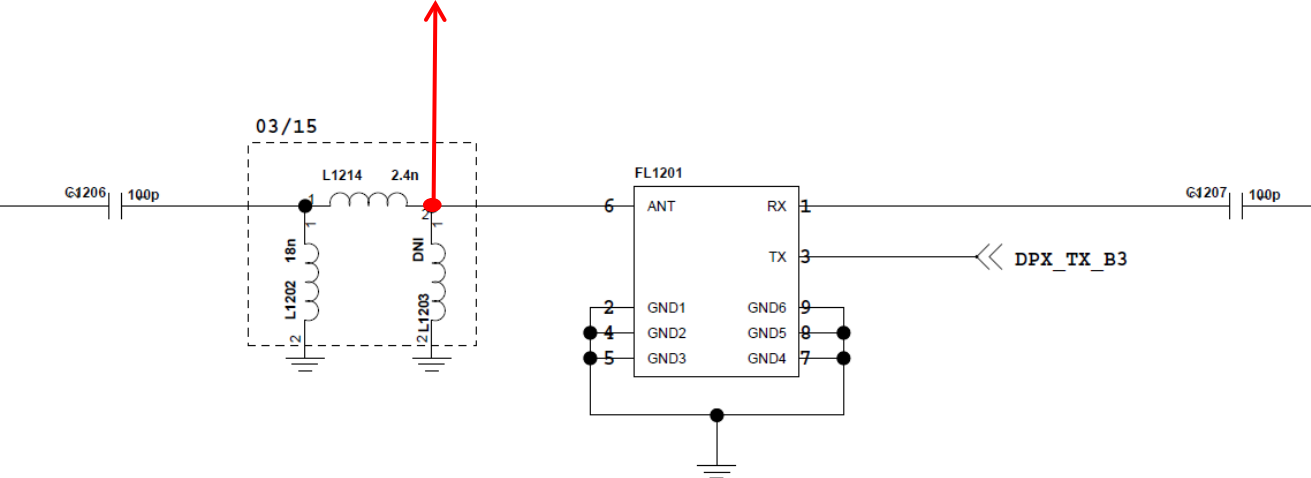


9-1. TX POWER가 나오지 않을 때 (LTE Tx, SKT)

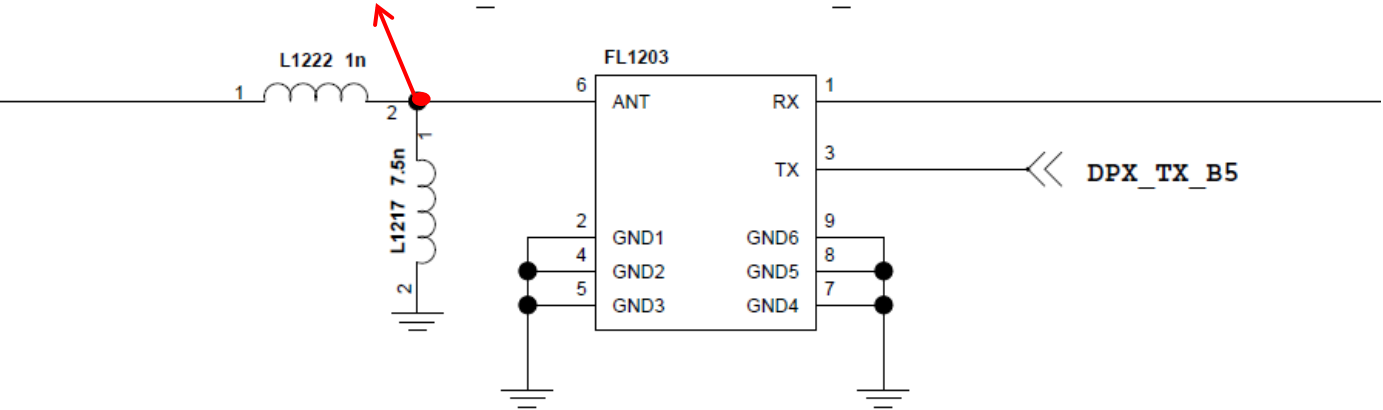


9-1. TX POWER가 나오지 않을 때 (LTE Tx, SKT)

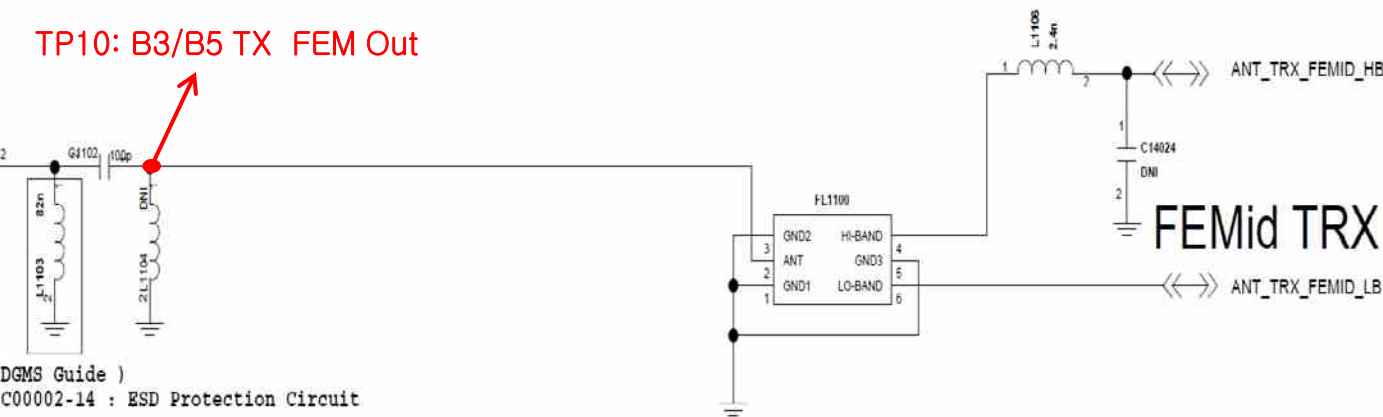
TP8: B3 TX DPX Out



TP9: B5 TX DPX Out

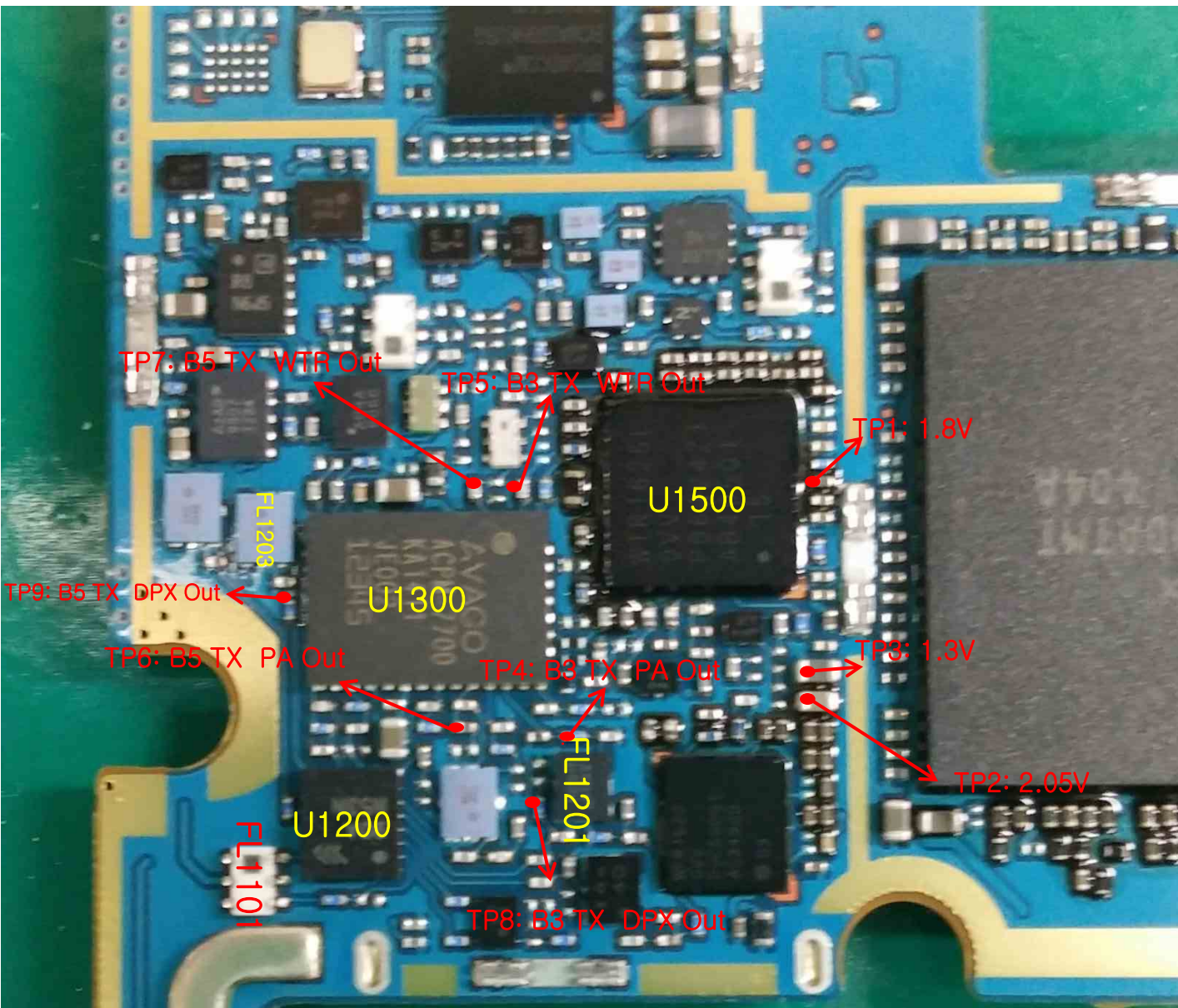


TP10: B3/B5 TX FEM Out



DGMS Guide)
.C00002-14 : ESD Protection Circuit

9-1. TX POWER가 나오지 않을 때 (LTE Tx, SKT)



9-2. TX POWER가 나오지 않을 때 (LTE Tx, KT)

TX test Setup(CMW500)

- Test Band : 3, 8 (KT)
- Test channel : 1600, 3750(up link)
- Bandwidth : 5MHz/10MHz
- Cell power : -85dBm
- Measure frequency domain waveform with Spectrum analyzer.
- Measure time domain waveform with Oscilloscope.

U4100 : PM8941(PMIC)

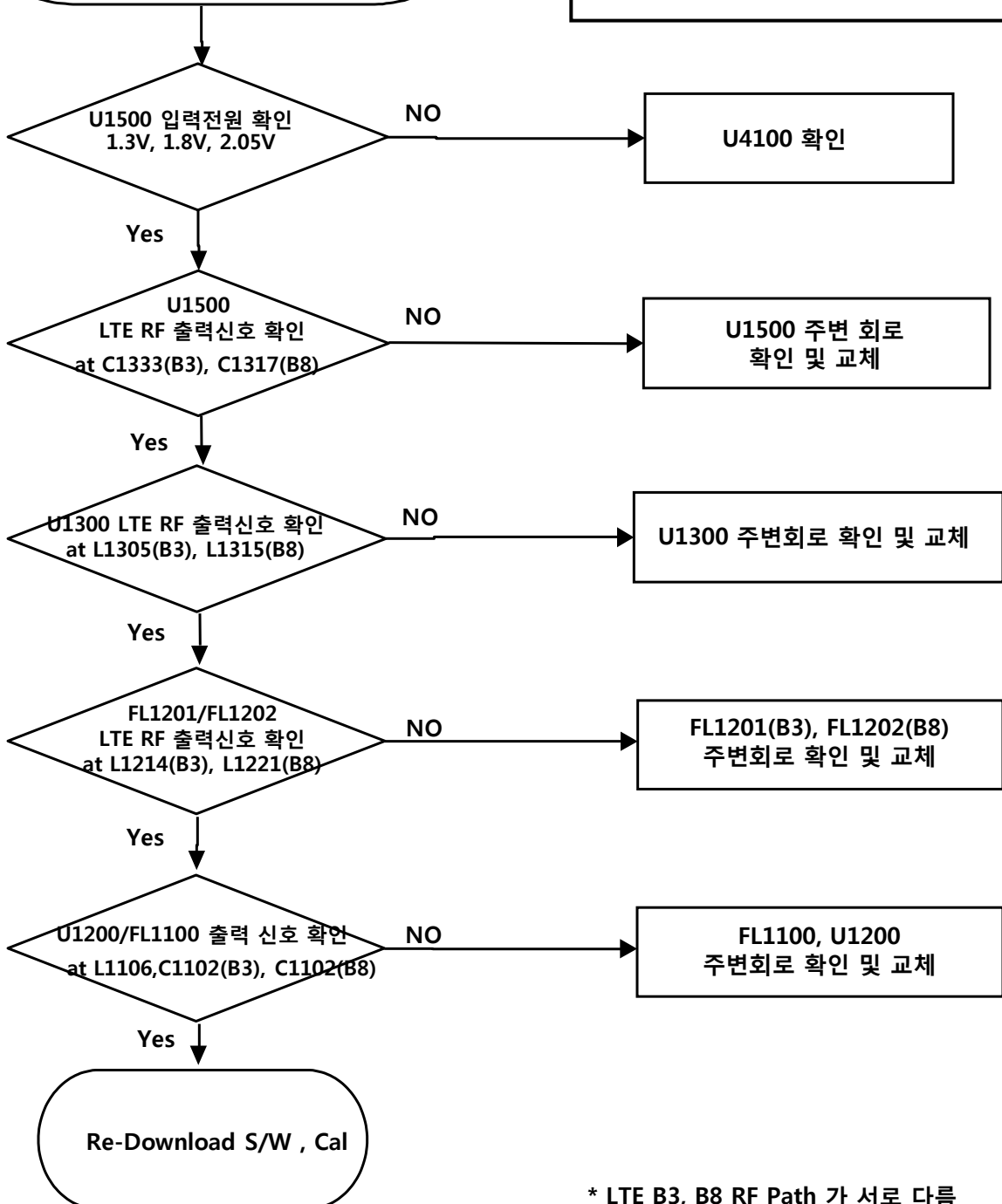
U1500 : WTR1625L (RF Transceiver)

U1300 : PAM

U1200/FL1201 : RF FEM/Duplexer(B3)

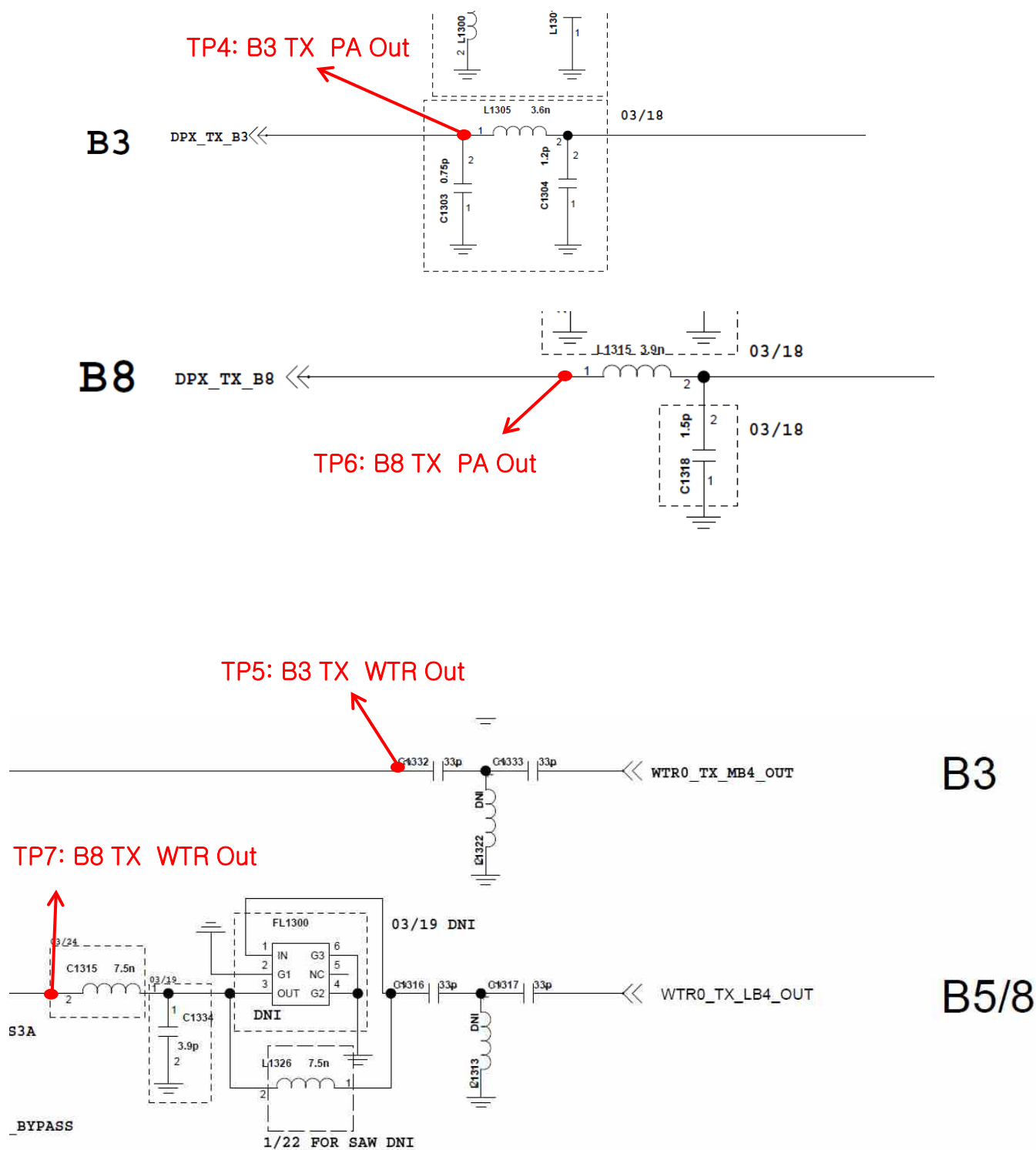
FL1202 : Duplexer(B8)

FL1100 : Diplexer



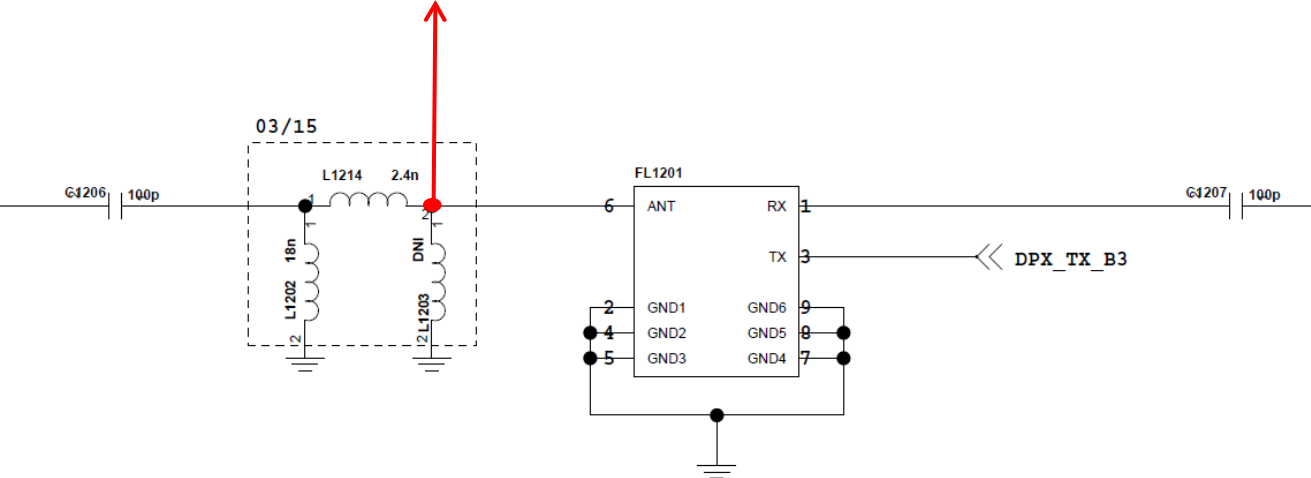
* LTE B3, B8 RF Path 가 서로 다름

9-2. TX POWER가 나오지 않을 때 (LTE Tx, KT)

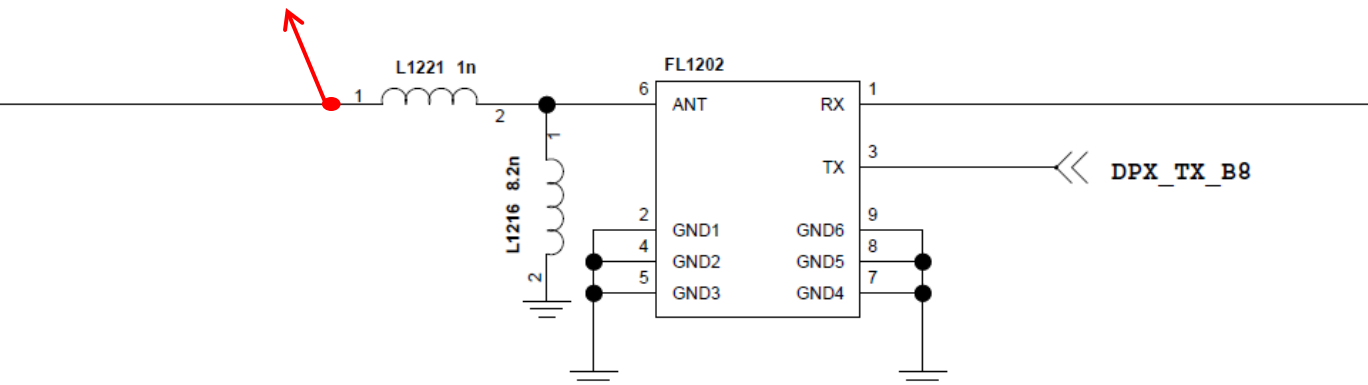


9-2. TX POWER가 나오지 않을 때 (LTE Tx, KT)

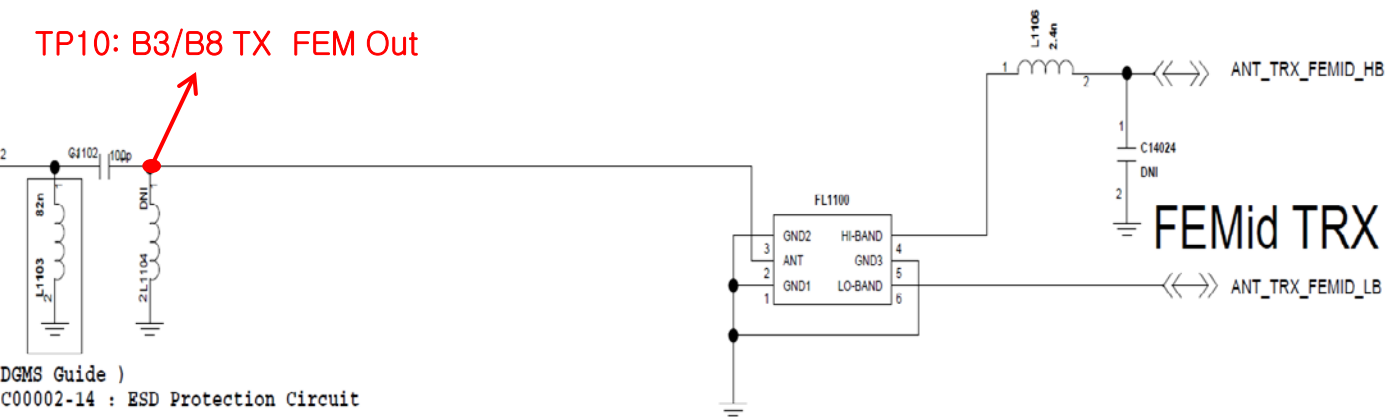
TP8: B3 TX DPX Out



TP9: B8 TX DPX Out

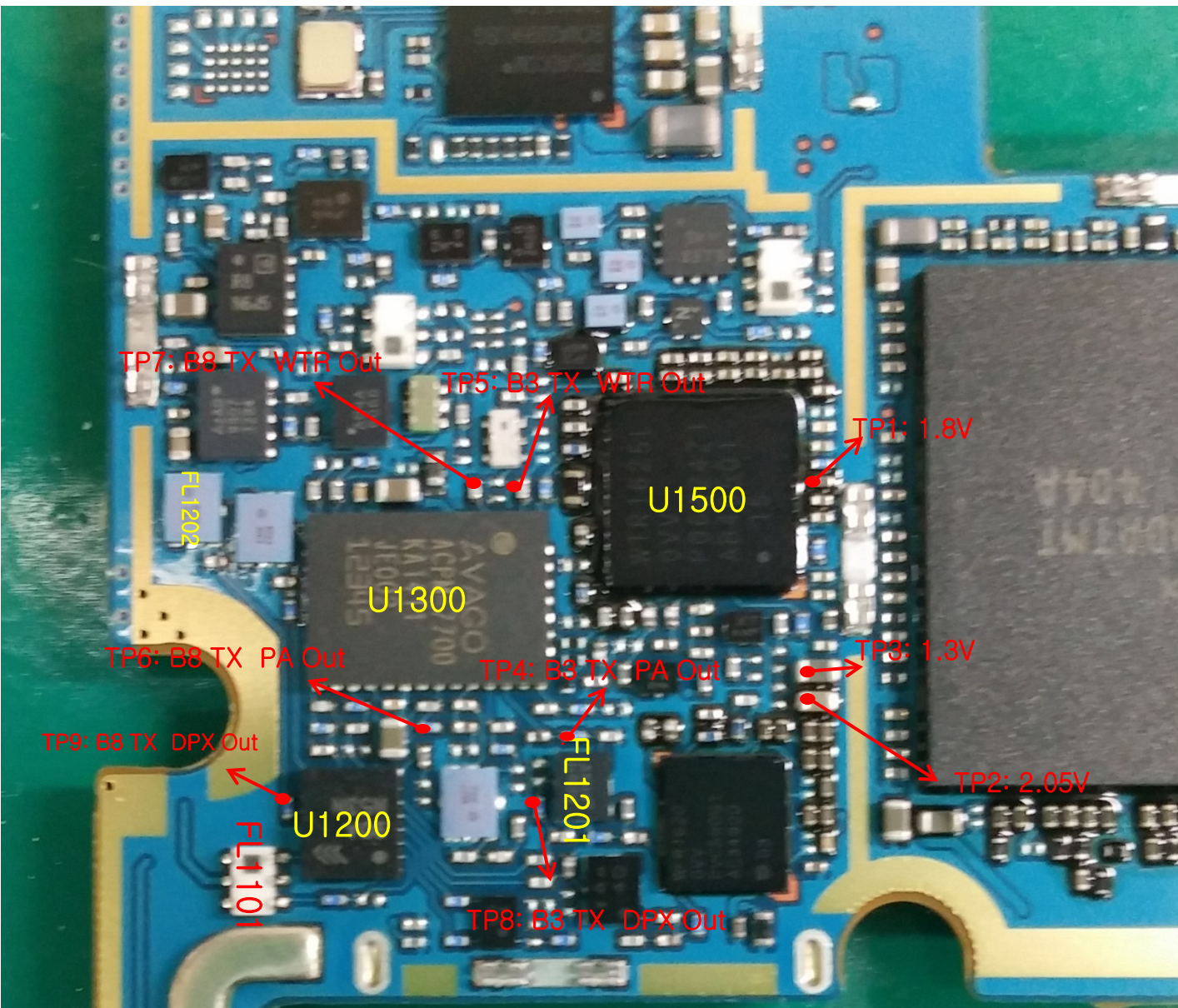


TP10: B3/B8 TX FEM Out

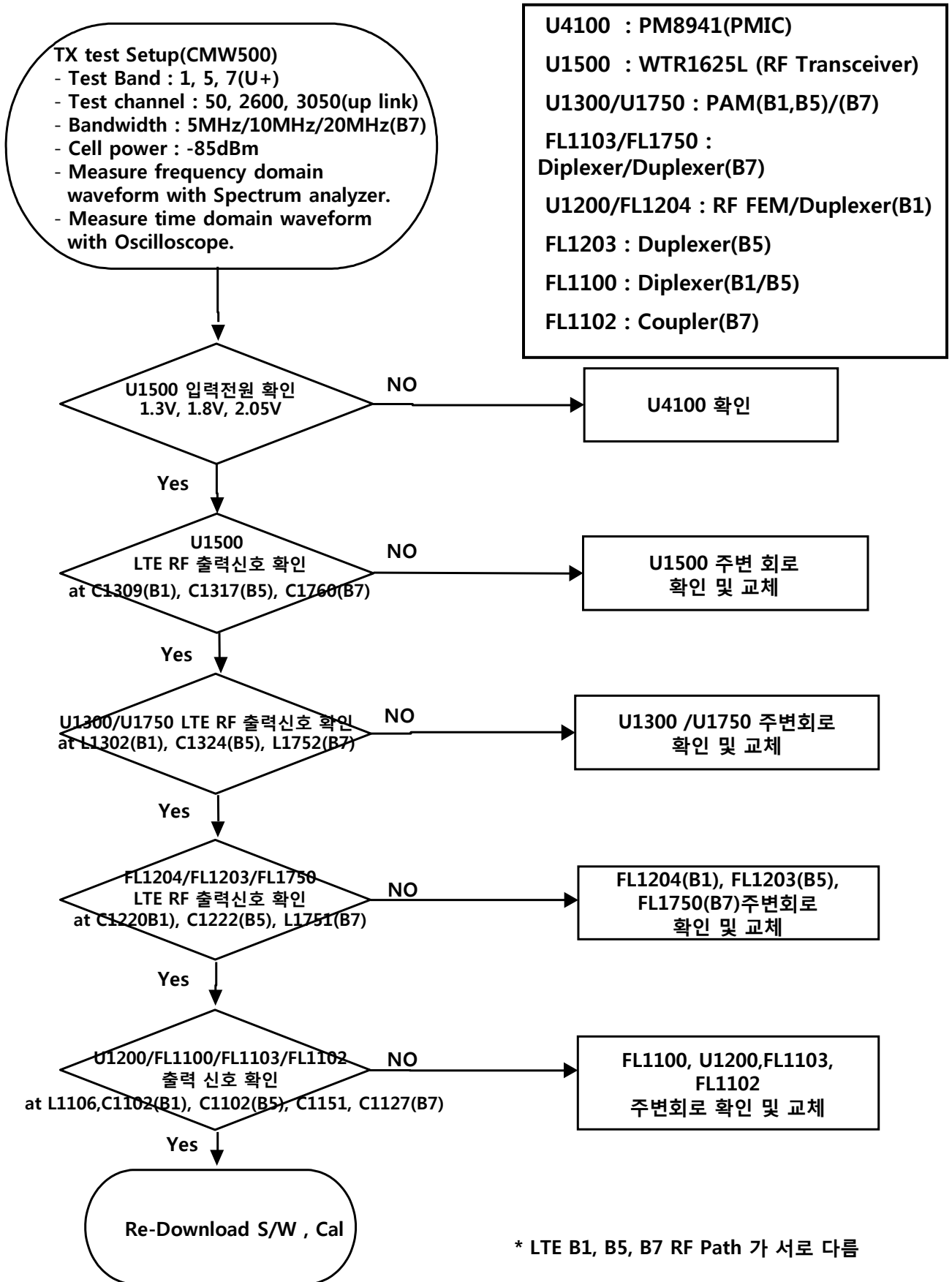


DGMS Guide)
C00002-14 : ESD Protection Circuit

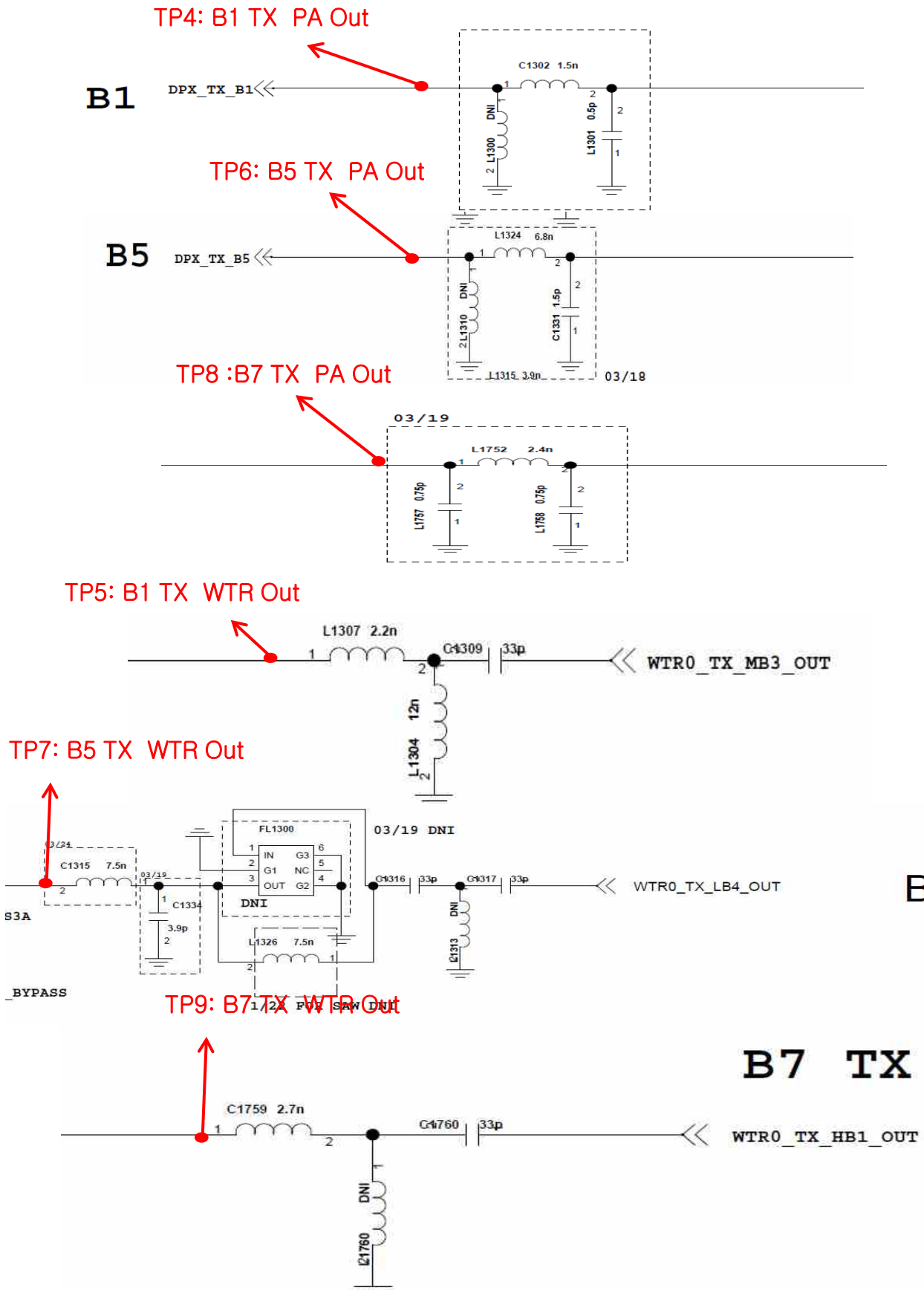
9-2. TX POWER가 나오지 않을 때 (LTE Tx, KT)



9-3. TX POWER가 나오지 않을 때 (LTE Tx, U+)

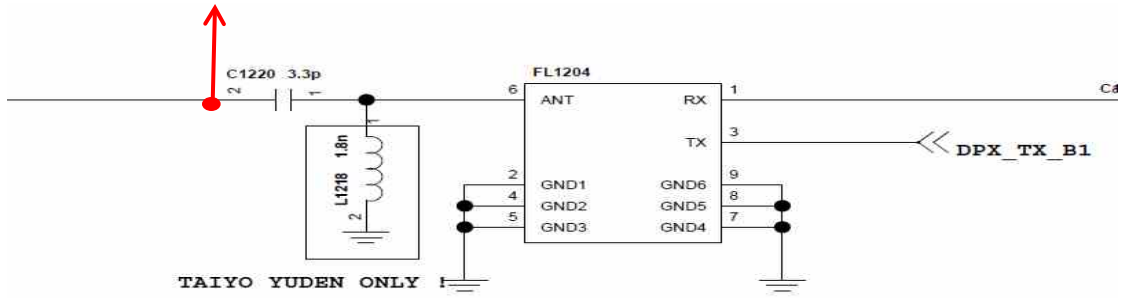


9-3. TX POWER가 나오지 않을 때 (LTE Tx, U+)

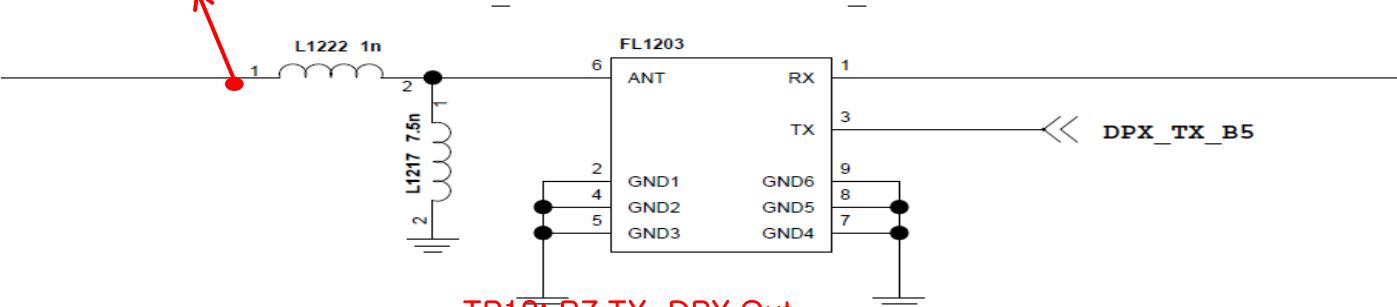


9-3. TX POWER가 나오지 않을 때 (LTE Tx, U+)

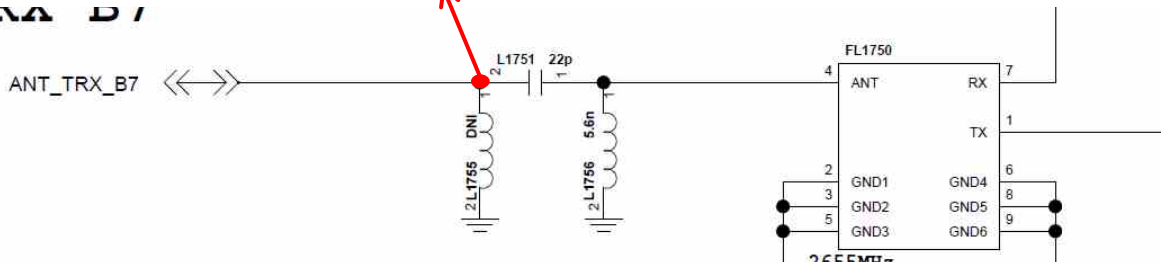
TP10: B1 TX DPX Out



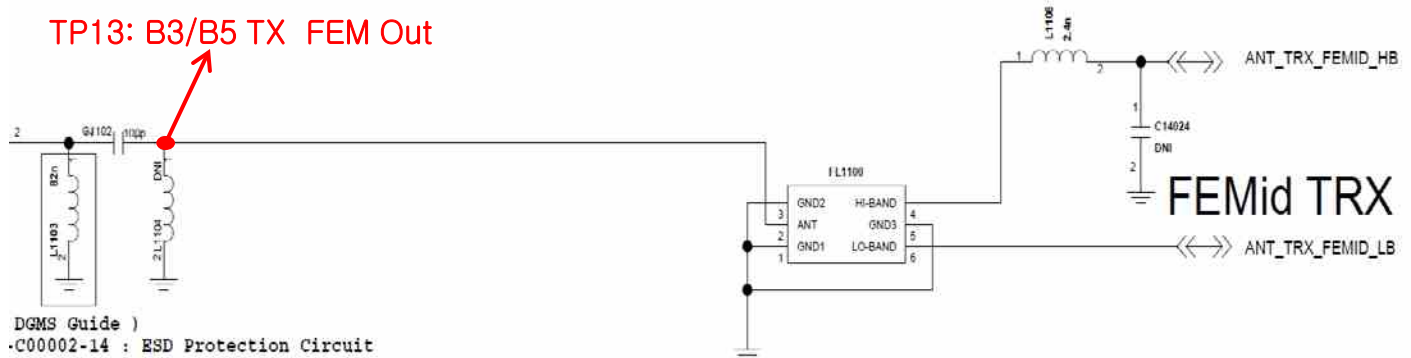
TP11: B5 TX DPX Out



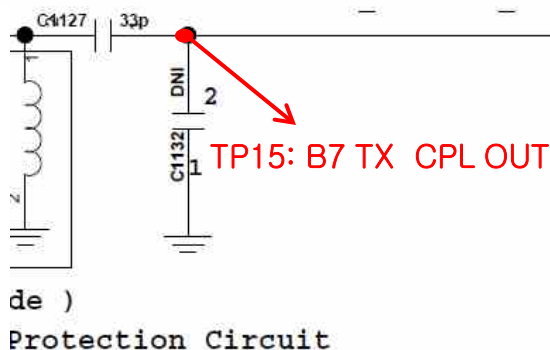
TP12: B7 TX DPX Out



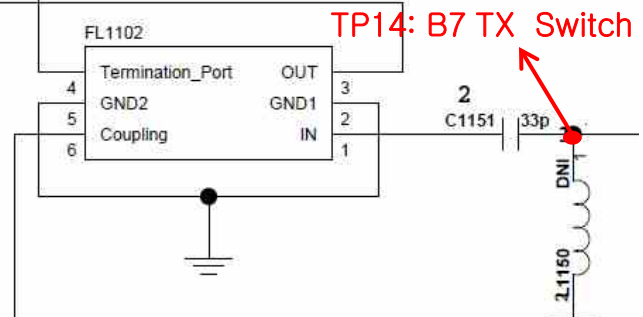
TP13: B3/B5 TX FEM Out



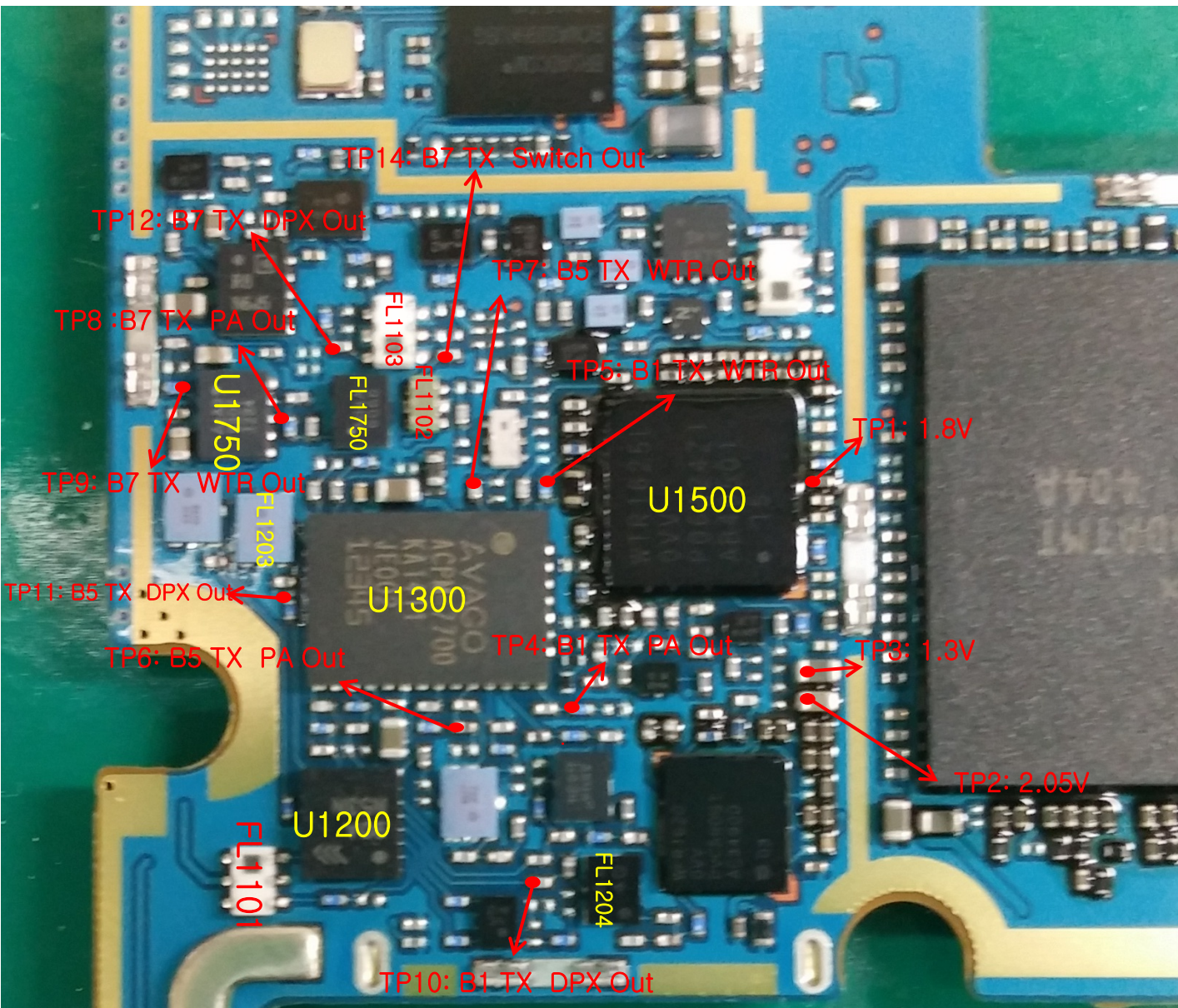
TP15: B7 TX CPL OUT



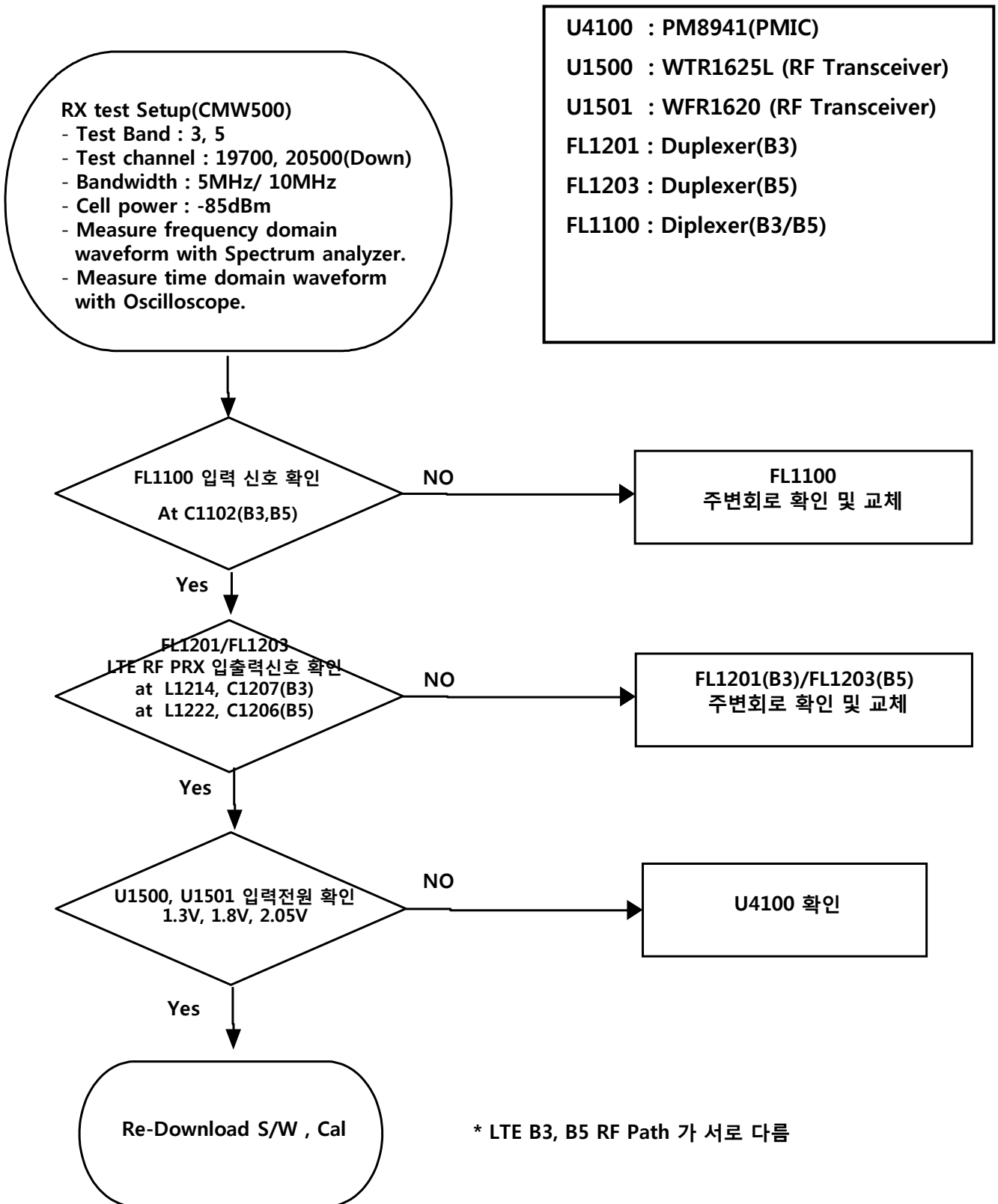
TP14: B7 TX Switch Out



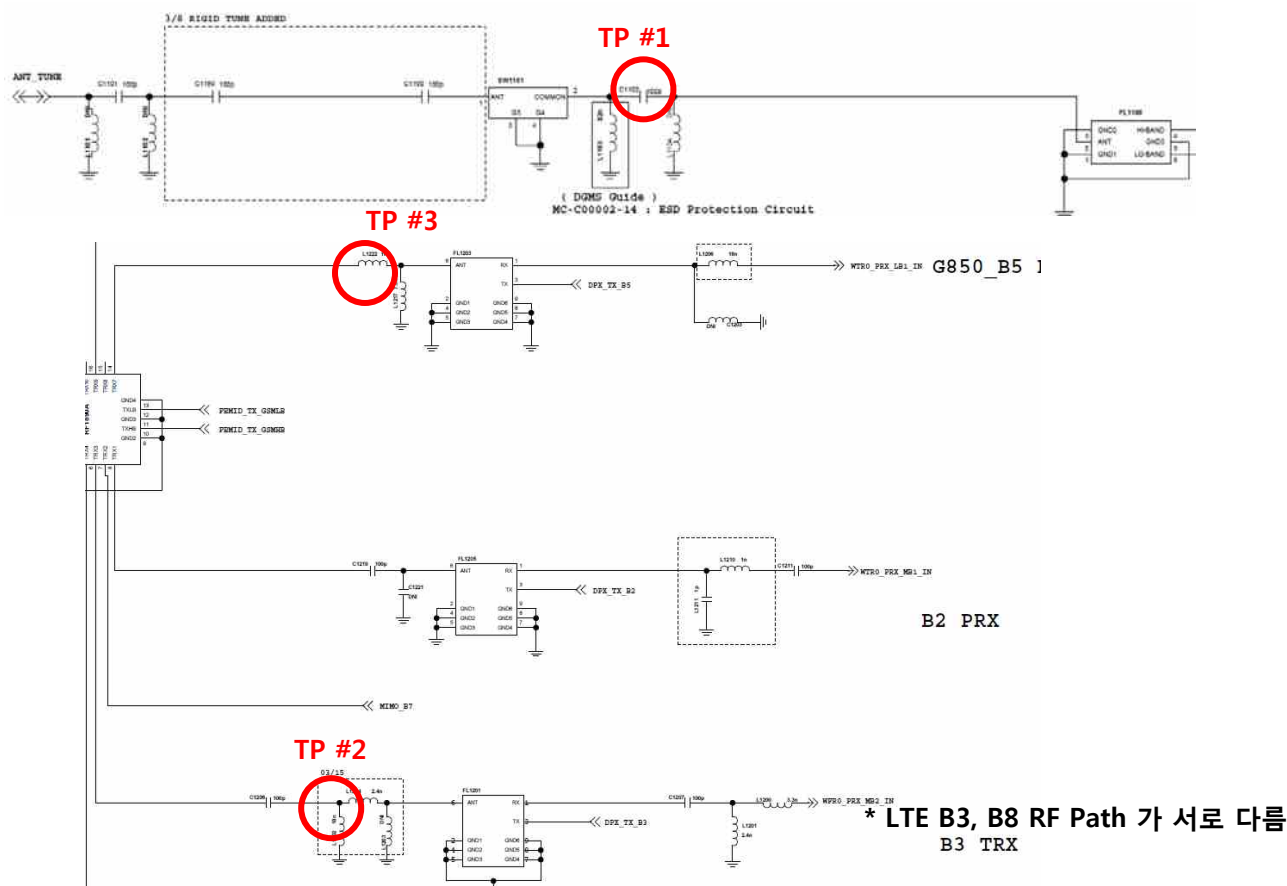
9-3. TX POWER가 나오지 않을 때 (LTE Tx, U+)



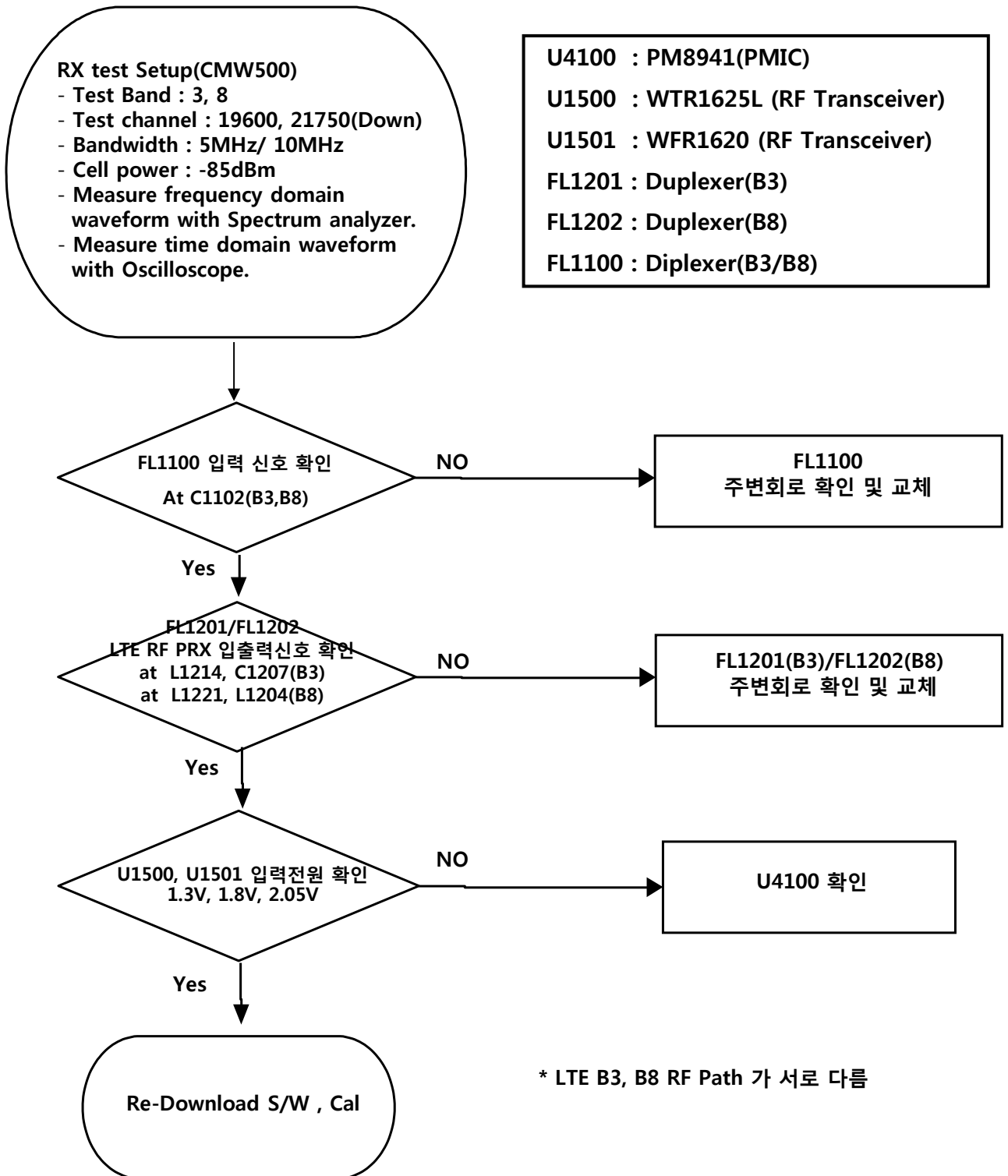
10-1. 수신감도가 좋지 않을 때 (LTE PRX, SKT)



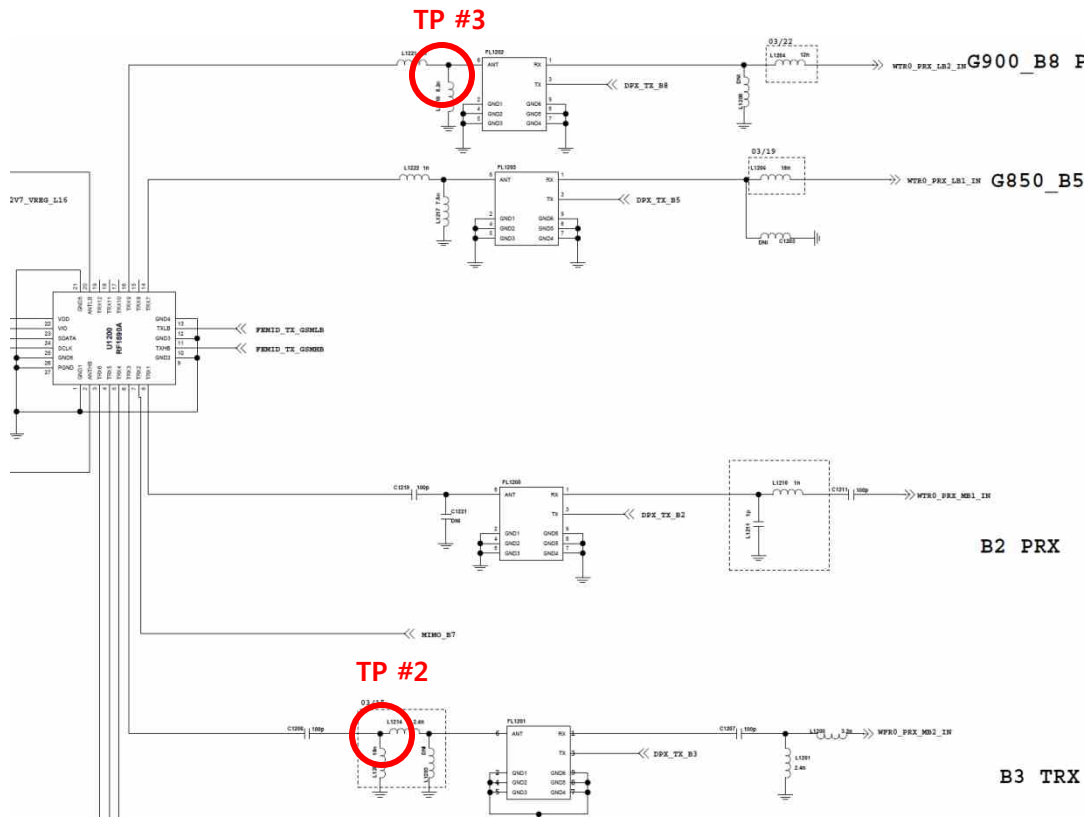
10-1. 수신감도가 좋지 않을 때 (LTE PRX, SKT)



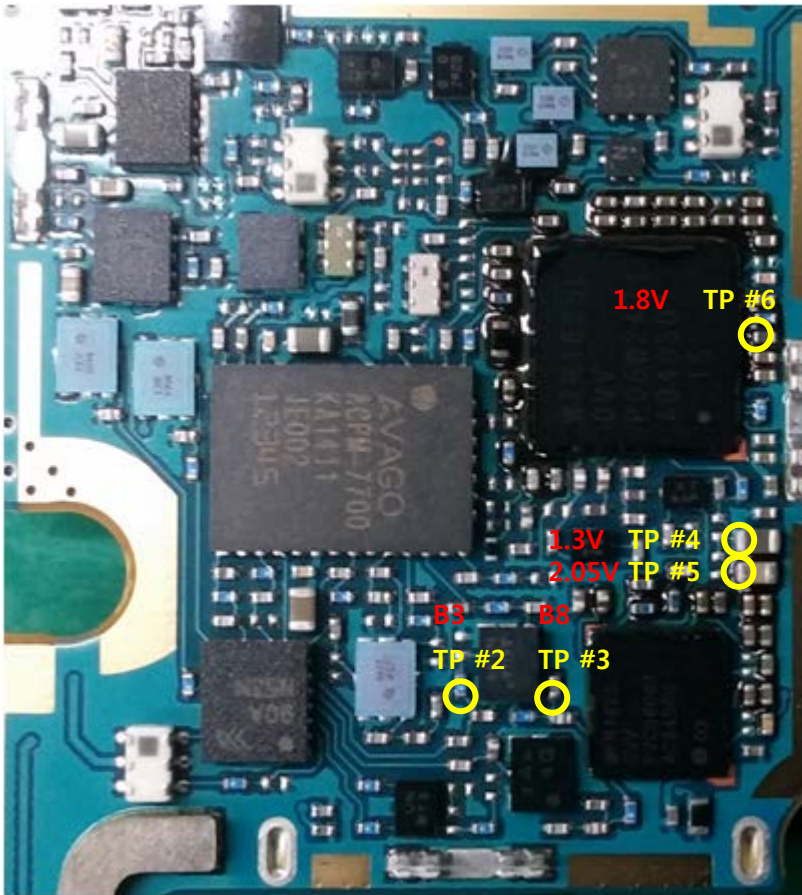
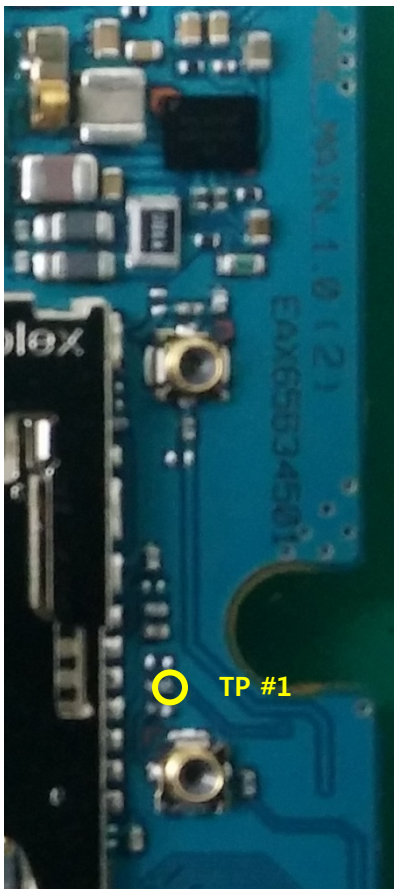
10-2. 수신감도가 좋지 않을 때 (LTE PRX, KT)



10-2. 수신감도가 좋지 않을 때 (LTE PRX, KT)



* LTE B3, B8 RF Path 가 서로 다름



10-3. 수신감도가 좋지 않을 때 (LTE PRX, U+)

RX test Setup(CMW500)

- Test Band : 1, 5, 7
- Test channel : 18050, 20600, 21050(Down)
- Bandwidth : 5MHz/10MHz/20MHz
- Cell power : -85dBm
- Measure frequency domain waveform with Spectrum analyzer.
- Measure time domain waveform with Oscilloscope.

U4100 : PM8941(PMIC)

U1500 : WTR1625L (RF Transceiver)

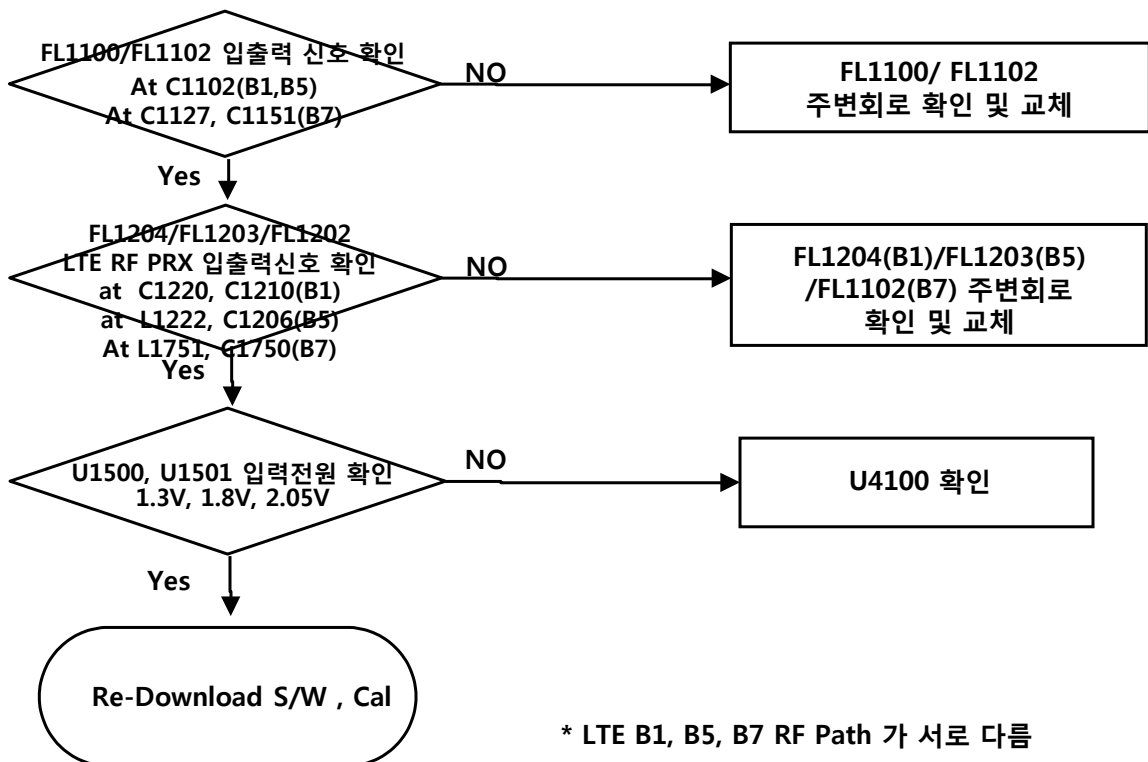
U1501 : WFR1620 (RF Transceiver)

FL1204/FL1203 : Duplexer(B1)/(B5)

FL1750 : Duplexer(B7)

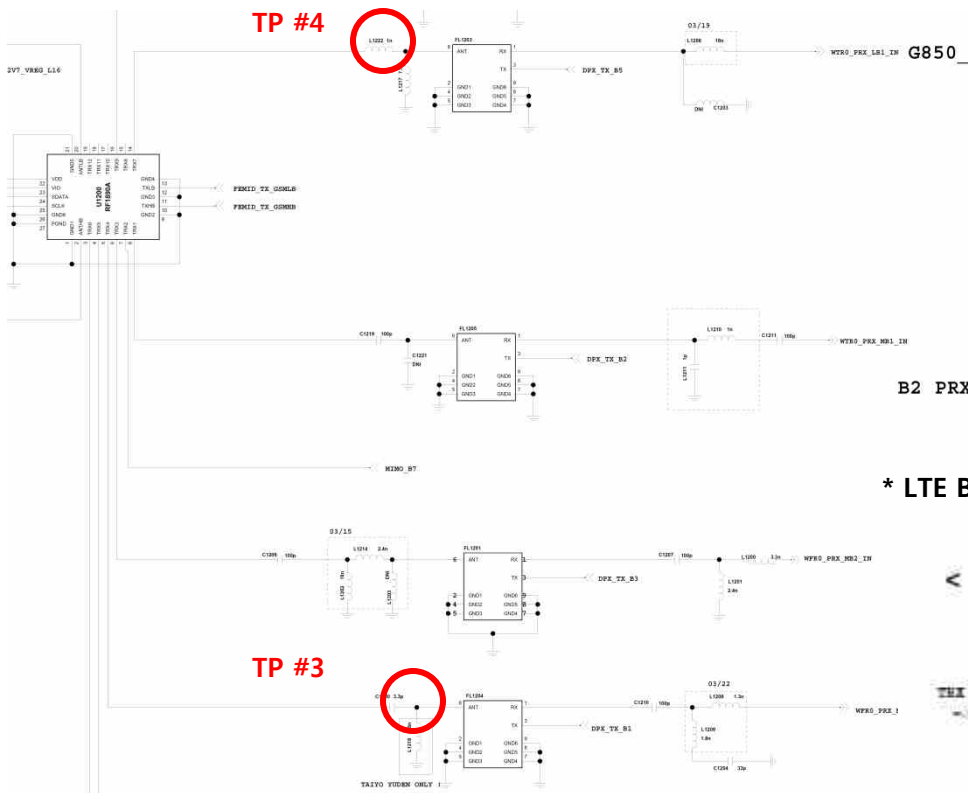
FL1100 : Diplexer(B1/B5)

FL1102 : Coupler(B7)



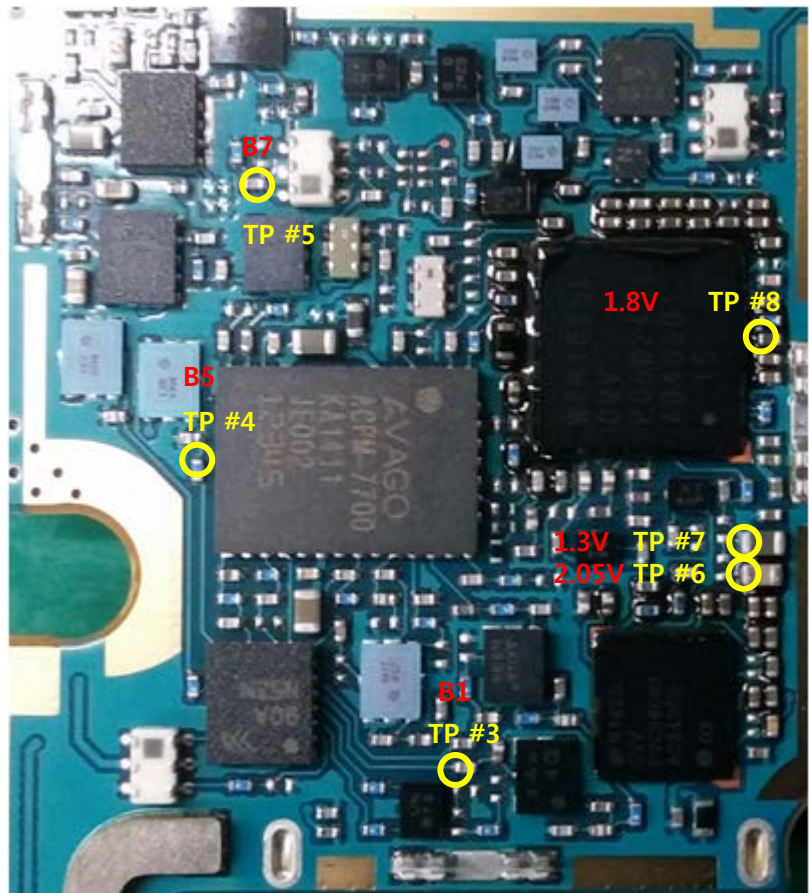
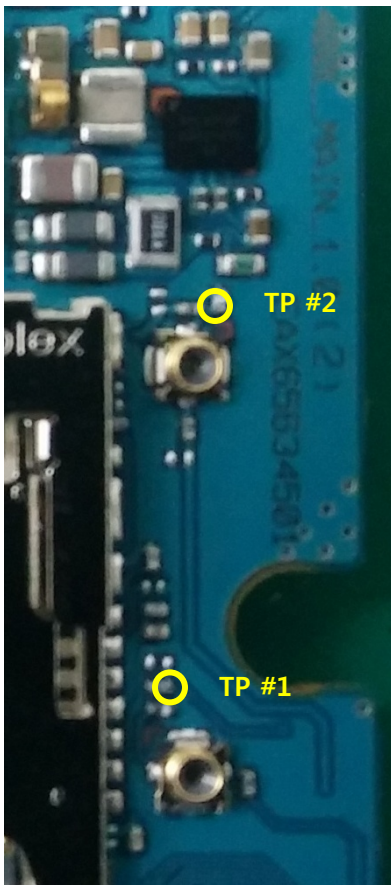
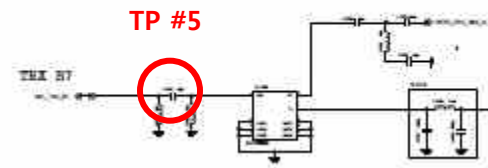
* LTE B1, B5, B7 RF Path 가 서로 다름

10-3. 수신감도가 좋지 않을 때 (LTE PRX, U+)

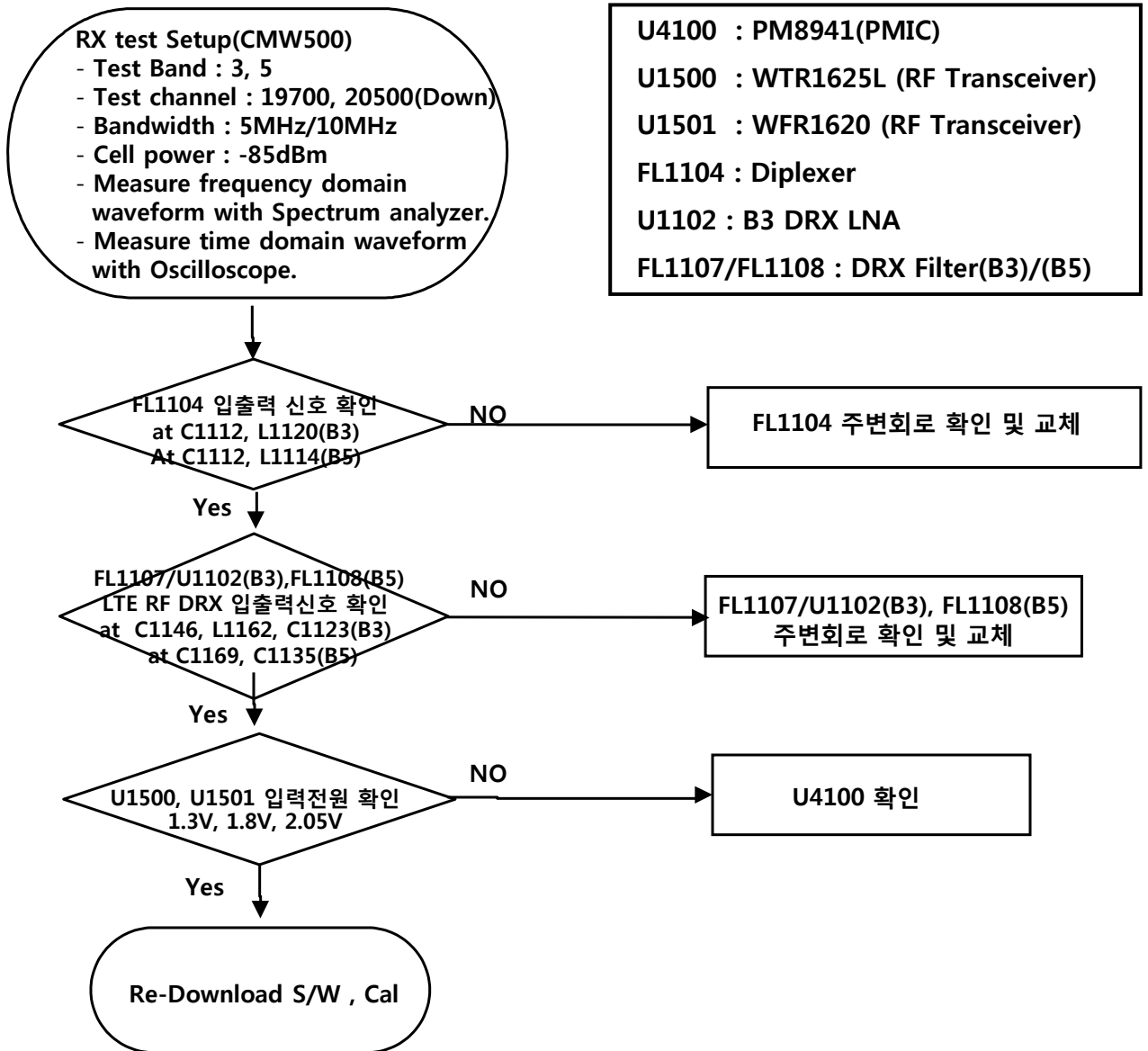


* LTE B1, B5, B7 RF Path 가 서로 다름

< 1-3-6-7_TRX_B7 >



11-1. 수신감도가 좋지 않을 때 (LTE DRX, SKT)

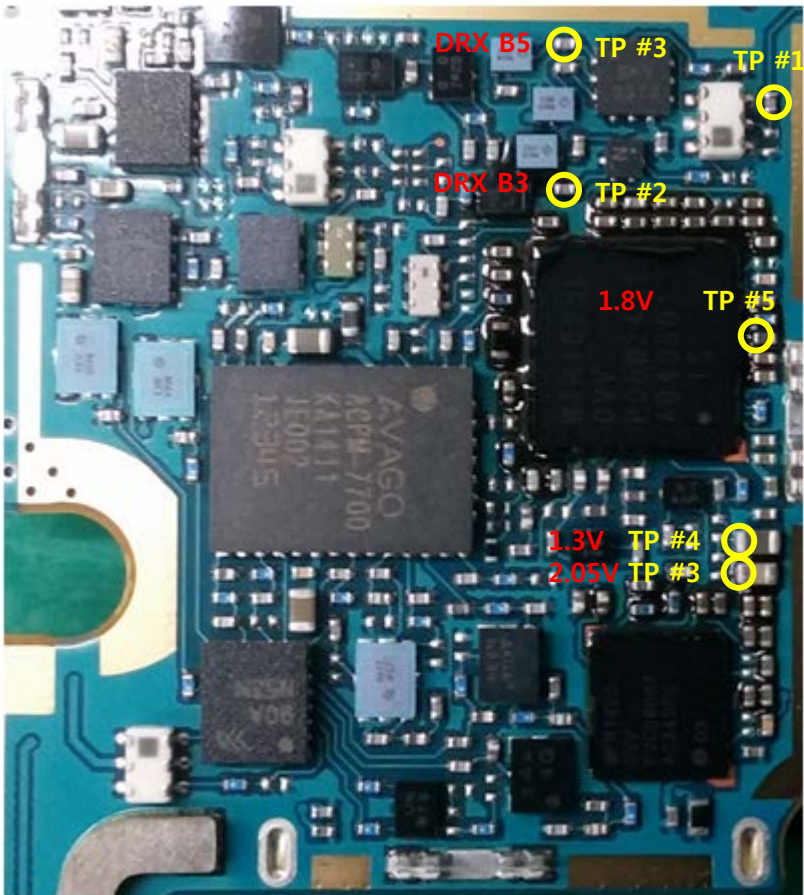
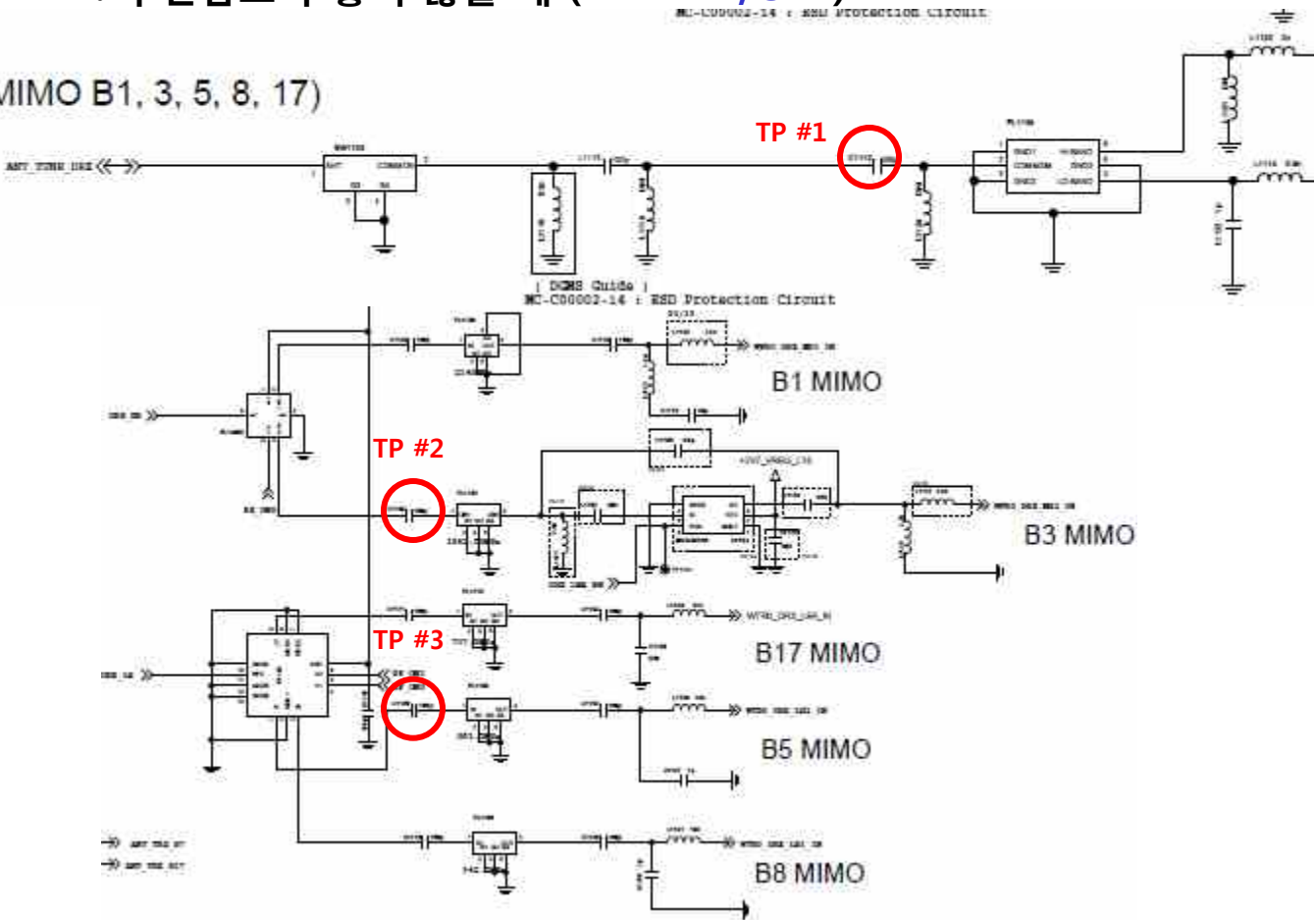


* LTE B1, B5 RF Path 가 서로 다름

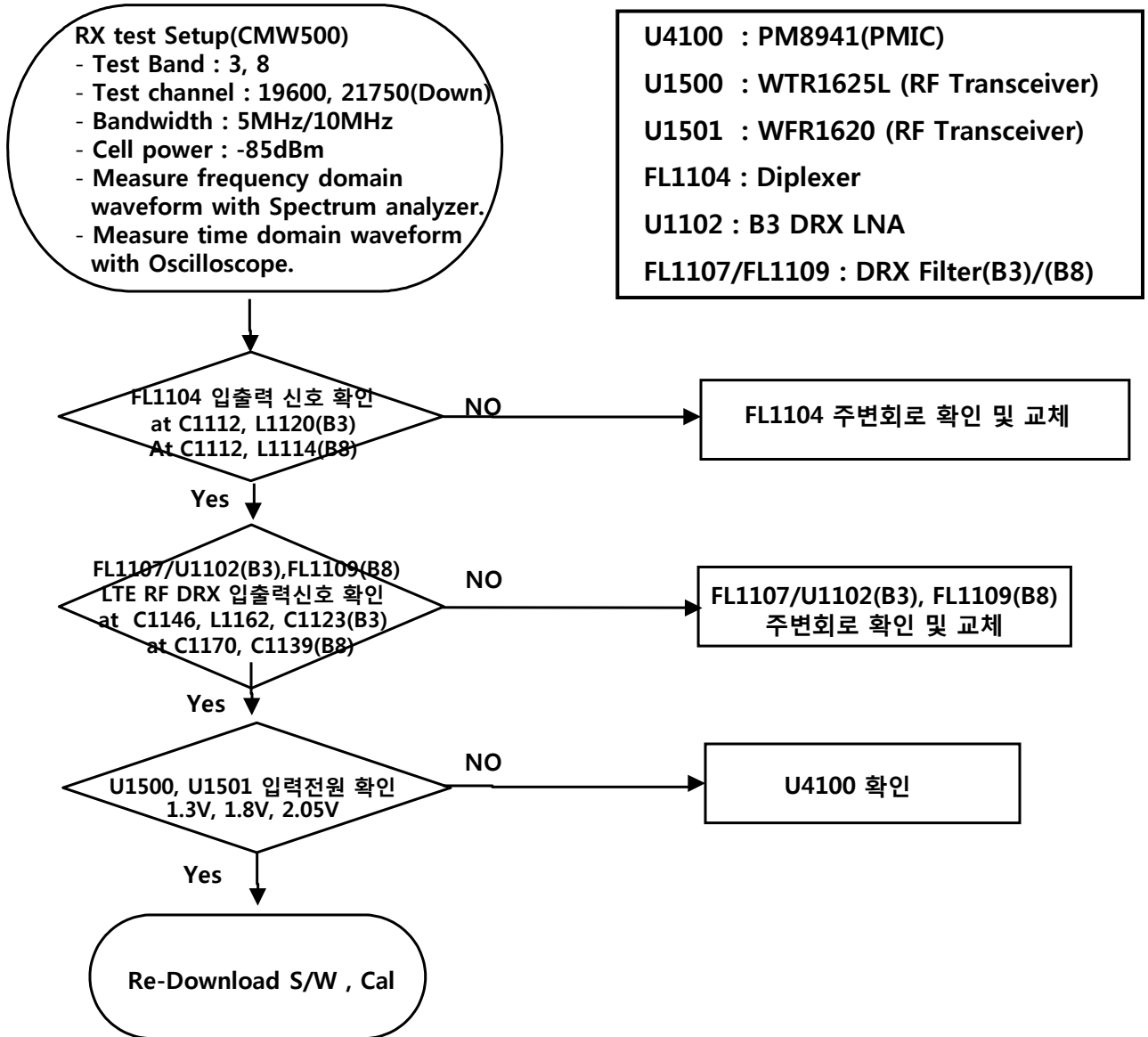
11-1. 수신감도가 좋지 않을 때 (LTE DRX, SKT)

MC-L29902-14 : ESD Protection Circuit

T3 (MIMO B1, 3, 5, 8, 17)



11-2. 수신감도가 좋지 않을 때 (LTE DRX, KT)

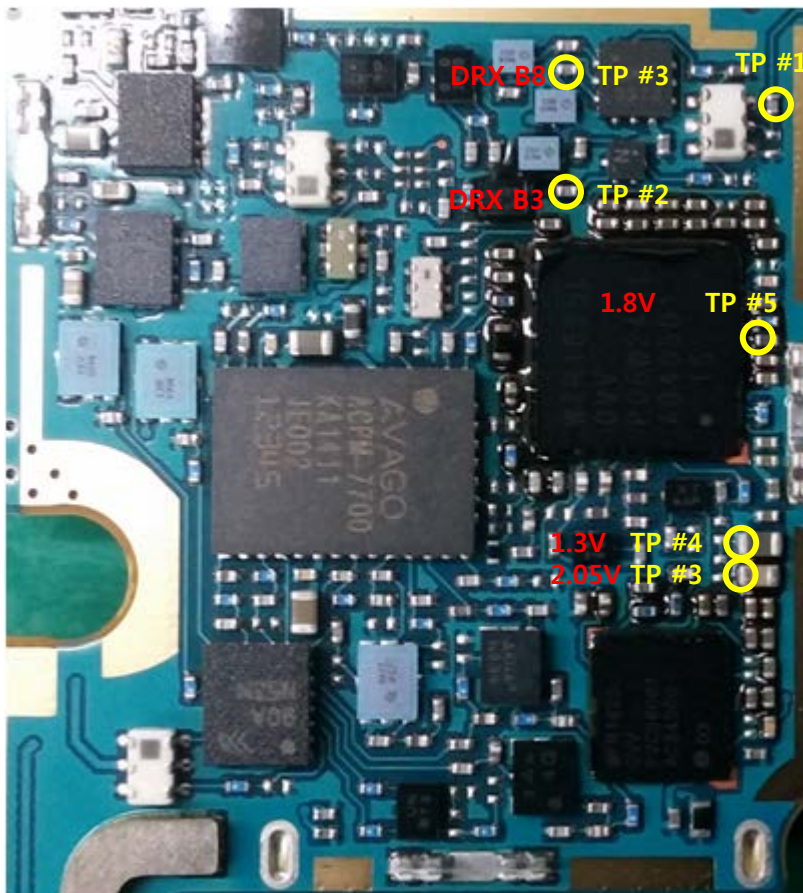
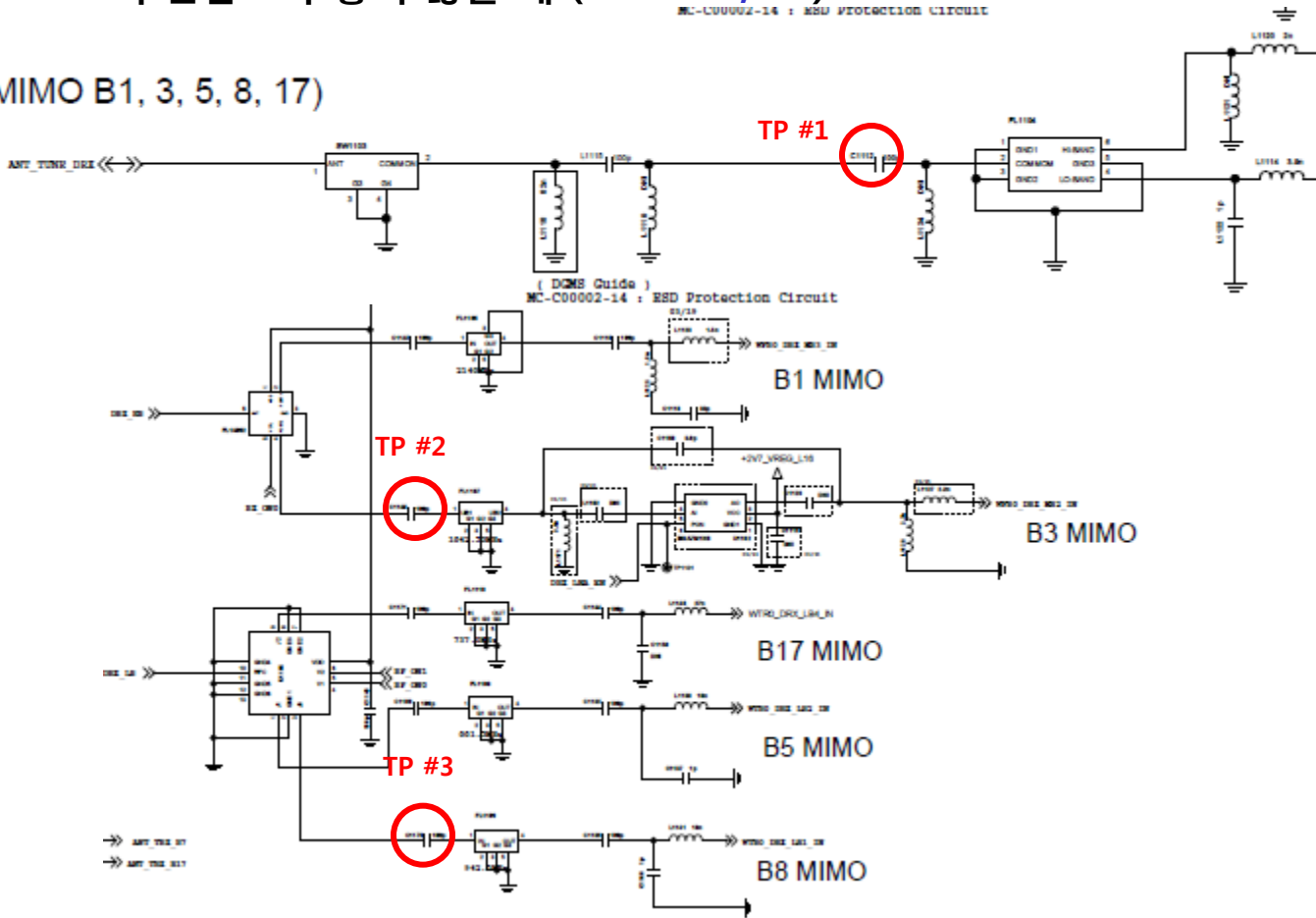


* LTE B1, B5 RF Path 가 서로 다름

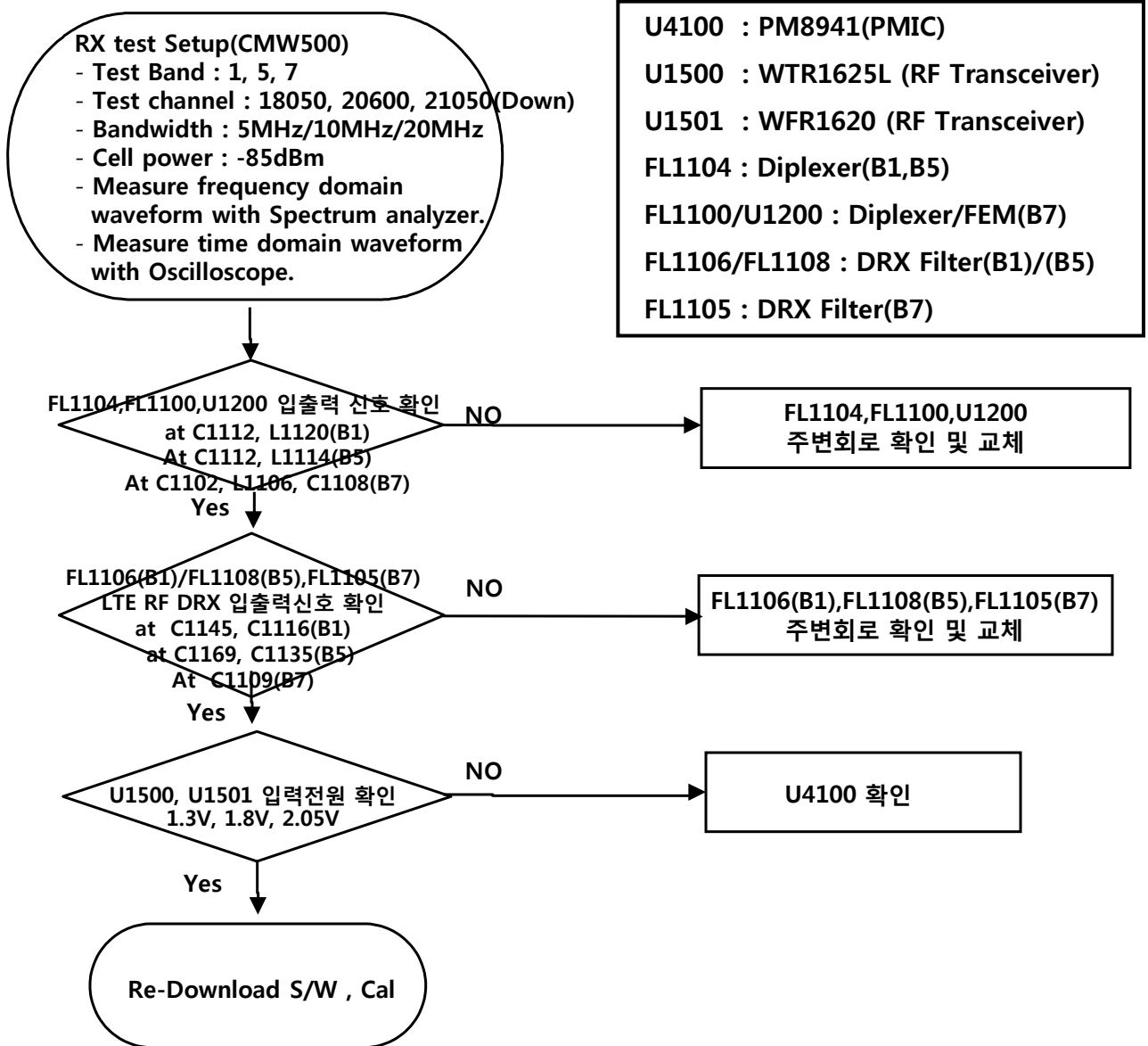
11-2. 수신감도가 좋지 않을 때 (LTE DRX, KT)

MC-C00002-14 : ESD Protection Circuit

T3 (MIMO B1, 3, 5, 8, 17)



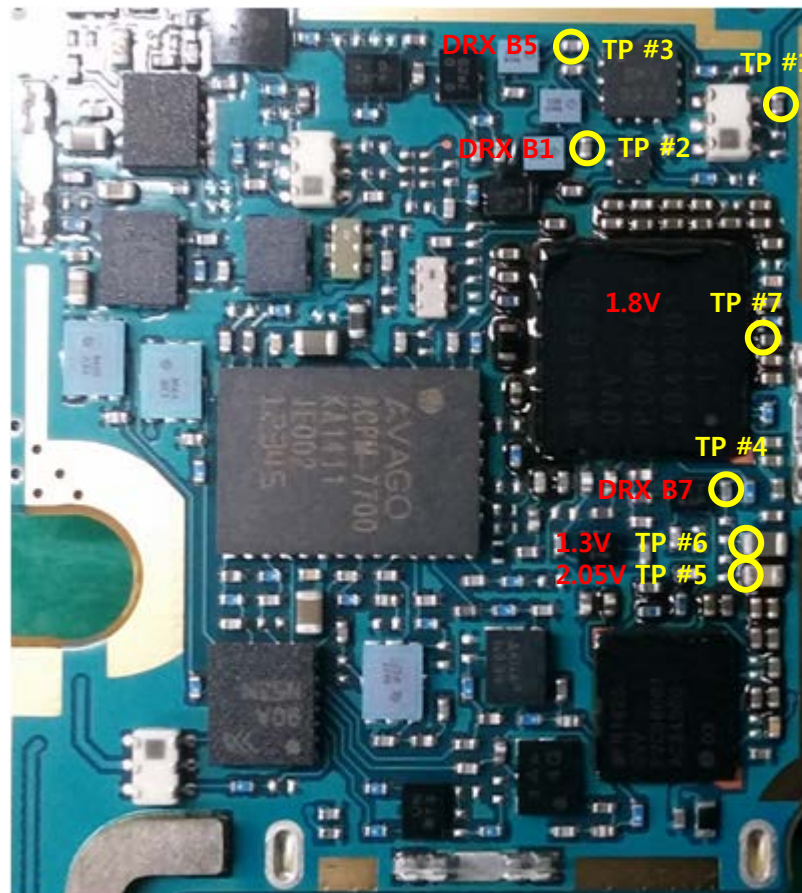
11-3. 수신감도가 좋지 않을 때 (LTE DRX, U+)



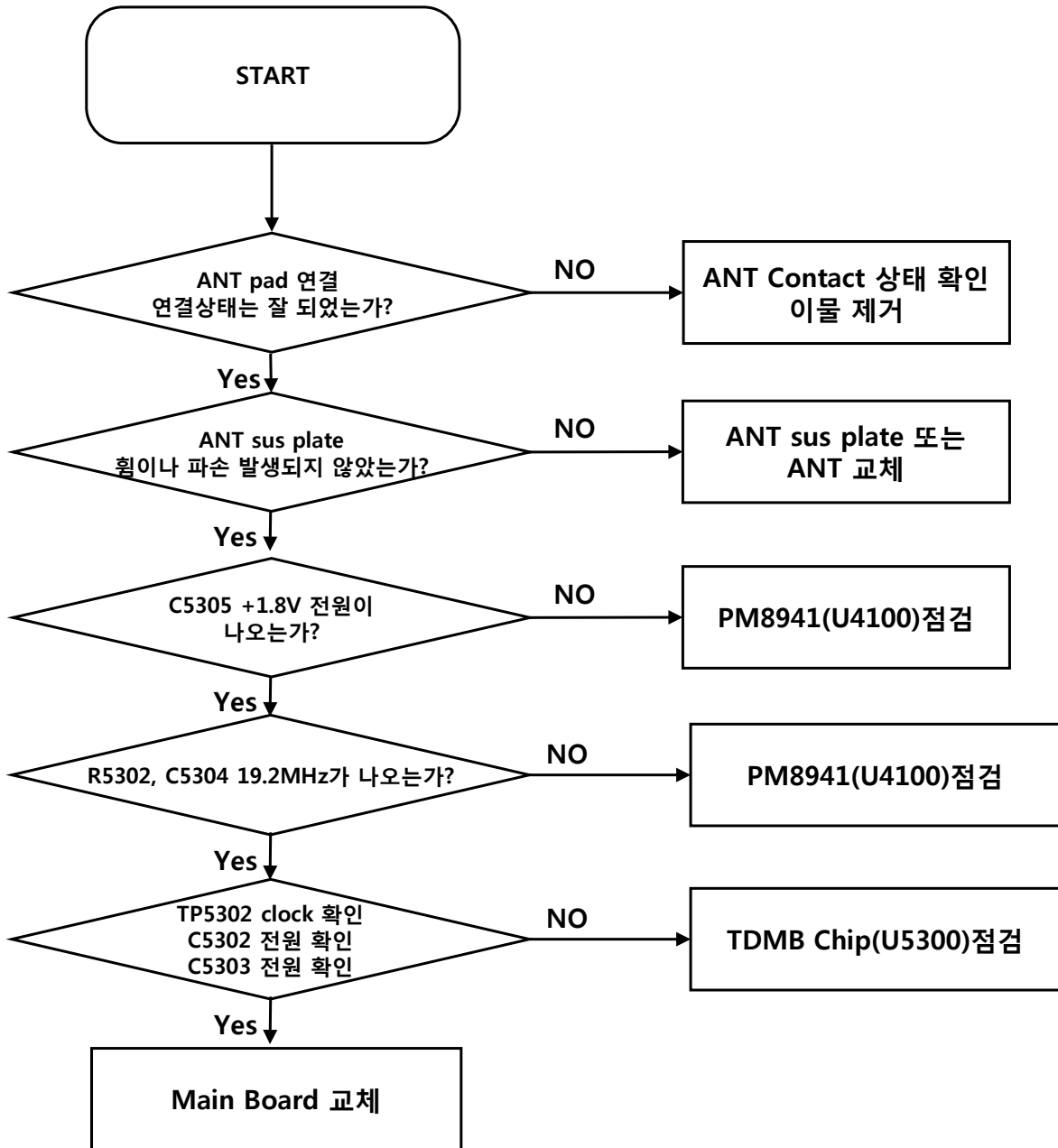
* LTE B1, B5, B7 RF Path 가 서로 다름

ML-CUUUUZ-14 : MSD PROTECTION CIRCUIT

TP #1



12. 지상파 DMB 가 동작하지 않을 때



U5300 FC8080

U5300 FC8080

TP5302

TP5303

TP5304

TP5305

TP5306

TP5307

TP5308

TP5309

TP5310

TP5311

TP5312

TP5313

TP5314

TP5315

TP5316

TP5317

TP5318

TP5319

TP5320

TP5321

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TP5409

TP5410

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TP5490

TP5491

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TP5495

TP5496

TP5497

TP5498

TP5499

TP5500

TP5501

TP5502

TP5503

TP5504

TP5505

TP5506

TP5507

TP5508

TP5509

TP5510

TP5511

TP5512

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TP5542

TP5543

TP5544

TP5545

TP5546

TP5547

TP5548

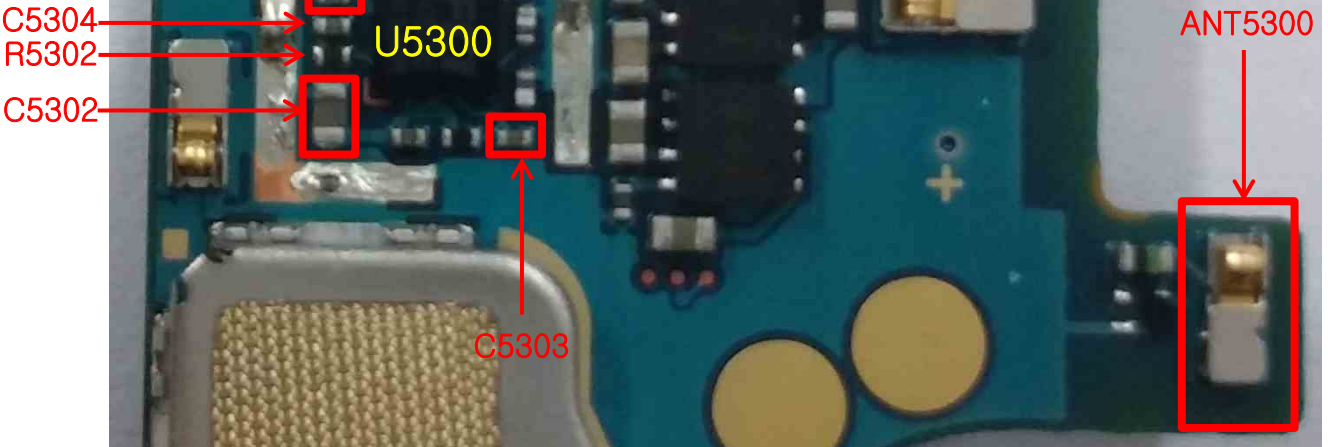
TP5549

TP5550

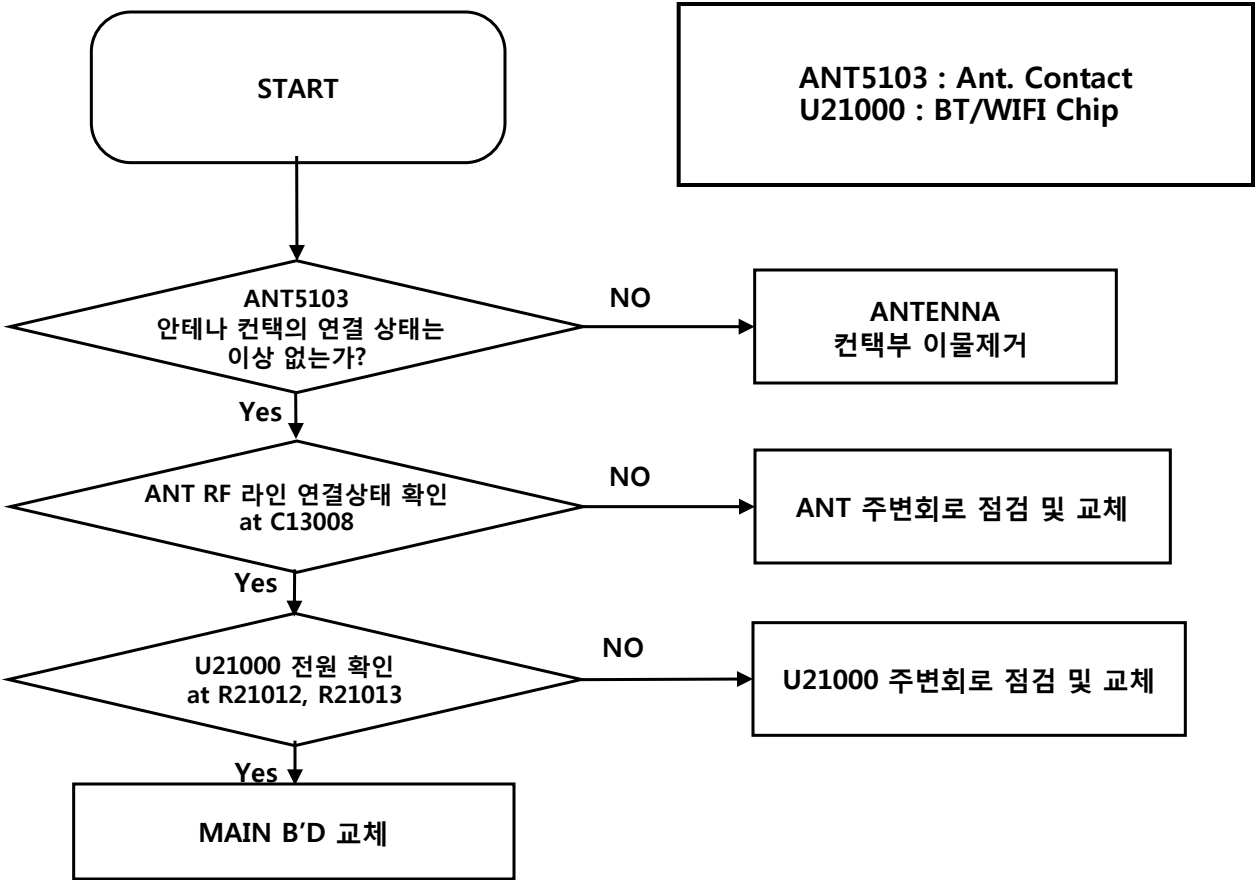
TP5551

TP5552

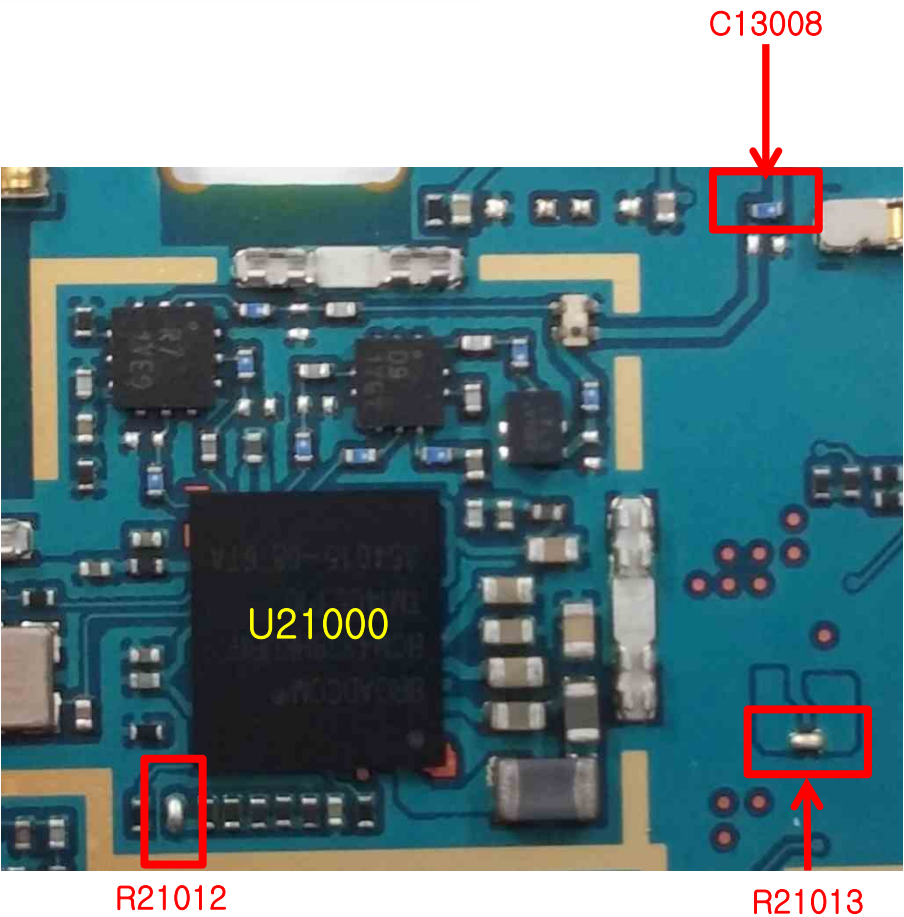
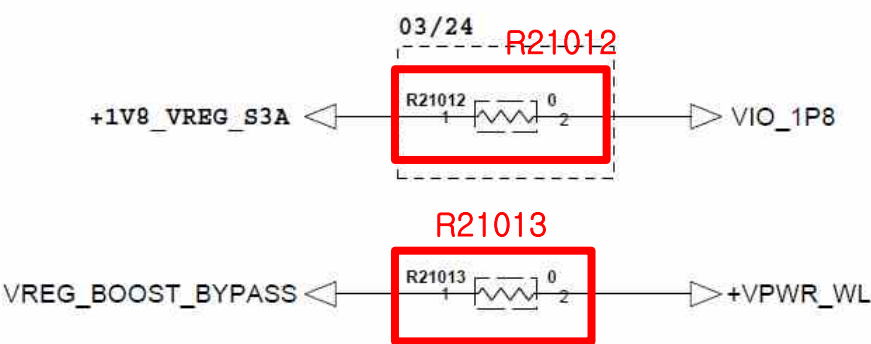
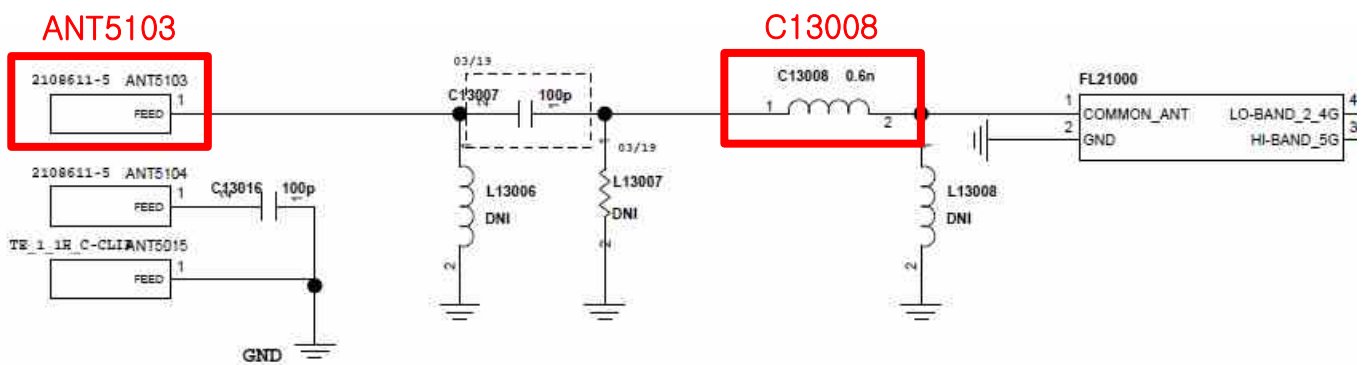
TP55



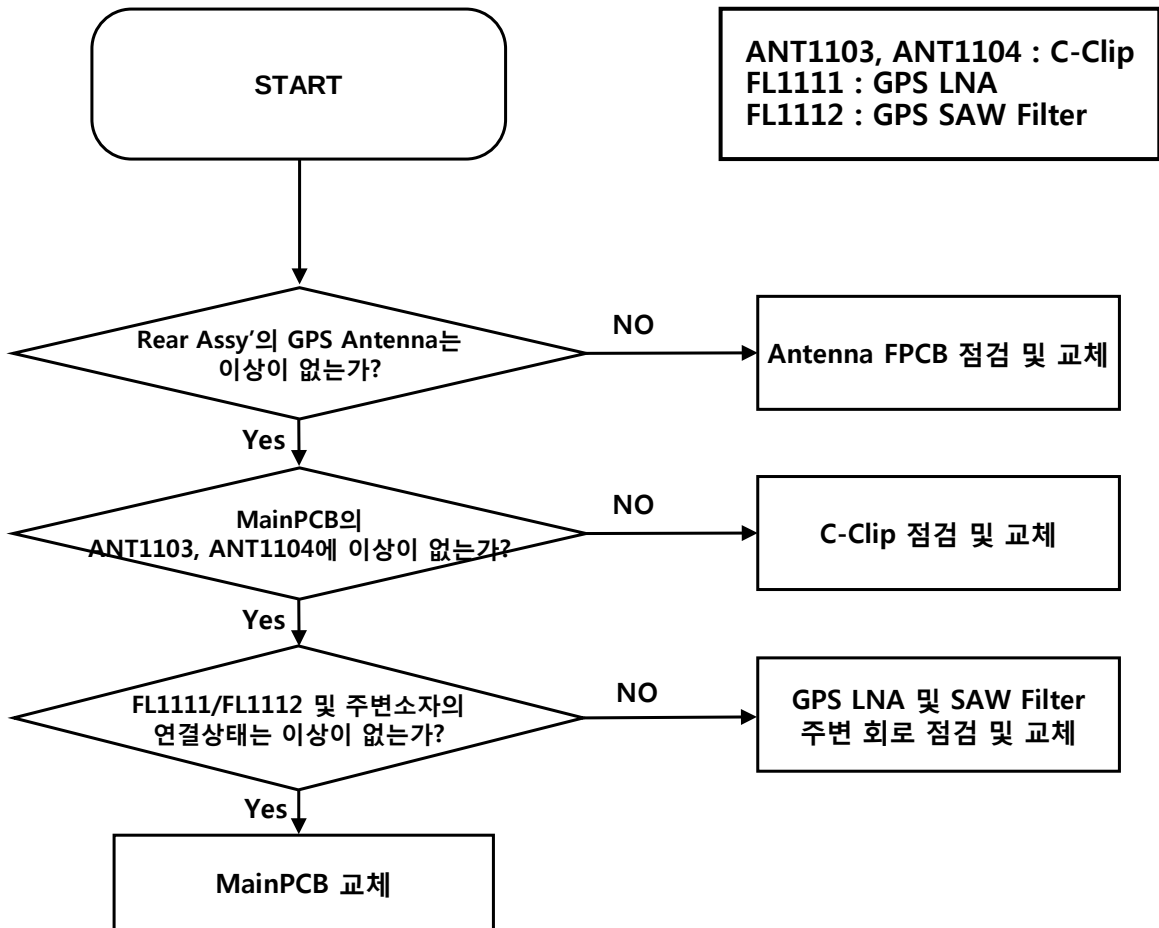
13. BT/WIFI 동작이 되지 않을 때



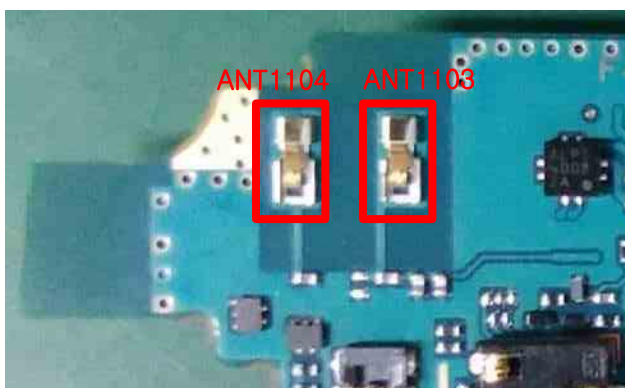
13. BT/WIFI 동작이 되지 않을 때



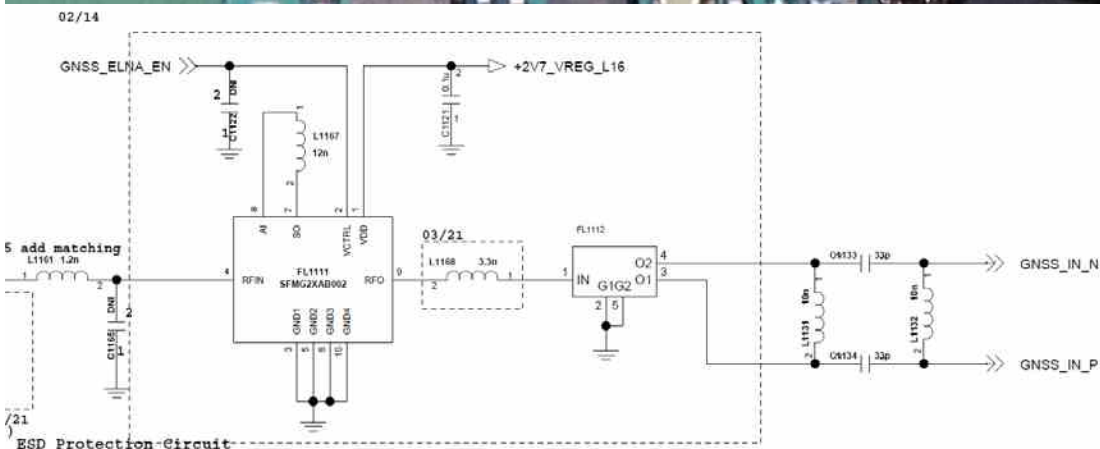
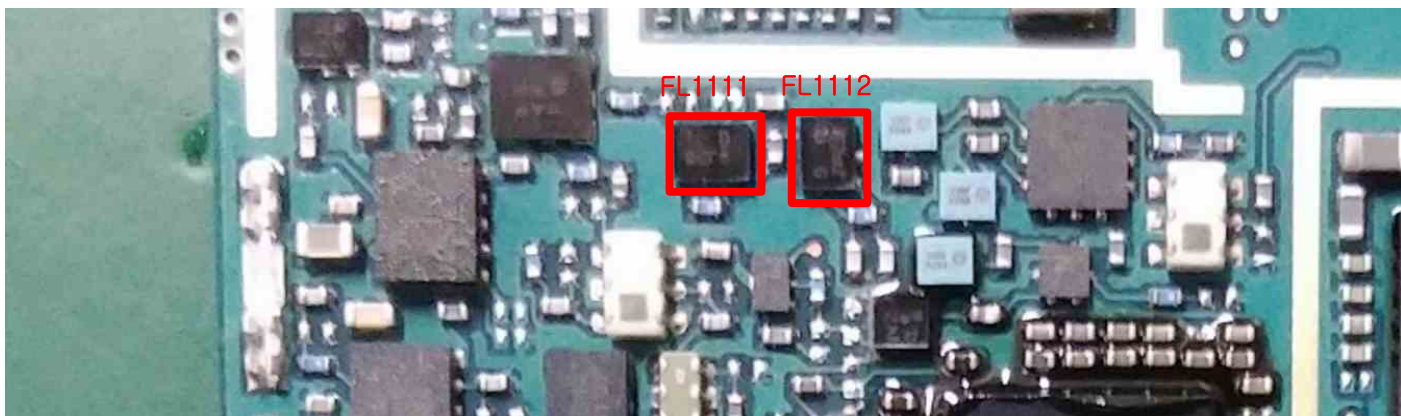
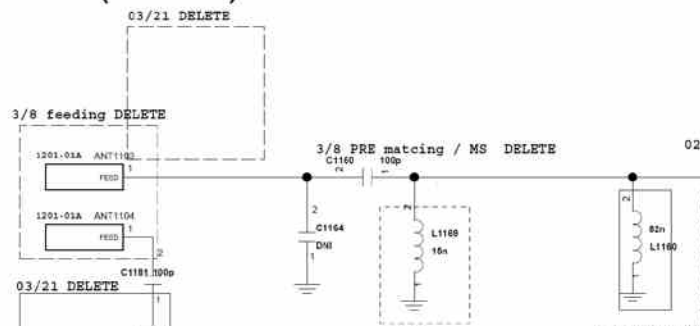
14. GPS 동작이 되지 않을 때



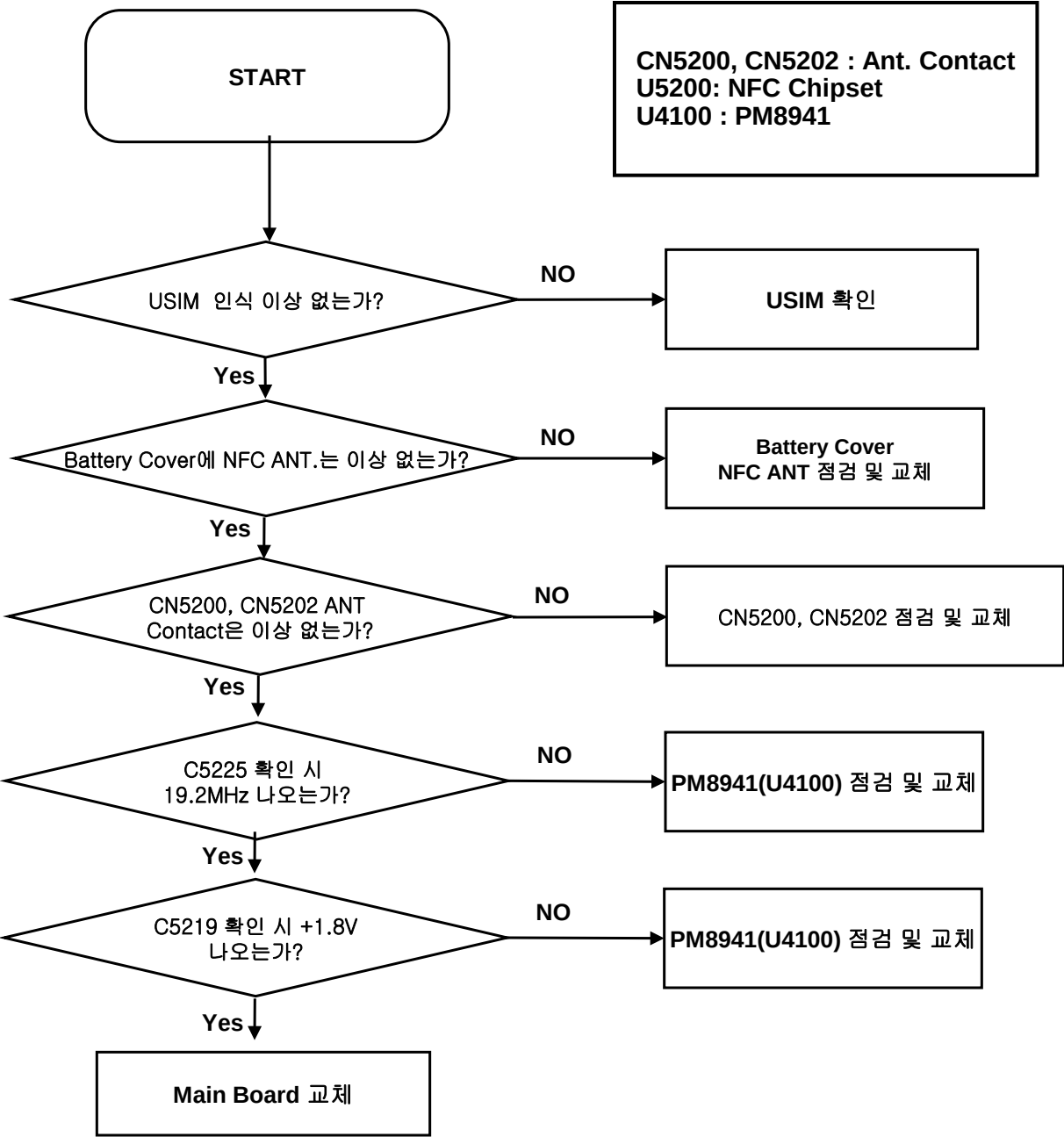
14. GPS 동작이 되지 않을 때



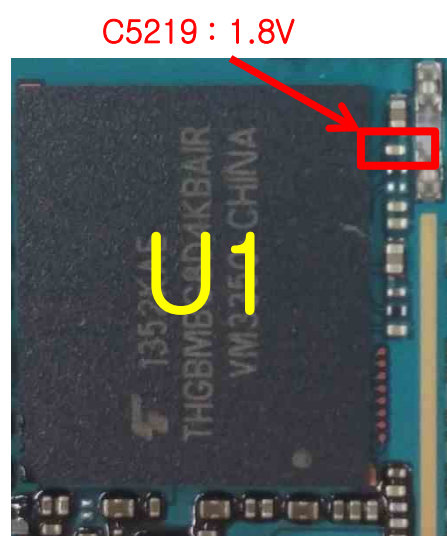
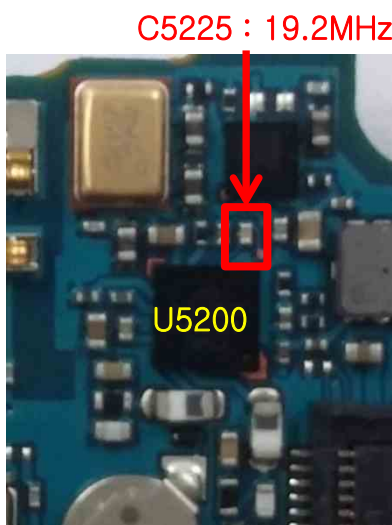
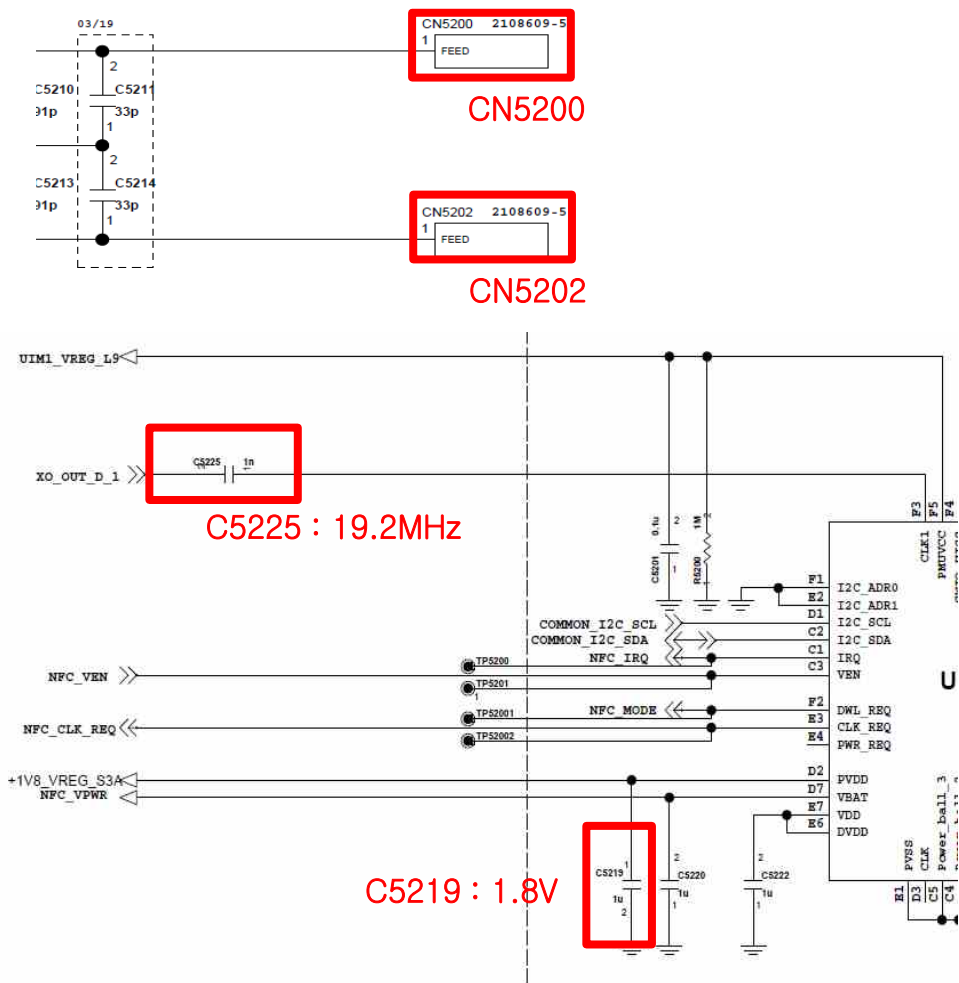
ANT5 (GNSS)



15. NFC 동작이 되지 않을 때



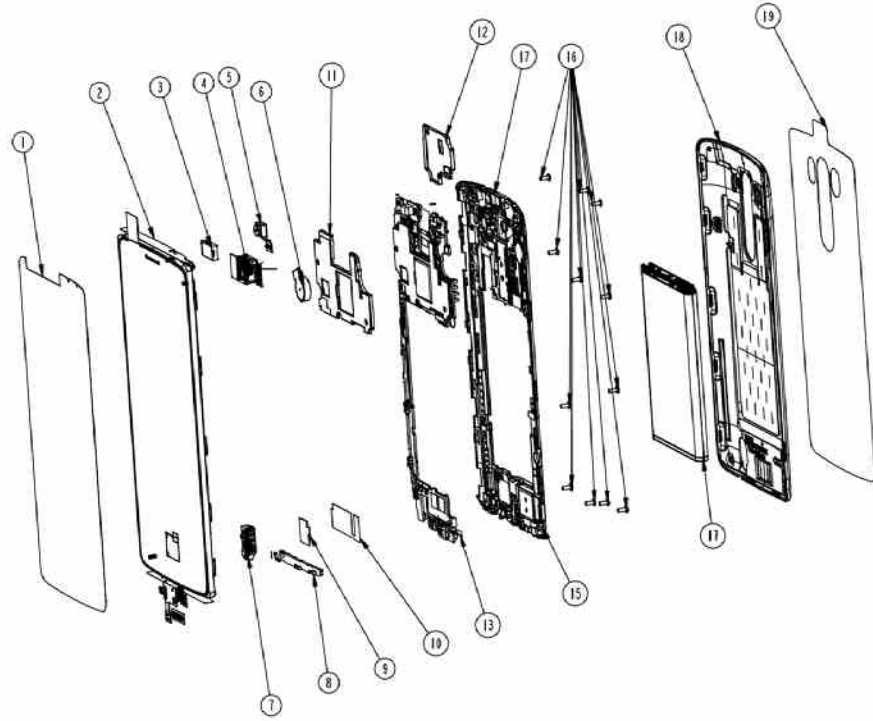
15. NFC 동작이 되지 않을 때



사. 제품 분해도 및 부품 목록



1.TOTAL_ASSY



1:2

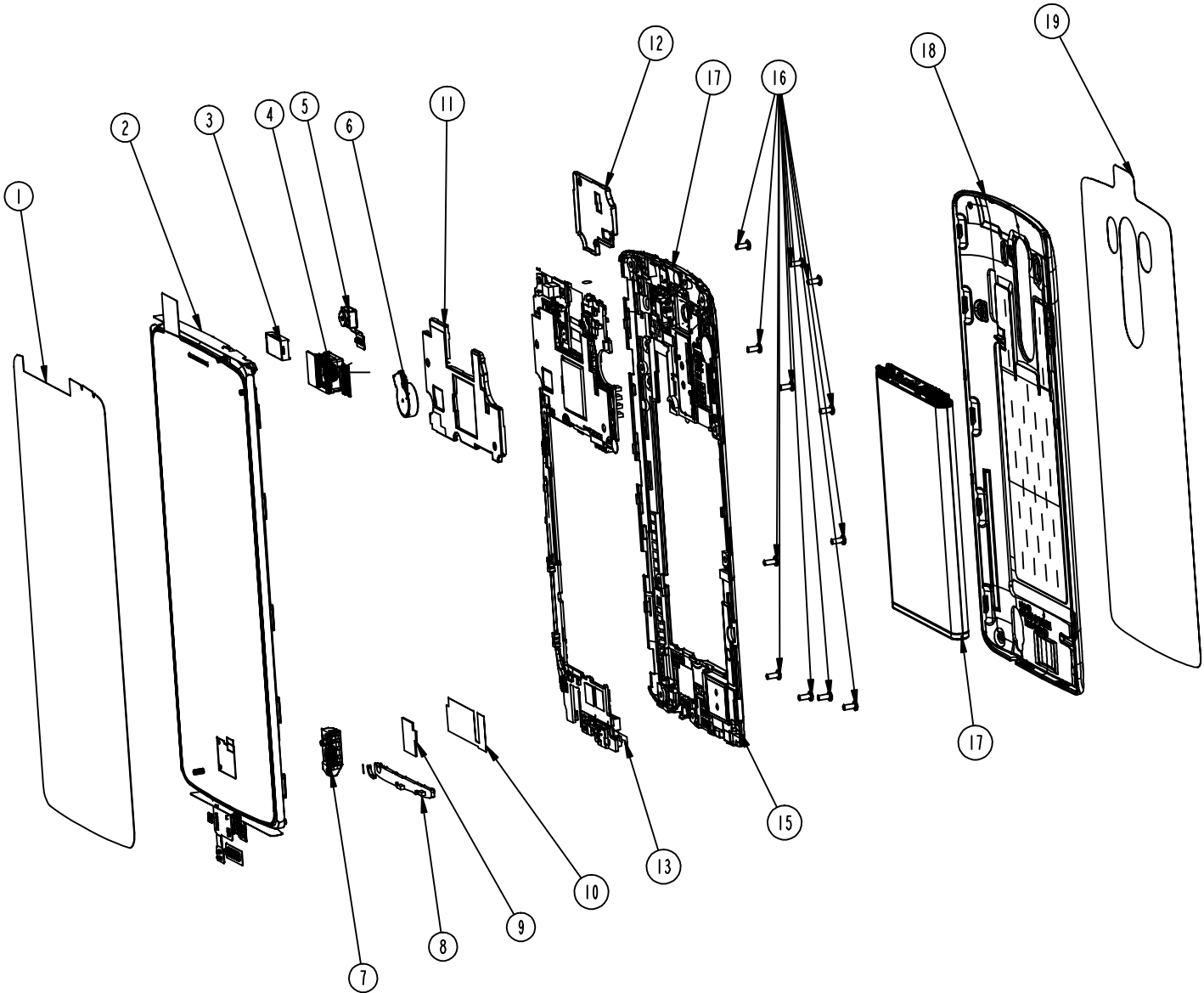
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Revision				
rev.	Contents	Engineer	Approver	Date
1.0	1.0	김영준	김영준	202140423

19	Tape Protection Cover Batt	MJN68927101	1	
18	Cover Batt Assy	ACQ87132801	1	
17	Battery	EAC62378705	1	
16	Screw	GMEY0014301	1	
15	Antenna Main	EAB63268801	1	
14	Rear Assy	ACQ87132601	1	
13	PCB Assy	EBR78684201	1	
12	Shieldcan BTM Assy	ABM74416101	1	
11	Shieldcan TOP Assy	ABM74396201	1	
10	Gasket touch ic	MDS64610901	1	
9	Pad LCD BTM	MCQ67705201	1	
8	Antenna Sosa	EBR78759401	1	
7	Earjack	EAG63849801	1	
6	Motor	EAU62004401	1	
5	Camera VT	EBP62061901	1	
4	Camera Main	EBP61801702	1	
3	Receiver	EAB63288401	1	
2	Bar Assy Sub	ACQ87112301	1	
1	Tape Protection Window	MJN68847901	1	
No	Part Name	Part Number	Q'ty	Remark
Material 材				
Unit	Scale	Size	Sheet	Designed By
mm	1:1	A3	1/1	김영준
				Date
				Model Name F400_PHONE
				Drawing Name F400_PHONE.ASM
				Drawing Number F400_PHONE.
				Approved By
				Date

Revision				
rev.	Contents	Engineer	Approver	Date
1.0	초도발행	유정운	최성우	202140423



배율 1:2

19	Tape Protection Cover Batt	MJN68927101	1	
18	Cover Batt Assy	ACQ87132801	1	
17	Battery	EAC62378705	1	
16	Screw	GMEY0014301	1	
15	Antenna Main	EAB63268801	1	
14	Rear Assy	ACQ87132601	1	
13	PCB Assy	EBR78684201	1	
12	Shieldcan BTM Assy	ABM74416101	1	
11	Shieldcan TOP Assy	ABM74396201	1	
10	Gasket touch ic	MDS64610901	1	
9	Pad LCD BTM	MCQ67705201	1	
8	Antenna Sosa	EBR78759401	1	
7	Earjack	EAG63849801	1	
6	Motor	EAU62004401	1	
5	Camera VT	EBP62061901	1	
4	Camera Main	EBP61801702	1	
3	Receiver	EAB63288401	1	
2	Bar Assy Sub	ACQ87112301	1	
1	Tape Protection Window	MJN68847901	1	
No	Part Name	Part Number	Q'ty	Remark

Material		거짓				
Unit	Scale	Size	Sheet	Designed By	Model Name	F400_PHONE
mm	1:1	A3	1 / 1	백성길 Date	Drawing Name	F400_PHONE.ASM
				Approved By	Drawing Name	F400_PHONE.ASM
				Date	Drawing Number	F400_PHONE.

LG-F400L MAIN PCB

Part No	Description	Qty	Designator
EBR78684501 194	PCB Assembly,Main,SMT Top	1	EBR071700
EAE62282201	Capacitor,Ceramic,Chip	3	C1321,C1327,C4119
EAE62286801	Capacitor,Ceramic,Chip	37	C1504,C1505,C1506,C1507,C1508,C1511,C1512,C1513,C1514,C1515,C1516,C1518,C1519,C1520,C1521,C1522,C1523,C1525,C1528,C1529,C1530,C1531,C1532,C1533,C1534,C1535,C1537,C1538,C1769,C1771,C4111,C4112,C4800,C5301,C5303,C5304,C5305
EAE62502901	Capacitor,Ceramic,Chip	8	C1324,C1328,C1524,C1663,C21017,C4100,C4704,C4710
EAE62505701	Capacitor,Ceramic,Chip	1	C4401
EAE62506501	Capacitor,Ceramic,Chip	8	C21016,C4101,C4102,C4703,C5302,C7307,C7311,C7312
EAE62522101	Capacitor,Ceramic,Chip	2	C4110,C4120
EAE62542701	Capacitor,Ceramic,Chip	1	C4706
EAE62685201	Capacitor,Ceramic,Chip	1	C4118
EAE62685301	Capacitor,Ceramic,Chip	1	C4151
EAE62762301	Capacitor,Ceramic,Chip	12	C21006,C21020,C2335,C2348,C4114,C4141,C4702,C4801,C7310,C7604,C7610,C7614
EAE62762401	Capacitor,Ceramic,Chip	1	C21009
EAE62767801	Capacitor,Ceramic,Chip	2	C4115,C4117
EAE62927201	Capacitor,Ceramic,Chip	1	C4116
EAE62945801	Capacitor(High Frequency),Ceramic,Chip	2	C21033,L1211
EAE62946401	Capacitor(High Frequency),Ceramic,Chip	1	C1220
EAE62946501	Capacitor(High Frequency),Ceramic,Chip	2	C1168,L1321
EAE62947001	Capacitor(High Frequency),Ceramic,Chip	3	C21002,C21011,C21030
EAE62962201	Capacitor(High Frequency),Ceramic,Chip	4	C1303,C1325,L1757,L1758
EAE62962301	Capacitor,Ceramic,Chip	4	C1661,C1768,C1770,C4105
EAE63043301	Capacitor(High Frequency),Ceramic,Chip	3	C1137,C1140,L1122
EAE63043401	Capacitor(High Frequency),Ceramic,Chip	1	C1304
EAE63043801	Capacitor(High Frequency),Ceramic,Chip	1	C1701
EAE63067401	Capacitor,Ceramic,Chip	2	C7500,C7503
EAE63162401	Capacitor,Ceramic,Chip	3	C7501,C7502,C7504
EAG63412001	Connector,RF	2	SW1003,SW1004
EAG63730501	C-Clip	5	ANT1111,ANT1112,ANT1114,ANT5015,ANT5300
EAG63910001	C-Clip	1	ANT1113
EAH61992801	Diode,Schottky	1	D4400
EAH61995401	Diode,TVS	1	ZD5500
EAM62070901	Filter,Bead	3	FB21000,FB4400,FB4401
EAM62451101	Filter,LCR	1	FL7600
EAM62632601	Filter,Duplexer	1	FL1201
EAM62672501	Filter,Separator	1	FL1100
EAM62673101	Filter,Saw	1	FL1112
EAM62690501	Filter,Ceramic	1	FL21001
EAM62691501	Filter,Ceramic	1	FL1300
EAM62692901	Filter,Duplexer	1	FL1203
EAM62730001	Filter,Separator,Switch	1	FL14002
EAM62732801	Filter,Saw	1	FL1108
EAM62732901	Filter,Saw	1	FL1109
EAM62771001	Filter,Saw	1	FL1106
EAM62811301	Filter,Saw,Dual	1	FL1206
EAM62870401	Filter,Duplexer	1	FL1700

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EAM62872101	Filter,Saw	1	FL1107
EAM62890101	Filter,Duplexer	1	FL1205
EAM62890201	Filter,Duplexer	1	FL1202
EAM62910201	Filter,Separator	2	FL1103,FL1104
EAM62930401	Filter,Saw	1	FL1110
EAM62930701	Filter,Saw	1	FL1105
EAM63050201	Filter,Separator,Switch	1	U1104
EAM63050301	Filter,Separator	1	FL21000
EAM63090101	Filter,Duplexer	1	FL1204
EAM63230001	Filter,Duplexer	1	FL1750
EAN62421601	IC,Fuel Gauge	1	U4800
EAN62568201	IC,Proximity	1	U8400
EAN62667101	IC,PMIC	1	U4100
EAN62682901	IC,Hall Effect Switch	1	U8700
EAN62697301	IC,RF Transceiver,4G	1	U1500
EAN62697601	IC,RF Transceiver,4G	1	U1501
EAN62698301	IC,WiFi	1	U21000
EAN62731801	IC,DC,DC Converter	1	U7500
EAN62735301	IC,LDO Voltage Regulator	1	U7602
EAN62768701	IC,LDO Voltage Regulator	1	U7600
EAN62776201	IC,Power Amplifier	1	U1300
EAN62867801	IC,Power Amplifier	1	U1750
EAN62909201	IC,TDMB	1	U5300
EAN62930601	IC,DC,DC Converter	1	U4400
EAN62957701	IC,Mobile SDRAM	1	U3100
EAN62972601	IC,Power Amplifier	1	U1650
EAN63049301	IC,Charger	1	U4700
EAN63106201	IC,Digital Baseband Processor,4G	1	U2100
EAN63106301	IC,RF Amplifier	1	FL1111
EAN63125901	IC,Signal Bridge	1	U7601
EAP61747501	Inductor,Multilayer,Chip	1	L1305
EAP61767701	Inductor,Multilayer,Chip	1	L21007
EAP61767801	Inductor,Multilayer,Chip	4	L1106,L1111,L1201,L1214
EAP61866701	Inductor,Multilayer,Chip	2	L5302,L5306
EAP61866801	Inductor,Multilayer,Chip	1	L1217
EAP61866901	Inductor,Multilayer,Chip	1	L1120
EAP61946101	Inductor,Multilayer,Chip	1	L1208
EAP61987001	Inductor,Multilayer,Chip	3	L1138,L1141,L1202
EAP62107901	Inductor,Multilayer,Chip	4	C1302,L1123,L1219,L1220
EAP62108001	Inductor,Multilayer,Chip	3	L1307,L1323,L21009
EAP62108401	Inductor,Multilayer,Chip	5	L1167,L1204,L1304,L1700,L1703
EAP62108601	Inductor,Multilayer,Chip	1	L1507
EAP62109001	Inductor,Multilayer,Chip	1	C13008
EAP62186301	Inductor,Multilayer,Chip	1	L1125
EAP62187401	Inductor,Wire Wound,Chip	1	L4101
EAP62225901	Inductor,Multilayer,Chip	6	L1212,L1221,L1222,L21005,L21010,L21011
EAP62226001	Inductor,Multilayer,Chip	3	L1161,L21004,L21013
EAP62226101	Inductor,Multilayer,Chip	6	L1114,L1171,L1315,L1316,L1325,L21008
EAP62226301	Inductor,Multilayer,Chip	2	C1306,L1756
EAP62226501	Inductor,Multilayer,Chip	2	L1131,L1132
EAP62226901	Inductor,Multilayer,Chip	1	L1210
EAP62227001	Inductor,Multilayer,Chip	1	L1754
EAP62227201	Inductor,Multilayer,Chip	2	L1209,L1218
EAP62227301	Inductor,Multilayer,Chip	1	L1127
EAP62227601	Inductor,Multilayer,Chip	1	L1324

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EAP62227701	Inductor,Multilayer,Chip	1	L1216
EAP62227901	Inductor,Multilayer,Chip	1	L1206
EAP62245901	Inductor,Multilayer,Chip	1	L1129
EAP62246001	Inductor,Multilayer,Chip	2	L1168,L1200
EAP62266401	Inductor,Multilayer,Chip	1	C1759
EAP62307701	Inductor,Multilayer,Chip	1	L21003
EAP62346801	Inductor,Multilayer,Chip	1	C1703
EAP62347901	Inductor,Multilayer,Chip	1	L1710
EAP62348201	Inductor,Multilayer,Chip	2	L1146,L1320
EAP62366601	Inductor,Wire Wound,Chip	1	L4700
EAP62367801	Inductor,Wire Wound,Chip	1	L4400
EAP62526601	Inductor,Wire Wound,Chip	3	L4100,L4103,L4104
EAP62526701	Inductor,Wire Wound,Chip	1	L4102
EAP62588001	Inductor,Wire Wound,Chip	1	L7500
EAP62807301	Inductor,Multilayer,Chip	1	L2300
EAP62906101	Inductor,Multilayer,Chip	1	L1752
EAT61794501	Module, FEM(Front End Module)	1	U1200
EAT61875701	Module, FEM(Front End Module)	1	U21002
EAT61875801	Module, FEM(Front End Module)	1	U21001
EAV62193601	LED,Chip	1	LD8900
EAV62251901	LED,Chip	1	LD4602
EAW61543401	Crystal	1	X7600
EAW61645401	Crystal	1	X21000
EAW61883401	Crystal	1	X4100
EAX65634501	PCB,Main	1	EAX010000
EBC61856101	Resistor,Chip	2	R21004,R8400
EBC61856201	Resistor,Chip	4	R2123,R2124,R2125,R2126
EBC61959901	Resistor,Chip	1	R2127
EBC62036001	Resistor,Chip	1	R21014
EBC62236701	Resistor,Chip	2	R1503,R1504
EBC62296301	Resistor,Chip	2	R7602,R7609
EBC62581901	Resistor,Chip	1	R2146
EBC62596701	Resistor,Chip	1	R4719
EBC62616701	Resistor,Chip	1	R4107
EBG61306601	Thermistor,NTC	1	PT1101
ECA30240001	Coupler,RF Directional	1	FL1102
ECCH0000182	Capacitor,Ceramic,Chip	3	C4103,C4708,C4709
ECCH0000198	Capacitor,Ceramic,Chip	9	C21008,C21022,C21024,C2371,C4135,C4136,C7313,C7600,C7602
ECCH0002002	Capacitor,Ceramic,Chip	1	C4705
ECCH0005603	Capacitor,Ceramic,Chip	2	C4104,C4106
ECCH0007804	Capacitor,Ceramic,Chip	1	C4400
ECCH0007805	Capacitor,Ceramic,Chip	2	C1526,C1527
ECCH0009101	Capacitor,Ceramic,Chip	18	C1121,C21003,C21004,C21005,C21026,C21039,C2110,C2111,C2360,C2373,C4144,C7601,C7603,C7605,C7607,C7609,C8700,C8701
ECCH0009103	Capacitor,Ceramic,Chip	26	C1108,C1109,C1112,C1116,C1135,C1139,C1144,C1145,C1146,C1149,C1169,C1170,C1171,C1200,C1201,C1206,C1207,C1210,C1211,C1219,C1502,C1517,C1662,C1700,C21032,L1753
ECCH0009104	Capacitor,Ceramic,Chip	10	C1118,C1133,C1134,C1151,C1204,C1705,C1711,C1712,C1750,C1760
ECCH0009109	Capacitor,Ceramic,Chip	1	C1664
ECCH0009208	Capacitor,Ceramic,Chip	2	C1305,L1301
ECCH0009216	Capacitor,Ceramic,Chip	2	C21028,L1751

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ECCH0009502	Capacitor,Ceramic,Chip	2	C1318,C1331
ECCH0009504	Capacitor,Ceramic,Chip	3	C21035,C7611,C7612
ECCH0009514	Capacitor,Ceramic,Chip	1	C7613
ECCH0009520	Capacitor,Ceramic,Chip	2	C1320,C1753
			C1500,C2300,C2301,C2302,C2307,C2313,C2314,C2315,C2316,C2317,C2318,C2319,C2320,C2321,C2322,C2323,C2324,C2325,C2326,C2327,C2328,C2329,C2330,C2336,C2337,C2338,C2339,C2340,C2341,C2342,C2343,C2344,C2345,C2346,C2347,C2349,C2351,C2353,C2355,C2356,C2357,C2358,C2361,C2362,C2363,C2364,C2365,C2366,C2367,C2368,C2369,C2370,C2372,C2374,C2375,C2376,C2377,C2378,C2379,C2380,C2381,C2382,C2383,C2384,C2385,C2386,C2387,C2388,C2389,C2390,C4122,C4123,C4125,C4126,C4128,C4130,C4131,C4132,C4134,C4142,C4145,C4146,C7606,C7608,C8400,C8401
ECCH0017301	Capacitor,Ceramic,Chip	86	C1501,C21021,C21023,C4121,C4124,C4133,C4137,C4138,C4139,C4140,C7314
ECCH0017601	Capacitor,Ceramic,Chip	11	C4113
ECZH0001210	Capacitor,Ceramic,Chip	1	C21018
ECZH0025905	Capacitor,Ceramic,Chip	1	C21010,C21034
ECZH0025911	Capacitor,Ceramic,Chip	2	C1309,C1315,C1316,C1317,C1323,C1332,C1333,C1706,L1704,R1500
ECZH0025916	Capacitor,Ceramic,Chip	10	C1540,C1541
ECZH0025920	Capacitor,Ceramic,Chip	2	D7600
EDSY0019602	Diode,Switching	1	L5307
ELCH0003843	Inductor,Multilayer,Chip	1	R21003,R4718
ERHY0009303	Resistor,Chip	2	R14013
ERHY0009311	Resistor,Chip	1	R2143,R2144,R2300,R2309,R2313,R5301
ERHY0009501	Resistor,Chip	6	R4102,R4700,R4701,R4702,R4703,R4704,R4707,R7610,R8401
ERHY0009505	Resistor,Chip	9	R2400,R2401
ERHY0009507	Resistor,Chip	2	R5302
ERHY0009511	Resistor,Chip	1	R21011
ERHY0009515	Resistor,Chip	1	R2108,R2109,R2110,R2111,R2114,R2115
ERHY0009516	Resistor,Chip	6	R4801,R7604,R7605,R7606
ERHY0009526	Resistor,Chip	4	R21005,R21006,R21007,R21008,R21009,R4118
ERHY0009527	Resistor,Chip	6	R4114,R7607
ERHY0009536	Resistor,Chip	2	R4105
ERHY0009537	Resistor,Chip	1	R7603
ERHY0009555	Resistor,Chip	1	R2139
ERHY0009584	Resistor,Chip	1	R21001,R21002,R4800
ERHY0017901	Resistor,Chip	3	R14014,R14015
ERHY0024601	Resistor,Chip	2	R4106
ERHZ0000204	Resistor,Chip	1	R7600
ERHZ0000205	Resistor,Chip	1	R2103,R2104
ERHZ0000235	Resistor,Chip	2	R4706
ERHZ0000242	Resistor,Chip	1	R7608
ERHZ0000350	Resistor,Chip	1	R3214
ERHZ0000401	Resistor,Chip	1	R4400
ERHZ0000406	Resistor,Chip	1	R7601,R7611
ERHZ0000438	Resistor,Chip	2	R2121
ERHZ0000463	Resistor,Chip	1	U7301,U7302
EUSY0407401	IC,LDO Voltage Regulator	2	SC1104,SC1105
MAZ63332001	Bracket	2	SC1101
MBK63435601	Can,Shield	1	SC130026
MBK63793401	Can,Shield	1	

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			SC13001,SC130010,SC130012,SC130013,SC130017,SC130018,SC13002,SC130021,SC130022,SC130023,SC13003,SC13004
MBV62321701	Clip	12	
MEZ65049701	Label	1	PID1
MGJ64064101	Plate	1	SC1102
MGJ64064201	Plate	1	SC1103
SAFP0000401	Wire Pad,Short	2	R21012,R21013
SFSY0041401	Filter,Saw	1	FL1701
EBR78684601 154	PCB Assembly,Main,SMT Bottom	1	EBR071600
EAB62909301	Microphone,Condenser	3	MIC6920,MIC6921,MIC6923
EAE62282201	Capacitor,Ceramic,Chip	1	C1403
EAE62286801	Capacitor,Ceramic,Chip	6	C11004,C2102,C2103,C4109,C5215,C5223
EAE62502901	Capacitor,Ceramic,Chip	4	C6202,C7100,C7102,C9101
EAE62505701	Capacitor,Ceramic,Chip	3	C4750,C7103,C7104
EAE62506501	Capacitor,Ceramic,Chip	2	C4902,C7309
EAE62522101	Capacitor,Ceramic,Chip	2	C4203,C4900
EAE62685201	Capacitor,Ceramic,Chip	2	C18001,C18012
EAE62685301	Capacitor,Ceramic,Chip	4	C18000,C18018,C4755,C4756
EAE62685601	Capacitor,Ceramic,Chip	2	C18004,C18013
EAE62685701	Capacitor,Ceramic,Chip	2	C18002,C18015
EAE62722601	Capacitor,Ceramic,Chip	2	C6201,C6926
EAE62762301	Capacitor,Ceramic,Chip	9	C4903,C5220,C5222,C6112,C6203,C6204,C6420,C6421,C8800
EAE62767801	Capacitor,Ceramic,Chip	5	C2391,C2392,C4205,C4206,C4207
EAE62884301	Capacitor(High Frequency),Ceramic,Chip	1	C1126
EAE62945801	Capacitor(High Frequency),Ceramic,Chip	1	L14017
EAE62946701	Capacitor(High Frequency),Ceramic,Chip	4	C9516,C9518,C9519,C9520
EAE62947001	Capacitor(High Frequency),Ceramic,Chip	5	C21001,C21012,C21013,C21014,C21015
EAE62962301	Capacitor,Ceramic,Chip	10	C10000,C4202,C4204,C4208,C4209,C4210,C4211,C4212,C4213,C6109
EAE62963401	Capacitor(High Frequency),Ceramic,Chip	1	C9521
EAE62966901	Capacitor,Ceramic,Chip	2	C18006,C18007
EAE63004901	Capacitor,Ceramic,Chip	1	C6930
EAE63069301	Capacitor,Ceramic,Chip	2	C18005,C18008
EAE63143201	Capacitor,Ceramic,Chip	2	C1400,C1402
EAE63144601	Capacitor,Ceramic,Chip	2	C5209,C5212
EAE63162301	Capacitor,Ceramic,Chip	2	C18003,C18014
EAE63162401	Capacitor,Ceramic,Chip	2	C18010,C18011
EAE63462901	Capacitor,Low ESL	10	FL2300,FL2301,FL2302,FL2303,FL2304,FL2305,FL2306,FL2307,FL2308,FL2309
EAE63582701	Capacitor,Ceramic,Chip	2	C2303,C2304
EAF61650701	Varistor	1	ZD7400
EAG63291501	Connector,BtoB	1	CN7100
EAG63310801	Card Socket	1	S9500
EAG63430401	Connector,I/O	1	CN9200
EAG63530401	Connector,Terminal Block	1	CN9100
EAG63613001	C-Clip	2	ANT14000,CN11005
EAG63651901	C-Clip	4	ANT4700,ANT4701,CN5200,CN5202
EAG63652201	C-Clip	2	ANT5103,ANT5104
EAG63730501	C-Clip	2	ANT1102,ANT1105
EAG63772101	Connector,RF	3	SW1101,SW1103,SW1104
EAG63789801	Connector,BtoB	1	CN7400
EAG63910001	C-Clip	4	ANT1103,ANT1104,ANT1107,ANT1108
EAH61892901	Diode,TVS	3	D4603,D7300,D7400
EAH62033201	Diode,TVS	4	D11000,D11001,D11002,D7800

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EAH62055601	Diode,TVS	1	D9200
EAM62451101	Filter,LCR	2	FL9201,FL9202
EAM62570201	Filter,EMI/Power	1	FL7302
EAM62630501	Filter,EMI/Power	7	FL7100,FL7101,FL7102,FL7103,FL7105,FL7300,FL7301
EAM62633801	Filter,Bead	3	FB6101,FB6201,FB6202
EAM62790301	Filter,EMI/Power	1	FL4750
EAN62632401	IC,NFC	1	U5200
EAN62648101	IC,Audio Codec	1	U6100
EAN62661801	IC,Microprocessors	1	U8900
EAN62667201	IC,PMIC	1	U4200
EAN62732201	IC,LDO Voltage Regulator	1	U10000
EAN62735901	IC,DC,DC Converter	1	U4900
EAN62768901	IC,LDO Voltage Regulator	1	U7300
EAN62773901	IC,Over Voltage Protection	2	U4750,U4751
EAN62788501	IC,DC,DC Converter	1	U1400
EAN62826901	IC,Gyro Sensor	1	U8100
EAN62972701	IC,Acceleration Sensor	1	U8500
EAN62986601	IC,Geomagnetic Sensor	1	U8200
EAN62986901	IC,Level Translator	1	U9500
EAN62992902	IC,MCP,eMMC	1	U1
EAN63029101	IC,Speaker Amplifier	1	U6200
EAN63086101	IC,Charger	1	U18000
EAN63149301	IC,Comparator	1	U6420
EAP61866701	Inductor,Multilayer,Chip	4	L1103,L1118,L1133,L1160
EAP62107901	Inductor,Multilayer,Chip	1	C14016
EAP62167601	Inductor,Multilayer,Chip	2	L5200,L5201
EAP62227001	Inductor,Multilayer,Chip	1	C1189
EAP62246301	Inductor,Multilayer,Chip	1	L1169
EAP62286801	Inductor,Wire Wound,Chip	1	L1400
EAP62327801	Inductor,Wire Wound,Chip	4	L4200,L4201,L4203,L4900
EAP62526501	Inductor,Wire Wound,Chip	1	L6201
EAP62526601	Inductor,Wire Wound,Chip	1	L4202
EAP62545901	Inductor,Multilayer,Chip	4	L4204,L4205,L4206,L4207
EAT62233201	Module,Assembly	1	LD4500
EBC61856101	Resistor,Chip	1	R9505
EBC62036001	Resistor,Chip	2	R6119,R6128
EBC62235801	Resistor,Chip	1	R8900
EBC62596701	Resistor,Chip	1	C6103
EBC62975901	Resistor,Chip	1	R6117
EBK61691601	FET	1	Q5200
ECCH0000115	Capacitor,Ceramic,Chip	1	C9200
ECCH0000122	Capacitor,Ceramic,Chip	2	C5208,C5216
ECCH0000143	Capacitor,Ceramic,Chip	2	C5204,C5218
ECCH0000146	Capacitor,Ceramic,Chip	1	C18009
ECCH0000155	Capacitor,Ceramic,Chip	2	C8101,C8801
ECCH0000161	Capacitor,Ceramic,Chip	1	C9105
ECCH0000198	Capacitor,Ceramic,Chip	9	C3201,C3203,C7300,C7301,C7303,C7304,C7305,C7400,C7401
ECCH0002001	Capacitor,Ceramic,Chip	4	C6111,C8200,C8201,C9401
ECCH0004904	Capacitor,Ceramic,Chip	2	C6113,C6116
ECCH0006201	Capacitor,Ceramic,Chip	1	C6106
ECCH0007804	Capacitor,Ceramic,Chip	1	C4901
ECCH0009101	Capacitor,Ceramic,Chip	19	C1401,C3202,C3204,C4200,C5201,C6101,C6102,C6104,C6105,C6107,C6108,C6210,C7206,C7302,C8100,C8102,C8501,C8901,C9510
ECCH0009103	Capacitor,Ceramic,Chip	12	C1101,C1102,C1124,C1125,C1160,C1181,C1190,C13007,C13016,C14023,L1115,L1170

LG-F400L MAIN PCB

ECCH0009104	Capacitor,Ceramic,Chip	3	C1127,C6117,C6118
ECCH0009514	Capacitor,Ceramic,Chip	2	C9511,C9512
ECCH0017301	Capacitor,Ceramic,Chip	27	C2305,C2306,C2308,C2309,C2310,C2311,C2312,C2350,C2352,C2354,C2359,C2399,C2400,C3200,C4127,C4143,C4201,C5207,C5219,C7101,C7200,C8902,C9501,C9503,C9504,C9508,C9509
ECCH0017501	Capacitor,Ceramic,Chip	4	C2331,C2332,C2333,C2334
ECCH0017601	Capacitor,Ceramic,Chip	8	C10001,C4129,C6209,C7306,C7308,C7402,C7403,C8900
ECTH0002703	Capacitor,TA,Conformal	1	C9102
ECZH0000813	Capacitor,Ceramic,Chip	2	C5210,C5213
ECZH0000830	Capacitor,Ceramic,Chip	2	C5211,C5214
ECZH0001217	Capacitor,Ceramic,Chip	2	C5200,C5217
ECZH0004402	Capacitor,Ceramic,Chip	1	C6931
ECZH0025916	Capacitor,Ceramic,Chip	14	C11003,C6207,C6208,C6615,C6616,C6618,C6920,C6921,C6922,C6923,C6924,C6925,C9100,C9502
ECZH0025917	Capacitor,Ceramic,Chip	1	C9103
ECZH0025920	Capacitor,Ceramic,Chip	1	C5225
EDTY0009101	Diode,TVS	2	ZD7401,ZD7402
EDTY0010002	Diode,TVS	1	D9100
EDTY0012101	Diode,TVS	1	D9502
EDTY0012102	Diode,TVS	1	ZD9100
ELCH0001444	Inductor,Multilayer,Chip	8	L6126,L6127,L6500,L6501,L8100,L8101,L8102,L8103
ENBY0040701	Connector,BtoB	1	CN7300
ENBY0051001	Connector,BtoB	1	CN7200
ENZY0025801	Terminal Block	2	CN11002,CN11003
ERHY0003601	Resistor,Chip	2	R5203,R5206
ERHY0009301	Resistor,Chip	1	R11006
ERHY0009306	Resistor,Chip	2	R2116,R2117
ERHY0009501	Resistor,Chip	9	R3230,R5207,R6100,R6101,R6102,R6106,R6109,R6620,R6621
ERHY0009505	Resistor,Chip	5	R3208,R4103,R4759,R8801,R8802
ERHY0009506	Resistor,Chip	6	R4755,R4903,R6421,R6422,R7101,R8901
ERHY0009507	Resistor,Chip	2	R5200,R9501
ERHY0009511	Resistor,Chip	1	R4753
ERHY0009516	Resistor,Chip	7	R2106,R2107,R2112,R2113,R2118,R2119,R6116
ERHY0009526	Resistor,Chip	1	R9107
ERHY0009527	Resistor,Chip	2	R11003,R11004
ERHY0009538	Resistor,Chip	1	R9105
ERHY0009543	Resistor,Chip	4	R4608,R4609,R4610,R9106
ERHY0009547	Resistor,Chip	1	R4752
ERHY0009550	Resistor,Chip	1	R4754
ERHY0009586	Resistor,Chip	1	R6423
ERHY0024601	Resistor,Chip	2	R11005,R11007
ERHY0035601	Resistor,Chip	1	R9102
ERHZ0000268	Resistor,Chip	1	R18006
ERHZ0000270	Resistor,Chip	2	R4111,R4112
ERHZ0000331	Resistor,Chip	1	R9104
ERHZ0000401	Resistor,Chip	1	R3211
ERHZ0000405	Resistor,Chip	4	R4905,R4906,R7200,R9500
ERHZ0000537	Resistor,Chip	1	R6424
ERHZ0000701	Resistor,Chip	1	R18007
ERHZ0003901	Resistor,Chip	1	R6111
EUSY0200803	IC,Motor Driver	1	U8800
MBV62321701	Clip	5	SC130015,SC130019,SC130020,SC130025,SC13005
MBV62562001	Clip	2	ANT10002,ANT10003
SAFP0000401	Wire Pad,Short	1	R6615

LG-F400L MAIN PCB

SEVY0005402	Varistor	11	D6600,D6601,D6602,D6603,VA6930,VA6931,VA6932,VA7103,VA7104,VA7105,VA7106
SEVY0008901	Varistor	2	VA6928,VA6929
SEVY0010501	Varistor	3	VA7101,VA7102,VA9100
SFBH0008102	Filter,Bead	4	FB6100,FB6920,FB6921,FB6922
SFBH0008103	Filter,Bead	3	FB6610,FB6611,FB6612
SFEY0015901	Filter,EMI/Power	1	FL7400
SMZY0023501	Capacitor Assembly	1	BAT4100

• F400 Main Board BOM 사업자별 상이 List

Location Number	SKT (F400S)	KT(F400K)	U+ (F400L)	Desc.
C1126	ECCH0009103	ECCH0009103	ECZH0025905	ANT2 B7 Matching 소자 변경
C1168	DNI	DNI	EAE62946501	LTE B3 적용
L1171	DNI	DNI	EAP62226101	LTE B3 적용
C1119	EAE62286801	EAE62286801	DNI	LTE B3 적용
C1146	EAP62227201	EAP62227201	ECCH0009103	LTE B3 적용
L1127	EAP61747501	EAP61747501	EAP62227301	LTE B3 적용
C1123	ECCH0009103	ECCH0009103	DNI	LTE B3 적용
L1162	EAP62226401	EAP62226401	DNI	LTE B3 적용
U1102	EAN63028601	EAN63028601	DNI	LTE B3 적용
L14017	EAE62945801	DNI	EAE62945801	KT MRD Matching value 변경
C14016	EAP62107901	EAP62266401	EAP62107901	KT MRD Matching value 변경
R3211	DNI	DNI	ERHZ0000401	사업자 저항
R3212	ERHZ0000401	DNI	DNI	사업자 저항
R3213	DNI	ERHZ000040	DNI	사업자 저항
R3214	DNI	DNI	ERHZ0000401	사업자 저항
R3215	ERHZ0000401	DNI	DNI	사업자 저항
R3216	DNI	ERHZ0000401	DNI	사업자 저항
L1170	EAP62227001	EAP62227001	ECCH0009103	LTE B1 U+ 무선 성능 개선
C1126	ECZH0025905	ECZH0025905	EAE62884301	LTE B1 U+ 무선 성능 개선
C1189	ECCH0009103	ECCH0009103	EAP62227001	LTE B1 U+ 무선 성능 개선

LG-F400L/S/K KEY FPCB

Part No	Description	Qty	Designator
EBR78743101 4	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
EAE62726601	Capacitor,Ceramic,Chip	2	C5000,C5002
EAH61995401	Diode,TVS	5	D5001,D5002,D5003,D5004,D5005
EAN63405901	IC,Proximity	1	U5000
ECCH0009101	Capacitor,Ceramic,Chip	1	C5001
EBR78782001	PCB Assembly,Flexible,SMT Top	1	EBR070300
EAX65783301	PCB,Flexible	1	EAX010700

LG-F400L/S/K SOSA PCB

Part No	Description	Qty	Designator
EBR78759601 3	PCB Assembly,Sub,SMT Bottom	1	EBR072600
EAG63730501	C-Clip	2	ANT100,ANT101
ECZH0000813	Capacitor,Ceramic,Chip	2	C100,C103
MGJ64044601	Plate	1	SC100
EBR78784101	PCB Assembly,Sub,SMT Top	1	EBR072700
EAX65783901	PCB,RF	1	EAX010300

